

# Four hundred thirty-nine conjectures in mathematics: the statements

## 1 Notation

$p, q$  denote primes;  $\gamma$  is Euler’s constant,  $e^\gamma = 1.781072\dots$ ;  $\varphi$  is Euler’s totient;  $\phi = (1 + \sqrt{5})/2$ ;  $F_n$  the Fibonacci numbers;  $p\#$  the primorial. For irreducible  $f_1, \dots, f_k \in \mathbb{Z}[x]$  with positive leading coefficients, the Bateman–Horn singular series is

$$C(f_1, \dots, f_k) = \prod_p \frac{1 - \omega(p)/p}{(1 - 1/p)^k}, \quad \omega(p) = \#\{n \bmod p : p \mid f_1(n) \cdots f_k(n)\},$$

and the corresponding main term is

$$I(N) = \int_2^N \frac{dt}{\prod_i \log f_i(t)},$$

which absorbs the usual  $1/\prod_i \deg f_i$ . The Bateman–Horn conjecture [2] asserts  $\#\{n \leq N : \text{all } f_i(n) \text{ prime}\} \sim C \cdot I(N)$  whenever  $C \neq 0$ ; each Bateman–Horn conjecture below is an instance with its constant evaluated. For nonlinear systems the product converges only conditionally; throughout, it is taken in the canonical ordering of increasing primes, in which convergence follows from the equidistribution of  $\omega(p)$  about its Chebotarev mean. Our numerical truncation wobble is an error bar for this canonically ordered product. We write  $\text{Li}_k(x) = \int_2^x (\log t)^{-k} dt$ . The twin singular series is  $\mathfrak{S}(d) = 2C_2 \prod_{p \mid d, p > 2} \frac{p-1}{p-2}$  for even  $d$ , with  $2C_2 = 1.3203236\dots$ ; the Goldbach series  $\mathfrak{S}(n)$  is the same product over odd  $p \mid n$ .

Eight of the conjectures below are built over the Bateman–Horn move—choose an admissible system, compute its singular series, state the count—but none is a bare instance of it: 10 and 15 are family laws with derived constant distributions, 23 and 25 carry structural classifications, 24 is a searched chain, and 18, 19, 14 are contamination-calculus predictions (two fresh races and the negative control) whose mechanism is the contamination calculus of Conjecture 1(v). Each carries its classical calibration instance as an attributed remark. The heuristic is uniform, so we give it once. Model the primality of the values  $f_i(n)$  as independent events of probability  $1/\log f_i(n)$ , corrected by the factor  $(1 - \omega(p)/p)/(1 - 1/p)^k$  at each prime  $p$  to account for the joint local behaviour; multiplying the corrections and integrating gives  $C \cdot I(N)$ .

For locally integrable  $F$  we write  $\mathcal{M}_x(F) = \frac{1}{\log x} \int_2^x F(t) \frac{dt}{t}$  for the logarithmic mean; all race conjectures below state their drift and null clauses through  $\mathcal{M}_x$ , so that “no persistent leader” and “systematic surplus” are claims about a defined functional rather than about pointwise behaviour. Drift clauses are stated at the *drift scale*—for a pair race,  $\mathcal{M}_x(D(t) \log^2 t / \sqrt{t})$  converging to an explicit constant—never as  $\mathcal{M}_x((D - T)/\sqrt{\pi}) \rightarrow 0$ , which is vacuous for the drift coefficient (both  $T/\sqrt{\pi}$  and its multiples have logarithmic mean  $\rightarrow 0$ ). Convergence of  $\mathcal{M}_x$  at drift scale is itself part of each such conjecture: a pure random-walk fluctuation model would make these means

divergent in variance, so their convergence encodes the Rubinstein–Sarnak-type almost-periodicity that the zero-oscillation hypothesis names.

Admissibility ( $\omega(p) < p$  for all  $p$ ) is checked for every constant computed. The natural-looking pair  $\{n^2 + n + 1, n^2 + n + 3\}$  has  $\omega(3) = 3$  and is rejected; the  $k = 4$  branch of Conjecture 25 and the naive iterated chain behind Conjecture 24 are inadmissible for the same reason, and both exclusions are part of the statements.

## Part I

# Conjectures from a local–global random model

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## 2 Principal conjectures

**Lemma 1** (Orientation elimination and class assignment). *In  $\psi_2(x) = \sum_{n \leq x} \Lambda(n)\Lambda(n+2)$ , the total contribution of terms in which  $n$  or  $n+2$  is a proper prime power is, apart from  $O(x^{1/3} \log^2 x)$  from cubes and higher powers, carried entirely by the patterns  $(q^2 - 2, q^2)$  with  $q$  prime and  $q^2 - 2$  prime: the mirror pattern  $(q^2, q^2 + 2)$  is annihilated by  $3 \mid q^2 + 2$  for every prime  $q > 3$ . Moreover, for every prime  $q \neq 5$ ,  $q^2 - 2 \equiv 2$  or  $4 \pmod{5}$  (never 1), and for every odd  $q$ ,  $q^2 - 2 \equiv 7 \pmod{8}$ . (Two bounded exceptions: at  $q = 5$  the term  $(23, 25)$  lands outside the twin-start classes altogether, and at  $q = 3$  the mirror term  $(9, 11)$  survives, the annihilation beginning only at  $q > 3$ ; each is a single term, absorbed in the bounded error.)*

**Conjecture 1** (Twin races modulo 5 and 8, and the prime-power contamination calculus; apparently new). *Twin starts  $p > 5$  lie in classes  $p \equiv 1, 2, 4 \pmod{5}$ ; write  $\pi_t(x; m, a)$  for the number of twin starts at most  $x$  in the class  $a$  modulo  $m$ . Let  $D_1(x) = \pi_t(x; 5, 1) - \frac{1}{2}(\pi_t(x; 5, 2) + \pi_t(x; 5, 4))$  and*

$$T(x) = \frac{1}{2 \log^2 x} \sum_{\substack{q \leq \sqrt{x} \\ q^2 - 2 \text{ prime}}} \log q \log(q^2 - 2).$$

*Then: (i) [drift law, at drift-scale normalization] the clause asserts two sub-claims, labelled separately because they can fail independently. (i-a) [mechanism] class 1 carries the systematic surplus  $T$ : one has the decomposition  $D_1(t) = T(t) + R(t)$  in which the normalized remainder  $R(t) \log^2 t / \sqrt{t}$  possesses a limiting logarithmic distribution with mean zero, so that the entire drift-scale deterministic content of  $D_1$  is the contamination term and its coefficient is thereby identified. (i-b) [averaging] at the drift-scale normalization the remainder averages away:*

$$\mathcal{M}_x \left( \frac{R(t) \log^2 t}{\sqrt{t}} \right) \rightarrow 0, \quad \text{equivalently} \quad \mathcal{M}_x \left( \frac{D_1(t) \log^2 t}{\sqrt{t}} \right) \rightarrow c_T,$$

where

$$c_T := \lim_{x \rightarrow \infty} \frac{T(x) \log^2 x}{\sqrt{x}} = \frac{1}{2} C(x, x^2 - 2)$$

is the half of the Bateman–Horn constant of the pair  $(q, q^2 - 2)$  (the weighted census sum is  $\sim C\sqrt{x}$ , so the limit exists and is explicit). A normalization at the noise scale,  $\mathcal{M}_x((D_1 - T)/\sqrt{\pi_t}) \rightarrow 0$ , would be vacuous for the coefficient of (i-a): since  $T/\sqrt{\pi_t} \asymp 1/\log t$ , one has  $\mathcal{M}_x(T/\sqrt{\pi_t}) \asymp \log \log x / \log x \rightarrow 0$ , so the condition holds with  $T$  replaced by 0,  $2T$ , or any fixed multiple, and does not express “exactly the removed mass.” The statement therefore normalizes at the drift scale, where the claimed limit is a nonzero constant. Convergence at this normalization is a Rubinstein–Sarnak-type regularity assertion: under a pure random-walk model the logarithmic mean would not converge (its variance grows like  $\log x$ ), so (i-b) asserts almost-periodic structure for the twin race and is the precise content of the zero-oscillation hypothesis, while (i-a) is what fixes the limit at  $c_T$  rather than 0 or  $2c_T$ , and is the calculus. The convergence asserted in (i-b) selects the rigid branch of the dichotomy registered at Conjecture 9(iii); the two statements stand or fall together at drift scale; (ii) [zero deterministic drift in the symmetric differences] the symmetric classes carry no deterministic term at the same normalization:  $\mathcal{M}_x((\pi_t(\cdot; 5, 2) - \pi_t(\cdot; 5, 4)) \log^2 t / \sqrt{t}) \rightarrow 0$ , and likewise for each of the three pairwise differences among classes  $\{1, 3, 5\} \bmod 8$  in clause (iv)—now a statement distinct from (i), since at drift scale “zero” and “ $c_T$ ” are different limits; (iii) [sign density, conditional] conditional on the Gaussian fluctuation model for the normalized remainder in (i)–(ii), the logarithmic density of  $\{D_1 > 0\}$  exists and equals  $\frac{1}{2}$ , approached from above with a finite- $x$  excess of order  $\log \log x / \log x$ : the pointwise drift-to-noise ratio is  $T/\sqrt{\pi_t} \asymp 1/\log t$ , and its logarithmic average  $\mathcal{M}_x(1/\log t) \sim \log \log x / \log x$  carries an extra  $\log \log$  (the scale is not  $1/\log x$ ). In contrast to Chebyshev’s races, the twin race has no persistent leader; (iv) [mod-8 companion] since  $q^2 - 2 \equiv 7 \pmod{8}$  always (Lemma 1), the entire deterministic term falls on the single class  $7 \pmod{8}$ : with  $D_7(x) = \frac{1}{3} \sum_{a \in \{1, 3, 5\}} \pi_t(x; 8, a) - \pi_t(x; 8, 7)$ ,

$$\mathcal{M}_x\left(\frac{D_7(t) \log^2 t}{\sqrt{t}}\right) \rightarrow 2c_T$$

(twice the mod-5 constant, undiluted—at this normalization the factor 2 is a falsifiable prediction, the mechanism and averaging claims separating here exactly as in (i-a)–(i-b)), while classes 1, 3, 5 are mutually symmetric. The mod-5 and mod-8 races are two projections of one mechanism, so their joint behaviour is a family test in the sense of the shape-of-agreement criterion; (v) [contamination calculus for balanced races] for a pattern  $(n, n + d)$  and modulus  $m$ , call the race balanced if the prime-prime singular series is identical across the competing residue classes (as CRT gives for the classes compared in (i)–(iv)) and no mechanism other than prime powers contributes at the  $\sqrt{x}$  scale under the zero-oscillation hypothesis. For balanced races the deterministic drift vector is computed by one operator. Define the contamination operator

$$\mathcal{C}_{d,m}(a; x) = \sum_{\text{orientations } o \in \{(q^2-d, q^2), (q^2, q^2+d)\}} \sum_{\substack{q \leq \sqrt{x}, \text{ } o \text{ prime-compatible} \\ \text{start}(o) \equiv a \pmod{m}}} \Lambda(q^2) \Lambda(\text{prime member of } o),$$

where prime-compatible means the non-square member is prime and no fixed prime annihilates the orientation. The provable layer is the orientation census and class assignment (which  $o$  survive, and where their starts land—Lemma 1 and its analogues); the conjectural layer is the transfer: the unweighted class counts are depressed by  $\mathcal{C}_{d,m}(a; x) / \log^2 x$  relative to the clean classes, in the drift-scale limiting-log-mean sense of (i), with the mechanism and averaging claims separated as in (i-a)–(i-b)—for each class  $a$ , the deficit obeys  $\mathcal{M}_x(\text{deficit}_a(t) \log^2 t / \sqrt{t}) \rightarrow c(a) := \lim \mathcal{C}_{d,m}(a; x) / \sqrt{x}$ , an explicit Bateman–Horn constant per class, so every coefficient in the vector is identified—i.e. “exactly the removed mass,” which requires the weighted identity, partial summation uniform in

classes, higher-power bounds (exponents  $r \geq 3$  contribute  $O(x^{1/3+\varepsilon})$  against the operator's  $\asymp \sqrt{x}$  and are negligible—the one easy layer of the transfer), and the zero-oscillation hypothesis, and is the calculus's single analytic conjecture rather than a rule. The drift is  $\asymp \sqrt{x}/\log^2 x$  against a count fluctuation  $\asymp \sqrt{x}/\log x$ , i.e. smaller by  $1/\log x$  in standard-deviation units at every height—so the calculus predicts occupation biases under long logarithmic aggregation, never a fixed nonzero standardized mean, and every verification in this family is designed around that fact. (Races with unequal primary constants, or with interfering orientations beyond those enumerated, are outside the calculus until those terms are separated.) In particular, for the cousin pattern  $d = 4$  the calculus makes two predictions with no free choices: the orientation  $(q^2 - 4, q^2)$  is dead apart from a single term ( $q^2 - 4 = (q - 2)(q + 2)$  is composite for  $q > 3$ , while at  $q = 3$  the value 5 is prime: one bounded term, whose start 5 lies outside the mod-5 cousin classes and contributes one bounded term to class  $5 \bmod 8$ ), the orientation  $(q^2, q^2 + 4)$  survives (no prime annihilates it),  $q^2 + 4$  prime forces  $q \equiv \pm 2 \pmod{5}$  for  $q \neq 5$  (at  $q = 5$  the value 29 is prime, but the start  $25 \equiv 0 \pmod{5}$  lies outside the cousin start classes  $\{2, 3, 4\}$ ), so the contaminated cousin-start class is  $4 \pmod{5}$  (of  $\{2, 3, 4\}$ ) and—since  $q^2 \equiv 1 \pmod{8}$ —the class  $1 \pmod{8}$  (of  $\{1, 3, 5, 7\}$ ): both deficits converge, in the drift-scale limiting-log-mean sense of (i)–(ii), to  $c^c = \lim_{x \rightarrow \infty} T^c(x) \log^2 x / \sqrt{x}$ , the Bateman–Horn constant of the pair  $(q, q^2 + 4)$ , where  $T^c(x) = \frac{1}{\log^2 x} \sum_{q \leq \sqrt{x}, q \neq 5, q^2 + 4 \text{ prime}} \log q \log(q^2 + 4)$ .

**Conjecture 2** (The pair-level Montgomery–Soundararajan reduction; apparently new). *For the window field of Conjecture 8(ii), the diagonal variance of the twin-pair count reduces to a pinned singular-series average. With  $R(h) = C_4(0, 2, h, h + 2)/\mathfrak{S}(2)^2$  for  $h \geq 1$  (zero when the offset set is degenerate or inadmissible—all odd  $h$ , and  $h = 2$  since  $\{0, 2, 4\}$  dies at 3; consecutive twin pairs cannot overlap beyond  $(3, 5, 7)$ , so no overlap term exists) and*

$$G(H) = \sum_{1 \leq h \leq H} \left(1 - \frac{h}{H}\right) (R(h) - 1),$$

the derived identity (clause (i)) is

$$\frac{\text{Var}_t \pi_2(t; H)}{\mathbb{E}_t \pi_2(t; H)} = 1 + \frac{\mathfrak{S}(2) (2G(H) - 1)}{\log^2 x} + o\left(\frac{1}{\log^2 x}\right) :$$

with  $p = \mathfrak{S}(2)/\log^2 x$  the per-site intensity, the variance is  $Hp(1 - p) + 2p^2HG(H)$ , so  $\text{Var}/\mathbb{E} = 1 - p + 2pG(H)$ —the Bernoulli diagonal term  $-p = -\mathfrak{S}(2)/\log^2 x$  is of exactly the order of the claimed error and cannot be dropped (it is the same diagonal that, in the single-prime case, merges into Montgomery–Soundararajan's constant  $\gamma + \log 2\pi - 1$ ). With that term retained, the entire deficit left open at Conjecture 8(ii) is the single computable function  $G$  plus the explicit Bernoulli correction—the pair analogue of Montgomery–Soundararajan's reduction of the prime-count variance to  $\sum_{h \leq H} \mathfrak{S}(h)(H - h)$  [101]. Clause (i) is a conditional proposition, not an independent conjecture: given quantitative Hardy–Littlewood for 4-tuples, the reduction follows. The conjectural clause is (ii):  $-G(H)/\log H$  diverges—the pinned average grows strictly faster than the single-prime secondary term  $\frac{1}{2}(\log H + \gamma + \log 2\pi - 1)$ , so pair counts in windows are more sub-Poisson than prime counts—with the caveat that data to  $H = 3000$  cannot distinguish divergence from a pure log with a large constant (nor  $c \log H \log \log H$  from  $(\log H)^\alpha$ ): what the computation establishes is only that the local slope exceeds the single-prime value sevenfold across the computable range; the divergence and its exact form are registered, not evidenced beyond that.

**Conjecture 3** (Least primes in progressions; part (i) previously stated [102, 103, 112, 111], part (ii) apparently new). *Let  $p(a, q)$  be the least prime  $\equiv a \pmod{q}$  and  $U(a, q) = \text{Li}(p(a, q))/\varphi(q)$ . Then: (i)  $U \Rightarrow \text{Exp}(1)$  marginally as  $q \rightarrow \infty$ ; and, under the additional hypothesis that the class processes are asymptotically jointly independent in the extreme-value sense (which marginal convergence alone does not supply),  $\max_a U - H_{\varphi(q)}$  converges to a Gumbel law ( $H_n = \sum_{k \leq n} 1/k$ ), the order-statistics form of Wagstaff's  $\max_a p(a, q) \sim \varphi(q) \log^2 q$ ; (ii) [canonical deficit, dyadic-average form] define, with no reference to any control model,  $\Theta(q) = (1 - \mathbb{E}_a[U(a, q)]) \log q$ . The conjecture is stated for dyadic averages over prime moduli—pointwise convergence along every prime  $q$  may be too strong, since arithmetic fluctuations in  $q$  could persist:*

$$\frac{1}{\#\{q \in [Q, 2Q] \text{ prime}\}} \sum_{\substack{Q \leq q \leq 2Q \\ q \text{ prime}}} \Theta(q) \rightarrow \Theta > 0 \quad (Q \rightarrow \infty).$$

*(The formal claim is model-free: existence and strict positivity of the dyadic-average limit. The stronger reading—that  $\Theta$  exceeds the discreteness level any parity-aware control produces—is the diagnostic layer below, not part of the display, since a control constant has no place in a model-free statement.) The limiting distribution of  $\Theta(q)$  over the dyadic block is left open, and in particular we do not assert  $\text{sd}(\Theta(q)) \rightarrow 0$ : fluctuations driven by the arithmetic of  $q$  and of  $q - 1$  (through the singular-series pair terms  $\mathfrak{S}(kq)$  and the class structure  $\text{mod } q$ ) may well persist at bounded size, so  $\Theta(q)$  need not concentrate around its dyadic mean.*

**Conjecture 4** (The Goldbach lane race; apparently new). *For  $n \equiv 2 \pmod{4}$  let  $R_1(n)$ ,  $R_3(n)$  be the ordered Goldbach representation counts in the lanes  $p \equiv q \equiv 1$  and  $p \equiv q \equiv 3 \pmod{4}$  (the lane convention of the Martin benchmark recorded at Conjecture 5), and  $D(n) = R_3(n) - R_1(n)$ . Prime-square terms  $n = q^2 + p$  ( $q$  odd prime) land exclusively in the (1, 1) lane, since  $q^2 \equiv 1 \pmod{4}$  and then  $n - q^2 \equiv 1 \pmod{4}$  automatically. Consequently the (3, 3) lane leads on average. Define*

$$D_{\text{sys}}(n) = \frac{2}{\bar{\ell}(n)} \sum_{\substack{q \text{ odd prime, } q \leq \sqrt{n} \\ n - q^2 \text{ prime}}} \log q \log(n - q^2), \quad \bar{\ell}(n) = \frac{n - 6}{\int_3^{n-3} \frac{dt}{\log t \log(n - t)}},$$

*(the analytic mean of  $\log p \log(n - p)$  under the lane profile; the empirical lane mean is its estimator). Write  $\mathbb{E}_x$  for the average over  $n$  drawn from  $\{n \equiv 2 \pmod{4}, n \leq x\}$  with weight proportional to  $1/n$  (the logarithmic sampling measure, now explicit). Then: (i) [drift law]  $\mathbb{E}_x[D - D_{\text{sys}}] = o(\mathbb{E}_x[D_{\text{sys}}])$  as  $x \rightarrow \infty$ : the (3, 3) lane leads on average by exactly the square contamination; (ii) [weighted mean and internal null lane]  $\mathbb{E}_x[D_{\text{sys}}]$  is given by the Hardy–Littlewood (Conjecture H) local model for  $n - q^2$ , whose local factors vary strongly with  $n$ —most drastically at  $p = 3$ : for  $n \equiv 1 \pmod{3}$ ,  $3 \mid n - q^2$  for every prime  $q > 3$ , so the contamination collapses to a single boundary family, the  $q = 3$  term (when  $n - 9$  is prime): the complementary possibility  $n - q^2 = 3$  is empty on this ensemble, since  $n \equiv 2 \pmod{4}$  forces  $n - 3 \equiv 3 \pmod{4}$ , never a square. The family is negligible in logarithmic mean, so the race carries no drift at the systematic scale on the subprogression  $n \equiv 10 \pmod{12}$ : with the positive comparison scale  $A_G(x) := \mathbb{E}_x^{(\neq 1(3))}[D_{\text{sys}}]$ , the logarithmic-ensemble mean of the systematic term over the contaminated lane  $n \not\equiv 1 \pmod{3}$  (a comparison against this subprogression's own systematic term, which vanishes, would be empty), the claim is  $\mathbb{E}_x[D] = o(A_G(x))$  on  $n \equiv 10 \pmod{12}$ , while the drift concentrates on  $n \not\equiv 1 \pmod{3}$ : an internal null lane, predicted by the same mechanism*



that predicts the lead; (iii) [sign density] the per- $n$  drift-to-noise ratio  $\kappa(n) = D_{\text{sys}}(n)/\sqrt{\mathfrak{S}(n)J(n)}$  (where  $J(n) = \int_3^{n-3} dt/(\log t \log(n-t))$ , so that  $\mathfrak{S}(n)J(n)$  is the total ordered representation count) satisfies  $\kappa(n) \asymp c(n)/\log n \rightarrow 0$ ; under the Gaussian fluctuation model the logarithmic density of  $\{D > 0\}$  is therefore  $\frac{1}{2}$  in the limit, with the finite- $x$  sign fraction predicted to be  $\mathbb{E}_x[\Phi(\kappa)]$  ( $\Phi$  the standard normal distribution function)—decaying, but far from  $\frac{1}{2}$  at any accessible height.

**Conjecture 5** (The least Goldbach summand: exponential law and ordering deficit; the time-changed law and deficit constant apparently unstated). *Let  $s(n)$  be the least prime  $p \nmid n$  with  $n-p$  prime (the exclusion  $p \nmid n$  removes a trivial obstruction:  $p \mid n$  forces  $p \mid n-p$ , so  $n-p$  is composite except in the diagonal case  $n=2p$ , which would otherwise distort the hazard at one atypical point), and time-change by the expected-arrivals clock  $U(n) = \mathfrak{S}(n) \sum_{p \leq s(n), p \nmid n} 1/\log(n-p)$ . Here  $s(n) = +\infty$  when no admissible prime exists, which among even  $n$  happens only at  $n=4$  and  $n=6$  and conjecturally at no larger even  $n$ , so the law below runs over the even  $n$  with  $s(n)$  finite. The ensemble is fixed first, since  $U(n)$  is deterministic at each  $n$  and an expectation therefore needs a declared measure: for a scale  $X$ ,*

$$\mathbb{E}_X U = \frac{\sum_{X < n \leq 2X, 2|n} U(n)/n}{\sum_{X < n \leq 2X, 2|n} 1/n}, \quad \Theta_G(X) = (1 - \mathbb{E}_X U) \log X.$$

Then: (i) [limit law] under  $\mathbb{E}_X$ -sampling,  $U \Rightarrow \text{Exp}(1)$  as  $X \rightarrow \infty$ ; (ii) [canonical ordering deficit]  $\Theta_G(X) \rightarrow \Theta_G > 0$ —the Goldbach sibling of the least-prime deficit of Conjecture 3(ii), produced by the same occupancy mechanism: the ordered primes fill the available representations faster than an exchangeable model, so the first arrival comes early and  $\mathbb{E}_X[U] < 1$ .

**Conjecture 6** ( $n^2 + 2^n$ ; sequence is OEIS A064539). *Every prime of the form  $n^2 + 2^n$  with  $n > 1$  has  $n \equiv 3 \pmod{6}$  (elementary), and: (i) infinitely many such primes exist, with counting function  $\asymp \log N$ ; (ii) [candidate law, entanglement-aware form] for a finite set  $S$  of primes  $p \geq 5$  (the primes 2 and 3 are exact in the lane reduction,  $\delta_2 = \delta_3 = 0$ , and  $\text{ord}_2 2$  is undefined) let  $D_S$  be the exact joint density, within the lane  $n \equiv 3 \pmod{6}$ , of  $n$  with  $p \nmid n^2 + 2^n$  for all  $p \in S$ , computed over the full joint period  $\text{lcm}_{p \in S} \text{lcm}(6, p \cdot \text{ord}_p 2)$ , and set  $\kappa_S = 3 D_S / \prod_{p \in S} (1 - 1/p)$ : the factor  $3 = 2 \cdot \frac{3}{2}$  is the exact lane bonus at the primes 2 and 3 (within the lane  $\delta_2 = \delta_3 = 0$ , so each contributes its full  $(1 - 1/p)^{-1}$ ), included in the definition so that  $\kappa_S$  is the complete local constant of the count in (ii-c). The conjecture is stated along the canonical exhaustion  $S_z = \{p \text{ prime} : 5 \leq p \leq z\}$ , and its three layers are logically separate and are asserted separately: (ii-a) the sequence  $\kappa_{S_z}$  converges as  $z \rightarrow \infty$  (a general net over arbitrary finite  $S$  carries an order-of-exhaustion ambiguity we do not need), with limit  $\kappa$ ; (ii-b) whether the limit factors—i.e. whether the exact finite-level factorization observed below persists, so that entanglement is asymptotically absent—is an open question, expressly not counted among this paper’s conjectural assertions, on which (ii-a) takes no position; and (ii-c) the count is  $\sim \sum_{n \leq N, n \equiv 3 \pmod{6}} \kappa / \log(n^2 + 2^n)$ —a genuine further step, since a convergent local density does not automatically govern the global count. The working value  $\kappa^* = 4.2734\dots$  is a hybrid, not a computation of any single  $\kappa_{S_z}$ : the lane factor 3 times the CRT-exact core  $S_{19}$  times independent per-prime factors for  $19 < p \leq 300$ —so  $\kappa^* = \kappa_{S_{300}}$  exactly if and only if the observed exact factorization persists to 300.*

**Conjecture 7** (Fermat quotients: the multibase subtorus law and the Wieferich ledger; single-base heuristic previously stated [104], the vertical multibase law apparently new). *With  $q_p(a) = (a^{p-1} - 1)/p \pmod{p}$  (defined for primes  $p \nmid a$ ; all clauses below range over such  $p$  only) and*

$q_p = q_p(2)$ , clauses (0)–(v), stated multibase group first: (0) [structure, classical]  $q_p(ab) \equiv q_p(a) + q_p(b) \pmod{p}$  (the Eisenstein–Lerch homomorphism; verified exactly on every tested prime), so for multiplicatively dependent bases the vector  $(q_p(a_1), \dots, q_p(a_r))/p$  is confined to the rational subtorus cut out by the relations; (iv) [multibase equidistribution, the claimed law] for multiplicatively independent bases the vector equidistributes on the full torus as  $p$  varies—the vertical joint law, complementing the fixed- $p$ , varying-base statistics of Ostafe–Shparlinski and Cobeli–Zaharescu—with correlations vanishing and the same LIL calibration as (i) below; (v) [simultaneous Wieferich] the expected count of  $p \leq x$  with  $q_p(2) = q_p(3) = 0$  has convergent sum  $\sum 1/p^2$ : at most finitely many simultaneous Wieferich primes exist (the single-base folklore of this accounting is Conrad’s), and the empirical list is empty to  $10^7$ . Three further clauses on the base-2 quotient alone: (i) [global equidistribution]  $q_p/p$  equidistributes on  $[0, 1)$  with discrepancy at exactly the calibrated random-model scale: in its sharp form,

$$\limsup_{x \rightarrow \infty} \frac{\sqrt{\pi(x)} D_{\text{KS}}(x)}{\sqrt{\log \log \pi(x)}} = \frac{1}{\sqrt{2}},$$

the Chung–Smirnov law-of-the-iterated-logarithm constant for an i.i.d. uniform sequence—demanding both no more (a bound  $O(\pi(x)^{-1/2})$  would be stronger than the random model supports, the same over-demand that killed the Mertens conjecture) and no less than the model earns: the upper bound alone would be consistent with hidden rigidity, and the matching lower bound asserts the quotients fluctuate like genuine noise; (ii) [shrinking targets] for every fixed  $K \geq 1$ , and uniformly for  $K = K(x) \leq \log x$ ,  $\#\{p \leq x : q_p < K\} \sim \sum_{p \leq x} \min(1, K/p)$ —the fixed- $K$  case down to  $K = 1$  is what governs the zero event and does NOT follow from (i), since  $\{q_p = 0\}$  is a target of shrinking measure  $1/p$ ; (iii) [Wieferich count, fluctuation-calibrated, with the probability space declared]  $W(x) = \#\{\text{Wieferich } p \leq x\}$  is a deterministic function, so its fluctuation clauses are stated in the two legitimate forms, an unqualified CLT for a fixed sequence having no meaning: the deterministic envelope conjecture

$$\limsup_{x \rightarrow \infty} \frac{W(x) - \log \log x}{\sqrt{2 \log \log x \log \log \log x}} = 1$$

(the iterated-logarithm scale at variance  $V(x) \sim \log \log x$ ; a bound  $O_\varepsilon(V^{1/2+\varepsilon})$  would be weaker than the model’s exact prediction), and the almost-sure central limit law, at the correct logarithmic weighting: with  $u = \log \log x$  the variance-time of the count, sample  $u$  with density  $du/u$  on  $[e, L]$ ,  $L = \log \log X$  (equivalently:  $\log u = \log \log \log x$  uniform); then the sampled distribution of  $(W(x) - u)/\sqrt{u}$  converges to  $N(0, 1)$  as  $X \rightarrow \infty$ . The weighting is what makes the clause true, and the two natural alternatives both fail. A block-sampled version on  $[X, X^A]$  fails because only  $\log A = O(1)$  new events enter the block. A uniform-in- $u$  version on  $[1, L]$  fails too, by Brownian scaling: with  $W(e^u) - u \approx B(u)$  and  $u = Ls$ , the self-similarity  $B(Ls)/\sqrt{Ls} \stackrel{d}{=} \tilde{B}(s)/\sqrt{s}$  means the uniform- $u$  empirical distribution converges not to  $\Phi$  but to the random occupation functional  $\int_0^1 \mathbf{1}\{\tilde{B}(s)/\sqrt{s} \leq z\} ds$ —values at proportional times stay correlated, so uniform sampling never averages over enough independent scales. The  $du/u$  weighting is the classical almost-sure-CLT weighting (the continuous analogue of the  $1/k$  weights), for which the model does predict almost-sure convergence of the sampled law to  $\Phi$ . Both clauses are conjectured on the Crandall–Dilcher–Pomerance side of the model choice that clause (iv)’s discussion records.

**Conjecture 8** (Uniform quantitative de Polignac; the core is previously stated). (i)  $\pi_d(x) = \#\{p \leq x - d : p, p + d \text{ prime}\} = \mathfrak{S}(d) \text{Li}_2(x) (1 + o(1))$  uniformly for even  $d \leq (\log x)^A$ . (ii)

[residual field, with derived covariance kernel] The same primes participate in many pair counts, and the covariance of two of them is carried by triple configurations sharing one prime. For even  $d \neq d'$  define the overlap constant

$$K(d, d') = C_3(0, d, d') + C_3(0, d', d + d') + C_3(0, d, d + d') + C_3(0, |d - d'|, \max(d, d')),$$

where  $C_3$  denotes the Hardy–Littlewood triple constant of the offset set (degenerate or inadmissible triples contributing 0). The field is randomized canonically—covariances of the deterministic cumulative counts are otherwise undefined: for  $t$  uniform on  $[x, 2x]$  and a window length  $H$ , let  $\pi_d(t; H) = \#\{t < n \leq t + H : n, n + d \text{ prime}\}$  and let  $Z_d(t; H)$  be its studentization. The mean count per window is  $\asymp H/\log^2 x$ , and the regime is set by that ratio (a phase condition: a bare  $H \rightarrow \infty$  does not fix the regime):

$$\text{Cov}_t(\pi_d, \pi_{d'}) \sim \frac{K(d, d') H}{\log^3 x}, \quad \rho(d, d') \sim \frac{K(d, d')}{\sqrt{\mathfrak{S}(d)\mathfrak{S}(d')} \log x},$$

valid for  $\max(d, d') = o(H)$ , which is the standing restriction of this clause and holds throughout the window class fixed below. In general each overlap orientation  $o$ —the in-window displacements  $h_o \in \{0, d, -d', d - d'\}$  and their reflections—enters with multiplicity  $(H - |h_o|)_+$  rather than  $H$ , so that the shared-prime term reads

$$\frac{1}{\log^3 x} \sum_o (H - |h_o|)_+ K_3^{(o)}(d, d'), \quad \sum_o K_3^{(o)}(d, d') = K(d, d'),$$

and an orientation with  $|h_o| > H$  is absent from the window altogether; under  $\max(d, d') = o(H)$  one has  $(H - |h_o|)_+ = H(1 + o(1))$  for every orientation and the displayed form is recovered, so the uniform-in- $d$  statement carries that restriction throughout. Here the shared-prime (triple) term is the first covariance contribution but not the whole of it: the full expansion is

$$\text{Cov}_t(\pi_d, \pi_{d'}) = \frac{K(d, d') H}{\log^3 x} + \frac{1}{\log^4 x} \sum_{|h| \leq H} (H - |h|) K_4(d, d'; h) + \text{error},$$

with  $K_4$  the centred singular series of the off-index four-point sets  $\{0, d, h, h + d'\}$ —the cross analogue of the pinned diagonal average of Conjecture 2—where degenerate offset relations are handled inside the kernels: whenever the four-point set collapses (repeated entries, as at  $h = 0$ ,  $h = d$ ,  $h = -d'$ , or  $d' - d = \pm h$ ) the configuration is a triple or pair and its contribution belongs to the  $K$ -term (or to the mean), so  $K_4$  is defined as the centred singular series of genuinely four-element sets, with special pairs such as  $d' = 2d$  acquiring their extra overlap orientations in  $K(d, d')$ 's census rather than double-counted across kernels. The triple term dominates only where the pinned sum satisfies  $\sum(1 - |h|/H)K_4 = o(\log x)$ ; the dominance range is stated as an explicit function class: for  $H = \lambda(\log x)^\alpha$ , any fixed  $\lambda > 0$  and any  $\alpha \geq 1$  with  $\alpha > A$ —the inequality on the exponents that makes  $\max(d, d') = o(H)$  hold uniformly for all  $d, d' \leq (\log x)^A$ —the per-shift centred average grows only polylogarithmically in  $H$  (the Conjecture-2 scale), so  $\sum(1 - |h|/H)K_4/H = O((\log \log x)^{2+\varepsilon}) = o(\log x)$  and the triple term dominates throughout the polylogarithmic window class; the conjecture asserts the two-term form on exactly that class and takes no position outside it. When  $H/\log^2 x \rightarrow \lambda$  the window counts converge jointly to Poisson laws with these vanishing covariances, and when  $H/\log^2 x \rightarrow \infty$  every fixed finite collection  $\{Z_{d_1}, \dots, Z_{d_r}\}$  is asymptotically jointly normal with independent limiting coordinates: both kernels are finite- $x$  corrections to independence (test below), not nondegenerate



limiting kernels. Both distributional conclusions are asserted under an explicit hypothesis beyond the displayed covariances, namely uniform Hardy–Littlewood tuple estimates on the window class fixed above, strong enough to control all higher joint cumulants of the window counts: every joint cumulant of order at least three vanishes in the respective normalizations. The diagonal correction to  $\text{Var}_t(\pi_d)$  is Conjecture 2’s pinned average, stated there.

**Conjecture 9** (The sub-diffusivity of balanced prime races; apparently new). *Model a balanced race (Conjecture 1(ii)’s symmetric class differences) as the walk of its class-assignment steps, the process defined at event index: enumerate the pattern occurrences  $n_1 < n_2 < \dots$  in the two competing classes, set  $\xi_i = \pm 1$  by class membership of  $n_i$ , and let  $\rho_k(x)$  denote the lag- $k$  sample autocorrelation of  $(\xi_i)$  over the events with  $n_i \leq x$ . Then the walk is measurably sub-diffusive: (i) [decaying-repulsion law] the step autocorrelations are negative at small lags—the race analogue of the Lemke Oliver–Soundararajan consecutive-pattern repulsion [106], which supplies the mechanism—and, since that repulsion is scale-dependent rather than fixed, the conjectured form is a decay law*

$$\rho_k(x) = -\frac{c_k \log \log x + d_k}{\log x} (1 + o(1)),$$

with constants  $c_k, d_k$  in principle derivable from the LOS correlation series (underived here; and the measured values  $\rho_1 \approx -0.037$ ,  $\rho_2 \approx -0.009$  at  $10^9$ —universal across twin and cousin races mod 5 and 8 to within 0.004, each  $\sim 50$  standard errors from zero—are observations at one height, consistent with  $c_1 \approx 0.25$  there, not asymptotic constants); (ii) [long memory, measured] short lags do not exhaust the deficit: the running maxima  $M(x) = \max_{t \leq x} |D(t)|/\sqrt{\text{count}}$  of the four races land at quantiles 0.02–0.17 of the simulated independent-step null (median 2.34 at this span), and still only 0.03–0.23 after correcting for  $\rho_{1,2}$ : the negative correlation persists across lags, so the cumulative diffusivity  $\sigma^2(x) = 1 + 2 \sum_k \rho_k(x)$  sits well below 1 at this height. The model layer is declared:  $(\xi_i)$  is a deterministic, nonstationary family and the  $\rho_k(x)$  are empirical statistics of it, so the Green–Kubo identity  $\sigma^2 = 1 + 2 \sum_k \rho_k$  is invoked under an explicit local-stationarity model hypothesis—the step process restricted to a window  $[x, 2x]$  is modelled as a stationary sequence whose autocorrelations are the measured  $\rho_k(x)$ —and the variance statement can equivalently be taken as a direct definition of  $\sigma^2(x)$  through block averages of  $(S_{n+m} - S_n)^2/m$  over the events up to  $x$ , a form in which no stationarity is assumed at all. The Green–Kubo sum requires more than the fixed-lag law of (i): the per-lag decay controls each  $\rho_k(x)$  but not the sum unless the decay is uniform in  $k$  with summable tail, since the number of relevant lags may grow with  $x$ . We therefore state the needed hypothesis as part of the clause:  $\rho_k(x)$  obeys the law of (i) uniformly for  $k \leq K(x)$  with  $\sum_{k > K(x)} |\rho_k(x)| = o(\log \log x / \log x)$  for some  $K(x) \rightarrow \infty$ , and—since the tail bound alone leaves the growing head sum uncontrolled—the repulsion constants are summable,  $\sum_k c_k < \infty$  and  $\sum_k |d_k| < \infty$ , so that  $\sigma^2(x) = 1 - 2(\sum_k c_k \log \log x + \sum_k d_k) / \log x (1 + o(1))$  is a genuine consequence rather than an unjustified interchange of limits. The measured correlation length at  $10^9$  (a few tens of lags, with  $|\rho_k|$  decreasing geometrically in the observable range) is consistent with both hypotheses. Under them,  $\sigma^2(x)$  returns to 1 asymptotically, making sub-diffusivity a finite-height phenomenon of  $\log \log x / \log x$  size, like every other second-order law in this paper; (iii) [registered dichotomy, open] asymptotically the two possibilities are mutually exclusive, and which of them holds is the open question. Either the fixed-lag expansion of (i) extends uniformly in  $k$  with summable constants—in which case the hypotheses of (ii) hold, (ii) applies, and the race is asymptotically diffusive with  $\sigma^2(x) \rightarrow 1$  (running maxima  $\sim \sqrt{2 \log \log x}$ , Darling–Erdős class)—or that expansion fails at long lags, and only in that case can the race inherit the almost-periodic rigidity that Rubinstein–Sarnak-type structure [6] forces on classical

*races ( $\text{maxima} \asymp (\log \log \log x)^A$ , the Montgomery–Ng class). Rigidity therefore requires a failure of the hypotheses of (ii) at long lags, and cannot coexist with them. No explicit formula is known for twin patterns, the two branches differ only beyond any computable height, and we register the dichotomy without choosing in this conjecture. What is conjectured here is therefore the fixed-lag law (i) outright, and the variance law of (ii) conditionally on its stated uniformity and summability hypotheses, which belong to the finite-height model layer and are not asserted asymptotically: asserting them would silently select the diffusive branch of (iii). The pattern-specific selection is made elsewhere—Conjecture 1 asserts the rigid branch for its races, through (i-b) for the contaminated combinations and through the drift-scale nulls of its clause (ii) for the symmetric differences—and consistency between the two conjectures then requires exactly that the hypotheses of (ii) fail at long lags for those races.*

### 3 Family laws and second-order refinements

**Conjecture 10** (Uniform quadratic de Polignac; family law apparently unstated, the  $d = 2$  instance previously stated, cf. OEIS A080149, [114]). *For even  $d \geq 2$  let  $\pi_d^q(N) = \#\{n \leq N : n^2 + 1, n^2 + 1 + d \text{ both prime}\}$  and  $C(d) = C(x^2 + 1, x^2 + 1 + d)$ , with  $I_d(N)$  the integral  $I(N)$  of this pair. Then: (i) [uniform family law] for every fixed  $B > 0$ ,  $\pi_d^q(N) = C(d) I_d(N)(1 + o(1))$  uniformly over even  $d \leq (\log N)^B$ ; (ii) [registered rate, strictly stronger] there is  $\eta_B > 0$  with*

$$\sup_{\substack{2 \leq d \leq (\log N)^B \\ 2|d}} \left| \frac{\pi_d^q(N)}{C(d) I_d(N)} - 1 \right| \ll_B (\log N)^{-\eta_B}$$

*—an error clause that no version of Bateman–Horn supplies, registered separately from (i) since its analytic content is independent; (iii) [family statistic: the law of the constants] each local factor of  $C(d)$  depends only on  $d \bmod p$ , and the factors at any finite set of primes are exactly independent as  $d$  varies (CRT), with the passage to the infinite product controlled by the convergent tail-variance sum—which supplies the uniform integrability (via  $L^2$ -boundedness of the normalized partial products) that the moment identities require, since almost-sure convergence alone would not carry means through the limit; so the moments of  $C(d)$  over even  $d \leq D$  converge, as  $D \rightarrow \infty$ , to derived Euler products; in particular*

$$\frac{1}{\#\{2 \mid d \leq D\}} \sum_{2 \mid d \leq D} C(d) \rightarrow \bar{C} = 2.7456\dots, \quad \text{sd}(C(d)) \rightarrow 1.6840\dots,$$

*and the empirical distribution of  $C(d)$  converges to the law of the random product  $\prod_p f_p(U_p)$  with independent uniform residues  $U_p$ . In particular ( $d = 2$ , the classical instance):  $\#\{n \leq N : n^2 + 1, n^2 + 3 \text{ both prime}\} \sim C(2) I(N)$ ,  $C(2) = 2.954014\dots$*

**Conjecture 11** (First occurrences of prime gaps, liminf form; the family is classical, the quantified form apparently unstated). *Let  $\mathcal{G}$  be the set of even  $g$  that occur as gaps between consecutive primes (all even  $g$ , under Polignac’s conjecture), and for  $g \in \mathcal{G}$  let  $p(g)$  be the prime starting the first such gap. Then: (i)  $\log p(g)/\sqrt{g}$  is bounded between positive constants on  $\mathcal{G}$ ; (ii) [liminf law, stated as the dual of the maximal-gap limsup, conditional on the realization hypothesis below]*

$$\liminf_{\substack{g \rightarrow \infty \\ g \in \mathcal{G}}} \frac{\log p(g)}{\sqrt{g}} = \sqrt{e^\gamma/2} = 0.943682\dots$$

The inversion of Granville's  $\limsup G(X)/\log^2 X = 2e^{-\gamma}$  into first-occurrence coordinates is not automatic: the  $\limsup$  controls how large maximal gaps get, not which gap sizes are realized as first occurrences near the envelope. Clause (ii) is therefore conditional on the realization hypothesis: infinitely many  $g \in \mathcal{G}$  first occur inside a maximal-gap event at the Cramér–Granville envelope scale, i.e. with  $g = (2e^{-\gamma} + o(1)) \log^2 p(g)$ . Under that hypothesis the displayed value is forced by algebra; without it only the lower bound  $\liminf \log p(g)/\sqrt{g} \geq \sqrt{e^\gamma/2}$  follows from the envelope. The value is registered as that dual (against the slope-1 laws), not claimed as an independent phenomenon; (iii) [second-order, singular-series dependence] sampling  $g$  uniformly from the realized gaps  $\mathcal{G} \cap [G, 2G]$  and letting  $G \rightarrow \infty$ ,

$$\log p(g) = \sqrt{g} + \frac{1}{2} \log g - \frac{1}{2} \log \mathfrak{S}^*(g) + E_g, \quad \mathfrak{S}^*(g) = \prod_{\substack{p|g \\ p>2}} \frac{p-1}{p-2},$$

where the error  $E_g$  converges in distribution: for the sampled  $g$ ,

$$\Pr[E_g \leq t] \longrightarrow 1 - \exp(-e^{2(t-\mu)}) \quad (G \rightarrow \infty)$$

for some centring constant  $\mu$ —the min-type (reversed) Gumbel law at scale  $\frac{1}{2}$ , i.e. the law of  $\frac{1}{2} \log W$  for  $W$  a unit exponential. The scale is forced by the hazard computation: the cumulative intensity of gap- $g$  events is  $\Lambda_g(y) \asymp (\mathfrak{S}(g)/y^2) \exp(y - g/y)$  in  $y = \log x$  (the factor  $e^{-g/y}$  is the no-intervening-prime probability), so  $\frac{d}{dy} \log \Lambda_g = 1 + g/y^2 - 2/y = 2 + o(1)$  at the centring point  $y_0 \approx \sqrt{g}$ , and the first-passage identity  $\Lambda_g(y) = W$  gives  $E_g = y - y_0 = \frac{1}{2} \log W + o(1)$ . The scale is  $\frac{1}{2}$  and not 1 precisely because the slope of the log-intensity at  $y_0$  is 2 (both the  $y$ -term and the  $g/y$ -term contribute), and the same slope-2 expansion is what produces the  $\frac{1}{2}$  coefficients of the centring. (Gumbel-type modelling of gap extremes and first occurrences is Kourbatov–Wolf territory [97, 107]; the claim of this clause is only the  $\mathfrak{S}^*$ -centring and the scale- $\frac{1}{2}$  first-passage form, not the extreme-value framework. Sampling uniformly over the realized gaps in  $[G, 2G]$  conditions on realization, which the independent first-passage model does not represent; the clause conjectures that this selection does not shift the limiting law, and that assumption is part of what a refutation would refute.) Smooth gaps, whose tuple constant is large, appear earlier by exactly half a logarithm of that constant—the waiting-time first-passage computation makes the coefficient  $-\frac{1}{2}$ , not a free parameter, while  $\mu$  is left as the one unresolved constant of the clause. (The restriction to  $\mathcal{G}$  is needed:  $p(g)$  is not known to be defined for every even  $g$ , and quantifying over all  $g$  would smuggle Polignac's conjecture into the statement. The  $\mathfrak{S}^*$ -clause records that the Hardy–Littlewood factor of  $g$  materially shifts waiting times; it does not move the  $\liminf$  constant, since  $\log \mathfrak{S}^*(g) = O(\log \log g)$ .)

**Conjecture 12** (The Stern lane race; apparently new). In Stern's representation problem  $n = p + 2k^2$  ( $n$  odd), race the  $k$ -even and  $k$ -odd lanes. Prime-square contamination of the  $\Lambda$ -weighted lane counts comes from  $n = q^2 + 2k^2$ —the norm form  $x^2 + 2y^2$  of  $\mathbb{Q}(\sqrt{-2})$  with prime  $x$ —and since  $q^2 \equiv 1 \pmod{8}$  while  $2k^2 \equiv 0$  or  $2 \pmod{8}$  by the parity of  $k$ : (i) [class assignment, direct congruence] contamination sits in the  $k$ -even lane iff  $n \equiv 1 \pmod{8}$ , in the  $k$ -odd lane iff  $n \equiv 3 \pmod{8}$ , and for  $n \equiv 5, 7 \pmod{8}$  no square contamination exists at all:  $q^2 + 2k^2 \equiv 1$  or  $3 \pmod{8}$  always, a two-line congruence (genus theory enters only for the converse question of which  $n$  are represented), so the null classes are provable, not conjectured; (ii) [drift law, in the ensemble form of Conjecture 4] Fix the ensemble: for a scale  $X$ , let  $\mathbb{E}_X$  average over odd  $n \in (X, 2X]$  of the given residue class mod 8 with weight proportional to  $1/n$

(the logarithmic sampling measure, as at Conjectures 5 and 4). Define, per class, with  $R_o(n)$  and  $R_e(n)$  the numbers of representations  $n = p + 2k^2$  with  $k$  odd and  $k$  even respectively, the clean-minus-contaminated difference with explicit sign:  $D(n) = R_o - R_e$  for  $n \equiv 1 \pmod{8}$  (even lane contaminated),  $D(n) = R_e - R_o$  for  $n \equiv 3 \pmod{8}$ , and  $D(n) = R_e - R_o$  (sign immaterial) on the null classes  $n \equiv 5, 7 \pmod{8}$ . Set  $D_{\text{sys}}(n) = [\sum_{(q,k): q^2+2k^2=n, q \text{ prime}, k \geq 1} \log q] / \lambda_{\text{an}}(n)$ —the weight is  $\Lambda(q^2) = \log q$ , each representation counted once in the  $k \geq 1$  convention (the ordered-representation factor 2 of Conjecture 4 has no counterpart here, and doubling would be a normalization error)—where  $\lambda_{\text{an}}(n)$  is the analytic lane mean of  $\log p$  over the candidate profile, defined with no reference to which candidates are prime:

$$\lambda_{\text{an}}(n) = \frac{\int_0^{\sqrt{(n-2)/2}} du}{\int_0^{\sqrt{(n-2)/2}} \frac{du}{\log(n-2u^2)}},$$

the mean of  $\log(n-2u^2)$  under the density proportional to  $1/\log(n-2u^2)$  on the candidate interval (the upper limit keeps  $n-2u^2 \geq 2$ , the convention of  $I(N)$ , avoiding the nonintegrable point  $n-2u^2 = 1$ ), which is the profile a Bateman–Horn heuristic assigns to the prime representations. The definition is out-of-sample: the verification’s empirical lane mean over the observed prime representations is an estimator of  $\lambda_{\text{an}}$ , used as a consistency check, not part of the definition. The law, coefficient-identifying as at Conjecture 4(i): on the contaminated classes

$$\mathbb{E}_X[D - D_{\text{sys}}] = o(\mathbb{E}_X[D_{\text{sys}}]) \quad (X \rightarrow \infty),$$

and on the null classes—where  $D_{\text{sys}}$  vanishes identically, so that a comparison against it would be empty—the statement is made against the positive comparison scale

$$A(X) := \mathbb{E}_X^{(1)}[D_{\text{sys}}],$$

the logarithmic-ensemble mean of the systematic census taken over the contaminated class  $n \equiv 1 \pmod{8}$ : writing  $D_{\text{null}}$  for the difference  $D$  on a null class  $n \equiv 5$  or  $7 \pmod{8}$ , the claim is  $\mathbb{E}_X[D_{\text{null}}] = o(A(X))$ .

**Conjecture 13** (Microscopic variance law; classical family, the constant is Montgomery–Soundararajan’s [101]). Fix  $\lambda > 0$  and  $c \in (0, 1]$ , and let  $X$  count primes in  $[t, t + \lambda \log x)$  for  $t$  uniform on  $[x, (1+c)x]$ . Then, under the Hardy–Littlewood  $k$ -tuple conjectures,  $X \Rightarrow \text{Poisson}(\lambda)$  (Gallagher’s argument [100], which is conditional on exactly those conjectures), and—conditional on the quantitative hypothesis spelled out after the statement (uniform Hardy–Littlewood with  $o(h)$  control of the averaged singular series), without which the display below is a registered extrapolation rather than a consequence—at finite  $x$

$$\frac{\text{Var } X}{\mathbb{E} X} = 1 - \frac{\log(\lambda \log x) + \gamma + \log 2\pi - 1}{\log x} + o\left(\frac{1}{\log x}\right),$$

with  $\gamma + \log 2\pi - 1 = 1.41509\dots$ , the leading correction being independent of the sampling constant  $c$ . [Interpolation clause] More generally, for windows of length  $H = \lambda(\log x)^\alpha$  the same formula with  $\log H$  in place of  $\log(\lambda \log x)$ ,

$$\frac{\text{Var } X}{\mathbb{E} X} = 1 - \frac{\log H + \gamma + \log 2\pi - 1}{\log x} + o\left(\frac{1}{\log x}\right),$$

holds uniformly for  $1 \leq \alpha \leq A$  (any fixed  $A$  with  $H \leq x^{o(1)}$ ): one expression interpolates from Gallagher's Poisson boundary ( $\alpha = 1$ , deficit  $\asymp \log \log x / \log x$ ) through the mesoscopic Montgomery–Soundararajan regime (deficit  $\asymp \alpha \log \log x / \log x$ ), with no transition other than the slow growth of  $\log H$ .

**Conjecture 14** (The null-mechanism race; apparently new). *For the quadratic twin pairs of Conjecture 10 at  $d = 2$ , square contamination is algebraically impossible:  $n^2 + 1 = q^2$  has no solution with  $n \geq 1$ , and  $n^2 + 3 = q^2$  only at  $(n, q) = (1, 2)$ . The surviving classes of  $n \bmod 5$  are  $\{0, 1, 4\}$  (the single escape  $n = 2$ , the pair  $(5, 7)$  in the killed class 2, is one bounded term), and the classes 1 and 4 have identical singular series (elementary, via CRT). Let  $\pi_a^q(x)$  count solutions with  $n \equiv a \pmod{5}$ ,  $D(x) = \pi_1^q(x) - \pi_4^q(x)$ , and define the normalized race process  $Y(x) = D(x) / \sqrt{\pi_1^q(x) + \pi_4^q(x)}$ . Then, in contrast to the twin race of Conjecture 1, this race is driftless: (i)  $\mathcal{M}_x(D(t) \log^2 t / \sqrt{t}) \rightarrow 0$  at the drift-scale normalization of Conjecture 1(i)—the scale at which the contaminated races converge to nonzero constants—so this is the genuine negation of Conjecture 1(i) for this pair (the weaker normalization  $\mathcal{M}_x(Y) \rightarrow 0$  with  $Y = D / \sqrt{\pi^q}$  would not exclude a drift at the contamination scale, hence would not make this race a control; the form stated excludes exactly that); (ii) [occupation law, conjectured directly, with a separately falsifiable local hypothesis] the process is defined precisely at event index: let  $n_1 < n_2 < \dots$  enumerate the  $n$  with  $n^2 + 1$  and  $n^2 + 3$  both prime and  $n \equiv 1$  or  $4 \pmod{5}$ , set  $\xi_i = +1$  if  $n_i \equiv 1$  and  $-1$  if  $n_i \equiv 4$ , and let  $S_N = \xi_1 + \dots + \xi_N$  (so  $D(x) = S_{N(x)}$  with  $N(x)$  the number of such events to  $x$ ). The sequence  $(\xi_i)$  is deterministic, so the hypothesis is stated for a randomly sited window rather than for  $S_N$  itself. Fix a growth scale  $N_0$  and a window length  $N = N(N_0)$  with  $N = o(N_0)$ , and choose the starting index  $M$  of the window according to the logarithmic measure on  $[1, N_0]$ :*

$$\Pr[M = m] = \left( \sum_{1 \leq m' \leq N_0} \frac{1}{m'} \right)^{-1} \frac{1}{m}, \quad 1 \leq m \leq N_0.$$

*The hypothesis is an invariance principle of ASIP type, stated in triangular-array form, for the induced random path  $(S_{M+j} - S_M)_{0 \leq j \leq N}$  under that sampling law: on a probability space carrying  $M$  and a standard Brownian motion  $B$ ,*

$$S_{M+j} - S_M = B(j) + O(N^{1/2-\delta}) \quad \text{uniformly for } 0 \leq j \leq N,$$

*with probability tending to one as  $N_0 \rightarrow \infty$ , for some  $\delta > 0$ ; the sampling law of  $M$  changes with  $N_0$ , so the coupling is asserted per  $N_0$  with vanishing exceptional probability rather than almost surely on a single space. The logarithmic sampling law is the same clock in which the occupation statement below is read, and randomizing the window start is what supplies the deterministic step sequence with a probability space. The weak functional limit theorem— $S_{\lfloor N\tau \rfloor} / \sqrt{N}$  converging weakly to standard Brownian motion on  $\tau \in [0, 1]$ —is retained as the weaker, separately falsifiable consequence of the windowed ASIP. The occupation law itself, however, is not a consequence of the windowed hypothesis, and is conjectured directly. The windowed path  $S_{M+j} - S_M$  is recentred at the window start, and recentring erases exactly the absolute level that occupation measures: a deterministic bias  $b(n) = \sqrt{n} / \log n$  added to  $S_n$  shifts the level by  $\sqrt{n} / \log n$  while contributing window increments of size  $\sim j / (2\sqrt{n} \log n) \ll \sqrt{j}$ , invisible to the windowed ASIP at every window scale, yet it would drive the occupation fraction to 1. A global coupling  $D(t) = B(N(t)) + o(\sqrt{N(t)})$  with  $N(t) \asymp t / \log^2 t$  would suffice, but for the deterministic sequence  $(\xi_i)$  no randomization is available that makes such a coupling well posed, so we do not hypothesize it. The Brownian computation is instead the motivation that identifies the limit and the constant: for a coupled walk,*



in logarithmic time  $u = \log t$  the Lamperti reduction  $Z(s) = e^{-s/2} B(e^s)$  is a stationary, ergodic Ornstein–Uhlenbeck process, and the logarithmic occupation measure of leadership converges almost surely to  $\frac{1}{2}$ . The conjectured law is that conclusion itself:

$$\frac{1}{\log x} \int_2^x \mathbf{1}_{\{D(t) > 0\}} \frac{dt}{t} \rightarrow \frac{1}{2},$$

with fluctuations of the Gaussian scale  $\sqrt{\log 2 / \log x}$  (the occupation-indicator covariance  $\frac{1}{2\pi} \arcsin e^{-u/2}$  integrates to  $\frac{1}{2} \log 2$ ; note that the integrated covariance for the sign average  $2\mathbf{1} - 1$  is four times this, its standard deviation twice, the occupation fraction itself carrying  $\sqrt{\log 2 / \log x}$ ); the classical arcsine spread survives only in the natural event clock, i.e. for occupation fractions of  $\{B(m) > 0, m \leq N(x)\}$ . A persistent systematic leader—a log-occupation stuck near 0 or 1 beyond the stated Gaussian scale—is the refutation.

**Conjecture 15** (The cubic shift family: the distribution of its constants; instance-level law apparently unstated). For non-cube  $a \geq 1$  let  $C(a) = C(x^3 + a)$ . Only  $p \equiv 1 \pmod{3}$  moves the product, and the local root count there takes three values as  $a$  varies mod  $p$ :  $\omega = 3$  with probability  $(p-1)/3p$  ( $-a$  a nonzero cube),  $\omega = 1$  with probability  $1/p$  (the case  $p \mid a$ , easy to overlook and essential to the mean-value identity below), and  $\omega = 0$  with probability  $2(p-1)/3p$ ; for  $p \equiv 2 \pmod{3}$  the cube map is a bijection and  $\omega = 1$  identically. Then: (i) [mean-value one, elementary lemma]  $\mathbb{E}_a[\omega(p)] = 3 \cdot \frac{p-1}{3p} + \frac{1}{p} = 1$  exactly at every  $p$ , so the derived mean of  $C(a)$  is exactly 1—the cubic analogue of the Korevaar–de Riele mean-value-one theorem for linear prime-pair constants, here a one-line computation at each finite level; passing the mean through the infinite product needs more than almost-sure convergence, and the needed ingredient is on hand: the convergent variance sum makes the partial products an  $L^2$ -bounded martingale (each factor has mean 1 and is independent of the earlier ones), so they are uniformly integrable and the limit law has mean exactly 1, not merely limit-of-means 1; (ii) [limit law] the empirical law of  $\{C(a) : a \leq A, a \text{ a non-cube}\}$  converges, as  $A \rightarrow \infty$ , to the law of the random Euler product  $\prod_{p \equiv 1(3)} f_p(\xi_p)$  with the three-state local variables  $\xi_p$  above—independent at any finite set of primes exactly, by CRT, with the infinite product’s tail controlled by the convergent variance sum (unlike the quadratic family of Conjecture 10(iii), no normalization is needed)—with derived standard deviation  $0.2762\dots$  computed from the full three-state law; (iii) [uniformity]  $\#\{n \leq N : n^3 + a \text{ prime}\} = C(a) I_a(N) (1 + o(1))$  uniformly over non-cube  $a \leq (\log N)^B$ .

**Conjecture 16** (Conjecture F as a family; the core is previously stated, [1], cf. [105]). For odd  $A$  let  $Q_A(N) = \#\{n \leq N : n^2 + n + A \text{ prime}\}$  and  $C(A) = C(x^2 + x + A)$ . Then: (i) for every fixed  $B > 0$ ,  $Q_A(N) = C(A) I_A(N) (1 + o(1))$  uniformly over odd  $1 \leq A \leq (\log N)^B$ ; (ii) [window field] the residual field obeys the analogue of Conjecture 8(ii), kernel derived and probability space specified: for  $t$  uniform on  $[N, 2N]$  and window length  $H \leq N^{o(1)}$ , let  $Q_A(t; H) = \#\{t < n \leq t + H : n^2 + n + A \text{ prime}\}$  (mean  $\asymp H/2 \log N$ , which sets the regime). Two members share the index window, so with  $C(A, A') = C(x^2 + x + A, x^2 + x + A')$ ,

$$\text{Cov}_t(Q_A, Q_{A'}) \sim [C(A, A') - C(A)C(A')] \frac{H}{4 \log^2 N},$$

with  $C(A, A') = 0$  (hence negative correlation) for the locally exclusive pairs. This is the same-index contribution only: the full covariance adds the off-index pinned sum

$$\frac{1}{4 \log^2 N} \sum_{0 < |h| \leq H} (H - |h|) [C(f_A(x), f_{A'}(x + h)) - C(A)C(A')],$$



the two-parameter analogue of Conjecture 2's  $G(H)$ , whose evaluation is this family's open component. (The order of this sum matters: the joint event here is two single prime values at distinct indices, of probability  $\asymp 1/4 \log^2 N$ , unlike Conjecture 8's pair-of-pairs events at  $1/\log^4 x$ . At that order the off-index sum, which grows like a pinned  $G$ -average over  $\asymp H$  shifts against the same-index term's single  $H$ , is generically the larger component, so the numerical cancellation of the same-index kernel establishes nothing about the full cross-member covariance, and the localization of the observed profile deficit to the diagonal is an open question rather than an inference.) For  $H/\log N \rightarrow \lambda$  the counts are jointly Poisson with local exclusions, and for  $H/\log N \rightarrow \infty$  each fixed finite collection of studentized counts is asymptotically jointly normal with independent limiting coordinates, all correlations of size  $\asymp 1/\log N$  making both kernels finite- $N$  corrections to independence. As at Conjecture 8(ii), both distributional conclusions are asserted under an explicit hypothesis beyond the displayed covariances—uniform Hardy–Littlewood tuple estimates over the family, strong enough that every joint cumulant of order at least three of the studentized counts vanishes in the respective normalizations—since the covariance kernels alone do not determine the limit laws. (Uniformity over a range growing with  $N$  is the substantive claim; uniformity over any fixed finite set of  $A$  is logically automatic from the individual asymptotics.) Among odd  $A \leq 199$ , Euler's  $A = 41$  has the largest constant,  $C(41) = 6.6395463\dots$  (Cohen's high-precision value [113]).

**Conjecture 17** (Twin-member Goldbach, orientation-resolved; the basis conjecture and the integral kernel are Dubner's [9]; the orientation decomposition apparently new). *For even  $n$  let  $R_T(n)$  count ordered  $(a, b)$ ,  $a + b = n$ , with  $a$  and  $b$  both members of twin pairs, each ordered role (lower/upper) counted. Each of the four role assignments is a 4-form linear system in  $t = a$  with root set  $\{0, \pm 2\} \cup \{n, n \pm 2\}$ -type mod each  $p$ , hence its own  $n$ -dependent singular series  $\mathfrak{S}_4^{(o)}(n)$ , and*

$$R_T(n) = \left[ \sum_o \mathfrak{S}_4^{(o)}(n) \right] \int_5^{n-5} \frac{dt}{\log^2 t \log^2(n-t)} (1 + o(1)),$$

the four constants computed exactly per  $n$  from the coincidence pattern of the root sets (which of  $n, n \pm 2, n \pm 4$  each small prime divides). The error term is stated as  $(1 + o(1))$ , not  $(1 + O(1/\log n))$ : our own measurements show a second-order deficit whose  $1/\log n$  coefficient is not yet pinned down (below). Two precision notes: the asymptotic runs over varying  $n$ , so what is assumed is a Hardy–Littlewood estimate for 4-form systems uniform in the  $n$ -dependent coefficients—strictly stronger than the fixed-system conjecture—and the number 5, the unique prime that is simultaneously a lower twin member (of  $(5, 7)$ ) and an upper one (of  $(3, 5)$ ), is double-counted consistently by the ordered-role convention, affecting finitely many representations per  $n$  and no asymptotic.

## 4 Instances and structural companions

**Conjecture 18** (Contamination in prime triplets; apparently new). *The contamination calculus of Conjecture 1(v) extends to  $k$ -tuples, and the triplet pattern  $(n, n + 2, n + 6)$  is its first instance beyond pairs. Of the three configurations placing a prime square inside the pattern, two die algebraically for  $q > 3$ — $(q^2, q^2 + 2, q^2 + 6)$  by  $3 \mid q^2 + 2$ , and  $(q^2 - 6, q^2 - 4, q^2)$  because  $q^2 - 4 = (q - 2)(q + 2)$  is composite (at  $q = 3$  the factorization is trivial,  $q^2 - 4 = 5$  is prime, and the single bounded exceptional configuration  $(3, 5, 9)$  occurs—one term, harmless to every asymptotic)—leaving the single doubly-thinned survivor  $(q^2 - 2, q^2, q^2 + 4)$ , which requires*

$q^2 - 2$  and  $q^2 + 4$  simultaneously prime and forces  $q \equiv \pm 2 \pmod{5}$  for  $q \neq 5$  (else  $5 \mid q^2 + 4$ ; at  $q = 5$  all three conditions do hold, the configuration being  $(23, 25, 29)$ , but its start  $23 \equiv 3 \pmod{5}$  lies outside the triplet start classes  $\{1, 2\}$ , so the single term is harmless). Triplet starts lie in classes  $n \equiv 1, 2 \pmod{5}$  (single bounded exception  $(5, 7, 11)$ , start  $\equiv 0$ ), and the surviving configuration has start  $q^2 - 2 \equiv 2$ : class 2 is contaminated and class 1 leads, in the drift-scale form of Conjecture 1(i):

$$\mathcal{M}_x \left( \frac{(\pi_3(t; 5, 1) - \pi_3(t; 5, 2)) \log^3 t}{\sqrt{t}} \right) \rightarrow c_3 := \lim_{x \rightarrow \infty} \frac{T_3(x) \log^3 x}{\sqrt{x}}, \quad T_3(x) = \frac{1}{\log^3 x} \sum_{\substack{q \leq \sqrt{x}, q \neq 5 \\ q^2-2, q^2+4 \text{ prime}}} \log(q^2-2) \log q \log$$

( $c_3$  exists and is the Bateman–Horn constant of the triple  $(q, q^2 - 2, q^2 + 4)$ , since the census sum is  $\sim c_3 \sqrt{x}$ ; convergence of the logarithmic mean at this normalization is the same Rubinstein–Sarnak-type structure hypothesis as at Conjecture 1(i); a normalization at the noise scale,  $\mathcal{M}_x((D - T_3)/\sqrt{\pi_3})$  with  $D$  the class difference displayed above and  $\pi_3$  the total triplet count to  $t$ , would be vacuous for the coefficient, exactly as at the pair races), with drift  $\asymp \sqrt{x}/\log^3 x$ , one logarithm weaker than the pair races, and drift-to-noise  $\asymp \log^{-3/2} x$  against their  $\log^{-1}$ , a scale the calculus itself predicts and the verification must respect. The mechanism clause and the averaging clause are asserted separately, as at Conjecture 1(i).

**Conjecture 19** (The contamination matrix: sexy pairs with two surviving orientations; apparently new). For the pattern  $(n, n+6)$ , both prime-square orientations survive, on complementary classes of  $q \pmod{5}$ —the first matrix instance of the calculus (every pair race so far had exactly one): orientation  $A = (q^2 - 6, q^2)$  requires  $q^2 - 6$  prime, forcing  $q \equiv \pm 2 \pmod{5}$  for  $q \neq 5$  (at  $q = 5$  the value 19 is prime, but its start  $19 \equiv 4 \pmod{5}$  lies outside the sexy start classes  $\{1, 2, 3\}$ ), and lands in start class 3  $\pmod{5}$ , class 3  $\pmod{8}$ ; orientation  $B = (q^2, q^2 + 6)$  requires  $q^2 + 6$  prime, forcing  $q \equiv \pm 1 \pmod{5}$  (the term  $q = 5$  falls outside the start classes), and lands in class 1  $\pmod{5}$ , class 1  $\pmod{8}$ . Sexy starts occupy classes  $\{1, 2, 3\} \pmod{5}$  and  $\{1, 3, 5, 7\} \pmod{8}$ ; with  $T_A(x) = \frac{1}{\log^2 x} \sum_{q^2-6 \text{ prime}, q \neq 5} \log q \log(q^2 - 6)$ ,  $T_B(x) = \frac{1}{\log^2 x} \sum_{q^2+6 \text{ prime}, q \neq 5} \log q \log(q^2 + 6)$ , and  $c_A, c_B$  their drift-scale limits ( $c_A = \lim T_A \log^2 x / \sqrt{x}$ , the Bateman–Horn constant of  $(q, q^2 - 6)$ , likewise  $c_B$  for  $(q, q^2 + 6)$ ), the predicted drift vector, in the drift-scale limiting-log-mean sense of Conjecture 1(i) (an  $\mathcal{M}_x$ -over- $\sqrt{\pi}$  normalization would be coefficient-blind), is, writing  $\pi(a)$  for the number of sexy starts up to  $t$  in class  $a$  of the modulus in question: mod 5:  $\mathcal{M}_x((\pi(2) - \pi(3)) \log^2 t / \sqrt{t}) \rightarrow c_A$ ,  $\mathcal{M}_x((\pi(2) - \pi(1)) \log^2 t / \sqrt{t}) \rightarrow c_B$  (class 2 clean); mod 8:  $\mathcal{M}_x((\frac{1}{2}(\pi(5) + \pi(7)) - \pi(3)) \log^2 t / \sqrt{t}) \rightarrow c_A$ ,  $\mathcal{M}_x((\frac{1}{2}(\pi(5) + \pi(7)) - \pi(1)) \log^2 t / \sqrt{t}) \rightarrow c_B$  (classes 5, 7 clean, mutually symmetric, with  $\mathcal{M}_x((\pi(5) - \pi(7)) \log^2 t / \sqrt{t}) \rightarrow 0$ ). Two independently computable drift constants, one certified null class per modulus. The mechanism clause and the averaging clause are asserted separately, as at Conjecture 1(i).

**Conjecture 20** (Fibonacci–Lucas twins: the convergent side stress-tested; the object is OEIS A080327). (i) [rank-disjointness, elementary, with its exception stated] every odd prime factor  $r$  of  $L_p$  has rank of apparition exactly  $2p$  (from  $z(r) \mid 2p$ ,  $z(r) \nmid p$ , and  $z(r) = 2$  being impossible), and every prime factor of  $F_p$  ( $p \neq 5$  prime) has rank exactly  $p$ : the odd divisor pools of the two numbers are disjoint. The prime 2 is the unique exception: 2 divides both  $F_p$  and  $L_p$  precisely when  $p = 3$  ( $F_3 = 2$ ,  $L_3 = 4$ ;  $\gcd(F_p, L_p) = 2$  iff  $3 \mid p$ )—the restriction to odd divisors is essential, the clause being false at  $p = 3$  without it. Disjoint pools remove shared-prime correlation but not all correlation: dependence can still enter through the order structure of the recurrence, which is why (iii) matters; (ii) [finiteness] only finitely many primes  $p$  have  $F_p$  and  $L_p$

simultaneously prime; (iii) [calibration clause, quantitative] the naive joint accounting—probability  $(e^\gamma \log p / (p \log \phi))^2$  per index—assigns prior mass about  $1.4 \times 10^{-2}$  to the whole index range beyond  $10^4$ , in which the catalogued index  $p = 148091$  falls (OEIS A080327; both  $F_p$  and  $L_p$  are probable primes—numbers of roughly 30,950 digits, with Lucas numbers proved prime only up to index 56003—so everything recorded here is conditional on both probable-prime classifications, for which no Baillie–PSW pseudoprime is known: no unconditional refutation is claimed from uncertified numbers). That rare-event datum is recorded descriptively, with no significance claim attached to it: the model was examined after the event was known, and no test protocol was fixed in advance. A single surprising event is not a strict refutation, and it does not by itself estimate the size of the correction: an underestimated constant, genuine cross-dependence, and a mis-calibrated finite-range tail are competing explanations that one event cannot separate. What the datum does bear on (conditionally as above) is not the model but the list: it is fatal to any Goldilocks completeness list; the conjectural content of (ii) is that even the corrected accounting has convergent sum.

**Conjecture 21** (Factorial twins; the uniqueness core is previously stated, OEIS A088054; clauses (i) and (iii) are ours). (i) [window rigidity, elementary] for  $n \geq 4$  and  $2 \leq |a| \leq n$  the number  $n! + a$  is composite (any prime factor  $p$  of  $a$  satisfies  $p \leq n$ , so  $p \mid n! + a$ , and  $n! + a > n \geq p$ ); the two small anomalies  $2! - 2 = 0$  and  $3! - 3 = 3$  are the only exceptions, which is why the clause is restricted to  $n \geq 4$ . Hence among all offsets of bounded size the only candidates for primality near  $n!$  are  $a = \pm 1$ : the twin question below is the unique bounded-offset constellation question at  $n!$ , and every admissible offset set is a subset of  $\{-1, +1\}$ . (ii) [uniqueness; previously stated] Only finitely many  $n$  have  $n! - 1$  and  $n! + 1$  simultaneously prime, and the complete list is  $n = 3$ : i.e.  $6 = 3! = 3\#$  is the unique factorial lying between twin primes. (iii) [joint fluctuation model] the two single-sided counts  $F_\pm(N) = \#\{n \leq N : n! \pm 1 \text{ prime}\}$  are each  $\sim e^\gamma \log N$  and, under the independent-indices model stated below, asymptotically independent, with  $F_+ - F_-$  normalized by  $\sqrt{2e^\gamma \log N}$  asymptotically standard normal. The probability space is declared: the clause is about a random model in which the events  $E_n^\pm = \{n! \pm 1 \text{ prime}\}$  carry the Caldwell–Gallot marginal hazards  $e^\gamma/n$  with an unspecified dependence structure, and the Gaussian limit is asserted under three explicit hypotheses on that structure: the same-index bound  $\sum_{n \leq N} |\text{Cov}(\mathbf{1}_{E_n^+}, \mathbf{1}_{E_n^-})| = o(\log N)$ , which pins the variance at  $2e^\gamma \log N (1 + o(1))$  and which the joint law  $\sim (e^\gamma/n)^2$  derived below supplies with a convergent sum—without it the remaining hypotheses leave the second cumulant free, and a model with  $\Pr(E_n^+ \cap E_n^-) = e^\gamma/2n$  halves the limiting variance; the cross-index covariance bound  $\sum_{m < n \leq N} |\text{Cov}(\mathbf{1}_{E_m}, \mathbf{1}_{E_n})| = o(\log N)$  over the four sign pairs (the variance itself is  $\asymp \log N$ ), and—since pairwise control alone cannot preclude higher-order dependence—the higher-cumulant condition  $\kappa_r(F_+ - F_-) = o((\log N)^{r/2})$  for each fixed  $r \geq 3$ , the method-of-moments sufficient condition. None of the three is imported from an independence model; whether they hold for the arithmetic dependence across factorial indices is the open component of this clause.

**Theorem 1** (Cube obstruction). For  $k \geq 2$ , the cube  $k^3$  is representable as  $p + j^3$  with  $p$  prime,  $j \geq 1$ , if and only if  $3k^2 - 3k + 1$  is prime. Consequently the set of integers not representable as  $p + k^3$  is infinite: the non-representable cubes alone have counting function  $\sim x^{1/3}$ . (That the full exceptional set is  $\sim x^{1/3}$  follows only when combined with Conjecture 22(i), and is conjectural.)

**Conjecture 22** (The boundary trichotomy for polynomial ladders; the divisibility principle is classical and its cubic case is Cunningham’s cuban observation (OEIS A002407); the family classification is apparently new). For  $F \in \mathbb{Z}[x]$  with  $\deg F \geq 2$  and positive leading coefficient, and

$m > j \geq 1$ ,  $(m - j) \mid F(m) - F(j)$  (textbook), and the divided difference  $(F(m) - F(j))/(m - j)$ —whose leading form is then positive on the cone  $m > j \geq 1$ —exceeds 1 outside an effectively bounded region; so  $F(m) - F(j)$  prime forces  $m - j = 1$  apart from finitely many exceptional pairs (for the family below the exceptional set is provably empty). Both hypotheses are necessary: for  $\deg F = 1$  the cofactor can be identically 1 and the collapse fails, and for negative leading coefficient the divided difference tends to  $-\infty$  on the cone, so the “exceeds 1” clause would be false as stated (one may equivalently keep general sign and read the claim through  $|F(m) - F(j)|$ ). Every representation problem  $F(m) = p + F(j)$  in this range thus collapses to the boundary polynomial  $D_F(m) = F(m) - F(m - 1)$ . The content is the classification of the boundary lanes. For  $F = x^3 + cx$ ,  $c \geq 0$ : (i) [trichotomy, elementary]  $D_F = 3m^2 - 3m + 1 + c$  and the lane is dead-parity for  $c$  odd ( $D_F$  always even), dead-3-adic for even  $c \equiv 2 \pmod{3}$  ( $3 \mid D_F$  always), and admissible exactly for  $c \equiv 0, 4 \pmod{6}$ ; (ii) [uniform boundary law] over the admissible lanes,  $\#\{m \leq M : D_F(m) \text{ prime}\} \sim C(D_F) I(M)$  uniformly for  $c \leq (\log M)^B$ —a Bateman–Horn family whose members are indexed by the ladder classification; (iii) [attributed core] the case  $c = 0$  is the cuban ladder: Cunningham’s classical observation that a prime difference of cubes forces consecutive arguments, with the non-cube exceptional set of  $n = p + k^3$  finite (Hardy–Littlewood’s  $E_3(X) = O(1)$  [1]).

**Conjecture 23** (The power-obstruction ladder: an elementary structural proposition with a Bateman–Horn corollary; apparently new as a family). For  $k \geq 2$  and  $m \geq 2$ , write  $D_k(m) = m^k - (m - 1)^k$ , and call  $m^k$  representable if  $m^k = p + j^k$  for some prime  $p$  and  $j \geq 1$ . Then: (i) [theorem] for composite  $k$ , no  $k$ -th power is representable: if  $r$  is a proper divisor of  $k$  with  $r > 1$  (such  $r$  exists precisely because  $k$  is composite; the excluded value  $r = 1$  would allow only the useless factor  $m - j = 1$  at  $j = m - 1$ ), then for every  $1 \leq j < m$  the difference  $m^k - j^k$  has the proper factor  $m^r - j^r > 1$  with cofactor  $> 1$ ; (ii) [elementary] for prime  $k$  (including  $k = 2$ ),  $m^k$  is representable if and only if  $D_k(m)$  is prime; and  $D_k$  is irreducible for prime  $k$  (a root  $\alpha$  has  $(\alpha/(\alpha - 1))^k = 1$  with  $\alpha/(\alpha - 1) \neq 1$ , so  $\zeta = \alpha/(\alpha - 1)$  is a primitive  $k$ -th root of unity and  $\mathbb{Q}(\alpha) = \mathbb{Q}(\zeta)$  has degree  $k - 1 = \deg D_k$ , forcing irreducibility; explicitly  $D_k(x) = (x - 1)^{k-1} \Phi_k(x/(x - 1))$ . The forcing  $j = m - 1$  needs only  $k \geq 2$ ); (iii) [conjecture] for each prime  $k \geq 3$  the representable lane follows Bateman–Horn ( $k = 2$  is a theorem, the prime number theorem on the lane  $2m - 1$ ):  $\#\{m \leq M : D_k(m) \text{ prime}\} \sim C_k \int_2^M dt / \log D_k(t)$ ).

**Conjecture 24** (The alternating cyclotomic chain; apparently new). There are infinitely many primes  $p$  such that, writing  $u = \Phi_3(p) = p^2 + p + 1$ , the three integers

$$p, \quad u = p^2 + p + 1, \quad \Phi_6(u) = u^2 - u + 1 = p^4 + 2p^3 + 2p^2 + p + 1$$

are all prime, with  $\#\{p \leq x\} \sim C_6^{\text{ch}} I(x)$  for the Bateman–Horn system  $\{x, x^2 + x + 1, x^4 + 2x^3 + 2x^2 + x + 1\}$ ,  $C_6^{\text{ch}} = 3.6143 \pm 0.011$ . The first chains are  $(2, 7, 43)$  and  $(3, 13, 157)$ .

**Conjecture 25** (Twin cyclotomic bases: the complete quadratic family; no prior trace found). For the three cyclotomic polynomials of degree 2 (indices  $k$  with  $\varphi(k) = 2$ , i.e.  $k \in \{3, 4, 6\}$ ), consider  $\Phi_k(n)$  and  $\Phi_k(n + 1)$  simultaneously prime. Then: (i) [parity obstruction, elementary] for  $k = 4$  one member of the pair  $(n^2 + 1, n^2 + 2n + 2)$  is even for every  $n$ , so the only prime pair is  $n = 1$ , giving  $(2, 5)$ —the same obstruction that forbids consecutive  $n^2 + 1$  primes; (ii) [collapse, elementary]  $\Phi_6(x) = \Phi_3(x - 1)$  identically, so the  $k = 6$  pair at index  $n$  is the  $k = 3$  pair at index  $n - 1$ : the family contains exactly one nontrivial statement; (iii) [conjecture, the live instance] there are infinitely many  $n$  for which the length-3 repunits in bases  $n$  and  $n + 1$  are simultaneously

prime:  $\#\{n \leq N : \Phi_3(n) = n^2 + n + 1 \text{ and } \Phi_3(n+1) = n^2 + 3n + 3 \text{ both prime}\} \sim C_5 I(N)$ ,  
with  $C_5 = C(x^2 + x + 1, x^2 + 3x + 3)$ .

## Part II

# Structural conjectures in five programmes

*Added to the deposit: 3 August 2026.*

## 5 Programme I: connected prime-pattern fields

Let  $H \subset \mathbb{Z}$  be finite and admissible, and write

$$\Lambda_H(n) = \prod_{h \in H} \Lambda(n+h), \quad Y_H(x, L) = \sum_n w\left(\frac{n-x}{L}\right) (\Lambda_H(n) - \mathfrak{S}(H)),$$

where  $w$  is a fixed smooth compactly supported function and  $\mathfrak{S}(H)$  is the Hardy–Littlewood singular series  $\prod_p (1 - \nu_H(p)/p)(1 - 1/p)^{-|H|}$ , with  $\nu_H(p)$  the number of residue classes occupied by  $H$  modulo  $p$ . The window position  $x$  is sampled from  $[X, 2X]$  with density  $dx/(x \log 2)$  and  $X^\varepsilon \leq L \leq X^{1-\varepsilon}$ . For translated motifs  $H_i + t_i$ , their *exact overlap type* records every equality among shifted prime constraints. The connected singular series  $\mathfrak{S}^c(K_1, \dots, K_r)$  is obtained by Möbius inversion over set partitions of the indexed tuple  $(K_1, \dots, K_r)$ .

**Conjecture 26** (Connected motif generating functional). *Fix admissible motifs  $H_1, \dots, H_m$  and put*

$$\mathcal{Z}_{X,L}(\mathbf{z}) = \mathbb{E}_x \exp\left(\sum_{i=1}^m z_i Y_{H_i}(x, L)\right).$$

*For every fixed total degree  $R$ , the degree- $\leq R$  Taylor polynomial of  $\log \mathcal{Z}_{X,L}$  has a uniform asymptotic expansion indexed by connected exact-overlap diagrams of at most  $R$  translated motifs. A diagram whose coincidence hypergraph connects the  $r$  translated sets  $H_{i_1} + t_1, \dots, H_{i_r} + t_r$  contributes, at the leading order of its stratum,*

$$W_\tau(L; \mathbf{t}) \mathfrak{S}\left(\bigcup_{j=1}^r (H_{i_j} + t_j)\right) \prod_v (\log X)^{m_v-1},$$

*where  $W_\tau(L; \mathbf{t}) = L \int_{\mathbb{R}} \prod_{j=1}^r w(u + t_j/L) du$  is the archimedean window-overlap weight of the diagram and  $m_v$  is the multiplicity of the prime constraint at  $v$ : any set partition that separates a coincidence loses at least one factor of  $\log X$ , so the Möbius alternation leaves the single-block moment as the leading term of a connected diagram. The same alternation cancels the leading terms of disconnected diagrams, whose contribution is carried by the connected singular series of their disjoint blocks at strictly smaller logarithmic order, reducing for the fully disjoint type to  $\mathfrak{S}^c(H_{i_1} + t_1, \dots, H_{i_r} + t_r)$  with no logarithmic enhancement. The log powers are therefore partition dependent and are never factored across the alternation, and every prime-power counterterm is included in the local weight attached to its multiplicity. Ordering terms lexicographically by powers of  $L$  and  $\log X$ , the remainder after any fixed truncation is smaller than the last retained scale, uniformly in the mesoscopic range.*



**Conjecture 27** (Complete overlap-renormalization filtration). *Fix a finite motif family  $H_1, \dots, H_m$ . Apply the degree-two specialization of Conjecture 26 and order all of its connected covariance strata by their asymptotic scales  $s_1 \succ s_2 \succ \dots$  (lexicographically in powers of  $L$ ,  $\log X$ , and the renormalized disjoint scales). In the mesoscopic range scales sharing the same total power of  $L$  and of logarithms differ only by powers of  $\theta = \log L / \log X$  and merge into a single stratum whose limiting form is polynomial in  $\theta$ . The filtration, the graded convergence, and the eigenvalue asymptotics below are asserted for every fixed  $\theta \in (\varepsilon, 1 - \varepsilon)$ . Recursively subtract the earlier strata and let  $A_\nu = A_\nu(\theta)$  be the limiting covariance form at scale  $s_\nu$ , and put*

$$\mathcal{W}_\nu = \bigcap_{\mu < \nu} \ker A_\mu.$$

*Each  $A_\nu$  is asserted to induce a nondegenerate form on the graded quotient  $\mathcal{W}_\nu / \mathcal{W}_{\nu+1}$  for every such  $\theta$ , so that  $\mathcal{W}_{\nu+1} = \mathcal{W}_\nu \cap \ker A_\nu$  and the filtration does not collapse. For every fixed truncation of this ordered diagram expansion, the spectral projections of the covariance matrix converge to the associated graded spaces  $\mathcal{W}_\nu / \mathcal{W}_{\nu+1}$ , and every eigenvalue whose first nonzero term occurs within the truncation is asymptotic to  $s_\nu$  times a positive eigenvalue of the induced form  $A_\nu$ . Vectors annihilating all retained forms pass canonically to the next connected diagram scales of Conjecture 26.*

*Consequently every asymptotic variance scale of a finite motif statistic is generated by a connected overlap or renormalized disjoint diagram from Conjecture 26, and no scale external to that diagrammatic expansion occurs. The filtration is invariant under replacing the motif family by one with isomorphic exact-overlap incidence algebra and identical local Euler weights.*

**Conjecture 28** (Regularized mesoscopic local-spectral trace formula). *Fix motifs  $H_i, H_j$ , a Schwartz function  $\phi$  with  $\phi(0) = 0$ , an even cutoff  $\eta \in C_c^\infty(\mathbb{R})$  with  $\eta = 1$  near the origin, and an even mollifier  $\kappa \in C_c^\infty(\mathbb{R})$  of integral one. Let  $Q_{ij}^{\text{loc}}(\phi; L)$  be the exact-overlap-renormalized singular-series form*

$$\sum_{t \in \mathbb{Z}} \left[ L \int_{\mathbb{R}} w(u) w(u + t/L) du \right] \phi(t/L) \mathfrak{S}^c(H_i, H_j + t).$$

*Let  $\Lambda_{T,\eta}^\#$  be the Weil explicit-formula distribution in the logarithmic variable, with each nontrivial zero of ordinate  $\gamma$  multiplied by  $\eta(\gamma/T)$  and with the pole, trivial-zero, archimedean, and prime-power terms kept as separately named summands. For  $\delta > 0$ , put*

$$\Lambda_{T,\eta,\delta}^\# = \Lambda_{T,\eta}^\# *_{\log} \kappa_\delta, \quad \kappa_\delta(u) = \delta^{-1} \kappa(u/\delta).$$

*This is a smooth function, so products of its shifted copies are unambiguous. Define*

$$Q_{ij}^{\text{spec}}[\eta, \kappa] = Q_{ij}^{\text{spec}}(\phi; X, L; T, \delta, \eta, \kappa)$$

*by inserting these mollified copies into the smoothed motif covariance, applying the same exact-overlap Möbius projection as on the local side, and subtracting the explicitly specified pole, diagonal, trivial-zero, archimedean, and prime-power counterterms before the mollifier is removed. In normal form: expand each inserted copy into its separately named summands together with the mollified zero sum and expand the covariance multilinearly;  $Q_{ij}^{\text{spec}}$  is then the zero-zero block of that expansion with its diagonal term removed, the subtracted counterterms being exactly the cross terms containing a pole, trivial-zero, archimedean, or prime-power summand, together with the diagonal term of the zero-zero block, so that two readers form the same object.*

For every sufficiently large fixed  $A, B$ , take

$$T = (X/L)(\log X)^A, \quad \delta = (L/X)(\log X)^{-B}.$$

Then

$$Q_{ij}^{\text{spec}}(\phi; X, L; T, \delta, \eta, \kappa) - Q_{ij}^{\text{loc}}(\phi; L) = o(V_{ij}(X, L))$$

uniformly for  $X^\varepsilon \leq L \leq X^{1-\varepsilon}$  and for  $\phi$  in bounded Schwartz sets, where  $V_{ij}$  is the first nonzero disjoint covariance scale. The same limit holds jointly for every fixed matrix of motifs and tests. Changing  $A, B, \eta$ , or  $\kappa$  within the stated admissible class changes the renormalized spectral form by  $o(V_{ij})$ , so the limiting trace functional is canonical.

**Conjecture 29** (Anchored arithmetic polymer expansion for prime motifs). *Fix an admissible motif  $H$  of size  $k$ . A finite set  $A \subset \mathbb{Z}$  of occurrence starts is a polymer when  $0 \in A$ , the overlap graph on  $A$  is connected, and  $U_A = \bigcup_{t \in A} (H + t)$  is admissible. Define its anchored codimension*

$$d_H(A) = |U_A| - |H|;$$

*this is the number of new prime constraints beyond the occurrence anchored at 0. In the regularized spectral model of Conjecture 28, let  $Z$  be the limiting low-mode vector (the joint limit, asserted as part of this conjecture, of the mode integrals of the Conjecture 28 functional against frequencies below a fixed cutoff  $K$  on the  $L$ -scale, the Conjecture 28 limits taken first and  $K \rightarrow \infty$  last), let  $\lambda(t \mid Z)$  be the conditional one-point occurrence intensity, and set*

$$\lambda_H^0 = \frac{\mathfrak{S}(H)}{(\log X)^k}, \quad \tilde{\lambda}(t \mid Z) = \lambda(t \mid Z) / \lambda_H^0.$$

For every fixed polymer  $A$ , uniformly for bounded diameter,

$$\frac{(\log X)^{d_H(A)} \rho(|A|)(A \mid Z)}{\lambda(0 \mid Z) \prod_{t \in A \setminus \{0\}} \tilde{\lambda}(t \mid Z)} \longrightarrow \frac{\mathfrak{S}(U_A)}{\mathfrak{S}(H)}$$

in probability in the successive limits defining  $Z$ . Thus the conditional environment supplies only the one-point intensity, while every connected Palm activity is graded by  $(\log X)^{-d_H(A)}$  and weighted by the complete-union singular series.

If admissible overlapping translates exist, put

$$\delta(H) = \min_{\substack{t \neq 0 \\ H \cup (H+t) \text{ admissible}}} (|H \cup (H+t)| - |H|), \quad \mathcal{T}_H = \{t \neq 0 : H \cup (H+t) \text{ admissible}, |H \cup (H+t)| - |H| = \delta(H)\}$$

Then the cluster-start fraction has the first nontrivial expansion

$$\theta_X(H) = 1 - \frac{c_H}{(\log X)^{\delta(H)}} + o((\log X)^{-\delta(H)}), \quad c_H = \sum_{\substack{t \in \mathcal{T}_H \\ t < 0}} \frac{\mathfrak{S}(H \cup (H+t))}{\mathfrak{S}(H)}.$$

The set  $\mathcal{T}_H$  is symmetric under  $t \mapsto -t$ , and the cluster-start fraction counts, for each occurrence, only whether it has an overlapping predecessor, so the sum is restricted to  $t < 0$  and equals half the symmetric sum over  $\mathcal{T}_H$ . If no admissible overlap exists,  $\theta_X(H) = 1 + o((\log X)^{-M})$  for every fixed  $M$ . Higher corrections are the connected Ursell sums (the signed connected sums of cluster-expansion theory) of anchored polymers ordered by  $d_H(A)$ . These finite- $X$  Gibbs expansions are projectively consistent and subcritical, and after division by the low-zero intensity the cluster-start process is Poisson to each fixed order of this triangular expansion.

**Conjecture 30** (Topological expansion of non-Gaussian prime cumulants). *For every connected exact-overlap diagram in Conjecture 26, expand its connected singular series over the linking primes and form the linkage graph of each term, whose vertices are the motif occurrences and whose edges join two occurrences sharing a linking prime in that term. Let  $g$  be the first Betti number of the linkage graph. Once all larger coincidence strata have been subtracted, the total contribution of the terms of Betti number  $g$  is smaller than that of the spanning-tree terms with the same external motifs by at least  $(\log L)^{-g}$ . The complete leading coefficient at each logarithmic order is the sum of the linked-term weights with the corresponding Betti number. In particular, the first nonzero odd cumulant of a balanced motif statistic, a combination  $Y_c$  whose coefficient vector annihilates every coincidence-stratum form of Conjecture 27, is carried by incidence trees. For the single-prime field the fully spread connected part  $\kappa_{2m+1}^{\text{sp}}$ , defined by subtracting every coincidence stratum as in Conjecture 27, obeys*

$$\kappa_{2m+1}^{\text{sp}} = (c_m + o(1))L^m(\log L)^{m+1},$$

where  $c_m$  is the total connected tree weight. The raw cumulant adds the explicit coincidence diagrams of Conjecture 26, which carry powers of  $\log X$  and are computable term by term, so the tree law is a statement about the spread diagrams, exactly the regime of Kuperberg's singular-series sums.

## 6 Programme II: arithmetic first-arrival fields and class groups

For a prime modulus  $q$  and  $a \in \mathbb{F}_q^\times$ , let  $p(q, a)$  (argument order reversed from Part I's  $p(a, q)$ ) be the least prime in the class  $a$  and put

$$T_q(a) = \frac{\text{Li}(p(q, a))}{q-1}, \quad \mathcal{N}_{q,a} = \sum_{p \equiv a \pmod{q}} \delta_{\text{Li}(p)/(q-1)}.$$

The probability space chooses  $q$  uniformly among primes in  $[Q, 2Q]$  and then chooses  $a$  uniformly from  $\mathbb{F}_q^\times$ .

**Conjecture 31** (Connected first-arrival functional). *Let  $w_{r,Q}^c$  be the connected factorial-cumulant measure of the random point process  $\mathcal{N}_{q,a}$ . For every fixed  $r \geq 2$  there is a signed measure  $K_r$ , locally finite on the off-diagonal region  $\{u \in \mathbb{R}_+^r : u_i \text{ distinct}\}$ , such that, on that region (the diagonal strata carry strictly smaller normalizations, as in the stratification of Conjecture 26, and are excluded here),*

$$(\log Q)^{r-1} w_{r,Q}^c \implies K_r$$

vaguely, and  $K_1$  denotes the limiting intensity measure, which in this clock is Lebesgue measure on  $\mathbb{R}_+$ . The measure  $K_r$  is obtained by averaging, over the modulus aspect, the connected Hardy–Littlewood densities of  $r$  primes constrained to one residue class. For every nonnegative compactly supported  $f$  in a fixed uniform ball on which the Laplace functional is analytic,

$$\log \mathbb{E} \exp(-\langle f, \mathcal{N}_{q,a} \rangle) = \sum_{r \geq 1} \frac{(-1)^r}{r!} \int \prod_{j=1}^r (1 - e^{-f(u_j)}) dw_{r,Q}^c(\mathbf{u}),$$

and the asymptotic expansion obtained by inserting the limits  $K_r$  is uniform to every fixed connected order. In particular the entire survival function, not merely its first correction, is determined by the hierarchy  $(K_r)_{r \geq 2}$ .

**Conjecture 32** (Tested Gauss-polyspectral reciprocity for the least-prime field). *Set  $T_q(0) = \text{Li}(q)/(q-1)$  and centre the resulting field  $f_q$  on all of  $\mathbb{F}_q$ . Let  $\mathbf{A}_q$  and  $\mathbf{M}_q$  be its complete additive and multiplicative Fourier coefficient vectors, including the explicit rank-one coordinate needed to record the value at 0. There is then an explicit invertible Gauss matrix  $U_q$  with*

$$\mathbf{M}_q = U_q \mathbf{A}_q.$$

*Choose  $q$  uniformly among primes in  $[Q, 2Q]$ . For every finite family of degrees  $r_1, \dots, r_m$  and uniformly bounded fibrewise test tensors  $\Phi_{j,q}$ , define the scalar additive observables*

$$X_{j,Q}(q) = \langle \Phi_{j,q}, \mathbf{A}_q^{\otimes r_j} \rangle$$

*and the transported multiplicative observables*

$$Y_{j,Q}(q) = \left\langle (U_q^{-1})^{*\otimes r_j} \Phi_{j,q}, \mathbf{M}_q^{\otimes r_j} \right\rangle.$$

*For every modulus,  $X_{j,Q}(q) = Y_{j,Q}(q)$ , so all joint cumulants under the random-modulus law satisfy the exact finite identity*

$$\text{Cum}_Q(X_{1,Q}, \dots, X_{m,Q}) = \text{Cum}_Q(Y_{1,Q}, \dots, Y_{m,Q}).$$

*This is the probability-space-correct form of Gauss transport. It makes no reference to a cumulant of a deterministic coefficient vector.*

*The conjectural content is that, after normalization by the first nonzero connected scale, tests supported on additive zero-sum tensors converge to contractions of the Fourier transforms of the arrival kernels  $K_r$  from Conjecture 31. Transporting those tests fibrewise by the exact Gauss matrices and only then taking the modulus average gives the same limits as the corresponding multiple explicit-formula correlations of Dirichlet  $L$ -function zeros. The two descriptions agree test by test: every uniformly bounded fibrewise family of test tensors yields the same normalized limits along both routes, so the two limiting multilinear functionals coincide on the tested class itself, no completion being invoked.*

**Conjecture 33** (Rubinstein–Sarnak terminal chaos dichotomy). *Assume GRH and a  $q$ -aspect linear-independence/random-phase hypothesis for the low zeros of primitive Dirichlet  $L$ -functions. Let  $x_q$  satisfy  $\text{Li}(x_q)/(q-1) = \log(q-1)$  and let  $G_{q,A}(a)$  be the centred, variance-normalized low-zero field obtained from characters  $\chi \neq \chi_0$  and zeros  $|\gamma| \leq (\log q)^A$ , with the standard compensator  $-\frac{1}{2} \text{Var } G_{q,A}(a)$ . Define the random empirical chaos measure*

$$\mathcal{M}_{q,A}(dg) = \frac{1}{q-1} \sum_{a \in \mathbb{F}_q^\times} \exp\left(G_{q,A}(a) - \frac{1}{2} \text{Var } G_{q,A}(a)\right) \delta_{G_{q,A}(a)}(dg).$$

*Let  $b_Q$  be the smallest level with  $\mathbb{E}_q \#\{a : T_q(a) > b_Q\} \leq 1$ . In the successive limits  $Q \rightarrow \infty$  and then  $A \rightarrow \infty$ , the pair consisting of  $\mathcal{M}_{q,A}$  and*

$$\sum_a \delta_{(G_{q,A}(a), T_q(a) - b_Q)}$$

*converges jointly. Conditionally on the limiting chaos measure  $\mathcal{M}$ , the terminal cluster-start process is Poisson with directing measure*

$$e^{-x} \mathcal{M}(dx).$$

At finite modulus the cluster-start thinning enters through the terminal extremal index  $\theta_{\text{occ}}(q)$ , the analogue for the last classes of the cluster-start fraction  $\theta_X(H)$  of Conjecture 29, together with the terminal connected decorations supplied by the hierarchy of Conjecture 31. Both corrections enter at the first nonzero connected scale and vanish in the successive limit, so  $\theta_{\text{occ}}(q) \rightarrow 1$  and the limiting directing measure carries no cluster factor.

The nature of the directing measure is governed by the covariance-energy statistic

$$\mathfrak{E}_{q,A} = \frac{1}{(q-1)^2} \sum_{a,b \in \mathbb{F}_q^\times} \text{Cov}(G_{q,A}(a), G_{q,A}(b))^2.$$

If  $\mathfrak{E}_{q,A} \rightarrow 0$  along the successive limit,  $\mathcal{M}$  self-averages to the deterministic exponentially tilted Gaussian law and the terminal process is an ordinary decorated Poisson process. If  $\liminf \mathfrak{E}_{q,A} > 0$  and the Gaussian-chaos second moments are uniformly integrable,  $\mathcal{M}$  is nondegenerate and the limit is a genuine Cox process. No nondegeneracy is asserted without this criterion.

**Conjecture 34** (Nonlinear spectral response calculus for arithmetic first arrivals). *Let a sequence of modulus ensembles have a small explicit-formula perturbation of the arrival intensity*

$$V_Q(a, t) = \sum_{\chi \in \mathcal{C}_Q} v_{Q,\chi}(t) \chi(a), \quad \|V_Q\|_T \rightarrow 0,$$

where  $\mathcal{C}_Q$  has uniformly bounded cardinality and, on every fixed interval  $[0, T]$ , the canonical norm is  $\|V\|_T = \sup_{a, 0 \leq t \leq T} |V(a, t)|$ . Assume that the connected hierarchy of Conjecture 31 satisfies an exponential cluster bound there, so its Laplace functional is analytic in a fixed  $\|\cdot\|_T$ -ball. If  $S_Q(a, t)$  and  $S_Q^{(0)}(t)$  are the perturbed and ordinary survival functions, then for every fixed  $R$

$$\log \frac{S_Q(a, t)}{S_Q^{(0)}(t)} = \sum_{m=1}^R \mathcal{R}_m[V_Q^{\otimes m}](a, t) + o(\|V_Q\|_T^R)$$

uniformly on compact  $t$ -ranges. The causal Volterra operator  $\mathcal{R}_m$  is the universal linked-cluster contraction of the full hierarchy  $(K_r)_{r \geq 1}$  with  $m$  marked perturbation insertions, and truncating the connected hierarchy at order  $J$  gives a convergent approximation as  $J \rightarrow \infty$ .

If the input character support is  $\mathcal{C}_Q$ , the  $m$ th response has Fourier support contained in the products of at most  $m$  input characters, with the reverse inclusion possibly failing through cancellation or the trivial character. In particular the linear response preserves character rank, while any new harmonics first appear at quadratic order with coefficients determined by connected three-point arrival correlations.

**Conjecture 35** (Local-information capacity of odd class groups). *Fix an odd prime  $\ell$ . Sample negative fundamental discriminants  $D \in [-2Y, -Y]$ , conditioned on a fixed coarse two-primary invariant. For a set of odd primes  $S_Y$ , put  $M_Y = \prod_{p \in S_Y} p$ , let  $\Sigma_Y(D) = ((D/p))_{p \in S_Y}$  be the attainable sign vector, and let  $H_Y$  be the logarithm of the number of attainable cells. Let  $\delta_Y(\sigma)$  be the exact CRT density of the sign cell after imposing quadratic reciprocity and the fixed two-primary data (for a single odd unramified prime the density of each sign is  $p/(2(p+1))$ ), and an unconditioned cell density is the corresponding product), and let  $\mathcal{D}_Y$  be the ambient discriminant family. Write*

$$\Delta_Y(S_Y) = \sup_{\sigma} \left| \frac{\#\{D \in \mathcal{D}_Y : \Sigma_Y(D) = \sigma\}}{|\mathcal{D}_Y| \delta_Y(\sigma)} - 1 \right|$$



over attainable cells with  $|\mathcal{D}_Y|\delta_Y(\sigma) \geq \log Y$ .

There is a sequence  $\varepsilon_Y \rightarrow 0$ , independent of  $S_Y$  in the subcritical range, such that uniformly over those cells

$$d_{\text{TV}}(\mathcal{L}(\text{Cl}(D)[\ell^\infty] \mid \Sigma_Y = \sigma, 2\text{-data}), \mathcal{L}_{\text{CL}, \ell}) \leq \varepsilon_Y + O(\Delta_Y(S_Y)),$$

and the analogous bound holds for every fixed Cohen–Lenstra moment. In particular, for every  $\varepsilon > 0$ , the conclusion is uniform whenever

$$H_Y \leq (1 - \varepsilon) \log Y, \quad \log M_Y \leq (\tfrac{1}{2} - \varepsilon) \log Y, \quad \Delta_Y(S_Y) = o(1).$$

Since every odd prime contributes at most  $\log 3 \leq \log p$  of cell entropy,  $H_Y \leq \log M_Y$  throughout, so the conductor constraint is the binding one in the displayed range, while the entropy governs the impossibility clause below. At  $H_Y \geq (1 + \varepsilon) \log Y$  no uniform statement is possible. Up to the conductor scale at which  $\Delta_Y$  becomes macroscopic, every first-order failure of local independence factors through this discriminant-cell discrepancy, and there is no earlier odd-class-group obstruction.

## 7 Programme III: finite logarithms, regulators, and algebraic tori

For  $p \nmid a$ , write

$$q_p(a) = \frac{a^{p-1} - 1}{p} \pmod{p}.$$

For a generating tuple  $g = (g_1, \dots, g_s)$ , let  $R(g)$  be its multiplicative relation lattice and let  $\mathbb{T}_g \subset (\mathbb{R}/\mathbb{Z})^s$  be the identity component of its annihilator. Its dimension is the saturated multiplicative rank. Write  $e(x) = e^{2\pi i x}$ .

**Conjecture 36** (Kummer–Haar law on relation tori). *Let  $\Gamma \leq \mathbb{Q}^\times$  be finitely generated and torsion-free, saturated modulo torsion (if  $x^n \in \Gamma \cdot \{\pm 1\}$  for some  $n \geq 1$  then  $x \in \Gamma \cdot \{\pm 1\}$ ), and let  $g$  be any ordered generating tuple. As  $p$  varies in every fixed compatible cyclotomic–Kummer Chebotarev class, the vectors*

$$\frac{(q_p(g_1), \dots, q_p(g_s))}{p} \in (\mathbb{R}/\mathbb{Z})^s$$

*equidistribute with Haar measure on  $\mathbb{T}_g$ . More strongly, for every finite collection of compatible Chebotarev restrictions, the conditional empirical measures converge to the Haar disintegration on the corresponding components, and the support is contained in no proper closed coset of  $\mathbb{T}_g$ . The law is invariant under changing generators and depends on  $\Gamma$  only through its saturation and the fixed Kummer conditioning data.*

**Conjecture 37** (Rank-dimensional large sieve for finite logarithms). *Choose a Smith-normal-form basis of the character lattice  $X^*(\mathbb{T}_g) = \mathbb{Z}^s/R(g)^{\text{sat}}$ . There is a constant  $\delta_\Gamma > 0$  such that, for every  $\varepsilon > 0$ , every  $M \leq X^{\delta_\Gamma}$ , and all complex weights  $a_p$  supported on  $X < p \leq 2X$ ,*

$$\sum_{\substack{m \in X^*(\mathbb{T}_g) \\ \|m\|_\infty \leq M}} \left| \sum_{X < p \leq 2X} a_p e\left(\frac{\langle m, q_p(g) \rangle}{p}\right) \right|^2 \ll_{\Gamma, \varepsilon} (M^{\text{rank } \Gamma} + \pi(X)) X^\varepsilon \sum_p |a_p|^2.$$

The same inequality holds after every fixed compatible cyclotomic–Kummer restriction, and for matrix-valued weights with the Hilbert–Schmidt norm. The exponent  $\delta_\Gamma$  is invariant under changing the generating tuple.

**Conjecture 38** (Mesoscopic-to-lattice shrinking-target transition). *Let  $\Gamma \leq \mathbb{Q}^\times$  be torsion-free of rank  $r$  and saturated modulo torsion, and choose a basis of  $\Gamma$ , identifying its relation torus with  $(\mathbb{R}/\mathbb{Z})^r$ . For integers  $w_p$  with  $1 \leq w_p = o(p)$ , let  $E_p(w)$  be the event that every coordinate of the Fermat-quotient vector has a representative of absolute value at most  $w_p$ .*

*Mesoscopic universality. If  $w_p \rightarrow \infty$ , then in every range in which the Haar mass*

$$\lambda(X) = \sum_{p \leq X} \left( \frac{2w_p + 1}{p} \right)^r$$

*tends to infinity, the count of  $E_p(w)$  is asymptotic to  $\lambda(X)$ . On prime blocks with bounded Haar mass, the randomly translated event process converges to a Poisson process with that mass.*

*Lattice transition. For every fixed finite target set  $B \subset \mathbb{Z}^r$ , there is an arithmetic regulator intensity  $\kappa_{\Gamma,B} \geq 0$  such that*

$$\#\{p \leq X : q_p|_\Gamma \in B\} = (\kappa_{\Gamma,B} + o(1)) \sum_{p \leq X} p^{-r}$$

*whenever the sum diverges, while only finitely many such primes occur when  $r \geq 2$ . As a separately falsifiable genericity clause, for Kummer-generic  $\Gamma$  (the Kummer fields  $\mathbb{Q}(\zeta_m, \Gamma^{1/m})$  have the maximal degree  $\varphi(m)m^r$  for every  $m$ ) and  $B = \{0\}$  one has  $\kappa_{\Gamma,B} = 1$ . The transition from Haar universality to the arithmetic intensity occurs only when  $w_p$  remains bounded.*

For a  $p$ -adic unit  $a$ , write  $\langle a \rangle_p$  for the Teichmüller lift of its residue, and for a prime  $\mathfrak{p} \mid p$  of a number field write  $\tau_{\mathfrak{p}}$  for the Teichmüller lift in the completion at  $\mathfrak{p}$ . Define  $q_{p,k}(a) = p^{-1} \log(a \langle a \rangle_p^{-1}) \pmod{p^k}$ . The matrix entries below are the  $\mathfrak{p}$ -adic analogues of  $q_{p,k}$ , and on  $\mathbb{Q}$  with  $k = 1$  one has  $q_{p,1}(a) \equiv -q_p(a) \pmod{p}$ , the Fermat quotient up to sign.

**Conjecture 39** (Global-lattice horizontal  $p$ -adic regulator-matrix law). *Let  $K/\mathbb{Q}$  be a fixed Galois number field with group  $G$ , and let  $\Gamma \leq \mathcal{O}_K^\times$  be torsion-free and saturated modulo  $\mu(K)$ , with ordered basis  $\epsilon_1, \dots, \epsilon_s$ . Define the fixed global relation lattice*

$$\mathcal{R}_\Gamma = \left\{ (c_{\sigma j}) \in \mathbb{Z}^{G \times s} : \prod_{\sigma \in G} \prod_{j=1}^s \sigma(\epsilon_j)^{c_{\sigma j}} \in \mu(K) \right\}.$$

*Let  $\mathbb{T}_\Gamma \subset (\mathbb{R}/\mathbb{Z})^{G \times s}$  be its annihilator. This lattice contains the norm and Galois relations and is independent of the chosen basis up to the natural integral change of coordinates.*

*For a rational prime  $p$  splitting completely in  $K$ , label the primes above  $p$  by  $G$ , use the Teichmüller lift in each completion, and form the depth- $k$  matrix*

$$L_{p,k}(\Gamma) = \left( p^{-1} \log_{\mathfrak{p}}(\epsilon_j \tau_{\mathfrak{p}}(\bar{\epsilon}_j)^{-1}) \pmod{p^k} \right)_{\mathfrak{p} \mid p, 1 \leq j \leq s}.$$

*It lies in the annihilator of  $\mathcal{R}_\Gamma \otimes \mathbb{Z}/p^k \mathbb{Z}$ . As  $p$  varies through any fixed compatible Frobenius class,  $L_{p,1}/p$  equidistributes with Haar measure on  $\mathbb{T}_\Gamma$ , and the depth- $k$  matrices are Haar at every fixed depth  $k$ , with the depth- $(k+1)$  law pushing forward to the depth- $k$  law under reduction (the*

moduli vary with  $p$ , so this is a per-depth statement with compatible marginals, not Haar on one fixed group).

Let  $\mathcal{D}$  be a basis-invariant determinantal or Smith stratum in this projective module, and let  $\mu_{p,k}(\mathcal{D})$  be its exact local Haar mass. The primes for which  $L_{p,k} \in \mathcal{D}$  have counting law governed by  $\sum_{p \leq X} \mu_{p,k}(\mathcal{D})$ . For a determinantal or Smith stratum of codimension  $c$ , whose point counts are polynomial in  $p$ ,  $\mu_{p,1}(\mathcal{D}) = p^{-c} + O(p^{-c-1})$ , and codimension one has a Poisson point process in  $\log \log p$  after uniform random translation, while codimension at least two has a summable large-prime tail together with the exact small-prime atoms.

**Conjecture 40** (Functorial horizontal finite logarithms on algebraic tori). *Let  $T/\mathbb{Q}$  be an algebraic torus, let  $\mathcal{T}$  be its smooth integral model away from a finite set, and let  $\Gamma \leq T(\mathbb{Q})$  be finitely generated. Let  $S$  be the identity component of the Zariski closure of  $\Gamma$ . At a good unramified prime  $p$ , let  $\tau_p(\bar{P})$  be the canonical prime-to- $p$  torsion lift of the reduction of  $P \in \Gamma$  and define*

$$\ell_p(P) = p^{-1} \log_T(P \tau_p(\bar{P})^{-1}) \pmod{p} \in \text{Lie}(S)(\mathbb{F}_p).$$

Here  $\text{Lie}(S)(\mathbb{F}_p)$  is identified with  $(\mathbb{Z}/p)^{\dim S}$  through a fixed integral basis of the Lie algebra of  $\mathcal{T}$ , with representatives in  $[0, p)$ ; a different basis changes the matrix by a fixed element of  $\text{GL}_{\dim S}(\mathbb{Z})$ , which preserves the Haar law and the shrinking-target intensities. For any generators  $P_1, \dots, P_s$  of  $\Gamma$ , the normalized matrix  $(\ell_p(P_i)/p)_i$  equidistributes, in every compatible Frobenius class, with Haar measure on the real relation subtorus cut out jointly by the algebraic subgroup  $S$  and the integral relations among the  $P_i$ . For every morphism of tori  $\phi : T \rightarrow T'$ ,

$$\ell_p(\phi(P)) = d\phi(\ell_p(P)),$$

and the horizontal Haar measures push forward under the induced map. Isogenies preserve the shrinking-target intensities after the explicit finite kernel correction.

## 8 Programme IV: arboreal arithmetic dynamics

For a degree- $d$  map  $f$  and basepoint  $a$ , write  $K_n = \mathbb{Q}(f^{-n}(a))$ ,  $G_n = \text{Gal}(K_n/\mathbb{Q})$ , and let  $\mu_n(C) = |C|/|G_n|$  on conjugacy classes. For primes  $p \leq X$  unramified in  $K_n$ , let  $\hat{\mu}_{n,X}$  be the empirical Frobenius measure.

**Conjecture 41** (Entropy-conductor profile of arboreal resolution). *For a degree- $d$  map  $f$  and level  $n$ , let  $G_{n,f}$  be the arboreal quotient and  $\mu_{n,f}$  its Haar measure on conjugacy classes. For  $N$  independent samples from  $\mu_{n,f}$  define the exact sampling obstruction*

$$\mathcal{S}_{n,f}(N) = \mathbb{E} d_{\text{TV}} \left( N^{-1} \sum_{j=1}^N \delta_{Z_j}, \mu_{n,f} \right).$$

For each nontrivial irreducible Artin representation  $\rho$  of  $G_{n,f}$ , let  $q(\rho)$  be its Artin conductor and define the GRH conductor energy

$$\mathcal{A}_{n,f}(X) = \frac{\log X}{\sqrt{X}} \left( \sum_{\rho \neq 1} (\dim \rho)^2 (\log q(\rho) + \dim(\rho) \log X)^2 \right)^{1/2}.$$

This is a canonical character-theoretic replacement for a single discriminant proxy. For a parameter family  $\mathcal{F}(B)$  put

$$n_{\mathcal{F}}^*(B, X; \eta) = \max \left\{ n : \mathbb{E}_{f \in \mathcal{F}(B)} [\mathcal{S}_{n,f}(\pi(X)) + \mathcal{A}_{n,f}(X)] \leq \eta \right\}.$$

Assume the generic arboreal image is open, exceptional parameters have density zero, Artin holomorphy and GRH hold at the levels considered, and the family is in the large-sieve range of Conjecture 42 below. Then the actual empirical resolution depth, defined by the same total-variation threshold with prime Frobenius samples, differs from  $n_{\mathcal{F}}^*(B, X; \eta)$  by  $O_{\mathcal{F}, \eta}(1)$ .

More precisely, suppose, uniformly for a density-one set of parameters, that the class measures have a limiting complete Rényi profile

$$\psi_{\mathcal{F}}(\alpha) = \lim_{n \rightarrow \infty} d^{-n} \log \sum_C \mu_{n,f}(C)^\alpha$$

uniformly for  $\alpha$  in a fixed interval containing  $[1/2, 2]$ , and the corresponding Artin-conductor energies have finite positive exponential profiles on the same  $d^n$  scale. Then

$$n_{\mathcal{F}}^*(B, X; \eta) = \log_d \log X + O_{\mathcal{F}, \eta}(1).$$

Two families of the same degree with boundedly different complete Rényi and conductor profiles have resolution depths differing by  $O(1)$ , and in particular a fixed-index open subgroup changes only the bounded shift.

**Conjecture 42** (Arboreal family large sieve and cumulant independence). *Let  $\mathcal{F} \rightarrow \mathbb{A}^1$  be a generically finite-index arboreal family. For each level  $n$  and irreducible representation  $\rho$  of the generic quotient  $G_n$ , let  $\mathcal{V}_{n,\rho}$  be the associated middle-extension sheaf on the good parameter locus (the canonical extension of the corresponding local system across the bad parameters). Define the geometric conductor budgets*

$$\mathfrak{C}_{n,r} = \sum_{\rho_1, \dots, \rho_r} \left( \prod_{j=1}^r \dim \rho_j \right)^2 \prod_{j=1}^r \text{cond}(\mathcal{V}_{n,\rho_j})^2, \quad \mathfrak{C}_n = \mathfrak{C}_{n,1},$$

where each conductor includes rank, the number of singular points, and the tame-drop and Swan terms (the tame and wild ramification invariants at those points); for  $r \geq 2$  the budget is attached to the product trace function on the CRT parameter set, whose factors live over distinct characteristics, so its complexity is by definition this product of local conductors rather than the conductor of an external tensor product over a single base.

Square-root tier. For every  $\varepsilon > 0$  and all complex coefficients  $\alpha_{p,\rho}$ ,

$$\sum_{|c| \leq B} \left| \sum_{p \leq X} \sum_{\rho \neq 1} \alpha_{p,\rho} \text{tr } \rho(\text{Frob}_{p,c}) \right|^2 \ll_{\varepsilon, \mathcal{F}} (B + X^2 \mathfrak{C}_n^{1+\varepsilon}) \sum_{p, \rho} |\alpha_{p,\rho}|^2,$$

uniformly when  $X^2 \mathfrak{C}_n^{1+\varepsilon} \leq B^{1-\varepsilon}$ , after deleting the singular fibres.

Connected tier. For every fixed  $r \geq 3$ , centred trace functions at distinct primes have family-averaged connected cumulants with square-root cancellation in the parameter-family count, uniformly in the range  $X^r \mathfrak{C}_{n,r}^{1+\varepsilon} \leq B^{1-\varepsilon}$ , and equivalently, after normalization, every fixed connected Frobenius cumulant tends to zero. An exponential connected-moment bound makes this convergence uniform in  $r$  over the range needed for occupancy functionals.

Consequently, whenever the analytic Chebotarev error is  $o(1)$  but the iid sampling functional of  $\mu_n$  is nonvanishing, the complete conjugacy-cylinder count vector, averaged over parameters, has the nonuniform multinomial occupancy law of independent samples from  $\mu_n$ , including its missing-mass, collision, and total-variation profile. The multinomial conclusion is derived from the connected tier, not from the  $L^2$  bound alone.

**Conjecture 43** (Coloured dynatomic Galois-factor process). *For  $f_c(x) = x^d + c$ , let the critical exact-period polynomial factor as  $\Psi_n(c) = \prod_{j=1}^{s_n} F_{n,j}(c)$  over  $\mathbb{Z}$ . Sample  $c$  from a dyadic height interval away from postcritically finite, discriminant, and fixed-resultant loci.*

*For every fixed  $n$ , condition on any fixed finite local valuation state and normalize the prime factors of each  $F_{n,j}(c)$  by that component's residual logarithmic mass. The component-coloured factor processes converge jointly to independent Frobenius-marked Poisson–Dirichlet PD(1) processes, with the mark in colour  $j$  given by the fixed-point-size-biased conjugacy law in the splitting field of  $F_{n,j}$ . The total logarithmic mass of ramified factors tends to zero.*

*Uniformly for  $n = n(H)$  with  $\deg \Psi_n = o(\log H)$ , every fixed correlation functional whose support lies inside the available level of distribution converges to the corresponding restricted-support correlation of these coloured marked processes. No full-process claim is made at growing degree without level-one distribution.*

**Conjecture 44** (Complexity-uniform primitive valuation process for dynatomic values). *For the critical exact-period polynomial  $\Psi_n(c)$ , remove its canonical greatest common divisor in  $\mathbb{Q}[c]$  with the earlier critical-orbit product  $\prod_{m < n} f_c^m(0)$ , in the notation of Conjecture 43, and denote the result by  $\Psi_n^{\text{new}}$ . Define the logarithmic arithmetic complexity*

$$\mathfrak{H}_n = \deg \Psi_n^{\text{new}} + \log^+ H_{\text{coeff}}(\Psi_n^{\text{new}}) + \log^+ |\text{Disc}(\Psi_n^{\text{new}})| + \sum_{m < n} \log^+ |\text{Res}(\Psi_n^{\text{new}}, f_c^m(0))|,$$

*where zero resultants have already been removed by the canonical gcd. For each prime  $p$ , let  $\nu_{n,p}$  be the exact Haar law on  $\mathbb{Z}_p$  of the primitive valuation*

$$V_{n,p}(c) = v_p(\Psi_n^{\text{new}}(c)) \mathbf{1}_{p \nmid f_c^m(0) \ \forall m < n}.$$

*All discriminant, resultant, and collision primes remain inside these local laws.*

*For every control function  $\omega(H) = o(\log H)$ , uniformly over levels with  $\mathfrak{H}_n \leq \omega(H)$ , the finite-prime valuation vectors for  $c$  in a dyadic height interval converge to the corresponding marginals of  $\otimes_p \nu_{n,p}$ . In addition, for every  $\eta > 0$ ,*

$$\lim_{z \rightarrow \infty} \limsup_{H \rightarrow \infty} \sup_{n: \mathfrak{H}_n \leq \omega(H)} \mathbb{P} \left( \sum_{p > z} (\log p) (V_{n,p} - 1)_+ > \eta \log |\Psi_n^{\text{new}}(c)| \right) = 0,$$

*and the analogous unnormalized tail  $\sum_{p > z} \mathbb{P}(V_{n,p} \geq 2)$  tends to zero whenever the local squarefree product is positive. These two clauses give convergence in the topology generated by bounded cylinder functions and the logarithmic squarefull-mass functional.*

*Consequently*

$$\mathbb{P}(\Psi_n^{\text{new}}(c) \text{ is squarefree}) = \prod_p \tilde{\nu}_{n,p}(\{0, 1\}) + o(1)$$

*uniformly in the stated complexity range, where  $\tilde{\nu}_{n,p} = \nu_{n,p}$  except at the finitely many collision primes dividing one of the resultants  $\text{Res}(\Psi_n^{\text{new}}, f_c^m(0))$ , at which  $\tilde{\nu}_{n,p}$  is the law of the raw valuation  $v_p(\Psi_n^{\text{new}}(c))$ , and the logarithmic squarefull mass converges to the law determined by the complete local product.*



**Conjecture 45** (Divisor-sensitive classification of dynamical gcd height). *Let  $F = (f, g)$  be a split map on  $(\mathbb{P}^1)^2$ , with  $f, g$  disintegrated (in the sense of Medvedev and Scanlon [116], so neither is linearly conjugate to a power map, a Chebyshev map, or a close relative of these) of the same degree, and let  $P = (a, b)$  have positive canonical heights. Let  $D_1, D_2$  be the two coordinate-zero divisors and let  $E$  be the exceptional divisor of the blow-up at  $D_1 \cap D_2$ . Then*

$$\limsup_{n \rightarrow \infty} \frac{h_E(F^n(P))}{\max\{h(f^n(a)), h(g^n(b))\}} > 0$$

*if and only if there exist  $m \geq 0$  and an irreducible  $F$ -periodic curve  $Z$  containing  $F^m(P)$ , with  $Z$  not contained in  $D_1 \cup D_2$  (automatic when both coordinates have positive canonical height), such that, on the normalization of  $Z$ , the pullbacks of  $D_1$  and  $D_2$  have a common nonzero effective component. In the absence of such eventual entry into a divisor-compatible periodic curve, the normalized gcd height tends to zero.*

*The converse is unconditional in every setting where the required Vojta inequality for the blow-up is known, and otherwise is explicitly conditional on that inequality. The same criterion extends to split maps on  $(\mathbb{P}^1)^r$  by replacing a common component with positive codimension-one intersection multiplicity among the pulled-back divisors.*

## 9 Programme V: adelic factorization and sieve flow

Throughout this programme  $P^+(m)$  and  $P^-(m)$  denote the largest and smallest prime factors of  $m$ , the constant  $\gamma$  is Euler's,  $\rho_f(p)$  counts the roots of  $f$  modulo  $p$ , and every sample  $n \in [N, 2N]$  is drawn uniformly.

**Conjecture 46** (Frobenius-marked Poisson–Dirichlet process). *Let  $f \in \mathbb{Z}[x]$  be irreducible with splitting field  $L$  and transitive Galois group  $G$ . Sample  $n \in [N, 2N]$ . Mark each unramified prime factor  $p$  of  $f(n)$  by its conjugacy class  $C \subset G$  and give it mass  $u = \log p / \log |f(n)|$ . The macroscopic marked process converges to a marked PD(1) process with conditional mark law*

$$\mathbb{P}(C \mid u) = \frac{|C|}{|G|} \text{fix}(C), \quad \text{fix}(C) \text{ the number of fixed roots of an element of } C,$$

*independent of  $u$ . Given the masses and every fixed finite local valuation state, the marks of finitely many macroscopic factors are asymptotically independent. Under a normal quotient equipped with the induced permutation representation, the marked process pushes forward functorially.*

**Conjecture 47** (Three-scale factorization of polynomial values). *Let  $f \in \mathbb{Z}[x]$  be irreducible, sample  $n \in [N, 2N]$ , choose  $y = y(N) \rightarrow \infty$  with  $\log y = o(\log N)$ , and write  $R_y = f(n)/S_y(f(n))$ , where  $S_y(m) = \prod_{p \leq y} p^{v_p(m)}$  is the  $y$ -smooth part. Jointly with the complete small-prime valuation field  $(v_p(f(n)))_{p \leq y}$ , the residual logarithmic factor process*

$$\sum_{p \mid R_y} v_p(R_y) \delta_{(\log p / \log |R_y|, \text{Frob}_p)}$$

*converges to the following conservation-corrected product object: the small valuations have their polynomial Kubilius law (independent exact local limit laws for the small-prime valuations), and conditional on their consumed logarithmic mass the residual process is a scale-invariant Poisson process of intensity  $du/u$ , marked by Conjecture 46, conditioned to have total mass*

one. The convergence is in total variation for the complete valuation field, the range  $\log y = o(\log N)$  being the classical Kubilius range for the field at  $f(x) = x$ , and in the sense of finite-dimensional distributions for the residual process jointly with the field. The small field and the unconditioned residual Poisson process are asymptotically independent, and all leading dependence after conditioning is the single residual-mass constraint.

For disjoint normalized size bands bounded away from zero, the unconditioned residual process has independent increments. Its mesoscopic restriction and its ranked macroscopic atoms yield respectively the scale-invariant factor process and the Frobenius-marked PD(1) partition. Thus the local Kubilius field is the boundary variable fixing the available mass, while the mesoscopic and macroscopic laws are two projections of one residual process, and no additional cross-scale coupling survives.

**Conjecture 48** (Adelic Gibbs gluing for reducible polynomial values). *Let  $f = \prod_{i=1}^s g_i$  be squarefree with irreducible components, sample  $n \in [N, 2N]$ , and put  $t = n/N$ . For each prime  $p$ , let  $\nu_p$  be the exact Haar law on  $\mathbb{Z}_p$  of the valuation vector  $(v_p(g_1(x)), \dots, v_p(g_s(x)))$ . For  $y \rightarrow \infty$  with  $\log y = o(\log N)$ , consider stable convergence jointly with the archimedean coordinate  $t$  and condition on  $\mathcal{L}_y = (v_p(g_i(n)))_{p \leq y, i \leq s}$ . After normalizing each component by its own residual logarithmic mass, the coloured Frobenius-marked factor processes converge to conditionally independent copies of the three-scale law of Conjecture 47, with the marks of Conjecture 46.*

Unconditionally, the joint process is the projective Gibbs mixture obtained by integrating those conditional product laws against Lebesgue measure in  $t$  and the adelic local law  $\bigotimes_p \nu_p$ , in the iterated order  $N \rightarrow \infty$  and then  $y \rightarrow \infty$ . Every bounded finite-dimensional Laplace functional is determined by

$$Z_p(\mathbf{z}) = \int_{\mathbb{Z}_p} \prod_i z_i^{v_p(g_i(x))} dx$$

together with the archimedean size profile  $(\log |g_i(Nt)|)_i$ . No additional colour coupling survives after this adelic–archimedean mixture.

**Conjecture 49** (Full multivariate adelic saddle law for joint smoothness). *Let  $f_1, \dots, f_s \in \mathbb{Z}[x]$  be pairwise coprime, primitive, and jointly admissible, meaning that for every prime  $p$  some residue  $x$  has  $p \nmid \prod_i f_i(x)$ , sample  $n \in [N, 2N]$ , and put  $t = n/N$ . For complex variables  $\mathbf{s}$  and smoothness bounds  $\mathbf{y}$ , define*

$$Z_{p,\mathbf{y}}(\mathbf{s}) = \int_{\mathbb{Z}_p} p^{-\sum_{i: p \leq y_i} (s_i - 1) v_p(f_i(x))} dx, \quad \widehat{Z}_{p,\mathbf{y}}(\mathbf{s}) = \frac{Z_{p,\mathbf{y}}(\mathbf{s})}{Z_{p,\mathbf{y}}(+\infty)}.$$

The denominator is the exact Haar probability that  $p$  divides none of the relevant  $f_i(x)$ . Put

$$\Phi_t(\mathbf{s}) = \sum_p \log \widehat{Z}_{p,\mathbf{y}}(\mathbf{s}) + \sum_i (s_i - 1) \log |f_i(Nt)|.$$

For  $f(x) = x$ , this gives the usual truncated zeta factor  $(1 - p^{-s})^{-1}$ . Define the joint local probability by the Perron-calibrated multivariate inverse Mellin integral

$$\mathcal{P}_t^{\text{joint}} = \frac{1}{(2\pi i)^s} \int_{\boldsymbol{\sigma}_t + i\mathbb{R}^s} e^{\Phi_t(\mathbf{s})} \prod_{i=1}^s \frac{ds_i}{s_i},$$

with the standard truncated-contour limit, where  $\boldsymbol{\sigma}_t$  is the unique real saddle in the admissible region. The factors  $s_i^{-1}$  are part of the canonical cumulative smoothness event, and for one variable they recover the Hildebrand–Tenenbaum Perron normalization [115].

Uniformly when every  $u_i(t) = \log |f_i(Nt)| / \log y_i$  lies in a fixed compact saddle-point region,

$$\mathbb{P}(P^+(f_i(n)) \leq y_i \ \forall i) \sim \int_1^2 \mathcal{P}_t^{\text{joint}} dt.$$

The integral has the full multivariate saddle expansion. Writing  $H_t = \nabla^2 \Phi_t(\boldsymbol{\sigma}_t)$ , its leading term includes the joint action, the amplitude  $\prod_i \sigma_{i,t}^{-1}$ , and  $(\det H_t)^{-1/2}$ , with the usual lattice correction when the valuation span is proper.

If  $\mathcal{P}_{i,t}^{\text{marg}}$  are the identically calibrated one-polynomial probabilities, then  $\mathcal{P}_t^{\text{joint}} / \prod_i \mathcal{P}_{i,t}^{\text{marg}}$  is the complete ratio of joint and marginal actions, Perron amplitudes, Hessian determinants, and lattice corrections. Its Euler-product component is the ratio of the exact  $p$ -adic partition functions evaluated at the joint and marginal saddles. None of these factors may be replaced by a scalar Euler correction at the marginal saddle.

**Conjecture 50** (Buchstab–Bateman–Horn component flow). *Let  $f \in \mathbb{Z}[x]$  be primitive, irreducible, and admissible ( $\rho_f(p) < p$  for every prime  $p$ ), and sample  $n \in [N, 2N]$ . For  $y$  with  $u_n = \log |f(n)| / \log y$  in a fixed compact subset of  $(1, \infty)$ , let  $\Omega_y(f(n))$  be the number of prime factors of  $f(n)$  exceeding  $y$ , counted with multiplicity, on the event  $P^-(f(n)) > y$ . Put*

$$V_f(y) = \prod_{p \leq y} \left(1 - \frac{\rho_f(p)}{p}\right).$$

Define the Buchstab component functions by

$$\omega_1(u) = \frac{1}{u}, \quad \omega_j(u) = \frac{1}{u(j-1)!} \int_{\substack{t_1, \dots, t_{j-1} \geq 1 \\ t_1 + \dots + t_{j-1} \leq u-1}} \frac{dt_1 \cdots dt_{j-1}}{t_1 \cdots t_{j-1}} \quad (j \geq 2),$$

so that  $\sum_{j \geq 1} \omega_j(u) = \omega(u)$ . Then for every fixed  $j$ ,

$$\mathbb{P}(P^-(f(n)) > y, \Omega_y(f(n)) = j) = V_f(y) e^\gamma \mathbb{E}_{n \in [N, 2N]} \omega_j(u_n) + o(V_f(y)),$$

uniformly on compact  $u$ -ranges. Conditional on the ordered logarithmic sizes of the  $j$  remaining factors, their unramified Frobenius marks are independent with the fixed-point-size-biased law of Conjecture 46. Summing over  $j$  gives the rough-value Buchstab law,  $j = 1$  gives the Bateman–Horn prime density, and  $j = 2$  gives the semiprime transition. Whenever  $\max_{N \leq n \leq 2N} |f(n)| < y^2$ , every rough value has exactly one prime factor above  $y$ , so the  $j = 1$  component exhausts the rough set, while the density formula itself remains conjectural.

## Part III

# Six conjectures across fields

Added to the deposit: 4 August 2026.

## 10 Relative polynomial pretentiousness

Let  $P \in \mathbb{Z}[x]$  be primitive (the greatest common divisor of its coefficients is 1) and irreducible, with positive leading coefficient. For a prime  $p$  write

$$\rho_P(p) = \#\{a \in \mathbb{Z}/p\mathbb{Z} : P(a) \equiv 0 \pmod{p}\}.$$

The weight  $\rho_P(p)/p$  is the first-order frequency with which  $p$  divides a polynomial value  $P(n)$ . A function  $f$  on the positive integers is completely multiplicative if  $f(mn) = f(m)f(n)$  for all  $m, n \geq 1$ . For completely multiplicative  $f, g$  with values in the closed complex unit disc, define the truncated  $P$ -relative pretentious distance

$$\mathbb{D}_P(f, g; X)^2 = \sum_{p \leq X} \frac{\rho_P(p)}{p} (1 - \operatorname{Re}(f(p)\overline{g(p)})), \quad \mathbb{D}_P(f, g)^2 = \lim_{X \rightarrow \infty} \mathbb{D}_P(f, g; X)^2 \in [0, \infty].$$

Every summand is nonnegative, so the truncated distance is nondecreasing in  $X$  and the limit exists. Throughout, a primitive Dirichlet character  $\chi$  of conductor  $c$  is regarded as the  $c$ -periodic completely multiplicative function on  $\mathbb{Z}$  that vanishes on integers not coprime to  $c$ , so that expressions such as  $\chi(P(a))$  are defined for every residue  $a$  modulo any multiple of  $c$ . For  $t \in \mathbb{R}$ , the comparison function  $n \mapsto \chi(n)n^{it}$  is completely multiplicative with values in the closed unit disc.

Let  $P_1, \dots, P_m \in \mathbb{Z}[x]$  be pairwise distinct primitive irreducible polynomials with positive leading coefficients and no fixed prime divisor, meaning  $\rho_{P_j}(p) < p$  for every  $j$  and every prime  $p$ . Assume multiplicative independence modulo constants:

$$\prod_{j=1}^m P_j(x)^{e_j} \in \mathbb{Q}^\times, \quad e_j \in \mathbb{Z} \implies e_1 = \dots = e_m = 0.$$

Write  $\mathbb{S}^1$  for the unit circle in the complex plane.

**Conjecture 51** (Relative polynomial pretentiousness inverse principle). *Let  $f_1, \dots, f_m : \{1, 2, 3, \dots\} \rightarrow \mathbb{S}^1$  be completely multiplicative. If*

$$\limsup_{X \rightarrow \infty} \left| \frac{1}{\log X} \sum_{\substack{n \leq X \\ P_j(n) > 0 \ \forall j}} \frac{\prod_{j=1}^m f_j(P_j(n))}{n} \right| > 0,$$

*then there exist primitive Dirichlet characters  $\chi_1, \dots, \chi_m$  and real numbers  $t_1, \dots, t_m$  such that the following three conditions hold simultaneously:*

- (i)  $\mathbb{D}_{P_j}(f_j, \chi_j(n)n^{it_j}) < \infty$  for every  $1 \leq j \leq m$ ,
- (ii)  $\sum_{j=1}^m t_j \deg P_j = 0$ ,
- (iii) for some common multiple  $q$  of the conductors of  $\chi_1, \dots, \chi_m$ ,

$$\frac{1}{q} \sum_{a \bmod q} \prod_{j=1}^m \chi_j(P_j(a)) \neq 0,$$

*the characters being extended by 0 on nonunits as above.*

## 11 Monotone threshold purification of vector paths

Let  $d, m \geq 1$ . Let  $v_1, \dots, v_m \in \mathbb{R}^d$  satisfy  $\|v_i\|_2 \leq 1$ , where  $\|\cdot\|_2$  is the Euclidean norm. Let  $a_i : [0, 1] \rightarrow [0, 1]$  be continuous nondecreasing functions with  $a_i(0) = 0$  and  $a_i(1) = 1$ , called a monotone activation schedule, and define the fractional path

$$F(t) = \sum_{i=1}^m a_i(t)v_i.$$

A threshold rounding of  $F$  is a choice of pairwise distinct  $\tau_1, \dots, \tau_m \in (0, 1]$  together with the pure path

$$G(t) = \sum_{i: \tau_i \leq t} v_i.$$

**Conjecture 52** (Monotone threshold purification of vector paths). *There exists a universal constant  $C < \infty$  such that for every  $d, m \geq 1$ , every  $(v_i)_{i \leq m}$ , and every monotone activation schedule  $(a_i)_{i \leq m}$  as above, some threshold rounding satisfies*

$$\sup_{0 \leq t \leq 1} \|G(t) - F(t)\|_2 \leq C\sqrt{d}.$$

## 12 Entropy dimension of random-free graphons

A graphon is a symmetric measurable function  $W : [0, 1]^2 \rightarrow [0, 1]$ , and  $W$  is random-free if it takes values in  $\{0, 1\}$  almost everywhere. Define the row pseudometric

$$d_W(x, y) = \int_0^1 |W(x, z) - W(y, z)| dz.$$

Identifying points at zero distance and completing yields a metric-measure space  $(\Omega_W, d_W, \rho_W)$ , the row space, where  $\rho_W$  is the pushforward of Lebesgue measure. The row space is Ahlfors  $s$ -regular if it is compact and there exist  $c, C, r_0 > 0$  such that

$$cr^s \leq \rho_W(B(x, r)) \leq Cr^s$$

for every  $x \in \text{supp } \rho_W$  and every  $0 < r < r_0$ . Let  $G(n, W)$  be the random labelled graph on  $[n] = \{1, \dots, n\}$  obtained from independent uniform  $U_1, \dots, U_n$  on  $[0, 1]$  by declaring  $ij$  an edge exactly when  $W(U_i, U_j) = 1$ , and let  $H(G(n, W))$  be its Shannon entropy with natural logarithms.

**Conjecture 53** (Entropy dimension of random-free graphons).

(i) [nonvacuity] *For every  $s \in (0, 2]$  there exists a random-free graphon whose row space is Ahlfors  $s$ -regular.*

(ii) [refined entropy-dimension law] *If the row space of a random-free graphon  $W$  is Ahlfors  $s$ -regular for some  $s > 0$ , then*

$$H(G(n, W)) = sn \log n + O_W(n).$$

**Proposition 1** (Entropy-dimension limit law). *If the row space of a random-free graphon  $W$  is Ahlfors  $s$ -regular for some  $s > 0$ , then  $H(G(n, W)) = sn \log n + O_W(n \log \log n)$ , and in particular  $\lim_{n \rightarrow \infty} H(G(n, W))/(n \log n) = s$ .*

**Remark 1.** *The refined assertion  $H(G(n, W)) = sn \log n + O_W(n)$  is a strictly stronger open refinement of clause (ii), and nothing in the mechanism below reaches it. The type-coding mechanism by itself yields only  $H(G(n, W)) = sn \log n + O_W(n \log \log n)$ : encoding a latent type for each vertex at metric resolution  $1/n$  costs about  $s \log n$  nats per vertex and predicts all but on the order of  $n$  of the edges, but recording which edges are mispredicted costs order  $n \log n$  if done naively, since each such edge must be located by a pair of labels. A two-scale argument, which first codes coarse types and then corrects within coarse classes, lowers the correction cost to  $O(n \log \log n)$  and stops there.*

### 13 A Hessian principle for isolated microcanonical graph phases

Fix a partition

$$[0, 1] = I_1 \sqcup \cdots \sqcup I_q, \quad |I_a| = \alpha_a > 0.$$

A coloured graphon is a symmetric measurable  $U : [0, 1]^2 \rightarrow [0, 1]$  considered together with this fixed colour partition. Let  $p(U) \in [0, 1]^D$  with  $D = q(q+1)/2$  denote the block-average vector

$$p_{ab}(U) = \frac{1}{|I_a||I_b|} \int_{I_a \times I_b} U(x, y) \, dx \, dy, \quad 1 \leq a \leq b \leq q.$$

A coloured finite graph is a finite simple graph  $F$  with a colour  $c(v) \in \{1, \dots, q\}$  attached to each vertex, and its coloured homomorphism density in  $U$  is

$$t(F, U) = \int_{[0, 1]^{V(F)}} \prod_{v \in V(F)} \alpha_{c(v)}^{-1} \mathbf{1}_{I_{c(v)}}(x_v) \prod_{uv \in E(F)} U(x_u, x_v) \prod_{v \in V(F)} dx_v.$$

Fix coloured finite graphs  $F_1, \dots, F_k$  and put  $\mathbf{t}(U) = (t(F_1, U), \dots, t(F_k, U))$ . Define the graphon entropy

$$\mathcal{S}(U) = \frac{1}{2} \int_{[0, 1]^2} [-U \log U - (1 - U) \log(1 - U)] \, dx \, dy,$$

with  $0 \log 0 = 0$ , and for a feasible constraint vector  $\mathbf{a}$  the contracted entropy

$$\Psi_{\mathbf{a}}(p) = \sup\{\mathcal{S}(U) : \mathbf{t}(U) = \mathbf{a}, p(U) = p\}.$$

The finite graphs are coloured with pinned classes. For each  $n$  set  $n_a = \lfloor \alpha_a n \rfloor$  and  $N_n = \sum_a n_a$ , and fix a partition of the vertex set  $[N_n]$  into colour classes  $V_1, \dots, V_q$  with  $|V_a| = \lfloor \alpha_a n \rfloor$  exactly, so that  $N_n = n + O(q)$ . For a graph  $G$  on  $[N_n]$  with these colour classes, partition each  $I_a$  into  $n_a$  consecutive intervals of length  $|I_a|/n_a$  assigned to the vertices of  $V_a$ , let  $U_G$  be the  $\{0, 1\}$ -valued coloured graphon equal to 1 on a product of two vertex intervals exactly when the corresponding pair is an edge, and put  $\mathbf{t}(G) = \mathbf{t}(U_G)$  and  $\hat{p}(G) = p(U_G)$ .

Fix  $0 < \eta < 1$  and assume the following regularity hypotheses.

(H1) The supremum of  $\mathcal{S}(U)$  over  $\mathbf{t}(U) = \mathbf{a}$  is attained at a unique coloured graphon  $U_*$ , up to measure-preserving maps preserving each colour class.

(H2) Writing  $p_* = p(U_*)$ , the feasible block-average set  $\{p(U) : \mathbf{t}(U) = \mathbf{a}\}$  is, near  $p_*$ , a  $C^3$  embedded manifold  $M \subset \mathbb{R}^D$ .

(H3) The restriction  $\Psi_{\mathbf{a}}|_M$  is  $C^3$  near  $p_*$  and has there a unique local maximum.

(H4) The quadratic form  $A = -D_M^2 \Psi_{\mathbf{a}}(p_*)$  is positive definite on the tangent space  $T = T_{p_*} M$ .

(H5) [transversally Lipschitz constraint correspondence] There exist  $\delta_0 > 0$  and  $L < \infty$  such that every coloured graphon  $U$  with  $\|p(U) - p_*\| \leq \delta_0$  and  $\|\mathbf{t}(U) - \mathbf{a}\|_{\infty} = \delta \leq \delta_0$  satisfies  $\text{dist}(p(U), M) \leq L\delta$ . In particular the conditioning window below confines  $\hat{p}_n$ , on the event  $\|\hat{p}_n - p_*\| \leq \delta_0$ , to an  $o(n^{-1})$  thickening of  $M$ .

(H6) [nonemptiness] For every sufficiently large  $n$  the conditioning set  $\{G : \|\mathbf{t}(G) - \mathbf{a}\|_{\infty} \leq n^{-1-\eta}\}$  over graphs  $G$  on  $[N_n]$  with the pinned colour classes is nonempty.

Let  $G_n$  be uniform on the conditioning set of (H6), let  $\hat{p}_n = \hat{p}(G_n)$ , and let  $\Pi_T$  be orthogonal projection of  $\mathbb{R}^D$  onto  $T$ .

**Conjecture 54** (Hessian principle for an isolated microcanonical phase). *Under hypotheses (H1)–(H6),*

$$n \Pi_T(\hat{p}_n - \mathbb{E} \hat{p}_n) \xrightarrow{d} N_T(0, A^{-1}),$$

where  $N_T(0, A^{-1})$  denotes the centred Gaussian law on  $T$  whose precision form is  $A$ .



## 14 Sharp stability of entropy idempotence on finite abelian groups

Let  $G$  be a finite abelian group and  $\mu$  a probability measure on  $G$ . Write  $H(\mu)$  for Shannon entropy with natural logarithms,  $\mu * \mu$  for the convolution, the law of  $X + Y$  with  $X, Y$  independent of law  $\mu$ , and

$$\Delta(\mu) = H(\mu * \mu) - H(\mu)$$

for the convolution entropy defect. If  $H \leq G$  is a subgroup and  $a \in G$ , let  $u_{a+H}$  be the uniform measure on the coset  $a + H$ . Total variation distance is

$$\|\mu - \nu\|_{\text{TV}} = \frac{1}{2} \sum_{x \in G} |\mu(x) - \nu(x)|.$$

**Conjecture 55** (Sharp stability of entropy idempotence). *For every finite abelian group  $G$  and every probability measure  $\mu$  on  $G$ , with entropies in natural logarithms,*

$$\inf_{\substack{H \leq G \\ a \in G}} \|\mu - u_{a+H}\|_{\text{TV}}^2 \leq \frac{2}{\log 2} (H(\mu * \mu) - H(\mu)),$$

and the constant  $2/\log 2$  is sharp.

**Remark 2.** *The equality case is verified. With  $X, Y$  independent of law  $\mu$ ,*

$$\Delta(\mu) = H(X + Y) - H(X) = H(X + Y) - H(X + Y | Y) = I(X + Y; Y) \geq 0,$$

where  $I$  denotes mutual information. If  $\Delta(\mu) = 0$  then  $X + Y$  is independent of  $Y$ , which forces the translates of  $\mu$  by the elements of its support to coincide, and hence forces  $\mu$  to be uniform on a coset of a subgroup. The right-hand side therefore vanishes exactly on the proposed extremizers.

## 15 Stationary convolution-entropy rigidity

Let  $K$  be a finite abelian group and  $\Omega = K^{\mathbb{Z}}$  with coordinatewise addition and the left shift  $\sigma$ . Let  $\mu$  be a  $\sigma$ -invariant ergodic Borel probability measure on  $\Omega$ , and write  $h_{\sigma}(\mu)$  for its Kolmogorov–Sinai entropy. Define the translational stabilizer

$$H_{\mu} = \{h \in K^{\mathbb{Z}} : (\tau_h)_* \mu = \mu\}, \quad \tau_h(x) = x + h,$$

a closed shift-invariant subgroup of  $K^{\mathbb{Z}}$ , and let  $\pi_{\mu} : K^{\mathbb{Z}} \rightarrow K^{\mathbb{Z}}/H_{\mu}$  be the quotient homomorphism. Let  $\mu * \mu$  be the law of  $X + Y$  for independent  $X, Y \sim \mu$ .

**Conjecture 56** (Stationary convolution-entropy rigidity). *For every shift-ergodic  $\mu$ ,*

$$h_{\sigma}(\mu * \mu) = h_{\sigma}(\mu) \iff h_{\sigma}((\pi_{\mu})_* \mu) = 0.$$

**Remark 3.** *The binary specialization deserves prominence. Take  $K = \mathbb{Z}/2\mathbb{Z}$ . By Kitchens' structure theory of group shifts [117], every closed shift-invariant subgroup of  $(\mathbb{Z}/2\mathbb{Z})^{\mathbb{Z}}$  is a group shift of finite type, and every proper one is the solution set of a nontrivial linear recurrence and hence finite. If  $H_{\mu}$  is finite the quotient map has finite fibres and  $h_{\sigma}((\pi_{\mu})_* \mu) = h_{\sigma}(\mu)$ , while if  $H_{\mu}$  is everything then  $\mu$  is the Haar measure, the uniform Bernoulli process, with entropy  $\log 2$ . Since  $h_{\sigma}(\mu * \mu) \geq h_{\sigma}(\mu)$  always, by conditioning on one summand, the conjecture therefore asserts: every ergodic binary process with  $0 < h_{\sigma}(\mu) < \log 2$  gains entropy under independent XOR, the coordinatewise sum modulo 2 with an independent copy.*

**Proposition 2** (Easy direction, conditional). Suppose  $h_\sigma((\pi_\mu)_*\mu) = 0$ . Assume the following two ingredients.

(a) The Abramov–Rokhlin entropy addition formula holds for the Haar-fibred compact group extension  $\pi_\mu : (\Omega, \mu, \sigma) \rightarrow (\Omega/H_\mu, (\pi_\mu)_*\mu, \sigma)$  and for the corresponding extension of  $\mu * \mu$ , so that each total entropy splits as the quotient entropy plus the common Haar fibre entropy, the conditional measures of  $\mu$  and of  $\mu * \mu$  over the quotient being Haar measures on cosets of  $H_\mu$ .

(b) The entropy of a factor of an independent joining is bounded by the sum of the entropies of the factors, so that  $(\pi_\mu)_*(\mu * \mu) = ((\pi_\mu)_*\mu) * ((\pi_\mu)_*\mu)$  has entropy at most  $2h_\sigma((\pi_\mu)_*\mu) = 0$ . Then  $h_\sigma(\mu * \mu) = h_\sigma(\mu)$ .

## Part IV

# The class-counting polynomial of graphical Lie algebras

*Added to the deposit: 5 August 2026.*

The twenty conjectures below concern a single canonical object, the class-counting polynomial of a finite graph, approached through the two-step nilpotent Lie algebras that realize it as a rank generating function. They form three programmes. A rigidity and reconstruction programme (Conjectures 57 to 66) asserts that coarse specializations of the polynomial already determine it and that the polynomial in turn determines classical invariants. A log-concavity programme (Conjectures 67 to 71) locates the exact strength of positivity that the coefficient arrays support, with the boundary calibrated by explicit counterexamples one level higher. A tree extremal programme (Conjectures 72 to 76) places the path and the star at the two ends of every ordering the polynomial induces on trees. The proved layer is separated from the conjectural one: the rank formula (Proposition 3) and the two lower bounds on the number of rank values in a tree (Lemma 2) are theorems with full proofs, joined by the attainment of those bounds by the path and the star (Proposition 4) and the reduction of the star majorization to a single anti-concentration bound (Proposition 5), and every conjecture is stated against that floor. Each statement is held to the standard of the rest of the collection: a canonical object, a stated mechanism, the nearest literature boundary, a first decisive theorem where one exists, and an honest failure mode. Two statements, Conjectures 63 and 71, have since been resolved false by explicit counterexamples recorded with them, a seven-vertex unicyclic pair and a connected eight-vertex graph respectively, and both are retained with their resolutions rather than renumbered.

For a finite simple graph  $G$  on vertex set  $V(G)$  with edge set  $E(G)$  and a field  $F$ , the *graphical Lie algebra*  $L_G(F)$  is the two-step nilpotent Lie algebra over  $F$  with basis  $\{v_i : i \in V(G)\}$  together with the central basis  $\{z_e : e \in E(G)\}$ , the bracket being  $[v_a, v_b] = z_{ab}$  for each edge  $\{a, b\}$  with  $a < b$ ,  $[v_b, v_a] = -z_{ab}$ , and all other brackets of basis vectors zero. For  $S \subseteq V(G)$  write  $N_G[S]$  for the closed neighbourhood of  $S$ , the set of vertices lying in  $S$  or adjacent to it, and  $c(G[S])$  for the number of connected components of the induced subgraph  $G[S]$ , and set

$$\rho_G(S) = |N_G[S]| - c(G[S]), \quad \rho_G(\emptyset) = 0.$$

**Proposition 3** (Rank formula). Let  $G$  be a graph on  $n$  vertices, let  $F$  be a field, and let  $x \in L_G(F)$  have vertex part supported on  $S$ , all coordinates  $x_i$  with  $i \in S$  being nonzero. Then

$$\text{rank}(\text{ad}_x) = \rho_G(S),$$

independently of the field and of the nonzero values.

The *class-counting polynomial* of  $G$  is

$$C_G(X, Y) = \sum_{S \subseteq V(G)} (X - 1)^{|S|} Y^{\rho_G(S)}.$$

The polynomial is Rossmann's [118]: he introduced it, proved that its specializations enumerate the conjugacy classes of graphical groups over  $\mathbb{F}_q$  by size, proved the multiplicativity  $C_{G \sqcup H} = C_G C_H$  over disjoint unions, and showed that the cardinalities of the connected dominating sets of  $G$  can be read off from it. Set

$$H_G(U, Y) = C_G(1 + U, Y), \quad F_G(X, Y) = X^{|E(G)|} C_G(X, X^{-1}Y), \quad f_G(X) = F_G(X, 1),$$

so that the coefficient of  $U^k Y^r$  in  $H_G$  counts the sets  $S$  with  $|S| = k$  and  $\rho_G(S) = r$ . At  $X = q$  the polynomial records the distribution of  $\text{rank}(\text{ad}_x)$  over  $L_G(\mathbb{F}_q)$ , by Proposition 3. The *subgraph-component polynomial* of Tittmann–Averbouch–Makowsky [119] is

$$Q_G(U, W) = \sum_{S \subseteq V(G)} U^{|S|} W^{c(G[S])}.$$

For a tree  $T$  on  $n$  vertices let  $s(T)$  be the number of distinct values taken by  $\rho_T$ , let  $\ell(T)$  be the number of leaves, and let  $\text{diam}(T)$  be the diameter.

**Lemma 2** (Two lower bounds on the number of rank values). *For every tree  $T$  on  $n \geq 2$  vertices,*

$$s(T) \geq n - \ell(T) + 2 \quad \text{and} \quad s(T) \geq \text{diam}(T) + 1.$$

**Proposition 4** (Attainment of the bounds: path and star). The path  $P_n$  has  $s(P_n) = n$  for every  $n \geq 2$ , and the star  $K_{1,n-1}$  has  $s(K_{1,n-1}) = 3$  for every  $n \geq 3$ . Both trees attain equality in each bound of Lemma 2.

## 16 Rigidity and reconstruction

**Conjecture 57** (Irreducibility and connectivity). *For every graph  $G$  with at least one vertex,  $H_G(U, Y)$  is irreducible in  $\mathbb{Z}[U, Y]$  if and only if  $G$  is connected.*

**Conjecture 58** (Recovery of the subgraph-component polynomial). *For all graphs  $G$  and  $H$ ,  $C_G = C_H$  implies  $Q_G = Q_H$ .*

**Conjecture 59** (Two-field rigidity). *If  $C_G(2, Y) = C_H(2, Y)$  and  $C_G(3, Y) = C_H(3, Y)$ , then  $C_G = C_H$ . Equivalently, the adjoint-rank distributions of the graphical Lie algebras over  $\mathbb{F}_2$  and  $\mathbb{F}_3$  determine them over every finite field.*

**Conjecture 60** (Hybrid rigidity). *If  $C_G(2, Y) = C_H(2, Y)$  and  $Q_G = Q_H$ , then  $C_G = C_H$ .*

**Conjecture 61** (Binary edge-deck reconstruction). *For graphs  $G$  and  $H$  with at least four edges each, if the multisets  $\{C_{G-e}(2, Y) : e \in E(G)\}$  and  $\{C_{H-e}(2, Y) : e \in E(H)\}$  coincide, then  $G$  and  $H$  are isomorphic.*

**Conjecture 62** (Bipartite binary rigidity). *For bipartite graphs  $G$  and  $H$ ,  $C_G(2, Y) = C_H(2, Y)$  implies  $C_G = C_H$ .*

**Conjecture 63** (Connected-pseudoforest binary rigidity, resolved false). *For  $G$  and  $H$  each a tree or a connected unicyclic graph,  $C_G(2, Y) = C_H(2, Y)$  implies  $C_G = C_H$ .*

*Resolution.* False: an explicit seven-vertex unicyclic pair shares the binary specialization of  $F_G$  but has distinct degree multisets, the unique violating class through nine vertices; the tree case survives exhaustively through fourteen vertices, the first decisive theorem the statement named.

**Remark 4** (Resolution). *The conjecture is false as stated: the connected unicyclic graphs on seven vertices with edge sets  $\{\{0, 3\}, \{0, 4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{3, 6\}, \{4, 5\}\}$  and  $\{\{0, 1\}, \{0, 3\}, \{0, 4\}, \{1, 2\}, \{2, 5\}, \{2, 6\}\}$  share  $C(2, Y)$  while their class-counting polynomials differ. The tree case survives exhaustively through fourteen vertices and remains open.*

**Conjecture 64** (Chordal binary rigidity). *For chordal graphs  $G$  and  $H$ ,  $C_G(2, Y) = C_H(2, Y)$  implies  $C_G = C_H$ .*

**Conjecture 65** (Recovery of the domination number). *For all graphs  $G$  and  $H$ ,  $C_G = C_H$  implies  $\gamma(G) = \gamma(H)$ , where  $\gamma$  is the domination number.*

**Conjecture 66** (Recovery of the independent domination number). *For all graphs  $G$  and  $H$ ,  $C_G = C_H$  implies  $i(G) = i(H)$ , where  $i$  is the independent domination number, the least cardinality of a set that is simultaneously independent and dominating.*

## 17 Log-concavity and support

**Conjecture 67** (Global double log-concavity). *For every graph  $G$  the coefficient sequence  $a$  of  $f_G(1 + Z)$  is log-concave, and so is its transform  $L(a)$  defined by  $L(a)_i = a_i^2 - a_{i-1}a_{i+1}$ , out-of-range terms being zero.*

**Conjecture 68** (Strict log-concavity for connected graphs). *For connected  $G$  with at least one edge,  $a_i^2 > a_{i-1}a_{i+1}$  at every interior position of the support of  $a$ , where  $a$  is the coefficient sequence of  $f_G(1 + Z)$ .*

**Conjecture 69** (Strict second-order log-concavity). *For connected  $G$  with at least one edge, the transform  $L(a)$  of Conjecture 67 is strictly log-concave at every interior position of its support.*

**Conjecture 70** (Rank-slice strict log-concavity). *For connected  $G$  and every attainable rank  $r$ , the slice  $A_{G,r}(Z) = [Y^r] F_G(1 + Z, Y)$ , when it is not a monomial, has strictly log-concave coefficients at every interior position of its support.*

**Conjecture 71** (Interval support at fixed rank, resolved false). *For every graph  $G$  and every attainable rank  $r$ , the set  $K_r(G) = \{|S| : \rho_G(S) = r\}$  is an interval of integers.*

*Resolution.* False: a connected eight-vertex graph has support sizes attaining rank five that skip size two, the unique violating order, while the transposed statement was already false; interval support still holds for all graphs through seven vertices.

**Remark 5** (Resolution). *The conjecture is false as stated: the connected eight-vertex graph with edge set  $\{\{0, 3\}, \{0, 6\}, \{0, 7\}, \{1, 4\}, \{1, 6\}, \{1, 7\}, \{2, 5\}, \{2, 6\}, \{3, 6\}, \{3, 7\}, \{4, 6\}, \{4, 7\}, \{5, 7\}\}$  attains rank five at support sizes  $\{1, 3, 4, 5, 6\}$ , skipping size two, and an exhaustive scan of all graphs on eight vertices finds exactly two such graphs, both connected, and none smaller.*

## 18 Tree extremality and rigidity

**Conjecture 72** (Coefficientwise tree extremality). *For every tree  $T$  on  $n$  vertices,*

$$f_{P_n}(1 + Z) \leq f_T(1 + Z) \leq f_{K_{1,n-1}}(1 + Z)$$

*coefficientwise, with equality only for the path and the star respectively.*

**Conjecture 73** (Star rank-majorization). *For every  $n \geq 5$  and every prime power  $q$ , the decreasingly ordered adjoint-rank distribution of  $L_{K_{1,n-1}}(\mathbb{F}_q)$  majorizes that of  $L_T(\mathbb{F}_q)$  for every  $n$ -vertex tree  $T$ .*

**Proposition 5** (Reduction of the star majorization). Let  $T$  be a tree on  $n \geq 3$  vertices and  $q$  a prime power, and write  $\pi_{T,q}(r) = q^{-n} \sum_{\rho_T(S)=r} (q-1)^{|S|}$  for the adjoint-rank distribution of a uniformly random element of  $L_T(\mathbb{F}_q)$ . The decreasingly ordered distribution of the star  $K_{1,n-1}$  majorizes that of  $T$  if and only if  $\max_r \pi_{T,q}(r) \leq 1 - \frac{1}{q}$ .

**Conjecture 74** (Tree rank-variance extremality). *For every  $n \geq 4$  and every prime power  $q$ , among  $n$ -vertex trees the variance of  $\text{rank}(\text{ad}_x)$  over uniformly random  $x \in L_T(\mathbb{F}_q)$  is uniquely minimized by the path and uniquely maximized by the star.*

**Conjecture 75** (Leaf-count rigidity). *For a tree  $T$  on  $n \geq 2$  vertices, equality holds in the bound  $s(T) \geq n - \ell(T) + 2$  of Lemma 2 exactly when  $T$  is a path or a star.*

**Conjecture 76** (Diameter rigidity). *For a tree  $T$  on  $n \geq 2$  vertices, equality holds in the bound  $s(T) \geq \text{diam}(T) + 1$  of Lemma 2 exactly when  $T$  is a path or a star.*

## Part V

# Pattern Lie algebras of finite posets

*Added to the deposit: 6 August 2026.*

The twenty conjectures below concern the pattern Lie algebras of finite posets, studied through two rank-weight enumerators that record how the bracket contracts against an element and against a functional. They form four programmes. A determination programme (Conjectures 77 to 82) asserts that the adjoint enumerator alone determines the coadjoint enumerator and a list of classical invariants. A characteristic-and-field programme (Conjectures 83 to 87) identifies what in these distributions is independent of the ground field and how the field affects them when it does. A centre-one programme (Conjectures 88 to 90) studies the incidence quotient that

measures the difference between the two enumerators when the centre is a line. A Hasse-forest and unitriangular programme (Conjectures 91 to 96) relates coadjoint structure to the shape of the cover graph and treats the unitriangular chain as a separate extremal case. The proved results are kept separate from the conjectural ones: the incidence identity (Proposition 6) is a theorem with a full proof, the standard adjoint–coadjoint duality of this family, included for calibration and not claimed as new, and every conjecture is stated relative to it. Fifteen of the twenty are supported throughout the tested corpus. Three more, Conjectures 79, 81, and 85, are stated in greater generality than the computations reach, and each carries a remark giving the restricted case its evidence covers. Two, Conjectures 87 and 90, have since been refuted by explicit seven-point posets and are recorded as resolved false with their counterexamples.

For a finite poset  $P$  on a set of points and a prime power  $q$ , the *pattern Lie algebra* over  $\mathbb{F}_q$  is

$$L_P(\mathbb{F}_q) = \text{span}\{e_{ij} : i <_P j\}$$

inside the strictly upper triangular matrices under a fixed linear extension of  $P$ , with the matrix-unit bracket

$$[e_{ij}, e_{kl}] = \delta_{jk} e_{il} - \delta_{li} e_{kj}.$$

It is nilpotent, and  $\dim L_P$  is the number of strict order relations of  $P$ . Write  $\text{ad}_x(y) = [x, y]$  for the adjoint map of  $x$ , and for a functional  $f \in L_P(\mathbb{F}_q)^*$  let  $B_f(u, v) = f([u, v])$  be the alternating Kirillov form. The two natural contractions of the bracket tensor give two rank-weight enumerators over a single finite field,

$$A_{P,q}(T) = \sum_{x \in L_P(\mathbb{F}_q)} T^{\text{rank}(\text{ad}_x)}, \quad K_{P,q}(T) = \sum_{f \in L_P(\mathbb{F}_q)^*} T^{\text{rank}(B_f)},$$

the *adjoint enumerator* and the *Kirillov enumerator*, each a distribution over one finite field. Write  $b_A(P, q) = \max_x \text{rank}(\text{ad}_x)$  for the *adjoint breadth* and  $b_K(P, q) = \max_f \text{rank}(B_f)$  for the *maximal Kirillov rank*, the largest coadjoint-orbit rank. Write  $Z(L_P)$ ,  $\text{Cent}(L_P)$ , and  $\text{Der}(L_P)$  for the centre, the centroid (the linear maps commuting with every  $\text{ad}_x$ ), and the derivation algebra, and  $\gamma_i(L_P)$  for the lower central series. Say  $P$  is a *Hasse forest* if the undirected Hasse cover graph of  $P$  is a forest. The chain on  $n$  points gives  $L = \text{ut}_n(\mathbb{F}_q)$ , the strictly upper triangular Lie algebra.

**Proposition 6** (Incidence identity and calibration). Let  $L = L_P(\mathbb{F}_q)$  and let

$$I_L = \{(x, f) \in L \times L^* : f([x, L]) = 0\}$$

be the incidence variety. Then

$$|I_L| = q^{\dim L} A_{L,q}(q^{-1}) = q^{\dim L} K_{L,q}(q^{-1}).$$

Consequently  $A_{L,q}(1) = K_{L,q}(1) = q^{\dim L}$  and  $A_{L,q}(q^{-1}) = K_{L,q}(q^{-1})$ , so  $(1 - T)(1 - qT)$  divides  $A_{L,q}(T) - K_{L,q}(T)$  in  $\mathbb{Z}[T]$ .

The identity is the standard adjoint–coadjoint duality of this family, a Knuth-type contraction of one bracket tensor, and is recorded here as calibration rather than as a new result. For the centre-one conjectures define

$$R_{P,q}(T) = \frac{K_{P,q}(T) - A_{P,q}(T)}{(1 - T)(1 - qT)},$$

a polynomial in  $\mathbb{Z}[T]$  by Proposition 6.



## 19 Determination by the adjoint enumerator

**Conjecture 77** (Adjoint–Kirillov determination). *For all finite posets  $P$  and  $Q$  and every prime power  $q$ , if  $A_{P,q}(T) = A_{Q,q}(T)$  then  $K_{P,q}(T) = K_{Q,q}(T)$ .*

**Conjecture 78** (Lower-central recovery). *For every finite poset  $P$  and every prime power  $q$ , the enumerator  $A_{P,q}(T)$  determines the vector of lower-central factor dimensions  $(\dim \gamma_i(L_P)/\gamma_{i+1}(L_P))_i$ .*

**Conjecture 79** (Ordinary Poincaré recovery). *For every finite poset  $P$  and every prime power  $q$ , the enumerator  $A_{P,q}(T)$  determines the ordinary cohomological Poincaré polynomial  $\sum_i \dim H^i(L_P, \mathbb{F}_q) U^i$ .*

**Conjecture 80** (Derivation recovery). *For every finite poset  $P$  and every prime power  $q$ , the enumerator  $A_{P,q}(T)$  determines  $\dim \text{Der}(L_P)$ .*

**Conjecture 81** (Adjoint Poincaré recovery). *For every finite poset  $P$  and every prime power  $q$ , the enumerator  $A_{P,q}(T)$  determines the adjoint-cohomology dimensions  $\dim H^i(L_P, L_P)$  for every  $i$ .*

**Conjecture 82** (Centroid-dimension recovery). *For every finite poset  $P$  and every prime power  $q$ , the enumerator  $A_{P,q}(T)$  determines  $\dim \text{Cent}(L_P)$ .*

## 20 Characteristic and field behaviour

**Conjecture 83** (Adjoint spectrum characteristic independence). *For every finite poset  $P$ , the set  $\{\text{rank}(\text{ad}_x) : x \in L_P(\mathbb{F}_q)\}$  of attained adjoint ranks is independent of  $q$ .*

**Conjecture 84** (Kirillov spectrum characteristic independence). *For every finite poset  $P$ , the set  $\{\text{rank}(B_f) : f \in L_P(\mathbb{F}_q)^*\}$  of attained Kirillov ranks is independent of  $q$ .*

**Conjecture 85** (Two-field arithmetic rigidity). *For all finite posets  $P$  and  $Q$ , if  $A_{P,2} = A_{Q,2}$  and  $A_{P,3} = A_{Q,3}$  then  $A_{P,q} = A_{Q,q}$  for every prime power  $q$ .*

**Conjecture 86** (Adjoint stochastic field monotonicity). *For every finite poset  $P$  and prime powers  $q < q'$ , the distribution of  $\text{rank}(\text{ad}_x)$  for uniformly random  $x \in L_P(\mathbb{F}_{q'})$  first-order stochastically dominates its distribution over  $\mathbb{F}_q$ .*

**Conjecture 87** (Kirillov likelihood-ratio field monotonicity, resolved false). *For every finite poset  $P$  and prime powers  $q < q'$ , the normalized even-rank distribution defined by  $K_{P,q'}$  dominates that defined by  $K_{P,q}$  in monotone likelihood-ratio order.*

*Resolution.* False: a seven-point poset has likelihood ratio from  $\mathbb{F}_2$  to  $\mathbb{F}_3$  already decreasing across the even ranks, refuting the monotone-likelihood-ratio strengthening; the first-order Kirillov dominance of Conjecture 86 stands.

**Remark 6** (Resolution). *The conjecture is false as stated: the seven-point poset with cover relations  $1 \prec 2, 2 \prec 3, 2 \prec 4, 1 \prec 5, 5 \prec 7, 1 \prec 6, 6 \prec 7$  has Kirillov even-rank coefficients 128, 384, 128, 384 over  $\mathbb{F}_2$  and 2187, 17496, 4374, 34992 over  $\mathbb{F}_3$ , whose likelihood ratio decreases from rank 2 to rank 4, so monotone likelihood-ratio order already fails at  $q = 2, q' = 3$ . The first-order dominance of Conjecture 86 stands.*

## 21 Centre-one incidence quotient

**Conjecture 88** (Centre-one reverse determination). *For every finite poset  $P$  with  $\dim Z(L_P) = 1$  and every prime power  $q$ , the enumerator  $K_{P,q}(T)$  determines  $A_{P,q}(T)$ .*

**Conjecture 89** (Centre-one Laplace positivity). *For every finite poset  $P$  with  $\dim Z(L_P) = 1$  and every prime power  $q$ , the polynomial  $R_{P,q}(T)$  has nonnegative integer coefficients.*

**Conjecture 90** (Centre-one quotient shape, resolved false). *For every finite poset  $P$  with  $\dim Z(L_P) = 1$  and every prime power  $q$ , the coefficient sequence of  $R_{P,q}(T)$  has interval support and is unimodal.*

*Resolution.* False: a seven-point poset has quotient  $R_{P,q}$  with a strict interior valley, so positivity does not force unimodality; the positivity and interval support of Conjecture 89 stand.

**Remark 7** (Resolution). *The conjecture is false as stated: the seven-point poset with cover relations  $1 \prec 2, 2 \prec 3, 2 \prec 4, 3 \prec 6, 4 \prec 5, 5 \prec 6, 6 \prec 7$  has a one-dimensional centre, and its incidence quotient  $R_{P,2}$  has nonnegative coefficients with interval support but is not unimodal, the coefficients 206336, 198656, 225280 at degrees 8, 9, 10 forming a strict interior valley. The positivity and interval support of Conjecture 89 stand.*

## 22 Hasse-forest and unitriangular structure

**Conjecture 91** (Hasse-forest reverse determination). *For every finite poset  $P$  whose Hasse cover graph is a forest and every prime power  $q$ , the enumerator  $K_{P,q}(T)$  determines  $A_{P,q}(T)$ .*

**Conjecture 92** (Hasse-forest coadjoint saturation). *For every finite poset  $P$  whose Hasse cover graph is a forest and every prime power  $q$ , with  $b_K = b_K(P, q)$  the maximal Kirillov rank,*

$$\text{supp } K_{P,q} = \{0, 2, 4, \dots, b_K\},$$

*so every even rank up to  $b_K$  is attained.*

**Conjecture 93** (Hasse-forest Kirillov unimodality). *For every finite poset  $P$  whose Hasse cover graph is a forest and every prime power  $q$ , the even-rank sequence  $([T^{2j}]K_{P,q}(T))_j$  is unimodal.*

**Conjecture 94** (Hasse-forest breadth-orbit bound). *For every finite poset  $P$  whose Hasse cover graph is a forest and every prime power  $q$ ,*

$$b_K(P, q) \leq 2b_A(P, q).$$

**Conjecture 95** (Unitriangular Kirillov strict log-concavity). *For  $L = \text{ut}_n(\mathbb{F}_q)$  and every prime power  $q$ , the even-rank sequence  $([T^{2j}]K_{L,q}(T))_j$  is strictly log-concave at every nontrivial interior index.*

**Conjecture 96** (Adjoint-coadjoint incidence anticorrelation). *For every nonabelian pattern Lie algebra  $L = L_P$  and every prime power  $q$ , sampling  $(x, f)$  uniformly on the incidence variety  $I_L = \{(x, f) : f([x, L]) = 0\}$  gives*

$$\text{Cov}(\text{rank}(\text{ad}_x), \text{rank}(B_f)) < 0,$$

*with covariance zero exactly when  $L_P$  is abelian.*

## Part VI

# Local structure of primes and arithmetic functions

*Added to the deposit: 10 August 2026.*

The twenty conjectures below return to the local–global aesthetic of Part I, but read it off objects other than residue classes of consecutive primes. Each takes a familiar arithmetic quantity, looks at it through a local lens that ought to have looked symmetric, and takes the residual asymmetry seriously. They form five programmes. A compass programme (Conjectures 97 to 102) measures the direction of the nearest prime to a polynomial centre and matches it to a local-sieve hazard. An order-pattern programme (Conjectures 103 and 104) studies the order patterns of  $\text{rad}(n)/n$  on short windows. An order programme (Conjectures 105 to 109) measures how often abundancy and totient ratios move monotonically. A shock programme (Conjectures 110 to 115) tracks the prime-factor count of combinatorial sequences that drop many prime factors in a single step, a *shock*. A boundary programme (Conjecture 116) reads the prime-factor anatomy of the four integers bracketing a prime gap. The proved layer is kept separate. Proposition 7 gives the factorial-ratio jump identities, Proposition 8 gives short-window exchangeability, Proposition 9 proves that every fixed radical-pattern density exists, Proposition 10 proves exact finite-sieve equivalence for triangular and hexagonal centres, Proposition 14 proves the  $k = 2$  case of the dyadic multinomial slope, Proposition 11 settles the three shock laws, Conjectures 110 to 112, by an elementary forcing construction, and Proposition 12 evaluates the prime-truncated lag-one covariance of the central-binomial slice at exactly  $-\frac{5}{3}$ , and Proposition 13 shows the lag-one shock covariance converges to a rational number in every family. Six of the twenty state a bias as a numerical band rather than a derived constant, where they occur, the compass and order constants of Conjectures 98, 99, 100, 105, 106, and 108. Two, Conjectures 102 and 113, extrapolate in a parameter rather than in the index and are the boldest.

For a composite integer  $a$  let  $p^-(a) < a < p^+(a)$  be its neighbouring primes and set  $L(a) = a - p^-(a)$  and  $U(a) = p^+(a) - a$ . The *prime compass* of  $a$  points right when  $U(a) < L(a)$ , left when  $L(a) < U(a)$ , and is tied when the two distances agree. For a sequence  $A_1 < A_2 < \dots$  of composite centres write

$$\theta_A(x) = \frac{\#\{n : A_n \leq x, U(A_n) < L(A_n)\}}{\#\{n : A_n \leq x, U(A_n) \neq L(A_n)\}}$$

for the right-closer frequency among the untied centres, and  $\theta_A = \lim_{x \rightarrow \infty} \theta_A(x)$  when it exists; we call  $\theta_A$  the *compass constant* of  $A$ . The polygonal numbers are  $P_s(n) = \frac{1}{2}((s-2)n^2 - (s-4)n)$  for  $s \geq 3$ , the pronic numbers are  $Q(n) = n(n+1)$ , and the squares are  $n^2$ . For the radical  $\text{rad}(n) = \prod_{p|n} p$  write  $R(n) = \text{rad}(n)/n$ , and for a window of length  $d$  call the permutation  $\pi_0 \dots \pi_{d-1}$  of  $\{0, \dots, d-1\}$  the *order pattern*—the ordinal pattern of the window—at  $n$  when  $R(n+\pi_0) < \dots < R(n+\pi_{d-1})$ , counting only windows with no ties. Write  $\delta_\pi$  for the density of the order pattern  $\pi$  among the untied windows, the limiting ratio of the two window counts, so that  $\sum_\pi \delta_\pi = 1$ ; ties have positive density, adjacent squarefree integers already tying at  $R = 1$ , so an unconditional normalization would be deficient. For the abundancy index  $\sigma(n)/n$  and the totient ratio  $\phi(n)/n$  call an adjacent triple *monotone* when it is strictly increasing or strictly decreasing, and for a stride  $W$  let  $\mu(W)$  be the density of  $n$  for which  $\sigma(n)/n, \sigma(n+W)/(n+W), \sigma(n+W)$

$2W)/(n+2W)$  is monotone. Write  $\Omega(m)$  for the number of prime factors of  $m$  with multiplicity, and set the binomial slice  $A_{k,n} = \binom{kn}{n}$ , the central multinomial  $M_{k,n} = (kn)!/(n!)^k$ , and the Fuss–Catalan number  $F_{r,n} = \frac{1}{rn+1} \binom{(r+1)n}{n}$ , with single-step jump  $J_{k,n} = \Omega(A_{k,n+1}) - \Omega(A_{k,n})$ . Finally, for consecutive odd primes  $p < q$  set

$$D(p, q) = \Omega(p+1) + \Omega(q-1) - \Omega(p-1) - \Omega(q+1),$$

the inward-minus-outward prime-factor balance of the four composites bracketing the gap.

**Proposition 7** (Factorial-ratio jump identities). For all  $k \geq 2$  and  $n \geq 1$ ,

$$J_{k,n} = \sum_{j=1}^k \Omega(kn+j) - \Omega(n+1) - \sum_{j=1}^{k-1} \Omega((k-1)n+j).$$

In particular  $\Omega\binom{2n+2}{n+1} - \Omega\binom{2n}{n} = 1 + \Omega(2n+1) - \Omega(n+1)$ , and for the Catalan numbers  $C_n = \frac{1}{n+1} \binom{2n}{n}$ ,  $\Omega(C_{n+1}) - \Omega(C_n) = 1 + \Omega(2n+1) - \Omega(n+2)$ . Consequently the central binomial jump  $J_{2,n}$ , the Catalan jump  $\Omega(C_{n+1}) - \Omega(C_n)$ , and the central multinomial jump  $\Omega(M_{k,n+1}) - \Omega(M_{k,n})$  for every  $k \geq 2$  are unbounded below, while the binomial slice  $J_{k,n}$  for  $k \geq 3$  is settled by the forcing construction of Proposition 11.

**Proposition 8** (Short-window radical exchangeability). For every window length  $d \leq 4$  the strict order-pattern densities of  $R(n) = \text{rad}(n)/n$  exist and are uniform, each equal to  $1/d!$ .

**Proposition 9** (Existence of radical-pattern densities). For every fixed window length  $d$ , each unconditional strict order-pattern count for  $R(n) = \text{rad}(n)/n$  has a natural density. The untied windows have positive density. Consequently every conditional density  $\delta_\pi$  exists.

**Proposition 10** (Triangular–hexagonal finite-sieve equivalence). For every modulus  $m \geq 1$  and residue  $c \bmod m$ ,

$$\frac{1}{2m} \# \{n \bmod 2m : P_3(n) \equiv c \pmod{m}\} = \frac{1}{m} \# \{k \bmod m : P_6(k) \equiv c \pmod{m}\}.$$

Consequently the triangular and hexagonal local compass frequencies are exactly equal for every finite set of sieving primes.

## 23 Prime compasses at polynomial centres

**Conjecture 97** (Local-sieve limit for polygonal compasses). *For every fixed  $s \geq 3$  the limit  $\theta_s = \theta_{P_s}$  exists. Moreover, if the nearest-prime race is replaced by the first offset on each side surviving every prime at most  $y$ , the resulting local compass frequency tends to  $\theta_s$  as  $y \rightarrow \infty$ .*

**Conjecture 98** (Pentagonal and pronic compasses). *The pentagonal compass satisfies  $0.56 < \theta_{P_5} < 0.62$ , while the pronic compass satisfies  $0.42 < \theta_Q < 0.48$ .*

**Conjecture 99** (Opposite polygonal orders). *The 60-gonal compass satisfies  $\theta_{60} > 0.58$  and the 69-gonal compass satisfies  $\theta_{69} < 0.49$ .*

**Conjecture 100** (Triangular and hexagonal equality). *The triangular and hexagonal compass constants are equal,  $\theta_{P_3} = \theta_{P_6} \in (0.52, 0.56)$ .*

**Conjecture 101** (Square compass by root class). *For each residue  $r \bmod 6$  the conditional square compass  $\theta_{\square, r}$  over roots  $n \equiv r$  exists, with  $\theta_{\square, r} < 0.45$  for  $r \in \{1, 5\}$  and  $\theta_{\square, r} > 0.54$  for  $r \in \{0, 2, 3, 4\}$ .*

**Conjecture 102** (Two-sided order spectrum). *The polygonal compass spectrum straddles one half in the limit of many sides,*

$$\liminf_{s \rightarrow \infty} \theta_s < 0.49 < 0.58 < \limsup_{s \rightarrow \infty} \theta_s,$$

*so infinitely many polygonal orders are left-biased and infinitely many are right-biased.*

## 24 Order patterns of the radical ratio

**Conjecture 103** (Permanent nonuniformity from length five). *For every fixed window length  $d \geq 5$  the strict order-pattern densities of  $R(n) = \text{rad}(n)/n$  exist, and their distribution is not uniform.*

**Conjecture 104** (Thirteen-fold pattern separation). *At window length five the strict densities satisfy  $12 < \delta_{30241}/\delta_{13240} < 15$ , with 30241 in the dominant class and 13240 in the suppressed class.*

## 25 Order laws for multiplicative ratios

**Conjecture 105** (Abundancy order constant). *The density of  $n$  for which  $\sigma(n)/n, \sigma(n+1)/(n+1), \sigma(n+2)/(n+2)$  is monotone exists and lies in  $(0.094, 0.098)$ .*

**Conjecture 106** (Totient order constant). *The density of  $n$  for which  $\phi(n)/n, \phi(n+1)/(n+1), \phi(n+2)/(n+2)$  is monotone exists and lies in  $(0.019, 0.022)$ .*

**Conjecture 107** (Five-term totient barrier, resolved false). *No five consecutive values of  $\phi(n)/n$  are strictly increasing, and no five are strictly decreasing.*

*Resolution.* False, by a theorem of Martin [14]: strictly monotone runs of  $\phi(n)/n$  of every length occur on a set of positive lower density, though none below  $4 \times 10^9$ , whereas the abundancy analogue already fails at  $n = 36,721,681$ ; retained for the contrast it draws.

**Remark 8** (Resolution). *The conjecture is false. By a theorem of Martin [14], for every  $k$  and every prescribed strict ordering of  $\phi(n+1), \dots, \phi(n+k)$  with successive ratios exceeding any fixed  $C$ , the realizing  $n$  have positive lower density; taking  $k = 5$  and ratios exceeding 2 forces strictly monotone runs of  $\phi(n)/n$  of length five in either direction, on positive lower density. The runs are astronomically rare, none below  $4 \times 10^9$ , whereas the abundancy analogue already fails at  $n = 36,721,681$ .*

**Conjecture 108** (Primorial-stride order constant). *Along primorial strides  $W_y = \prod_{p \leq y} p$  the abundancy order density  $\mu(W_y)$  has a limit as  $y \rightarrow \infty$ , and  $\lim_{y \rightarrow \infty} \mu(W_y) \in (0.16, 0.19)$ .*

**Conjecture 109** (Dyadic completion overshoot). *For every even squarefree  $W$  divisible by six the limit  $\lim_{e \rightarrow \infty} \mu(2^e W)$  exists and exceeds  $1/3$ , so  $\mu(2^e W) > 1/3$  for all large  $e$ .*

## 26 Prime-factor shocks in combinatorial sequences

**Conjecture 110** (Binomial-slice negative shocks, resolved true). *For every fixed  $k \geq 3$  the jump  $J_{k,n} = \Omega\binom{k(n+1)}{n+1} - \Omega\binom{kn}{n}$  is unbounded below.*

*Resolution.* The conjecture is true: Proposition 11 forces arbitrarily deep negative jumps for every  $k \geq 3$  by a Chinese-remainder construction that loads the denominator forms  $(k-1)n+j$  with many prime factors while a first-moment bound keeps the numerator forms typical.

**Conjecture 111** (Binomial-slice positive shocks, resolved true). *For every fixed  $k \geq 2$  the jump  $J_{k,n}$  is unbounded above.*

*Resolution.* The conjecture is true: Proposition 11 applies the same forcing with the two form families exchanged, loading the numerator forms  $kn+j$ ; the second negative block that resisted the power-of-two argument is controlled on average rather than pointwise.

**Conjecture 112** (Fuss–Catalan two-sided shocks, resolved true). *For every fixed  $r \geq 1$  the jump  $\Omega(F_{r,n+1}) - \Omega(F_{r,n})$  is unbounded both below and above.*

*Resolution.* The conjecture is true in both directions for every  $r \geq 1$ : Proposition 11 applies to the Fuss–Catalan ratio’s two families of forms, which are pairwise nonproportional since  $(r+1)j' = rj$  has no admissible solution.

**Proposition 11** (Two-sided shock forcing). *For every fixed  $k \geq 2$  the jump  $J_{k,n}$  is unbounded below and unbounded above, and for every fixed  $r \geq 1$  the Fuss–Catalan jump  $\Omega(F_{r,n+1}) - \Omega(F_{r,n})$  is unbounded below and unbounded above. In particular Conjectures 110, 111, and 112 are true.*

*Proof.* Proposition 7 and  $\Omega(kn+k) = \Omega(k) + \Omega(n+1)$  give  $J_{k,n} = \Omega(k) + \sum_{j=1}^{k-1} (\Omega(kn+j) - \Omega((k-1)n+j))$ , and cancelling  $rn+1$  against the denominator block and  $(r+1)(n+1)$  against  $n+1$  in the exact ratio  $F_{r,n+1}/F_{r,n}$  gives  $\Omega(F_{r,n+1}) - \Omega(F_{r,n}) = \Omega(r+1) + \sum_{j=1}^r \Omega((r+1)n+j) - \sum_{j=2}^{r+1} \Omega(rn+j)$ . Each claim is therefore an instance of the following: for affine forms  $L_i(n) = a_i n + b_i$ ,  $1 \leq i \leq s$ , and  $L'_j(n) = a'_j n + b'_j$ ,  $1 \leq j \leq s$ , with positive integer coefficients and no  $L_i$  proportional to any  $L'_j$ , the difference  $\sum_i \Omega(L_i(n)) - \sum_j \Omega(L'_j(n))$  is unbounded above. Nonproportionality holds in all four applications:  $kj' = (k-1)j$  forces  $k \mid j$  with  $1 \leq j \leq k-1$ , and  $(r+1)j' = rj$  forces  $r+1 \mid j$  with  $1 \leq j \leq r$ , both impossible.

For the claim, extract the contents  $\gcd(a'_j, b'_j)$ , whose contribution to  $\Omega(L'_j)$  is constant, so the  $L'_j$  may be taken primitive; let  $R$  bound the coefficients and the cross-resultants  $|a_i b'_j - a'_j b_i|$ , fix  $m$ , and choose  $sm$  distinct primes exceeding  $R$ , one block  $Q_i$  of  $m$  primes for each  $L_i$ . Each congruence  $L_i(n) \equiv 0 \pmod{Q_i}$  is a single residue class since the primes exceed  $a_i$ , and the classes combine by the Chinese remainder theorem into  $n \equiv \rho \pmod{Q}$  with  $Q = Q_1 \cdots Q_s$ , on which  $\Omega(L_i(n)) \geq m$  for every  $i$ . No forced prime  $p$  divides any  $L'_j(n)$ : together with  $p \mid L_i(n)$  it would divide  $a_i L'_j(n) - a'_j L_i(n) = a_i b'_j - a'_j b_i$ , a nonzero integer of absolute value below  $p$ . Now let  $n$  run over  $\rho + tQ$  for  $1 \leq t \leq Q^3$ , so that every value  $L'_j(n)$  is at most a constant times  $Q^4$ . A prime power  $p^v \leq Q^3$  with  $p \nmid Q$  divides  $L'_j(n)$  along at most  $Q^3/p^v + 1$  values of  $t$ , primes dividing  $a'_j$  divide no value of the primitive form, and summing  $1/p^v$  over prime powers gives an average of  $\log \log Q + O(1)$  per form, the  $+1$  terms totalling  $O(Q^3/\log Q)$ . Prime powers exceeding  $Q^3$  add  $O(1)$  per form on average: a prime above  $Q^3$  divides a value at most once and to the first power, since  $(Q^3)^2$  exceeds every value, while prime powers  $p^v > Q^3$  with  $v \geq 2$  are squarefull, number  $O(Q^2)$  below the value bound, and each catches at most two values of  $t$ . By Markov some  $t \leq Q^3$  has  $\sum_j \Omega(L'_j(n)) \leq 2s(\log \log Q + O(1))$ , whence  $\sum_i \Omega(L_i(n)) - \sum_j \Omega(L'_j(n)) \geq$



$sm - 2s \log \log Q - O(s) \rightarrow \infty$  as  $m \rightarrow \infty$ , since  $\log \log Q = O(\log(sm \log m))$ . Explicit witnesses are cheap: six primes per form already force  $J_{3,n} \leq -20$  and  $J_{4,n} \leq -30$  at heights near  $10^{21}$  and  $10^{35}$ , and  $J_{3,n} \geq 21$  on the positive side, beyond every scanned extreme.  $\square$

**Conjecture 113** (Negative shock covariance). *In every fixed binomial-slice ( $k \geq 2$ ), central-multinomial ( $k \geq 2$ ), and Fuss–Catalan ( $r \geq 1$ ) family the lag-one covariance of successive jumps has a finite strictly negative limit.*

**Proposition 12** (Exact truncated covariance for the central-binomial slice). For a cutoff  $y \geq 2$  write  $\Omega_y(m) = \sum_{p \leq y} v_p(m)$  and  $J_n^{(y)} = 1 + \Omega_y(2n+1) - \Omega_y(n+1)$ . The lag-one covariance of the truncated central-binomial jump over  $n \leq N$  converges as  $N \rightarrow \infty$ , to  $-1$  at  $y = 2$  and to

$$\lim_{N \rightarrow \infty} \text{Cov}(J_n^{(y)}, J_{n+1}^{(y)}) = -\frac{5}{3} \quad \text{for every } y \geq 3.$$

**Proposition 13** (Existence of the shock covariance). In each fixed binomial-slice ( $k \geq 2$ ), central-multinomial ( $k \geq 2$ ), and Fuss–Catalan ( $r \geq 1$ ) family, the lag-one covariance of successive  $\Omega$ -jumps over  $n \leq N$  converges as  $N \rightarrow \infty$  to a rational number.

**Conjecture 114** (Dyadic multinomial slope). *For every fixed  $k \geq 2$  the dyadic jump of the central multinomial obeys*

$$\frac{\Omega(M_{k,2^m}) - \Omega(M_{k,2^m-1})}{m} \longrightarrow -(k-1).$$

**Proposition 14** (The dyadic multinomial slope for  $k = 2$ ). The conclusion of Conjecture 114 holds for  $k = 2$ .

**Conjecture 115** (Eventual sign balance, resolved true). *For every fixed binomial-slice and Fuss–Catalan family the positive-jump and negative-jump densities each tend to  $\frac{1}{2}$ , while the zero-jump density tends to 0.*

*Resolution.* The conjecture is true. By the identities of Propositions 7 and 11, each jump is, up to a bounded term, a signed sum of  $\Omega$  over  $2(k-1)$ , respectively  $2r$ , pairwise nonproportional affine forms, whose primitive parts are pairwise coprime linear polynomials. The multivariate Erdős–Kac theorem of El-Baz, Loughran, and Sofos [280], whose one-variable case recovers Tanaka’s theorem for pairwise coprime irreducible polynomials, makes the vector of  $\omega$ -values asymptotically Gaussian with identity covariance, so the signed sum, normalized by  $\sqrt{2(k-1) \log \log n}$ , is asymptotically standard normal; replacing  $\omega$  by  $\Omega$  shifts each coordinate by a quantity of bounded mean, negligible at this scale. The diverging variance swamps the bounded centring, giving sign densities  $\frac{1}{2}$ , and the zero-jump density vanishes because the limit law assigns every shrinking central window vanishing mass.

## 27 The arithmetic anatomy of prime-gap boundaries

**Conjecture 116** (Boundary Euler sum). *For each prime  $\ell$  let  $r$  be uniform on  $\mathbb{Z}/\ell^a \mathbb{Z}$  conditioned on  $r(r+g) \not\equiv 0 \pmod{\ell}$ , and set*

$$m_{\ell,a}(g) = \mathbb{E}[v_\ell(r+1) + v_\ell(r+g-1) - v_\ell(r-1) - v_\ell(r+g+1)].$$

*Then for every fixed even  $g$  occurring infinitely often as a consecutive-prime gap,*

$$\lim_{x \rightarrow \infty} \mathbb{E}[D(p, q) \mid p \leq x, q - p = g] = \sum_{\ell} \lim_{a \rightarrow \infty} m_{\ell,a}(g).$$

## Part VII

# Topological invariants of geometric families

*Added to the deposit: 12 August 2026.*

The twenty conjectures below leave arithmetic for algebraic topology while keeping the same discipline. Each takes a canonical family of spaces, computes an exact classical invariant across the whole family, and takes the resulting shape law seriously. They form four programmes. A Calabi–Yau programme (Conjectures 117 to 121) studies the Euler indices of smooth complete intersections in projective space, ending in an exact dimension-wise gcd law. A link programme (Conjectures 122 to 126) studies the middle Betti numbers of Brieskorn–Pham links under degree caps and common scaling. A moment-angle programme (Conjectures 127 to 131) studies the Hochster strands of flag moment-angle complexes, their support, shape, and zero locations. A monodromy programme (Conjectures 132 to 136) studies torus mapping tori with signed-permutation monodromy over the signed-orbit calibration proved below. The calibration layer is classical and exact: the Chern-class formula for complete intersections, the Milnor–Orlik formula for Brieskorn links, Hochster’s formula for bigraded Betti numbers, and the Wang sequence for mapping tori. Six adversarial controls are retained in place, each refuting a tempting strengthening. Two of the twenty, the limit clause of Conjecture 117 and Conjecture 135, are the strongest statements of the part. Two more, Conjectures 120 and 124, lie close to classical methods. Since deposit, Conjectures 117 and 118 have been resolved true by Propositions 16 and 17, and Conjecture 135 has been resolved false by the Landau floor of Proposition 19, which fixes the compression exponent at 2. Conjecture 125 and the convexity clause of Conjecture 126 have since been proved by the Ehrhart model of Proposition 18.

For  $n \geq 2$  and a multidegree  $\mathbf{d} = (d_1, \dots, d_r)$  with every  $d_i \geq 2$  and  $\sum_i d_i = n + r + 1$ , let  $X_{\mathbf{d}}^{(n)} \subset \mathbf{CP}^{n+r}$  be a smooth complete intersection of multidegree  $\mathbf{d}$ , a Calabi–Yau  $n$ -fold, with Euler characteristic computed exactly by  $c(TX) = (1 + H)^{n+r+1} / \prod_i (1 + d_i H)$  and  $\chi = (\prod_i d_i) [H^n] c(TX)$ ; write  $E_n(\mathbf{d}) = (-1)^n \chi(X_{\mathbf{d}}^{(n)})$ ,  $D(\mathbf{d}) = \prod_i d_i$ , and let the *merge*  $M_{ij}\mathbf{d}$  replace  $d_i, d_j$  by  $d_i + d_j - 1$ , preserving both  $n$  and the Calabi–Yau condition. Write  $Q_n = E_n(2, \dots, 2)$  for the all-quadric configuration of  $n + 1$  quadrics. For  $\mathbf{a} = (a_1, \dots, a_N)$  with  $a_i \geq 2$ , let  $L(\mathbf{a})$  be the Brieskorn–Pham link  $\{z_1^{a_1} + \dots + z_N^{a_N} = 0\} \cap S^{2N-1}$ , of real dimension  $2N - 3$ , with middle Betti number  $\beta(\mathbf{a}) = b_{N-2}(L(\mathbf{a}); \mathbf{Q})$  given exactly by the Milnor–Orlik formula [15]

$$\beta(\mathbf{a}) = (-1)^N + \sum_{\emptyset \neq J \subseteq [N]} (-1)^{N-|J|} \frac{\prod_{j \in J} a_j}{\text{lcm}_{j \in J} a_j},$$

the *Fano* tuples being those with  $\sum_i 1/a_i > 1$ , with cap  $A = \max_i a_i$  and scale sequence  $\beta_{\mathbf{a}}(k) = \beta(ka_1, \dots, ka_N)$ . For a finite simple graph  $G$  on  $m$  vertices with clique complex  $K = \text{Cl}(G)$ , define the Hochster strands

$$h_{r,s}(G) = \sum_{\substack{I \subseteq V(G) \\ |I|=s}} \dim_{\mathbf{F}_2} \tilde{H}_r(K_I; \mathbf{F}_2), \quad H_{G,r}(z) = \sum_s h_{r,s}(G) z^s,$$

the bigraded Betti sectors of the Stanley–Reisner ring and of the moment-angle complex  $\mathcal{Z}_K$  by Hochster’s formula [16, 17]. Finally, for a signed permutation matrix  $A \in SL(d, \mathbb{Z})$  let  $M_A$  be the mapping torus of the induced automorphism of the torus  $T^d$ , set  $f_k(A) = \dim_{\mathbf{Q}} \ker(\wedge^k A - I)$ ,

so that the Wang sequence gives  $b_k(M_A; \mathbf{Q}) = f_k(A) + f_{k-1}(A)$  [22, 21], and let  $\tau(A) = \sum_k d(\text{Tor } H_k(M_A; \mathbb{Z}))$  count minimal torsion generators. The cap  $A$  of the link programme and the matrix  $A$  of the monodromy programme are unrelated symbols, each local to its own sections.

**Proposition 15** (Signed-orbit calibration). Let  $A$  be a signed permutation matrix on  $\mathbb{Z}^d$ . Then  $\wedge^k A$  permutes the wedge basis up to sign, and every orbit of the induced permutation of  $k$ -subsets carries a holonomy sign, the product of the signs collected around one period. If  $p_k$  and  $n_k$  are the numbers of positive and negative orbits at level  $k$ , then

$$f_k(A) = p_k, \quad \text{coker}(\wedge^k A - I) \cong \mathbb{Z}^{p_k} \oplus (\mathbb{Z}/2)^{n_k},$$

and consequently  $b_k(M_A; \mathbf{Q}) = p_k + p_{k-1}$ ,  $\text{Tor } H_k(M_A; \mathbb{Z}) \cong (\mathbb{Z}/2)^{n_k}$ , and  $\tau(A) = \sum_k n_k$ , while  $\sum_k b_k(M_A; \mathbf{Q}) = 2 \sum_k p_k$ .

## 28 Euler indices of Calabi–Yau complete intersections

**Conjecture 117** (Quadric-ladder ratio law, resolved true). *For every  $n \geq 5$ ,*

$$\frac{Q_{n+1}}{Q_n} < 8, \quad \frac{Q_{n+2}}{Q_{n+1}} > \frac{Q_{n+1}}{Q_n}, \quad \text{and} \quad \lim_{n \rightarrow \infty} \frac{Q_{n+1}}{Q_n} = 8.$$

*Resolution.* True, by Proposition 16 below, which proves all three clauses with the sharpening  $8 - Q_{n+1}/Q_n \sim 4/n$ .

**Proposition 16** (Resolution of the quadric-ladder ratio law). Conjecture 117 holds in full, with the sharpening  $8 - Q_{n+1}/Q_n \sim 4/n$ . The generating function  $A(z) = \sum_{n \geq 0} Q_n z^n$ , with  $Q_0 = 2$  and  $Q_1 = 0$ , is algebraic,

$$A = \frac{2(1-y)^3}{1-3y}, \quad z = \frac{y(1-2y)}{2(1-y)^2}, \quad y(0) = 0,$$

and the coefficients satisfy  $(n+1)(n+2)Q_{n+2} = (n+1)(7n+9)Q_{n+1} + 4(n+2)(2n+3)Q_n$ .

*Proof.* The Chern-class formula gives  $Q_n = [t^n] \varphi(t)^{n+1}$  for  $\varphi(t) = 2(1-t)^2/(1-2t)$ : indeed  $E_n(2, \dots, 2) = (-1)^n 2^{n+1} [H^n](1+H)^{2n+2}(1+2H)^{-n-1}$ , and the substitution  $t = -H$  converts one extraction into the other. For each small  $z$  the equation  $y = z\varphi(y)$  has exactly one root  $y(z)$  near the origin, and the residue form of Lagrange inversion, summing the geometric series  $\sum_n (z\varphi(t)/t)^n$  on a small contour and picking up the simple pole at  $y(z)$ , gives

$$A(z) = \sum_{n \geq 0} [t^n] \varphi(t)^{n+1} z^n = \frac{\varphi(y)}{1 - z\varphi'(y)};$$

inverting  $y = z\varphi(y)$  gives  $z = y(1-2y)/(2(1-y)^2)$ , and the residue value simplifies to  $2(1-y)^3/(1-3y)$ , the stated parametrization. Differentiating along it shows that  $(1+z)(1-8z)A'' - 9(1+4z)A' - 24A$  is a rational function of  $y$  that vanishes identically, a finite identity checked by exact expansion, and extracting the coefficient of  $z^n$  from this differential equation yields the recurrence.

Now the three clauses. Set  $q_n = Q_{n+1}/Q_n$ , so the recurrence reads  $q_{n+1} = \frac{7n+9}{n+2} + \frac{4(2n+3)}{(n+1)q_n}$ , strictly decreasing as a function of  $q_n$ . Write  $\delta_n = 8 - q_n$ , and call the two-sided bound  $4/(n+2) < \delta_n < 4/(n+1)$  the corridor at index  $n$ . The corridor holds at the base  $n = 5$ :

$Q_5 = 6912$  and  $Q_6 = 51,072$  give  $\delta_5 = 11/18 \in (4/7, 4/6)$ . For the step, the map carries the corridor at index  $n$  into the interval bounded by the images of its two walls, since it is decreasing, and the two inequalities placing that image inside the corridor at index  $n+1$  clear denominators to polynomials with positive coefficients in  $n-5$ ; so the corridor propagates to every  $n \geq 5$ . The corridor gives  $q_n < 8$ ; it gives  $q_{n+1} > 8 - 4/(n+2) > q_n$ , the strict increase; and both walls are asymptotic to  $4/n$ , so  $q_n \rightarrow 8$  with  $8 - q_n \sim 4/n$ . For  $2 \leq n \leq 4$  the ratios are  $16/3$ ,  $15/2$ , and  $36/5$ , all below 8.  $\square$

**Conjecture 118** (Normalized merge monotonicity, resolved true). *For every  $n \geq 3$ , every multidegree  $\mathbf{d}$  with  $r \geq 2$ , and every pair  $i \neq j$ ,*

$$\frac{E_n(M_{ij}\mathbf{d})}{D(M_{ij}\mathbf{d})} > \frac{E_n(\mathbf{d})}{D(\mathbf{d})}.$$

*Resolution.* True, by Proposition 17 below.

**Proposition 17** (Resolution of normalised merge monotonicity). Put  $a_\ell = d_\ell - 1$ , and let  $a = a_i$  and  $b = a_j$  for the entries being merged. Then

$$\frac{E_n(M_{ij}\mathbf{d})}{D(M_{ij}\mathbf{d})} - \frac{E_n(\mathbf{d})}{D(\mathbf{d})} = ab[u^{n-2}] \frac{1}{(1+u)^2(1-au)(1-bu)(1-(a+b)u)} \prod_{\ell \neq i,j} \frac{1}{1-a_\ell u} > 0.$$

Consequently Conjecture 118 holds in full. The same inequality also holds in dimension  $n = 2$ .

*Proof.* The Calabi–Yau condition gives  $\sum_\ell a_\ell = n+1$ . The Chern-class formula and the coefficient substitution  $t = u/(1+u)$  give

$$\frac{E_n(\mathbf{d})}{D(\mathbf{d})} = [t^n] \frac{(1-t)^{n+r+1}}{\prod_\ell (1-d_\ell t)} = [u^n] \frac{1}{(1+u)^2} \prod_\ell \frac{1}{1-a_\ell u}.$$

The merge replaces  $a, b$  by  $a+b$ . Subtracting the two rational series and using

$$\frac{1}{1-(a+b)u} - \frac{1}{(1-au)(1-bu)} = \frac{abu^2}{(1-au)(1-bu)(1-(a+b)u)}$$

gives the displayed coefficient formula. For every integer  $c \geq 1$  and  $q \geq 0$ ,

$$[u^q] \frac{1}{(1+u)(1-cu)} = \frac{c^{q+1} + (-1)^q}{c+1} \geq 0.$$

Pair the two factors  $(1+u)^{-1}$  with  $(1-au)^{-1}$  and  $(1-bu)^{-1}$ . Every remaining factor is a geometric series with nonnegative coefficients. The coefficient of degree  $n-2$  is strictly positive because  $(1-(a+b)u)^{-1}$  supplies the term  $(a+b)^{n-2}u^{n-2}$  while all other factors supply their constant terms. Since  $ab > 0$ , the merge increment is positive.  $\square$

**Conjecture 119** (Maximal-degree merge monotonicity). *For every  $n \geq 3$  and every  $\mathbf{d}$  with  $r \geq 2$ , if  $d_i = \max_j d_j$  then  $E_n(M_{ik}\mathbf{d}) > E_n(\mathbf{d})$  for every  $k \neq i$ .*

**Conjecture 120** (Extremal multidegrees). *For every  $n \geq 3$  and every multidegree  $\mathbf{d}$ ,*

$$E_n(\underbrace{2, \dots, 2}_{n+1}) \leq E_n(\mathbf{d}) \leq E_n(n+2),$$

*with equality only at the all-quadratic and hypersurface configurations.*

**Conjecture 121** (Dimension-wise Euler gcd law). *Let  $g_n$  be the greatest common divisor of  $\chi(X_{\mathbf{d}}^{(n)})$  over all multidegrees at dimension  $n$ . Then*

$$g_n = \frac{24}{\gcd(24, n)} \times \begin{cases} 2, & n \equiv 0, 2 \pmod{8}, \\ 1, & \text{otherwise.} \end{cases}$$

## 29 Middle Betti numbers of Brieskorn–Pham links

**Conjecture 122** (Five-link cap extremum). *For  $N = 4$  and every cap  $A \geq 9$ , every sorted Fano tuple  $\mathbf{a}$  with  $\max_i a_i = A$  satisfies  $b_2(L(\mathbf{a})) \leq A - 1$ , with equality only for  $\mathbf{a} = (2, 2, A, A)$ .*

**Conjecture 123** (Seven-link cap extremum). *For  $N = 5$  and every cap  $A \geq 13$ , every sorted Fano tuple with  $\max_i a_i = A$  satisfies  $b_3(L(\mathbf{a})) \leq (A - 1)(A - 2)$ , with equality only for  $(2, 2, A, A, A)$ .*

**Conjecture 124** (Connected gcd-graph positivity). *For  $N = 4$ , if the graph on  $\{1, 2, 3, 4\}$  joining  $i \sim j$  when  $\gcd(a_i, a_j) > 1$  is connected, then  $b_2(L(\mathbf{a})) \geq 1$ .*

**Conjecture 125** (Common-scale monotonicity, resolved true). *For every  $N \geq 4$  and every base tuple  $\mathbf{a}$ , the scale sequence satisfies  $\beta_{\mathbf{a}}(k + 1) \geq \beta_{\mathbf{a}}(k)$  for every  $k \geq 1$ .*

*Resolution.* The conjecture is true: Proposition 18 identifies the scale sequence as a sum of interior Ehrhart counting functions of  $N - 1$  integral weighted hypersimplices, whose forward differences of every order through  $N - 1$  are nonnegative, strictly more than monotonicity.

**Proposition 18** (Ehrhart model for common scaling). *For a base tuple  $\mathbf{a}$  and  $1 \leq m \leq N - 1$  let  $P_m(\mathbf{a}) = \{x \in \mathbf{R}^N : 0 \leq x_i \leq a_i, \sum_i x_i/a_i = m\}$ , a lattice polytope of dimension  $N - 1$ . Then*

$$\beta_{\mathbf{a}}(k) = \sum_{m=1}^{N-1} \#(\text{relint}(kP_m(\mathbf{a})) \cap \mathbb{Z}^N),$$

and consequently every forward difference of the scale sequence of order  $r \leq N - 1$  is nonnegative at every  $k \geq 1$ . In particular Conjecture 125 is true, and the discrete-convexity clause of Conjecture 126 is true.

*Proof.* The multiplication map  $\prod_{j \in J} \mu_{a_j} \rightarrow \mu_{\text{lcm}_{j \in J} a_j}$  on roots of unity is surjective with kernel of order  $\prod_{j \in J} a_j / \text{lcm}_{j \in J} a_j$ , so that quantity counts the tuples over  $J$  with product 1; by inclusion–exclusion over the coordinates forced to equal 1, the Milnor–Orlik sum counts the tuples  $(\zeta_1, \dots, \zeta_N)$  with each  $\zeta_i$  a nontrivial  $a_i$ -th root of unity and  $\prod_i \zeta_i = 1$ , the empty set contributing the term  $(-1)^N$ . For the scaled tuple  $k\mathbf{a}$  write  $\zeta_i = \exp(2\pi i r_i / (ka_i))$  with  $0 < r_i < ka_i$ ; the product is 1 exactly when  $\sum_i r_i/a_i = km$  for an integer  $m$ , and since each summand lies strictly between 0 and  $k$ , the integer satisfies  $1 \leq m \leq N - 1$ . These conditions say precisely that  $(r_1, \dots, r_N)$  is a lattice point of the relative interior of  $kP_m(\mathbf{a})$ , proving the displayed identity. Each  $P_m(\mathbf{a})$  is integral: the substitution  $y_i = x_i/a_i$  carries it to the hypersimplex  $\{0 \leq y_i \leq 1, \sum_i y_i = m\}$ , whose vertices are 0–1 vectors, so every vertex of  $P_m(\mathbf{a})$  has coordinates 0 or  $a_i$ . For a lattice polytope  $P$  of dimension  $d$  with  $h^*$ -vector  $(h_0^*, \dots, h_d^*)$ , Ehrhart reciprocity [281] turns the expansion  $L_P(k) = \sum_i h_i^* \binom{k+d-i}{d}$  into  $L_{P^\circ}(k) = \sum_i h_i^* \binom{k+i-1}{d}$ , and the  $h_i^*$  are nonnegative integers [282]; applying the forward difference  $r$  times gives  $\Delta^r L_{P^\circ}(k) = \sum_i h_i^* \binom{k+i-1}{d-r} \geq 0$  for  $0 \leq r \leq d$ . Summing over the  $N - 1$  polytopes  $P_m(\mathbf{a})$ , all of dimension  $N - 1$ , yields  $\Delta^r \beta_{\mathbf{a}}(k) \geq 0$  for every  $k \geq 1$  and  $r \leq N - 1$ ; the case  $r = 1$  is Conjecture 125, and the case  $r = 2$  is the convexity clause of Conjecture 126.  $\square$

**Conjecture 126** (Scale convexity and log-concavity). *For every  $N \geq 4$  and every base tuple, the scale sequence is simultaneously discretely convex and log-concave:*

$$\beta_{\mathbf{a}}(k+1) - 2\beta_{\mathbf{a}}(k) + \beta_{\mathbf{a}}(k-1) \geq 0, \quad \beta_{\mathbf{a}}(k)^2 \geq \beta_{\mathbf{a}}(k-1)\beta_{\mathbf{a}}(k+1).$$

### 30 Hochster strands of flag moment-angle complexes

**Conjecture 127** (Hochster interval support). *For every finite simple graph  $G$  and every  $r \geq 0$ , the support  $\{s : h_{r,s}(G) > 0\}$  is an interval of integers.*

**Conjecture 128** (Hochster-strand log-concavity). *For every  $G$ ,  $r$ , and  $s$ ,*

$$h_{r,s}(G)^2 \geq h_{r,s-1}(G)h_{r,s+1}(G).$$

**Conjecture 129** (Strand Hurwitz stability). *For every  $G$  and  $r$  with  $H_{G,r}$  nonconstant, every zero of  $H_{G,r}(z)/z^{\min \text{supp}}$ , the strand polynomial divided by its lowest monomial, satisfies  $\text{Re } z < 0$ .*

**Conjecture 130** (Cycle extremality at connectivity two). *If  $G$  is 2-vertex-connected on  $m \geq 4$  vertices, then  $h_{0,s}(G) \leq h_{0,s}(C_m)$  for every  $s$ , with equality in every coefficient only for the cycle  $C_m$ .*

**Conjecture 131** (Strict component-strand log-concavity). *For every connected noncomplete graph, every triple of positive consecutive component-strand coefficients is strictly log-concave:  $h_{0,s}^2 > h_{0,s-1}h_{0,s+1}$ .*

### 31 Torus mapping tori with signed monodromy

**Conjecture 132** (Parity-strand unimodality). *For every orientation-preserving signed permutation  $A$ , the sequences  $(f_0(A), f_2(A), f_4(A), \dots)$  and  $(f_1(A), f_3(A), f_5(A), \dots)$  are unimodal.*

**Conjecture 133** (Mapping-torus Betti unimodality). *For every orientation-preserving signed permutation  $A$  in dimension  $d$ , the Betti sequence  $(b_0(M_A), \dots, b_{d+1}(M_A))$  is unimodal.*

**Conjecture 134** (Sparse-cycle minimizer structure). *For every  $d \geq 8$ , some minimizer of the total Betti number  $\sum_k b_k(M_A; \mathbf{Q})$  over orientation-preserving signed permutations has no positive signed cycle when  $d$  is even, exactly one when  $d$  is odd, and pairwise distinct negative-cycle lengths.*

**Conjecture 135** (Square-root-two compression exponent, resolved false). *Let  $M_d$  be the minimum of  $\sum_k b_k(M_A; \mathbf{Q})$  over orientation-preserving signed permutations in dimension  $d$ . Then  $M_d^{1/d} \rightarrow \sqrt{2}$  as  $d \rightarrow \infty$ .*

*Resolution.* False, by the Landau floor of Proposition 19 below: the true compression exponent is 2, and the square-root-two fit was a finite-size illusion.

**Proposition 19** (Landau floor for the minimal total cohomology). *For every signed permutation  $A$  on  $\mathbb{Z}^d$ ,*

$$\sum_k b_k(M_A; \mathbf{Q}) \geq \frac{2^{d+1}}{\text{ord}(A)} \geq \frac{2^d}{g(d)},$$

where  $g(d)$  is Landau's function, the maximal order of a permutation of  $d$  letters. Consequently  $2^d/g(d) \leq M_d \leq 2^{d+1}$ , and since  $\log g(d) \sim \sqrt{d \log d}$ , the minimal total cohomology satisfies  $M_d^{1/d} \rightarrow 2$ .



*Proof.* By Proposition 15,  $\sum_k b_k(M_A; \mathbf{Q}) = 2P$  with  $P = \sum_k p_k = \sum_k \dim_{\mathbf{Q}} \ker(\wedge^k A - I)$ , the dimension of the fixed space of  $A$  on the full exterior algebra  $\Lambda^\bullet(\mathbf{Q}^d)$ . Since  $A$  has finite order  $m$ , that fixed space is the invariant subspace of the cyclic group generated by  $A$ , and averaging characters gives  $P = \frac{1}{m} \sum_{j=0}^{m-1} \text{tr } \Lambda^\bullet(A^j) = \frac{1}{m} \sum_{j=0}^{m-1} \det(I + A^j)$ . Every term is nonnegative:  $A^j$  is again a signed permutation, and a signed cycle of length  $L$  with holonomy  $\varepsilon$  contributes  $\prod_{\zeta^L=\varepsilon} (1+\zeta) = 1 - (-1)^L \varepsilon \in \{0, 2\}$  to the determinant. The term  $j = 0$  alone is  $2^d$ , so  $P \geq 2^d/m$ . If the underlying permutation has cycle lengths  $\ell_1, \dots, \ell_r$ , then  $A^{\text{lcm}(\ell_i)}$  is diagonal with entries  $\pm 1$ , so  $m \leq 2 \text{lcm}(\ell_1, \dots, \ell_r) \leq 2g(d)$ , giving the displayed floor. The identity monodromy gives  $M_d \leq 2 \cdot 2^d$ , and Landau’s asymptotic  $\log g(d) \sim \sqrt{d \log d}$  [283] makes  $(2^d/g(d))^{1/d} \rightarrow 2$ , so the two-sided bound forces  $M_d^{1/d} \rightarrow 2$ .  $\square$

**Remark 9** (Resolution). *The conjecture is false: by Proposition 19 the compression exponent is 2, not  $\sqrt{2}$ , with the whole deviation from  $2^{d+1}$  confined to a subexponential factor at most  $2g(d)$ . The finite scans could not separate the two laws, because  $\log_2 g(d) \approx d/2$  throughout the tested range— $\log_2 g(21) \approx 9.7$  against  $d/2 = 10.5$ —so the floor  $2^d/g(d)$  is numerically indistinguishable from  $2^{d/2}$  there, and the square-root fit was a finite-size illusion. The observed minima respect the floor with room to spare,  $M_{12} = 160$  against  $2^{12}/g(12) = 68$ , and the surviving question is the exact order of the subexponential correction  $\log_2(2^{d+1}/M_d)$ , now pinned between 0 and  $\log_2(2g(d)) = O(\sqrt{d \log d})$ . The torsion budget of Conjecture 136 is untouched: it is consistent with the true growth, and no incompatibility remains once the exponent is corrected.*

**Conjecture 136** (Torsion budget inequality). *For every signed permutation  $A \in SL(d, \mathbb{Z})$ ,*

$$\tau(A) \leq \sum_k b_k(M_A; \mathbf{Q}), \quad \text{equivalently} \quad \sum_k n_k \leq 2 \sum_k p_k$$

*in the orbit counts of Proposition 15.*

## Part VIII

# Shape laws for enumerative geometric invariants

*Added to the deposit: 12 August 2026.*

The twenty conjectures below continue the topological turn of Part VII with a second discipline: take a classical enumerative invariant of a geometric family, compute it exactly across the family, and conjecture the shape of the resulting array. They form five programmes. A hypersurface programme (Conjectures 137 to 140) studies the Euler indices of anticanonical hypersurfaces in products of projective spaces under an order dual to Part VII’s: where Part VII merged defining equations at fixed ambient, these statements split ambient factors at fixed dimension. A Hilbert-scheme programme (Conjectures 141 to 144) studies the Betti profiles of Göttsche’s product over a proved pair of exact moment identities. An orbifold programme (Conjectures 145 to 148) studies the twisted-sector age histograms of cyclic Calabi–Yau quotients. An entropy programme (Conjectures 149 to 152) studies the exterior torsion-entropy profile of hyperbolic toral automorphisms. A matroid programme (Conjectures 153 to 156) lifts the Hochster strand laws of Part VII from graphs to simple binary matroids. The calibration layer

is exact throughout: the Chern-class evaluation, Göttsche's product, the age grading of the McKay correspondence, the eigenvalue evaluation of torsion growth rates, and Hochster's formula. Retained controls again mark every calibrated boundary. The limit clauses of Conjectures 139 and 143 assert limits; Conjectures 146 to 148 are random-model statements resting on Monte Carlo evidence, with Conjectures 146 and 147 likely within reach of classical equidistribution machinery; Conjecture 145 lies close to classical methods. Since deposit, Conjectures 151 and 152 have been proved by convexity, while Conjectures 153 to 156 have been refuted by two explicit graphic matroids.

For a partition  $\lambda = (\lambda_1, \dots, \lambda_r)$  of  $d+1$  let  $A_\lambda = \prod_i \mathbf{CP}^{\lambda_i}$  and let  $X_\lambda \in |-K_{A_\lambda}|$  be a smooth anticanonical hypersurface, a Calabi–Yau  $d$ -fold; its Euler characteristic is evaluated exactly from  $c(TA_\lambda) = \prod_i (1 + H_i)^{\lambda_i+1}$  and  $[X_\lambda] = \sum_i (\lambda_i + 1)H_i$ , and we write  $E_d(\lambda) = (-1)^d \chi(X_\lambda)$ . A *split*  $S_{m,a}\lambda$  replaces one part  $m$  by  $a$  and  $m - a$ ; a *dominance move* transfers one unit from a smaller positive part to a weakly larger part. For  $b \geq 1$  define  $h_b(n, k)$  by

$$\sum_{n,k \geq 0} h_b(n, k) q^n z^k = \prod_{m \geq 1} \frac{1}{(1 - z^{m-1} q^m)(1 - z^m q^m)^b (1 - z^{m+1} q^m)},$$

so that  $h_{22}(n, k) = b_{2k}(\text{Hilb}^n(\mathbf{K3}))$  by Göttsche's theorem [23]. For  $r \geq 3$  and weights  $a_1, \dots, a_d \in (\mathbb{Z}/r)^\times$  with  $\sum_i a_i \equiv 0 \pmod{r}$ , the cyclic quotient  $\frac{1}{r}(a_1, \dots, a_d)$  has ages  $\text{age}(j) = \frac{1}{r} \sum_i (ja_i \bmod r)$  for  $1 \leq j < r$ , integers between 1 and  $d-1$ , with histogram  $N_k = \#\{j : \text{age}(j) = k\}$  grading twisted sectors in the McKay correspondence [26]. For a hyperbolic  $A \in SL(d, \mathbb{Z})$  with eigenvalue log-moduli  $x_1, \dots, x_d$  summing to zero, set

$$h_k(x) = \sum_{|S|=k} \left( \sum_{i \in S} x_i \right)_+, \quad 1 \leq k \leq d-1;$$

away from exterior resonances  $h_k$  is the exponential growth rate of  $|\det(\wedge^k A^n - I)|$ , the degree- $k$  torsion order of the mapping torus of  $A^n$ , since each factor  $|\lambda_S^n - 1|$  grows like  $|\lambda_S|^n$  or stays bounded [27, 30]. Finally, for a simple binary matroid  $M$  on ground set  $E$  with independence complex  $\Delta(M)$ , put  $\beta(A) = \dim \tilde{H}_{r(A)-1}(\Delta(M|_A); \mathbf{F}_2)$  and

$$H_{c,s}(M) = \sum_{\substack{A \subseteq E, |A|=s \\ |A|-r(A)=c}} \beta(A),$$

the aggregated multigraded Betti numbers of the Stanley–Reisner ring by Hochster's formula [16], graded by restriction size and nullity, with strand polynomial  $\sum_s H_{c,s}(M) z^s$  at each fixed nullity  $c$ . The profile  $h_b(n, k)$  and the spectral functional  $h_k(x)$  are unrelated symbols, each local to its own programme.

**Proposition 20** (Profile moment identities). *For every  $b \geq 1$  and  $n \geq 1$ , the probability measure on  $\{0, \dots, 2n\}$  proportional to  $h_b(n, \cdot)$  has mean exactly  $n$  and variance exactly  $2n/(b+2)$ .*

## 32 Euler indices of anticanonical hypersurfaces in products

**Conjecture 137** (Strict dominance law). *For every  $d \geq 3$  and every dominance move  $\lambda \rightarrow \mu$  on partitions of  $d+1$ ,*

$$E_d(\mu) > E_d(\lambda).$$

**Conjecture 138** (Strict refinement contraction). *For every  $d \geq 3$ , every  $\lambda$ , and every nontrivial split of one part,*

$$E_d(S_{m,a}\lambda) < E_d(\lambda).$$

**Conjecture 139** (The 8/9 refinement barrier). *Let  $S_d$  be the maximum of  $E_d(S_{m,a}\lambda)/E_d(\lambda)$  over all  $\lambda$  and all one-part splits. Then  $S_d \rightarrow 8/9$  as  $d \rightarrow \infty$ .*

**Conjecture 140** (Positive interaction of disjoint refinements). *For  $d \geq 6$  and splits  $S, T$  acting on distinct parts of  $\lambda$ ,*

$$E_d(\lambda) E_d(ST\lambda) > E_d(S\lambda) E_d(T\lambda).$$

### 33 Betti profiles of Hilbert schemes of surfaces

**Conjecture 141** (Strict Betti log-concavity). *For every  $b \geq 3$ ,  $n \geq 1$ , and  $0 < k < 2n$ ,*

$$h_b(n, k)^2 > h_b(n, k-1) h_b(n, k+1).$$

**Conjecture 142** (Cohomology-seed total positivity). *For every  $b \geq 1$ ,  $n \geq 1$ , and  $0 \leq k < n$ ,*

$$h_{b+1}(n, k+1) h_b(n, k) \geq h_{b+1}(n, k) h_b(n, k+1).$$

**Conjecture 143** (Kurtosis trichotomy). *Let  $\kappa_b(n)$  be the standardized fourth moment of the profile of Proposition 20. Then: for  $b = 3, 4, 5$ ,  $\kappa_b(n)$  increases strictly to  $21/5$ ; for  $b = 6$  it has a unique maximum at  $n = 9$  and decreases thereafter to  $21/5$ ; for  $b = 7$ ,  $\kappa_7(1) = \kappa_7(2)$  and it decreases strictly thereafter; for  $b \geq 8$  it decreases strictly to  $21/5$ .*

**Conjecture 144** (Fixed-charge log-concavity). *For every  $b \geq 3$  and fixed  $j \geq 0$ , the sequence  $n \mapsto h_b(n, n+j)$ ,  $n \geq \max(1, j)$ , is strictly log-concave.*

### 34 Twisted-sector ages of cyclic quotients

**Conjecture 145** (Sharp low-dimensional age unimodality). *For  $d = 4$  and  $d = 5$ , the age histogram  $(N_1, \dots, N_{d-1})$  is unimodal for every isolated cyclic Calabi–Yau quotient  $\frac{1}{r}(a_1, \dots, a_d)$ .*

**Conjecture 146** (Gaussian bulk age law). *Let  $r$  be prime and let distinct nonzero weights  $a_1, \dots, a_d$  be uniform conditioned on zero sum. If  $d, r \rightarrow \infty$  with  $d = o(r)$ , then in probability the empirical law of  $(\text{age}(j) - d/2)/\sqrt{d/12}$  over  $1 \leq j < r$  converges weakly to the standard Gaussian.*

**Conjecture 147** (Local central-sector law). *Under the sampling law of Conjecture 146 with  $d$  even,  $N_{d/2}/(r-1) \sim \sqrt{6/(\pi d)}$ , and uniformly for  $|k - d/2| = O(\sqrt{d \log d})$  the ratio  $N_k/(r-1)$  is asymptotic to the corresponding lattice Gaussian mass.*

**Conjecture 148** (High-sampling restoration of unimodality). *In the random prime-order model, if  $r/d^3 \rightarrow \infty$  then the entire age histogram is unimodal with probability tending to one.*

## 35 Exterior torsion entropy of toral automorphisms

**Conjecture 149** (Exterior torsion-entropy log-concavity). *For every nonzero zero-sum real spectrum  $x$  and  $1 < k < d - 1$ ,*

$$h_k(x)^2 \geq h_{k-1}(x) h_{k+1}(x).$$

**Conjecture 150** (Equality rigidity). *If an interior equality  $h_k(x)^2 = h_{k-1}(x)h_{k+1}(x)$  occurs, then the multiset  $\{x_1, \dots, x_d\}$  has exactly two distinct values.*

**Conjecture 151** (Sharp binomial upper envelope, resolved true). *For every zero-sum spectrum and every  $1 \leq k \leq d - 1$ ,*

$$h_k(x) \leq \binom{d-2}{k-1} h_1(x),$$

*with equality in every degree for spectra proportional to  $(1, 0, \dots, 0, -1)$ .*

*Resolution.* True, by Proposition 21 below.

**Proposition 21** (Resolution of the sharp binomial upper envelope). Conjecture 151 holds in full.

*Proof.* If  $h_1(x) = 0$ , then the zero-sum spectrum is zero and the result is immediate. Otherwise scale so that  $h_1(x) = 1$ . Fix the sets  $P$  and  $N$  of positive and negative coordinates. The positive coordinates lie in the simplex  $\sum_{i \in P} x_i = 1$ , while the magnitudes of the negative coordinates lie in the simplex  $\sum_{j \in N} (-x_j) = 1$ . On the closure of their product,

$$h_k(x) = \sum_{|S|=k} \left( \sum_{i \in S} x_i \right)_+$$

is a convex function, since it is a sum of positive parts of linear forms. A convex function on a compact polytope attains a maximum at a vertex. At such a vertex exactly one coordinate is 1, exactly one is  $-1$ , and every other coordinate is zero. A  $k$ -subset contributes 1 precisely when it contains the positive coordinate and avoids the negative coordinate. There are  $\binom{d-2}{k-1}$  such subsets. Rescaling proves the inequality, and the vertex calculation proves the asserted equality for spectra proportional to  $(1, 0, \dots, 0, -1)$ .  $\square$

**Conjecture 152** (Two-level lower envelope, resolved true). *Normalize  $h_1(x) = 1$ . For every  $d$  and  $k$ , the minimum of  $h_k$  over zero-sum spectra is attained by a two-level spectrum  $x^{p,q}$  with  $p$  entries  $1/p$ ,  $q$  entries  $-1/q$ , and  $d - p - q$  zeros.*

*Resolution.* True, by Proposition 22 below.

**Proposition 22** (Resolution of the two-level lower envelope). Conjecture 152 holds in full.

*Proof.* Let  $x$  be a zero-sum spectrum with  $h_1(x) = 1$ , and let  $P$  and  $N$  be its positive and negative index sets with sizes  $p$  and  $q$ . The function  $h_k$  is convex and invariant under permutations within  $P$  and within  $N$ . Replace two positive coordinates by their average. The new spectrum is the midpoint of the original spectrum and the spectrum obtained by swapping those coordinates, so convexity and symmetry show that  $h_k$  cannot increase. Repeating this operation makes every

positive coordinate equal to  $1/p$ . The same argument applied to the negative magnitudes makes every negative coordinate equal to  $-1/q$ . The zero coordinates remain zero. Thus

$$h_k(x^{p,q}) \leq h_k(x).$$

There are only finitely many pairs  $p, q \geq 1$  with  $p + q \leq d$ , and every  $x^{p,q}$  is admissible. The global minimum is therefore attained by one of these two-level spectra.  $\square$

## 36 Matroidal Hochster tables

**Conjecture 153** (Nullity-strand interval support, resolved false). *For every simple binary matroid  $M$  and every nullity  $c$ , the support  $\{s : H_{c,s}(M) > 0\}$  is an interval of integers.*

*Resolution.* False, by Proposition 23 below: the cycle matroid of  $K_{3,3}$  has nullity-one strand  $(9, 0, 6)$  at sizes four to six.

**Proposition 23** (A graphic nullity-strand certificate). Let  $M = M(K_{3,3})$  be the cycle matroid of the complete bipartite graph. Then

$$H_{1,4}(M) = 9, \quad H_{1,5}(M) = 0, \quad H_{1,6}(M) = 6.$$

The matroid  $M$  is simple, binary, and 3-connected.

*Proof.* In a nullity-one graphic restriction, the top reduced homology is nonzero exactly when the restricted edge set is a circuit. Indeed, any edge outside the unique circuit is a coloop, which makes the independence complex a cone. The circuits of  $M(K_{3,3})$  are the cycles of  $K_{3,3}$ . There are  $\binom{3}{2}^2 = 9$  cycles of length four, no cycles of length five by bipartiteness, and six Hamilton cycles of length six. This gives the displayed strand. The graph  $K_{3,3}$  is simple and 3-vertex-connected, so its cycle matroid is simple, binary, and 3-connected.  $\square$

**Remark 10** (Resolution). *Conjecture 153 is false. Proposition 23 gives nullity-one support  $\{4, 6\}$ , with an internal gap at size five.*

**Conjecture 154** (Row-and-column log-concavity, resolved false). *Both coordinate directions of the table are log-concave:  $H_{c,s}^2 \geq H_{c,s-1}H_{c,s+1}$  and  $H_{c,s}^2 \geq H_{c-1,s}H_{c+1,s}$ .*

*Resolution.* False: the same nullity-one strand  $(9, 0, 6)$  of  $K_{3,3}$  gives the horizontal failure  $0^2 < 9 \cdot 6$ ; the vertical direction remains a separate open question.

**Remark 11** (Resolution). *Conjecture 154 is false. For the matroid of Proposition 23, the horizontal inequality at  $(c, s) = (1, 5)$  reads  $0^2 \geq 9 \cdot 6$ .*

**Conjecture 155** (Nearest-rhombus log-concavity, resolved false). *Both diagonal directions are log-concave:  $H_{c,s}^2 \geq H_{c-1,s-1}H_{c+1,s+1}$  and  $H_{c,s}^2 \geq H_{c-1,s+1}H_{c+1,s-1}$ .*

*Resolution.* False, by Proposition 24 below: a ten-edge graphic matroid has  $H_{2,6} = 2$ ,  $H_{1,5} = 2$ ,  $H_{3,7} = 3$ , giving  $4 < 6$ .

**Proposition 24** (A graphic rhombus certificate). Let  $G$  have vertex set  $\{0, \dots, 6\}$  and edge set

$$\{03, 04, 05, 13, 14, 23, 24, 26, 34, 56\},$$

where  $ij$  denotes the edge  $\{i, j\}$ . For its cycle matroid  $M(G)$ ,

$$H_{1,5} = 2, \quad H_{2,6} = 2, \quad H_{3,7} = 3.$$

The graph  $G$  is simple and 2-connected.

*Proof.* For a restriction  $A$ , its contribution is the absolute reduced Euler characteristic of its independence complex,

$$\beta(A) = \left| -1 + \sum_{\substack{\emptyset \neq I \subseteq A \\ I \text{ independent}}} (-1)^{|I|-1} \right|.$$

Deletion and contraction on the ten displayed edges gives the following complete list of nonzero contributions to the three cells. At  $(1, 5)$  the two five-cycles  $\{03, 05, 23, 26, 56\}$  and  $\{04, 05, 24, 26, 56\}$  contribute one each. At  $(2, 6)$  the restriction  $\{03, 04, 13, 14, 23, 24\}$  contributes two. At  $(3, 7)$  that restriction together with 34 contributes three. Every other restriction of the required size and nullity has zero top reduced homology. Summing gives the displayed values. Deleting any one vertex leaves  $G$  connected, so  $G$  is 2-connected.  $\square$

**Remark 12** (Resolution). *Conjecture 155 is false. Proposition 24 gives*

$$H_{2,6}^2 = 4 < 6 = H_{1,5}H_{3,7}.$$

**Conjecture 156** (Hurwitz stability of nullity strands, resolved false). *For every simple binary matroid and every nonconstant nullity strand, every zero of the strand polynomial  $\sum_s H_{c,s}(M) z^{s-s_{\min}}$ , with  $s_{\min}$  the least size in its support, has strictly negative real part.*

*Resolution.* False: strict Hurwitz stability fails at the boundary of the left half-plane, linked to the same support gap that refutes Conjectures 153 and 154.

**Remark 13** (Resolution). *Conjecture 156 is false. The nullity-one polynomial of Proposition 23, after removing its lowest monomial, is  $9 + 6z^2$ . Its zeros are  $\pm i\sqrt{3}/2$  and have real part zero.*

## Part IX

# Spectra and filtrations of combinatorial geometries

*Added to the deposit: 12 August 2026.*

The twenty conjectures below extend the geometric turn of Parts VII and VIII to spectra and filtrations. They form four programmes. A toric programme (Conjectures 157 to 161) studies the  $h$ -polynomials of graph associahedra, the even-Betti spectra of their smooth toric varieties, through root location, interlacing, and a Mahler-measure order. A McKay programme



(Conjectures 162 to 166) filters the Cayley graph of an isolated cyclic Calabi–Yau quotient by orbifold age and conjectures one-crest laws for the low homology of the resulting flag complexes. A torsion programme (Conjectures 167 to 171) studies microcanonical ensembles of lens-space sine factors, their temperature-dependent sector shapes, and a  $\zeta(3)$  central parabola over the log-sine moment identities proved below. An entropy programme (Conjectures 172 to 176) orders the boundary-map pseudodeterminants of simplicial spheres across degree and under stellar refinement. The calibration layer is exact where the objects are: tubing  $f$ -vectors,  $\mathbf{F}_2$  homology, and coefficient convolutions are integer computations, while root locations and pseudodeterminants are checked spectrally in floating point and. Four retained controls mark calibrated boundaries. Conjectures 169 and 170 assert limits in a random or iterated-limit model and are the boldest; Conjectures 157, 161, 164, 169, and 170 sit closest to classical machinery; and Conjecture 170 deliberately strengthens the annealed Gaussian age law drawn from Conjecture 146 to a joint decoupling law.

For a connected simple graph  $G$  on  $n \geq 2$  vertices, a *tube* is a proper connected induced subgraph, two tubes being compatible when nested or disjoint without adjacency, and a *tubing* is a set of pairwise compatible tubes; the graph associahedron  $P_G$  is the simple  $(n - 1)$ -polytope whose codimension- $k$  faces are the tubings with  $k$  tubes [32]. Write  $h_G(t) = \sum_i h_i(G)t^i$  for its  $h$ -polynomial, so that  $h_i(G) = b_{2i}(X_G)$  for the smooth projective toric variety of the normal fan, a positive palindromic sequence; put  $\mathcal{M}(G) = \sum_{h_G(\rho)=0} \log \max(1, |\rho|)$ , the logarithmic Mahler measure, and  $\mu_G(i) = h_i(G)/h_G(1)$ . For an odd prime  $p$  and pairwise distinct weights  $a = (a_1, \dots, a_d) \in (\mathbb{F}_p^\times)^d$  with  $\sum_i a_i = 0$ , the quotient  $\frac{1}{p}(a_1, \dots, a_d)$  acts on the vertex set  $\mathbb{Z}/p$ , including 0, through the Cayley graph  $x \sim y$  when  $y - x \in \{\pm a_1, \dots, \pm a_d\}$ ; with  $\text{age}(j) = \frac{1}{p} \sum_i [ja_i]_p$  and  $\text{age}(0) = 0$ , let  $K_k$  be the flag complex of the induced graph on  $\{j : \text{age}(j) \leq k\}$  for  $0 \leq k \leq d - 1$ , and write  $c_k$  and  $b_{q,k}$  for its number of components and its  $q$ -th  $\mathbf{F}_2$  Betti number. For a prime  $p \geq 5$  let  $\Omega_{p,d,m}$  be the tuples  $x \in \{1, \dots, p - 1\}^d$  with  $\sum_i x_i = mp$ , let  $H_p(x) = \sum_i \log(2 \sin(\pi x_i/p))$ , the lens-space log-torsion energy [33], and set  $Z_{p,d,m}(s) = \mathbb{E}[e^{sH_p}]$  and  $\Phi_{p,d,m} = \mathbb{E}[H_p]$  over uniform  $\Omega_{p,d,m}$ . Finally, for a finite simplicial  $d$ -sphere  $K$  with boundary maps  $\partial_k$ , put  $D_k(K) = \text{pdet}(\partial_k \partial_k^*)$ , the product of nonzero eigenvalues,  $E_k(K) = \log D_k(K)$ ,  $r_k(K) = \text{rank } \partial_k$ , and  $A_k(K) = E_k(K)/r_k(K)$ , the entropy per boundary rank, with higher matrix-tree readings of these determinants in [34, 35].

The sector expectations  $Z_{p,d,m}$  and  $\Phi_{p,d,m}$  are defined only when  $\Omega_{p,d,m}$  is nonempty. This holds exactly when  $d \leq mp \leq d(p - 1)$ .

**Proposition 25** (Log-sine moment identities). For every integer  $p \geq 2$ ,

$$\prod_{j=1}^{p-1} 2 \sin \frac{\pi j}{p} = p, \quad \text{and} \quad \int_0^1 \left(u - \frac{1}{2}\right)^2 \log(2 \sin \pi u) du = -\frac{\zeta(3)}{2\pi^2}.$$

## 37 Toric spectra of graph associahedra

**Conjecture 157** (Universal negative real roots). For every connected simple graph  $G$ , every zero of  $h_G(t)$  is real and strictly negative.

**Conjecture 158** (Connected-deletion interlacing). If  $v$  is a vertex of  $G$  with  $G - v$  connected, the zeros of  $h_{G-v}$  weakly interlace the zeros of  $h_G$ .

**Conjecture 159** (Strict Mahler growth under edge addition). For every missing edge  $e$  of a connected graph  $G$ ,  $\mathcal{M}(G + e) > \mathcal{M}(G)$ .

**Conjecture 160** (Edge addition centralizes the spectrum). *For every missing edge  $e$  of a connected graph  $G$  and every symmetric outer tail that is nonempty and proper,  $\sum_{|i-c|\geq r} \mu_{G+e}(i) \leq \sum_{|i-c|\geq r} \mu_G(i)$ , where  $c = (n-1)/2$ .*

**Conjecture 161** (Path-star Mahler extremality). *For every tree  $T$  on  $n \geq 4$  vertices,  $\mathcal{M}(P_n) \leq \mathcal{M}(T) \leq \mathcal{M}(K_{1,n-1})$ , with equality on the left only for the path and on the right only for the star.*

## 38 Age-filtered McKay complexes

**Conjecture 162** (One-crest component law). *For every isolated cyclic quotient with pairwise distinct weights, the component sequence  $(c_0, \dots, c_{d-1})$  is unimodal.*

**Conjecture 163** (Early component crest). *The first index attaining  $\max_k c_k$  is at most  $\lfloor d/2 \rfloor$ .*

**Conjecture 164** (Half-age connectivity). *For  $d \geq 5$ , the sublevel complex  $K_{\lfloor d/2 \rfloor}$  is connected.*

**Conjecture 165** (One-crest first homology). *The sequence  $(b_{1,0}, \dots, b_{1,d-1})$  is unimodal.*

**Conjecture 166** (One-crest second homology). *The sequence  $(b_{2,0}, \dots, b_{2,d-1})$  is unimodal.*

## 39 Sine-torsion sectors of cyclic quotients

**Conjecture 167** (Positive-temperature log-concavity). *For every  $s > 0$  and  $2 \leq m \leq d-2$  such that  $\Omega_{p,d,q}$  is nonempty for  $q \in \{m-1, m, m+1\}$ ,  $Z_{p,d,m}(s)^2 > Z_{p,d,m-1}(s) Z_{p,d,m+1}(s)$ .*

**Conjecture 168** (Concavity of mean log torsion). *For  $2 \leq m \leq d-2$  such that  $\Omega_{p,d,q}$  is nonempty for  $q \in \{m-1, m, m+1\}$ ,  $2\Phi_{p,d,m} > \Phi_{p,d,m-1} + \Phi_{p,d,m+1}$ .*

**Conjecture 169** (The  $\zeta(3)$  central parabola). *Let  $m_d$  satisfy  $(m_d - d/2)/\sqrt{d/12} \rightarrow x$ . Then, taking  $p \rightarrow \infty$  through primes first and  $d \rightarrow \infty$  second,*

$$\lim_{d \rightarrow \infty} \lim_{p \rightarrow \infty} \left( \Phi_{p,d,m_d} - \frac{d \log p}{p-1} \right) = \frac{3\zeta(3)}{\pi^2} (1-x^2).$$

**Conjecture 170** (Age-torsion decoupling). *With weights uniform on distinct nonzero zero-sum tuples and  $J$  uniform on  $\mathbb{F}_p^\times$ , the standardized pair of  $\text{age}_A(J)$  and of the centred log-torsion sum converges to a standard bivariate Gaussian as  $p, d \rightarrow \infty$  with  $d = o(p)$ .*

**Conjecture 171** (Negative-temperature reversal). *For  $-1 \leq s < 0$  and  $2 \leq m \leq d-2$  such that  $\Omega_{p,d,q}$  is nonempty for  $q \in \{m-1, m, m+1\}$ ,  $Z_{p,d,m}(s)^2 < Z_{p,d,m-1}(s) Z_{p,d,m+1}(s)$ .*

## 40 Boundary-determinant entropy of simplicial spheres

**Conjecture 172** (Log-concavity of boundary entropy). *For every simplicial  $d$ -sphere and  $2 \leq k \leq d-1$ ,  $E_k(K)^2 \geq E_{k-1}(K) E_{k+1}(K)$ .*

**Conjecture 173** (Entropy per rank decreases). *For every simplicial  $d$ -sphere,  $A_1(K) \geq A_2(K) \geq \dots \geq A_d(K)$ .*

**Conjecture 174** (Simplex rigidity of equality). *An adjacent equality  $A_k(K) = A_{k+1}(K)$  occurs for some  $k$  if and only if  $K$  is the boundary of a simplex, in which case all adjacent equalities occur.*

**Conjecture 175** (Strict stellar growth). *If  $K'$  is obtained from  $K$  by stellar subdivision of a facet, then  $E_k(K') > E_k(K)$  for every  $1 \leq k \leq d$ .*

**Conjecture 176** (Log-concave stellar increments). *With  $\delta_k = E_k(K') - E_k(K)$  as in Conjecture 175,  $\delta_k^2 \geq \delta_{k-1}\delta_{k+1}$  for  $2 \leq k \leq d-1$ .*

## Part X

# Flux cohomology and protected index laws

*Added to the deposit: 12 August 2026.*

The twenty conjectures below carry the collection into string-geometric index theory. They form three programmes. A flux programme (Conjectures 177 to 184) studies the cohomology of wedge multiplication by an integral three-form on a torus, the invariant-forms model of  $H$ -twisted cohomology, through saturation laws for random flux, sparse witnesses, and an arithmetic torsion staircase for the complete flux. A profile programme (Conjectures 185 to 191) studies the signed  $\chi_y$ -coefficient profile of projective Calabi–Yau complete intersections along the merge order of Part VII, extending the proved Euler monotonicity of Conjecture 118 to every protected coefficient, with root, log-concavity, and Mahler laws. A signature programme (Conjectures 192 to 196) does the same for the Hirzebruch signature, ending in a quadric ratio law with limit 27 and a dimension-wise gcd law that is the signature sibling of Conjecture 121. The calibration layer is exact where the objects are: finite-field cohomology, Smith normal forms,  $\chi_y$ -coefficients, signatures, merge inequalities, ratios, and gcds are integer or rational computations, while characteristic-zero ranks beyond dimension nine are dual-prime proxies and the root, Mahler, and radius laws of Conjectures 188, 190, and 191 are checked spectrally in floating point and. Two retained controls mark calibrated boundaries, a falsified all-prime torsion precursor and a falsified elementary-torsion strengthening. Conjectures 177, 178, 180, and 184 are stated in a random model and Conjecture 181 is stated in greater generality than the computations reach, the part’s boldest statement; Conjectures 184, 189, and 195 sit closest to classical machinery; and Conjecture 190 returns the Mahler order of Conjecture 159 on a different polynomial.

For  $n \geq 3$  and a commutative ring  $R$  let  $E_n(R) = \Lambda^* R^n$ , and for an integral three-form  $H \in \Lambda^3 \mathbb{Z}^n$  let  $d_H$  be wedge multiplication by  $H$ , a differential because  $H$  has odd degree; write  $b_k(H; R)$  for the dimensions of the cohomology of  $(E_n(R), d_H)$ ,  $B(H; R) = \sum_k b_k(H; R)$ , and  $t_p(H)$  for the number of  $p$ -primary cyclic summands of the integral cohomology. This complex is the invariant-forms model of  $H$ -twisted cohomology [38], whose K-theoretic refinement classifies Ramond–Ramond charges in a topologically nontrivial B-field [39]. The *generic target*  $g_n$  is the cohomology profile on the Zariski-open locus of maximal wedge ranks, that of a generic principal ideal generated in degree three: for  $n = 2m$  it is  $(3^{m-1}, 2 \cdot 3^{m-1}, 3^{m-1})$  in degrees  $m-1, m, m+1$ ; for  $n \equiv 3 \pmod{4}$  it is  $3^{(n-1)/2}$  in each of the two middle degrees; for  $n \equiv 1 \pmod{4}$  it has inner pair  $3^{(n-1)/2}$  and outer pair 1, except that the outer pair is 3 at  $n = 9$ ; its total is  $G_n$ . Lundqvist and Nicklasson prove a lower bound for the generic Hilbert series in every odd degree, disproving the Moreno-Socias–Snellman conjecture above degree three, and establish

equality only in special cases, leaving the minimal series open in general and in degree three in particular [40]; the displayed profile is therefore the expected one and not a cited theorem, and the exceptional outer pair at  $n = 9$  is the four-dimensional kernel computed there. The *complete flux* is  $H_\Delta = \sum_{i < j < k} e_i \wedge e_j \wedge e_k$ , and the target totals are  $F_n = \binom{n+1}{(n+1)/2}$  for odd  $n$ ,  $F_n = 2\binom{n}{n/2}$  for even  $n$ , and  $P_n = 2^{n-1} + 2^{(n-1)/2}$  for odd  $n$ ,  $P_n = 2(2^{n-2} + 2^{(n-2)/2})$  for even  $n$ . Write  $t_{p,k}(n)$  for the number of  $p$ -primary cyclic summands of the integral cohomology of  $H_\Delta$  in degree  $k$ , and  $T_p(n) = \sum_k t_{p,k}(n)$  for their total. On the geometric side,  $X_{\mathbf{d}}^n \subset \mathbb{CP}^{n+r}$  is the Calabi–Yau complete intersection of multidegree  $\mathbf{d} = (d_1, \dots, d_r)$ ,  $d_i \geq 2$ ,  $\sum_i d_i = n + r + 1$ , with degree  $D(\mathbf{d}) = \prod_i d_i$ ; merging replaces  $d_i, d_j$  by  $d_i + d_j - 1$ , preserving dimension and the Calabi–Yau condition, exactly as in Part VII. Writing  $\chi_y = \sum_p \chi^p y^p$  for the Hirzebruch genus [42], put  $a_p(\mathbf{d}) = (-1)^{n-p} \chi^p(X_{\mathbf{d}}^n)$  and  $A_{\mathbf{d}}(y) = \sum_p a_p(\mathbf{d}) y^p$ , a palindromic integer polynomial with nonnegative coefficients throughout the tested range; for even  $n$  put  $S(\mathbf{d}) = (-1)^{n/2} \sigma(X_{\mathbf{d}}^n)$ . Let  $Q = (2, \dots, 2)$  be the all-quadric and  $Y = (n + 2)$  the hypersurface configuration.

**Proposition 26** (Coefficient and parity calibrations). For every  $H \in \Lambda^3 \mathbb{Z}^n$  and every prime  $p$ ,  $B(H; \mathbb{F}_p) = B(H; \mathbb{Q}) + 2t_p(H)$ . If every coefficient of  $H$  is odd, then the complexes of  $H$  and of  $H_\Delta$  over  $\mathbf{F}_2$  coincide, so  $B(H; \mathbb{F}_2) = B(H_\Delta; \mathbb{F}_2)$ .

## 41 Integral flux complexes on tori

**Conjecture 177** (Dense discrete flux saturation). Fix a finite symmetric set  $V \subset \mathbb{Z} \setminus \{0\}$ , for example  $\{\pm 1\}$ , and let every coefficient of  $H_n \in \Lambda^3 \mathbb{Z}^n$  be independent and uniform on  $V$ . Then  $\Pr[(b_k(H_n; \mathbb{Q}))_k = g_n] \rightarrow 1$ , and the failure probability decays exponentially in  $n$ .

**Conjecture 178** (Sparse-flux threshold at the coverage scale). Include each coordinate triple independently with probability  $p_n$  and give included triples independent coefficients from a fixed finite symmetric set of nonzero integers. If  $p_n n^2 / \log n \rightarrow \infty$  then  $\Pr[(b_k; \mathbb{Q})_k = g_n] \rightarrow 1$ .

**Proposition 27** (Coverage floor for the sparse threshold). If  $p_n = (c + o(1)) \log n / n^2$  with a fixed  $c < 2$ , then the saturation probability of Conjecture 178 tends to zero.

**Conjecture 179** (Linear-complexity sparse witnesses). Let  $\mu_n$  be the minimum support size of a  $\{\pm 1\}$ -valued three-form on  $\mathbb{Z}^n$  with rational profile  $g_n$ . Then for all sufficiently large  $n$  a saturating support of at most  $n + C$  triples can be chosen to form a linear 3-uniform hypergraph, for an absolute constant  $C$ ; in particular  $\mu_n = \Theta(n)$ .

**Conjecture 180** (Coefficient-law universality in the sparse window). Fix  $c > 0$  and let the support be sampled at  $p = c \log n / n^2$  as in Conjecture 178. For every finite symmetric coefficient set  $V \subset \mathbb{Z} \setminus \{0\}$ , the limiting saturation probability and the limiting joint law of the excess vector  $(b_k(H; \mathbb{Q}) - g_{n,k})_k$  exist and depend only on  $c$ , not on  $V$ .

**Proposition 28** (Support obstructions and coefficient rescaling). Let  $H \in \Lambda^3 \mathbb{Z}^n$  have support  $S$ , let  $u$  be the number of coordinates lying in no triple of  $S$ , and let  $A$  be the  $|S| \times n$  incidence matrix of  $S$ . (i) The Betti polynomial of  $(E_n(\mathbb{Q}), d_H)$  is  $(1+t)^u$  times that of the form read in the used coordinates, and  $H$  attains the generic profile only if  $g_n = (1+t)^u g_{n-u}$ ; for the deposited profile this forces  $u \leq 1$ , with  $u = 1$  possible only when  $n \equiv 0 \pmod{4}$ . (ii) Consequently  $\mu_n \geq \lceil (n-1)/3 \rceil$ , and if  $p_n n^2 / \log n \rightarrow 0$  then the saturation probability of Conjecture 178 tends to zero. (iii) If  $\text{rank}_{\mathbf{F}_2} A = |S|$  then the rational profile of  $H$  depends only on  $S$  and not on

the coefficients. Coefficient dependence therefore requires a nonempty  $S' \subseteq S$  covering every coordinate an even number of times, and then  $|S'|$  is even,  $|S'| \geq 4$ , and  $|V(S')| \leq \frac{3}{2}|S'|$ . Since  $A$  has  $n$  columns the criterion requires  $|S| \leq n$ , so it is vacuous in the window of Conjecture 180, where  $|S| = \Theta(n \log n)$  and the left kernel of  $A$  has dimension at least  $|S| - n$ .

**Conjecture 181** (Finite-field resonance power law). *Fix  $n \geq 5$ . There are a positive integer  $\delta_n$ , a finite Galois extension  $K_n/\mathbb{Q}$ , and nonnegative integers  $\kappa_{n,\sigma}$  indexed by the Frobenius conjugacy classes  $\sigma$  of  $K_n/\mathbb{Q}$ , positive for at least one class, such that, for  $H$  uniform on  $\Lambda^3 \mathbb{F}_p^n$ , the probability of the resonance event  $R_{n,p}$ , that the profile  $(b_k(H; \mathbb{F}_p))_k$  exceeds  $g_n$  in some degree, satisfies*

$$\Pr(R_{n,p}) = \kappa_{n,\text{Frob}_p} p^{-\delta_n} + O_n(p^{-\delta_n-1/2}) \quad \text{as } p \rightarrow \infty.$$

**Proposition 29** (Conditional Lang–Weil resonance law). Fix  $n \geq 5$  and suppose that the displayed  $g_n$  is the characteristic-zero profile on the Zariski-open locus of maximal wedge ranks. Then the conclusion of Conjecture 181 holds for every sufficiently large prime of good reduction. More precisely,  $\delta_n$  is the codimension of the largest geometric component of the resonance scheme, and  $\kappa_{n,\text{Frob}_p}$  is the number of its largest geometric components fixed by Frobenius.

**Conjecture 182** (Central-binomial law for the complete flux, resolved true). *For every  $n \geq 4$ ,  $B(H_\Delta; \mathbb{Q}) = F_n$  and  $B(H_\Delta; \mathbb{F}_2) = P_n$ .*

*Resolution.* The conjecture is true: Proposition 30 proves both formulas, reducing wedge multiplication by  $H_\Delta = \alpha \wedge \omega$  to Boolean inclusion matrices, whose rational ranks are full and whose characteristic-two complex is exact. The rational total is the binomial transform  $\sum_s \binom{g}{s} 2^s \binom{g-s}{\lfloor (g-s)/2 \rfloor} = \binom{2g+1}{g}$  and the characteristic-two total is  $4^g + 2^g$ ,  $g = \lfloor (n-1)/2 \rfloor$ ; the symmetric-group route anticipated at deposit is replaced by a symplectic normal form.

**Conjecture 183** (Prime torsion staircase for the complete flux, resolved false). *For every prime  $p$ , the integral cohomology of  $(E_n(\mathbb{Z}), d_{H_\Delta})$  has no  $p$ -primary torsion for  $n \leq 4p-2$  and exactly one  $p$ -primary cyclic summand, isomorphic to  $\mathbb{Z}/p$ , at  $n = 4p-1$ . Moreover, for every  $n$  the 2-primary subgroup is elementary abelian of rank  $t_2(H_\Delta) = (P_n - F_n)/2$ .*

*Resolution.* The staircase clause is true and the exponent clause is false, both by Proposition 30: the onset at  $n = 4p-1$  with a single  $\mathbb{Z}/p$  holds for every prime, while at  $n = 15$  the singleton-free sector carries the inclusion matrix  $U_{7,3}$ , whose Smith normal form  $1^{20}, 2^9, 6^5, 12$  contributes a summand  $\mathbb{Z}/4$  in degree nine. This is exactly the divisor exceeding two that the statement named as its own refutation, and  $n = 15$  is the least dimension at which it can occur, so the exhaustive range through  $n = 13$  could not have seen it; the surviving question is the exact 2-adic invariant factors, whose first departure from exponent one is now located.

**Proposition 30** (Complete-flux normal form). Write  $\alpha = e_1 + \cdots + e_n$  and  $\omega = \sum_{2 \leq i < j \leq n} e_i \wedge e_j$ , and put  $g = \lfloor (n-1)/2 \rfloor$ . Then  $H_\Delta = \alpha \wedge \omega$ , and the integral cohomology of  $(E_n(\mathbb{Z}), d_{H_\Delta})$  satisfies: (i)  $B(H_\Delta; \mathbb{Q}) = F_n$  and  $B(H_\Delta; \mathbb{F}_2) = P_n$ ; (ii) for every prime  $p$  there is no  $p$ -primary torsion for  $n \leq 4p-2$ , and exactly one  $p$ -primary cyclic summand, isomorphic to  $\mathbb{Z}/p$ , at  $n = 4p-1$ ; (iii) whenever  $p$ -primary torsion is present, the positive support of  $k \mapsto t_{p,k}(n)$  is the full interval  $2p+1 \leq k \leq n-2p+2$ ; (iv)  $T_p(n+1) > T_p(n)$  for every  $n \geq 4p-1$ ; (v) at  $n = 15$  the 2-primary subgroup is not elementary abelian: it contains a summand  $\mathbb{Z}/4$  in degree 9.

*Proof.* Comparing coefficients,  $\alpha \wedge \omega$  produces each triple meeting  $\{1\}$  once, and each triple  $\{a < b < c\} \subset \{2, \dots, n\}$  three times, from  $e_a \wedge (e_b \wedge e_c)$ ,  $e_b \wedge (e_a \wedge e_c)$  and  $e_c \wedge (e_a \wedge e_b)$  with

signs  $+, -, +$ ; so  $H_\Delta = \alpha \wedge \omega$ . The substitution  $(e_1, \dots, e_n) \mapsto (\alpha, e_2, \dots, e_n)$  is unimodular, so with  $W = \langle e_2, \dots, e_n \rangle$  we may write  $E_n(\mathbb{Z}) = \Lambda^* W \oplus \alpha \wedge \Lambda^* W$ . Since  $\alpha \wedge \alpha = 0$  the differential annihilates  $\alpha \wedge \Lambda^* W$  and carries  $x \in \Lambda^* W$  to  $\alpha \wedge Lx$ , where  $L = \omega \wedge -$  acts on  $\Lambda^* W$ ; as  $\alpha \wedge -$  is injective on  $\Lambda^* W$ , the cohomology is  $\ker L \oplus \alpha \wedge \operatorname{coker} L$ , so it is  $\operatorname{coker} L$  that carries every torsion summand.

The alternating form  $\omega$  has all elementary divisors equal to 1, so an integral change of basis of  $W$  writes it as  $x_1 \wedge y_1 + \dots + x_g \wedge y_g$ , together with one null basis vector when  $n$  is even. Split  $\Lambda^* W$  according to the state of each symplectic pair, empty,  $x_i, y_i$ , or  $x_i y_i$ . Then  $L$  annihilates every pair held in a singleton state, and on the remaining  $r = g - s$  pairs it sends the set of doubly occupied pairs to the sum of its one-element extensions; that is, on a sector with  $s$  singletons it is the 0–1 inclusion matrix  $U_{r,j}$  from  $j$ -subsets to  $(j+1)$ -subsets of an  $r$ -set. There are  $\binom{g}{s} 2^s$  such sectors, doubled by the null vector when  $n$  is even, and a monomial with  $s$  singletons and  $j+1$  doubly occupied pairs sits in cohomological degree  $s + 2j + 3$ .

Wilson's diagonal form for  $t$ -subset versus  $k$ -subset inclusion matrices [41], applied to  $U_{r,j}$  when  $2j+1 \leq r$  and to its transpose otherwise, gives diagonal entries  $h-i$  with multiplicity  $\binom{r}{i} - \binom{r}{i-1}$  for  $0 \leq i < h$ , where  $h = \min(j+1, r-j)$ . These number  $\binom{r}{h-1} = \min(\binom{r}{j}, \binom{r}{j+1})$ , so over  $\mathbb{Q}$  every  $U_{r,j}$  has full possible rank, a sector of  $r$  active pairs contributes  $2^{\lfloor r/2 \rfloor}$  to the total cohomology, and summing over sectors gives  $\sum_s \binom{g}{s} 2^s \cdot 2^{\lfloor (g-s)/2 \rfloor} = 2^{\binom{2g+1}{g}}$ , which is  $F_n$  for odd  $n$  and half of it for even  $n$  before the null vector doubles it back. Over  $\mathbf{F}_2$  the operator is multiplication by  $u_1 + \dots + u_r$  in  $\mathbf{F}_2[u_1, \dots, u_r]/(u_i^2)$ , hence conjugate under a linear change of variables to multiplication by  $u_1$  and therefore exact, so its ranks sum to  $2^{r-1}$  and such a sector contributes  $2^r$ , or 2 when  $r = 0$ ; summing gives  $4^g + 2^g$ , again doubled by the null vector in even dimension, which is  $P_n$ . This is (i).

A diagonal entry divisible by  $p$  forces  $h \geq p$  and hence  $r \geq 2p-1$ ; since  $r \leq g$ , no  $p$ -torsion occurs for  $n \leq 4p-2$ . At  $n = 4p-1$  we have  $g = 2p-1$ , the only qualifying sector is  $s = 0$ ,  $r = 2p-1$ ,  $j = p-1$ , and its diagonal contains the entry  $p$  exactly once, giving a single summand  $\mathbb{Z}/p$ ; this is (ii). In a sector with  $r$  active pairs,  $p$  divides a diagonal entry of  $U_{r,j}$  exactly when  $p-1 \leq j \leq r-p$ , so that sector realises the degrees  $s + 2j + 3$  for  $p-1 \leq j \leq g-s-p$ , together with their shifts by one when  $n$  is even. The sectors  $s = 0$  and  $s = 1$  already supply the two parities, running over  $2p+1, 2p+3, \dots, 2g-2p+3$  and  $2p+2, 2p+4, \dots, 2g-2p+2$  respectively, and the intervals for larger  $s$  are nested inside these; the union is therefore the single interval  $2p+1 \leq k \leq n-2p+2$ . This is (iii).

For (iv) let  $A_r$  count the  $p$ -primary summands contributed by one pure  $r$ -pair sector and put  $T_g = \sum_r \binom{g}{r} 2^{g-r} A_r$ , so that  $T_p(2g+1) = T_g$  while the null vector gives  $T_p(2g+2) = 2T_g$ . Pascal's rule yields  $T_{g+1} - 2T_g = \sum_{r \geq 1} \binom{g}{r-1} 2^{g+1-r} A_r$ , which is positive as soon as some  $A_r$  is, that is, after onset.

Finally let  $n = 15$ , so  $g = 7$ . The sector  $s = 0$ ,  $r = 7$ ,  $j = 3$  has  $h = 4$ , so Wilson's diagonal carries the entry 4: the Smith normal form of the  $35 \times 35$  matrix  $U_{7,3}$  is  $1^{20}, 2^9, 6^5, 12$ , its cokernel is  $(\mathbb{Z}/2)^{14} \oplus (\mathbb{Z}/3)^6 \oplus \mathbb{Z}/4$ , and the corresponding degree is  $s + 2j + 3 = 9$ . This is (v).  $\square$

**Conjecture 184** (Parity-locked torsion count for sign flux). *For dense  $\{\pm 1\}$  flux as in Conjecture 177, conditional on the rational profile equalling  $g_n$ , the number of 2-primary cyclic summands is exactly  $t_2(H_n) = (P_n - G_n)/2$ ; unconditionally, this count holds with probability tending to one.*



## 42 Protected profiles of complete intersections

**Conjecture 185** (Coefficientwise endpoint extremality). *For  $n \geq 3$  and every interior index  $1 \leq p \leq n - 1$ ,  $a_p(Q) \leq a_p(\mathbf{d}) \leq a_p(Y)$ , and equality across all interior indices holds only for the endpoint configuration itself. Dimension 2 is the equality calibration: every configuration has the K3 profile  $(2, 20, 2)$ .*

**Conjecture 186** (Degree-normalized merge monotonicity). *If  $\mathbf{d}'$  is obtained from  $\mathbf{d}$  by merging two entries, then  $a_p(\mathbf{d}')/D(\mathbf{d}') > a_p(\mathbf{d})/D(\mathbf{d})$  for every  $1 \leq p \leq n - 1$ .*

**Conjecture 187** (Maximal-degree raw monotonicity). *For  $n \geq 3$ , if a merge uses an entry equal to  $\max_i d_i$ , then  $a_p(\mathbf{d}') > a_p(\mathbf{d})$  for every  $1 \leq p \leq n - 1$ .*

**Conjecture 188** (Real-rooted protected polynomial). *For every configuration, the polynomial  $A_{\mathbf{d}}(y)$ , after removing in odd complex dimension the single factor  $y$  forced by  $a_0 = a_n = 0$ , has only simple, strictly negative real zeros.*

**Conjecture 189** (Strict ultra-log-concavity). *For every configuration and every interior triple of positive coefficients,*

$$(a_p/\binom{n}{p})^2 > (a_{p-1}/\binom{n}{p-1})(a_{p+1}/\binom{n}{p+1}).$$

**Conjecture 190** (Degree-normalized Mahler growth). *For every merge,  $M(A_{\mathbf{d}'})/D(\mathbf{d}') > M(A_{\mathbf{d}})/D(\mathbf{d})$ , where  $M$  is the multiplicative Mahler measure, the modulus of the leading coefficient times  $\prod_{\rho} \max(1, |\rho|)$  over the zeros  $\rho$ .*

**Conjecture 191** (Spectral-radius growth with low-dimensional rigidity). *Let  $\rho(\mathbf{d})$  be the largest modulus of a zero of the reduced polynomial of Conjecture 188, the polynomial  $A_{\mathbf{d}}$  with its forced factor removed. If a merge uses a maximal entry, then  $\rho(\mathbf{d}') \geq \rho(\mathbf{d})$ , with equality for every such merge in complex dimensions  $n = 2, 3, 5$  and strict inequality in every other dimension  $n \geq 4$ .*

## 43 Signature laws for complete intersections

**Conjecture 192** (Signature endpoint extremality). *In every even complex dimension  $n \geq 4$ ,  $S(Q) \leq S(\mathbf{d}) \leq S(Y)$ , with equality only at the corresponding endpoint. Dimension 2 is the calibration: every configuration is a K3 surface with  $S = 16$ .*

**Conjecture 193** (Degree-normalized signature monotonicity). *Every merge in even complex dimension satisfies  $S(\mathbf{d}')/D(\mathbf{d}') > S(\mathbf{d})/D(\mathbf{d})$ .*

**Conjecture 194** (Maximal-degree signature monotonicity). *In even complex dimension  $n \geq 4$ , a merge using a maximal entry strictly increases  $S$ .*

**Conjecture 195** (The quadric ratio law with limit 27, resolved true). *For even  $n$  let  $\Sigma_n = S(Q)$  for the all-quadric configuration of  $n + 1$  quadric equations in complex dimension  $n$ . Then  $R_n = \Sigma_{n+2}/\Sigma_n$  is strictly increasing in  $n$ ,  $R_n < 27$ , and*

$$R_n \rightarrow 27, \quad 27 - R_n \sim \frac{27}{n}.$$

*Resolution.* True, by Proposition 31 below, the quadric-ladder ratio law with limit 27.

**Proposition 31** (Resolution of the quadric ratio law). Conjecture 195 holds in full. With  $a_m = \Sigma_{2m}$  and  $a_0 = 2$ , the generating function  $A(w) = \sum_{m \geq 0} a_m w^m$  is algebraic,

$$A(w) = \frac{2}{(1+y)(1-3y)}, \quad 4w = y(1-y)^2, \quad y(0) = 0,$$

and the coefficients satisfy  $D_m a_{m+2} = (M_m - D_m) a_{m+1} + M_m a_m$  with  $D_m = (m+2)(2m+3)(14m+9)$  and  $M_m = 6(3m+2)(3m+4)(14m+23)$ .

*Proof.* For the all-quadric configuration in complex dimension  $n = 2m$  there are  $2m+1$  defining quadrics, so the ambient dimension is  $N = 4m+1$ , and the classical signature formula  $\sigma = [x^N] (x/\tanh x)^{N+1} \prod_i \tanh(d_i x)$  for complete intersections in  $\mathbb{CP}^N$  [42] gives  $\sigma = [x^{4m+1}] (x/\tanh x)^{4m+2} \tanh(2x)^{2m+1}$ . Setting  $f(x) = x \tanh(2x)/\tanh^2 x$ , an even power series with  $f(0) = 2$ , the expression under the coefficient equals  $x^{2m+1} f(x)^{2m+1}$ , so  $\sigma = [x^{2m}] f(x)^{2m+1}$ . Because  $f$  is even, replacing  $x^2$  by  $-z$  multiplies the coefficient of  $x^{2m}$  by  $(-1)^m$  and turns  $f$  into  $\psi(z) = \sqrt{z} \tan(2\sqrt{z})/\tan^2 \sqrt{z}$ , analytic on  $|z| < \pi^2/4$ ; hence  $a_m = (-1)^m \sigma = [z^m] \psi(z)^{2m+1}$ . For each small  $w > 0$  the equation  $z = w\psi(z)^2$  has exactly one root  $z_0(w)$  near the origin, and the residue form of Lagrange inversion, summing the geometric series  $\sum_m (w\psi(z)^2/z)^m$  on a small contour and picking up the simple pole at  $z_0$ , gives

$$A(w) = \sum_{m \geq 0} [z^m] \psi(z)^{2m+1} w^m = \frac{\psi(z_0)}{1 - w(\psi^2)'(z_0)}.$$

Substituting  $z = u^2$  and  $T = \tan u$  and using the double-angle formula  $\tan 2u = 2T/(1-T^2)$ , the root equation becomes  $4w = T^2(1-T^2)^2$  and the residue value becomes  $2/((1+T^2)(1-3T^2))$ , two exact identities; with  $y = T^2$  this is the stated parametrization.

Next, the recurrence. Write  $\theta = w d/dw$ , let  $D(\theta)$  and  $M(\theta)$  denote the polynomials  $D_m$  and  $M_m$  with  $m$  replaced by the operator  $\theta$ , and put  $B_s = w^{-s}(A - \sum_{i < s} a_i w^i)$ , the generating function of the shifted sequence  $(a_{m+s})_{m \geq 0}$ . Then

$$\sum_{m \geq 0} (D_m a_{m+2} - (M_m - D_m) a_{m+1} - M_m a_m) w^m = D(\theta) B_2 - (M(\theta) - D(\theta)) B_1 - M(\theta) B_0,$$

and under the parametrization, where  $\theta$  acts as  $\frac{y(1-y)}{1-3y} \frac{d}{dy}$ , the right side is a rational function of  $y$  that vanishes identically, a finite identity checked by exact expansion. So the recurrence holds for every  $m \geq 0$ .

Finally, the three clauses. Set  $r_m = a_{m+1}/a_m = R_{2m}$  and  $q_m = M_m/D_m$ , so the recurrence reads  $r_{m+1} = q_m - 1 + q_m/r_m$ . First,  $27 - q_m = 3(126m^2 + 271m + 118)/D_m > 0$  for every  $m$ , so  $q_m < 27$ . Second, the map  $g_m(r) = q_m - 1 + q_m/r$  is strictly decreasing on  $r > 0$  and fixes  $q_m$ , and the interlacing  $q_{m-1} < r_m < q_m$  holds for all  $m \geq 1$  by induction: the base is  $q_0 = \frac{184}{9} < 22 = r_1 < \frac{518}{23} = q_1$ ; if  $r_m < q_m$ , then, because  $g_m$  is decreasing,  $r_{m+1} = g_m(r_m) > g_m(q_m) = q_m$ ; and if  $r_m > q_{m-1}$ , then  $r_{m+1} < g_m(q_{m-1}) < q_{m+1}$ , the last inequality clearing denominators to a polynomial of degree seven in  $m-1$  with positive coefficients. The interlacing gives  $r_m < q_m < 27$  and  $r_{m+1} > q_m > r_m$ , the boundedness and monotonicity clauses; and since  $27 - q_m \sim 27/(2m)$ , the squeeze  $27 - q_m < 27 - r_m < 27 - q_{m-1}$  gives  $27 - R_n \sim 27/n$  for  $n = 2m$ , the limit clause.  $\square$

**Proposition 32** (Block ladders and their saddle constants). Let  $B = (d_1, \dots, d_s)$  be a multidegree block with  $A = \sum_j (d_j - 1)$  odd, and for  $m \geq 0$  let  $S_m(B)$  be the sign-normalized signature of

the  $(2m + 1)$ -fold repetition of  $B$ , a Calabi–Yau complete intersection of complex dimension  $n_m = A(2m + 1) - 1$ . Then

$$S_m(B) = [y^{Am+(A-1)/2}] \frac{C_B(y)^{2m+1}}{1+y}, \quad C_B(y) = \prod_{j=1}^s \frac{\tan(d_j \arctan \sqrt{y})}{\sqrt{y}},$$

where the factor with entry  $d$  equals  $\sum_i (-1)^i \binom{d}{2i+1} y^i / \sum_i (-1)^i \binom{d}{2i} y^i$ , a rational function, so the generating function  $\sum_m S_m(B) w^m$  is algebraic. The series  $C_B$  has nonnegative aperiodic coefficients; there is a unique  $y_* > 0$  below the first positive pole of  $C_B$  with  $y_* C'_B(y_*) / C_B(y_*) = A/2$ , and with  $\Lambda_B = C_B(y_*)^2 / y_*^A$ ,

$$\frac{S_{m+1}(B)}{S_m(B)} = \Lambda_B \left( 1 - \frac{1}{2m} + O(m^{-2}) \right),$$

so the ratios are eventually strictly increasing with limit  $\Lambda_B$ , and  $\Lambda_B - S_{m+1}/S_m \sim \Lambda_B A / n_m$ . For the equal-degree block  $B = (d)$  with  $d$  even,  $y_* = \tan^2 \frac{\pi}{2(d+1)}$  and

$$\Lambda_d = \cot^{2(d+1)} \left( \frac{\pi}{2(d+1)} \right),$$

so  $\Lambda_2 = 27$  recovers Proposition 31; and for the Euler ladders with  $r$  equations of equal degree  $d$ , the analogous saddle sits at  $t_* = 1/(d+1)$  and the ratio limit is  $d^{d+1}$ , recovering the limit  $8 = 2^3$  of Proposition 16 at  $d = 2$ .

**Conjecture 196** (Dimension-wise signature gcd law). *Let the gcd range over all Calabi–Yau complete-intersection multidegrees in even complex dimension  $n$ . Then*

$$\gcd_{\mathbf{d}} S(\mathbf{d}) = \begin{cases} 2, & n \equiv 0 \pmod{4}, \\ 2^{\max(4, 1+\nu_2(n+2))}, & n \equiv 2 \pmod{4}, \end{cases}$$

where  $\nu_2$  is the 2-adic valuation.

## Part XI

# Spectral entropies and multibase carry fields

*Added to the deposit: 13 August 2026.*

The twenty conjectures below pair a spectral programme on graphs with a return to the integers. An entropy programme (Conjectures 197 to 201) reads the Rényi entropies of the trace-normalized graph Laplacian, the density-matrix reading of a graph introduced in [44], across tree and unicyclic operations, asserting that a small family of local moves is entropy-monotone at every order at once. A metric programme (Conjectures 202 to 207) poses the same functionals for the distance signless Laplacian of [45], an object whose spectral theory is extensively surveyed in [46] but whose normalized entropy appears unstudied. A carry programme (Conjectures 208 to 216) returns to the local–global aesthetic of Part I on a new object: the carry fields  $V_p(n) = v_p \binom{2n}{n}$  at distinct primes, finite-state functionals of the base- $p$  digits by Kummer’s theorem, for which complete cross-base decoupling is conjectured at Gaussian, local, modular, large-deviation, and

cumulant resolution. The calibration layer is proved where it can be: the degree identity at Rényi order two with its tree and leaf-move corollaries and the complete-graph majorization law from Proposition 33, and the exact single-base carry-chain law forms Proposition 34. The adversarial controls marking the genuine boundaries of the entropy windows are retained in place. The spectral scans are floating-point eigensolver checks, with the smallest margins near  $10^{-4}$ ; the carry statements are limit laws, with Conjecture 216, the transducer principle, the part's boldest statement; and Conjectures 197 and 208 sit closest to [47] and [48].

For a finite graph  $G$  with  $m \geq 1$  edges, let  $L(G)$  be the combinatorial Laplacian and  $\rho_L(G) = L(G)/(2m)$  its trace normalization, a density matrix whose eigenvalues  $p_1, \dots, p_n$  are nonnegative with unit sum; for  $\alpha > 0$ ,  $\alpha \neq 1$ , the Rényi entropy is  $H_\alpha^L(G) = (1 - \alpha)^{-1} \log \sum_i p_i^\alpha$ , with the Shannon limit  $H_1^L = -\sum_i p_i \log p_i$  at  $\alpha = 1$  and min-entropy  $H_\infty^L = -\log \max_i p_i$  at  $\alpha = \infty$ , zero eigenvalues contributing nothing. For a connected  $G$ , let  $\mathcal{D}$  be the distance matrix,  $t(v)$  the transmission  $\sum_u d(u, v)$ ,  $W(G) = \frac{1}{2} \sum_v t(v)$  the Wiener index, and  $Q_D(G) = \text{diag}(t) + \mathcal{D}$  the distance signless Laplacian [45], positive definite for connected graphs on at least three vertices by the identity  $x^T Q_D x = \sum_{i < j} d(i, j)(x_i + x_j)^2$ , with trace  $2W$ ; write  $H_\alpha^Q$  for the Rényi entropy of  $Q_D/(2W)$ . On the arithmetic side, for an odd prime  $p$  put  $V_p(n) = v_p(\binom{2n}{n})$ , by Kummer's theorem the number of carries in the base- $p$  addition of  $n + n$ , a bounded-output finite-state functional of the digits that is not  $p$ -additive, since each output depends on the incoming carry state;  $\mu_p(N)$  and  $\sigma_p^2(N)$  denote the mean and variance of  $V_p$  over uniform  $n < N$ .

**Proposition 33** (Degree and majorization calibrations). (i) For every graph with  $m$  edges and degrees  $d_i$ ,  $H_2^L(G) = -\log((2m + \sum_i d_i^2)/(2m)^2)$ ; consequently, among trees of a fixed order the path uniquely maximizes  $H_2^L$ , and if a leaf  $x$  adjacent to  $u$  is moved to a neighbour  $v$  of  $u$  with  $d(v) \geq d(u)$ , then  $H_2^L$  strictly decreases. (ii) For every connected  $n$ -vertex graph  $G$  and every  $\alpha > 0$ ,  $H_\alpha^Q(G) \leq H_\alpha^Q(K_n)$ . (iii) For a tree  $T$  on  $n \geq 3$  vertices, joining two leaves or subdividing one edge strictly increases  $H_2^L$ , and each move also strictly increases  $H_\alpha^L$  for all sufficiently small  $\alpha > 0$ ; and among trees of a fixed order the path uniquely maximizes  $H_\infty^L$ .

**Proposition 34** (Single-base carry-chain law). For uniform base- $p$  digits, the carry sequence in the addition  $n + n$  is a two-state Markov chain with  $P(0 \rightarrow 1) = (p - 1)/(2p)$ ,  $P(1 \rightarrow 1) = (p + 1)/(2p)$ , stationary carry probability  $\frac{1}{2}$ , and second eigenvalue  $1/p$ ; the stationary variance per digit is  $(p + 1)/(4(p - 1))$ ; and over the exact block  $0 \leq n < p^k$ ,

$$\mathbb{E}V_p = \frac{k}{2} - \frac{1 - p^{-k}}{2(p - 1)}.$$

## 44 Rényi entropies of the Laplacian density matrix

**Conjecture 197** (All-order path maximality). *For every  $n \geq 3$ , every  $n$ -vertex tree  $T \neq P_n$ , and every  $\alpha > 0$ ,  $H_\alpha^L(T) < H_\alpha^L(P_n)$ .*

**Conjecture 198** (Leaf-pair closure law). *Let  $T$  be a tree on at least three vertices and  $u, v$  distinct leaves. Then  $H_\alpha^L(T + uv) > H_\alpha^L(T)$  for every  $\alpha > 0$ .*

**Conjecture 199** (Edge-subdivision law). *If  $T'$  is obtained by subdividing one edge of a tree  $T$ , then  $H_\alpha^L(T') > H_\alpha^L(T)$  for every  $\alpha > 0$ .*

**Conjecture 200** (Leaf-compression window). *Let  $x$  be a leaf adjacent to  $u$  in a tree  $T$ , let  $v \neq x$  be a neighbour of  $u$  with  $d(v) \geq d(u)$ , and let  $T'$  be the tree with  $xu$  deleted and  $xv$  added. If  $T' \not\cong T$ , then  $H_\alpha^L(T') < H_\alpha^L(T)$  for all  $1/2 \leq \alpha \leq 2$ .*

**Conjecture 201** (Unicyclic endpoint law). *Among connected unicyclic graphs on  $n \geq 4$  vertices, for every finite  $\alpha > 0$ , the graph  $S_n^+$  obtained from the star by joining two leaves uniquely minimizes  $H_\alpha^L$  and the cycle  $C_n$  uniquely maximizes it.*

## 45 Entropy of the distance signless Laplacian

**Conjecture 202** (Tree one-vertex extension law). *For every tree  $T$  and every  $\alpha > 0$ , attaching a pendant vertex anywhere, or subdividing any edge once, strictly increases  $H_\alpha^Q$ .*

**Conjecture 203** (Low-order extension law). *For every connected graph  $G$  and every  $0 < \alpha \leq 2$ , pendant attachment and single edge subdivision strictly increase  $H_\alpha^Q$ .*

**Conjecture 204** (Star maximality). *For every  $n \geq 3$ , every  $n$ -vertex tree  $T \neq K_{1,n-1}$ , and every  $\alpha > 0$ ,  $H_\alpha^Q(T) < H_\alpha^Q(K_{1,n-1})$ .*

**Conjecture 205** (Bipartite path minimality at the Shannon order, resolved false). *Among connected bipartite graphs on  $n$  vertices,  $P_n$  uniquely minimizes  $H_1^Q$ .*

*Resolution.* The conjecture is false. The deposited fourteen-vertex tree has a degree-five hub carrying five leaves at diameter ten, and its Shannon entropy lies  $3.1535496 \times 10^{-5}$  below the path. This gap is certified by interval arithmetic at one hundred digits. A further exhaustive enumeration of all 3,159 unlabelled trees on fourteen vertices found a sharper witness  $T_*$  on vertices  $0, \dots, 13$  with edge set

$$\{\{0, 1\}, \{1, 2\}, \{0, 5\}, \{2, 3\}, \{3, 4\}, \{5, 6\}\} \cup \{\{6, j\} : 7 \leq j \leq 13\}.$$

Its degree multiset is  $(8, 2, 2, 2, 2, 2, 1, 1, 1, 1, 1, 1, 1, 1)$ , and a sixty-digit computation gives

$$H_1^Q(P_{14}) - H_1^Q(T_*) = 9.10088461876217 \times 10^{-5}.$$

The interval-certified tree remains the formal certificate. The sharper computation shows that it is not extremal even among trees of the same order. The path may remain minimal in narrower classes such as bounded-degree trees or caterpillars. Identifying the true bipartite minimiser is the surviving problem.

**Conjecture 206** (Balanced-biclique window maximality). *Among connected bipartite graphs on  $n$  vertices, for every  $1 \leq \alpha \leq 2$ ,  $H_\alpha^Q(G) \leq H_\alpha^Q(K_{\lfloor n/2 \rfloor, \lfloor n/2 \rfloor})$ , with equality only for the balanced complete bipartite graph.*

**Conjecture 207** (Unicyclic cycle maximality above the Shannon order). *Among connected unicyclic graphs on  $n \geq 4$  vertices,  $C_n$  uniquely maximizes  $H_\alpha^Q$  for every  $\alpha \geq 1$ , including  $\alpha = \infty$ .*

## 46 Multibase carry fields

**Conjecture 208** (Cross-prime carry central limit law). *Fix distinct odd primes  $p_1, \dots, p_r$ . For  $n$  uniform on  $\{0, \dots, N-1\}$ , the vector with coordinates  $(V_{p_i}(n) - \mu_{p_i}(N))/\sigma_{p_i}(N)$  converges in distribution to a standard  $r$ -dimensional Gaussian, with  $\mu_p(N) = \frac{1}{2} \log_p N + O_p(1)$  and  $\sigma_p^2(N) = \frac{p+1}{4(p-1)} \log_p N + O_p(1)$ .*

**Conjecture 209** (Bounded cross-prime covariance). *For distinct odd primes  $p \neq q$ ,  $\text{Cov}_{n < N}(V_p(n), V_q(n)) = O_{p,q}(1)$ , so the correlation is  $O_{p,q}(1/\log N)$ .*

**Conjecture 210** (Cross-prime local limit factorization). *Fix distinct odd primes  $p_1, \dots, p_r$ . For every fixed  $C$ , uniformly for integers  $k_i$  with  $|k_i - \mu_{p_i}(N)| \leq C\sqrt{\log N}$ ,*

$$\Pr(V_{p_i}(n) = k_i \ \forall i) = (1 + o(1)) \prod_i \frac{1}{\sqrt{2\pi\sigma_{p_i}^2(N)}} \exp\left(-\frac{(k_i - \mu_{p_i}(N))^2}{2\sigma_{p_i}^2(N)}\right).$$

**Conjecture 211** (Modular equidistribution of carry vectors). *For distinct odd primes  $p_i$  and fixed moduli  $m_i \geq 2$ , the vector  $(V_{p_1}(n) \bmod m_1, \dots, V_{p_r}(n) \bmod m_r)$  equidistributes on  $\prod_i \mathbb{Z}/m_i\mathbb{Z}$  as  $n$  ranges uniformly below  $N \rightarrow \infty$ .*

**Conjecture 212** (Additive large deviations). *Let  $I_p$  be the large-deviation rate function per base- $p$  digit of the carry fraction of the chain of Proposition 34. For fixed distinct odd primes, the vector  $(V_{p_i}(n)/\log_{p_i} N)_i$  satisfies a large-deviation principle at speed  $\log N$  with rate  $J(x) = \sum_i I_{p_i}(x_i)/\log p_i$  throughout the interior of the attainable region.*

**Conjecture 213** (Bounded mixed cumulants). *For fixed distinct odd primes, every fixed joint cumulant of total order at least three involving at least two distinct primes remains  $O(1)$  as  $N \rightarrow \infty$ .*

**Conjecture 214** (Fixed-multiplier carry vectors). *Fix an integer  $a \geq 2$  and distinct primes  $p_1, \dots, p_r$  such that  $\text{Var}_{n < N} v_{p_i}(\frac{an}{n}) \rightarrow \infty$  and the accessible carry-state chain of each base- $p_i$  addition automaton is irreducible and aperiodic under independent uniform digits. Then the standardized vector of these valuations converges to a standard Gaussian vector.*

**Conjecture 215** (Balanced factorial-ratio vectors). *Let  $R(n) = \prod_j (a_j n)! / \prod_k (b_k n)!$  be integer-valued with  $\sum_j a_j = \sum_k b_k$ . For distinct primes  $p_i$  with  $\text{Var}_{n < N} v_{p_i}(R(n)) \sim c_{p_i} \log N$ ,  $c_{p_i} > 0$ , the standardized vector  $(v_{p_i}(R(n)))_i$  converges to a standard Gaussian vector.*

**Conjecture 216** (Multibase transducer principle). *Let  $q_1, \dots, q_r \geq 2$  be jointly multiplicatively independent bases, meaning  $\prod_i q_i^{a_i} = 1$  with integer exponents only when every  $a_i = 0$ , and let  $F_i(n)$  be the total output of a deterministic finite-state transducer reading the base- $q_i$  digits of  $n$ , with bounded transition outputs, such that (i) under independent uniform digits the accessible state chain is irreducible and aperiodic, (ii)  $\text{Var}_{n < N} F_i(n) \sim c_i \log N$  with  $c_i > 0$ , and (iii) no two of the centred functionals factor, up to a bounded coboundary, through a common nonconstant eventually periodic function of  $n$ . Then the standardized vector  $(F_i(n))_i$  converges to a standard Gaussian vector, and every fixed mixed cumulant involving at least two bases is  $O(1)$ .*

## Part XII

# Subdivision spectra, spanning-tree correlation, and information dimension

*Added to the deposit: 14 August 2026.*



The twenty conjectures below run four programmes at the common interface of spectral theory and information. A subdivision programme (Conjectures 217 to 221) asks what weak spectral universality under barycentric refinement, proved at the graph level by Knill [54, 55], in every Hodge degree for the barycentric rule by Knill [56], and for top-dimensional Laplacians of general inclusion-uniform rules by Mårte [57], does *not* see: exponentially rare high eigenvalues, organized by a factorial hierarchy of critical moment exponents. A spanning-tree programme (Conjectures 222 to 226) introduces the total correlation of the uniform-spanning-tree edge process, an invariant assembled from Kirchhoff marginals and the matrix-tree theorem in the tradition of transfer impedances [59], and conjectures its extremal graphs, led by a mesoscopic core-periphery shape principle at scale  $\sqrt{n/\log n}$ . A cyclic-entropy programme (Conjectures 227 to 231) starts beyond the prime-field entropy-doubling floor proved this year by Gavalakis, Goh, and Kontoyiannis [62] and conjectures the full inverse theory: localization at near-equality, Gaussian rigidity, a heat-kernel crossover profile at the wrap scale, and uniform many-summand monotonicity. A graphon programme (Conjectures 232 to 236) conjectures that the Shannon entropy of a sampled random-free graph, known to be subquadratic of essentially arbitrary growth [68], is governed at scale  $n \log n$  exactly by the information dimension of the latent row space. The calibration layer is proved where it can be: block additivity and the exact cycle law for the tree invariant form Proposition 35, and the threshold-graphon entropy bound that anchors the second-order programme forms Proposition 36. The adversarial controls marking genuine boundaries are retained in place: the bipartite crossover at seventeen vertices in Conjecture 223, the separated-bump doubling penalty in Conjecture 227, the onset of pre-wrap failures in Conjecture 231, and the threshold graphon itself as the permanent negative control of Conjecture 234. All graph and spectral scans are floating-point checks. Three statements carry elevated exposure or overlap with prior work and are recorded where they occur: Conjecture 220, whose large-deviation law has no direct numerical support at reachable depth and is the part's boldest statement; Conjecture 219, which interpolates two published theorems [57, 56] and is close to prior work; and Conjecture 230, whose integer-lattice analogue is a recent theorem [67], so that only the modular content is new.

Throughout,  $\text{sd}^m K$  denotes the  $m$ -fold barycentric subdivision of a finite pure  $d$ -dimensional simplicial complex  $K$ , realized as the order complex of the face poset;  $\Delta_k$  is the unweighted Hodge Laplacian on  $k$ -simplices,  $f_k$  the number of  $k$ -simplices, and for the graph Laplacian  $L_m$  of the 1-skeleton,  $M_\alpha(m) = \text{Tr}(L_m^\alpha)/|V(\text{sd}^m K)|$ ; we write  $A_d = (d+1)!$ ,  $B_{d,0} = d!$ , and  $B_{d,k} = (d-k+1)!$  for  $1 \leq k < d$ . For a connected simple graph  $G$ ,  $\tau(G)$  is the number of spanning trees,  $T$  a uniform spanning tree, and  $R_e$  the unit-conductance effective resistance across  $e$ , so that  $\Pr(e \in T) = R_e$  by Kirchhoff's theorem; with  $h(x) = -x \log x - (1-x) \log(1-x)$ , the *spanning-tree correlation* is

$$\mathcal{C}(G) = \sum_{e \in E(G)} h(R_e) - \log \tau(G),$$

the total correlation of the edge-indicator vector of  $T$ , nonnegative and zero exactly on trees, since  $H(T) = \log \tau(G)$ ;  $S_{n,r} = K_r \vee \overline{K}_{n-r}$  denotes the complete split graph. On the cyclic side, laws  $\mu_p$  on  $\mathbb{Z}/p\mathbb{Z}$  with  $p$  prime are called *mesoscopic* when  $H(\mu_p) \rightarrow \infty$  and  $\log p - H(\mu_p) \rightarrow \infty$ ; convolution powers are written  $\mu_p^{*m}$ . A graphon  $W$  is *random-free* when it is  $\{0,1\}$ -valued almost everywhere;  $G(n, W)$  is the sampled graph on  $n$  vertices,  $d_W(x, y) = \int_0^1 |W(x, u) - W(y, u)| du$  the neighborhood distance, and  $(\Omega_W, d_W, \rho)$  the *purified row space*, the completion of the quotient of  $[0, 1]$  by  $d_W$ -null pairs carrying the pushforward of Lebesgue measure. The measure  $\rho$  satisfies a *uniform local doubling bound* when there are  $C$  and  $\varepsilon_0 > 0$  with  $\rho(B(x, 2\varepsilon)) \leq C\rho(B(x, \varepsilon))$  for all  $x$

in the support and  $\varepsilon \leq \varepsilon_0$ , and is *exact-dimensional of dimension  $s$*  when  $\log \rho(B(x, \varepsilon)) / \log \varepsilon \rightarrow s$  for  $\rho$ -almost every  $x$ .

**Proposition 35** (Block additivity and the cycle law). (i)  $\mathcal{C}$  is additive over the blocks of a connected graph, and bridges contribute zero. (ii)  $\mathcal{C}(C_n) = (n-1) \log \frac{n}{n-1}$ , strictly increasing in  $n$ ; consequently, subdividing an edge that lies on exactly one cycle strictly increases  $\mathcal{C}$ .

**Proposition 36** (Threshold-graphon entropy bound). For the random-free graphon  $W(x, y) = \mathbf{1}\{x + y \geq 1\}$ , every realization of  $G(n, W)$  is a threshold graph, the law of  $G(n, W)$  is supported on at most  $n! 2^{n-1}$  labeled graphs, and hence

$$H(G(n, W)) \leq n \log n - (1 - \log 2) n + O(\log n),$$

although the purified row space is  $[0, 1]$  with the metric  $|x - y|$  and Lebesgue measure, whose scale- $1/n$  quantization entropy is  $\log n$  per point with vanishing linear correction.

## 47 Spectral tails of barycentric subdivision

**Conjecture 217** (Critical moment exponent for barycentric vertex Laplacians). *Let  $K$  be a finite pure  $d$ -dimensional simplicial complex,  $d \geq 2$ , with at least one  $d$ -simplex. For every  $\alpha > 0$ ,*

$$\lim_{m \rightarrow \infty} \frac{1}{m} \log M_\alpha(m) = \max\{0, \alpha \log d! - \log(d+1)!\},$$

*and at the critical order  $\alpha_c(d) = \log(d+1)! / \log d!$  the moment  $M_{\alpha_c}(m)$  grows linearly in  $m$  up to constants depending on  $K$ .*

**Conjecture 218** (Hodge incidence-tail hierarchy). *For  $0 \leq k \leq d$  and the Hodge Laplacians  $\Delta_k$  on  $\text{sd}^m K$ , the empirical  $\alpha$ -moments of  $\Delta_k$  stay bounded for  $\alpha < \alpha_c(d, k) = \log(d+1)! / \log B_{d,k}$ , grow polynomially at  $\alpha_c(d, k)$ , and grow at exponential rate  $\alpha \log B_{d,k} - \log(d+1)!$  above it; for  $k = d$  all moments stay bounded.*

**Conjecture 219** (All-degree inclusion-uniform universality). *For every nontrivial inclusion-uniform subdivision rule  $\text{div}$  on pure  $d$ -complexes and every  $0 \leq k \leq d$ , the empirical spectral measure of  $\Delta_k(\text{div}^m K)$  converges weakly to a limit law depending only on the rule,  $d$ , and  $k$ , and not on  $K$ , apart from asymptotically negligible homological zero modes.*

**Conjecture 220** (Spectral-tail large deviations from simplex age). *Fix  $0 \leq k < d$  and  $0 < \theta < 1$ , and let  $N_{m,k}(\theta)$  count eigenvalues of  $\Delta_k(\text{sd}^m K)$  that are at least  $B_{d,k}^{\theta m}$ . Then*

$$\lim_{m \rightarrow \infty} \frac{1}{m} \log \frac{N_{m,k}(\theta)}{f_k(\text{sd}^m K)} = -\theta \log(d+1)!,$$

*uniformly for  $\theta$  in compact subintervals of  $(0, 1)$ ; for  $k = d$  there is no exponential tail.*

**Conjecture 221** (Finite-state renormalization for barycentric Hodge limits). *For each  $d \geq 2$  there exist an integer  $N(d)$ , a rational map  $R_d$  on  $\mathbb{C}^{N(d)}$ , and rational functions  $\Phi_{d,k}$ ,  $0 \leq k \leq d$ , such that the Schur complement of one barycentric refinement onto a generating set of  $N(d)$  boundary Green functions is exactly  $R_d$ , the iteration of  $R_d$  has a unique attracting fixed point on the relevant domain, and the Stieltjes transforms of all limiting Hodge laws  $\nu_{d,k}$  are the  $\Phi_{d,k}$  of that fixed point; for  $d = 1$  the scheme reduces to the known quadratic spectral decimation. The claim fails if the minimal rank of an exact boundary Schur-complement closure grows without bound under refinement.*

## 48 Total correlation of uniform spanning trees

**Conjecture 222** (Complete-split extremality). *Among connected  $n$ -vertex graphs,  $\mathcal{C}(G)$  is maximized by a complete split graph  $S_{n,r}$  for some  $r = r_n$ , and*

$$r_n \sim \sqrt{\frac{n}{\log n}}, \quad \max_G \mathcal{C}(G) = n - (1 + o(1))\sqrt{n \log n}.$$

**Conjecture 223** (Triangle-free bipartite extremality). *Among connected triangle-free  $n$ -vertex graphs,  $\mathcal{C}$  is maximized by a complete bipartite graph  $K_{r,n-r}$  with  $r$  maximizing the exact bipartite formula; the maximizing  $r$  satisfies  $r \sim \sqrt{n/\log n}$  and the maximum is  $n - (1 + o(1))\sqrt{n \log n}$ .*

**Conjecture 224** (Cycle minimum for bridgeless dependence). *Every bridgeless connected  $n$ -vertex simple graph satisfies  $\mathcal{C}(G) \geq \mathcal{C}(C_n) = (n-1) \log \frac{n}{n-1}$ , with equality only for  $C_n$ .*

**Conjecture 225** (Core-periphery stability at the ceiling). *If connected graphs  $G_n$  satisfy  $\mathcal{C}(G_n) = n - (1 + o(1))\sqrt{n \log n}$ , then there are vertex sets  $A_n$  with  $|A_n| = (1 + o(1))\sqrt{n/\log n}$  such that all but  $o(n|A_n|)$  of the pairs between  $A_n$  and its complement are edges, while the complement spans  $o(n|A_n|)$  edges. No structure is asserted for  $G_n[A_n]$ .*

**Conjecture 226** (Complete-multipartite reduction). *Among complete multipartite graphs on  $n$  vertices,  $\mathcal{C}$  is maximized by one part of size  $n - r$  together with  $r$  singleton parts, that is, by the complete split graph  $S_{n,r}$  for the optimal  $r$ .*

## 49 Prime-cyclic entropy power and the wrap transition

**Conjecture 227** (Localization of near-minimizers on prime cycles). *Let  $p \rightarrow \infty$  through primes and let  $\mu_p$  be mesoscopic laws on  $\mathbb{Z}/p\mathbb{Z}$  with*

$$H(\mu_p * \mu_p) - H(\mu_p) \rightarrow \frac{1}{2} \log 2.$$

*Then there are affine automorphisms  $\phi_p$  of  $\mathbb{Z}/p\mathbb{Z}$  and cyclic intervals  $I_p$  of length  $o(p)$  with  $(\phi_p)_* \mu_p(I_p) \rightarrow 1$ .*

**Conjecture 228** (Gaussian rigidity after localization). *Under the conclusion of Conjecture 227, choose centered integer lifts  $X_p$  with variances  $\sigma_p^2 \rightarrow \infty$  and assume a uniform standardized  $(2 + \delta)$ -moment bound. If the doubling gain tends to  $\frac{1}{2} \log 2$ , then  $(X_p - \mathbb{E}X_p)/\sigma_p \Rightarrow N(0, 1)$  and  $H(X_p) - \frac{1}{2} \log(2\pi e \sigma_p^2) \rightarrow 0$ .*

**Conjecture 229** (Heat-kernel wrap crossover, resolved false). *Let  $\mu_p$  have centered span-one integer lifts with variances  $\sigma_p^2$  and characteristic functions obeying  $|\mathbb{E} e^{i\theta(X_p - \mathbb{E}X_p)}| \leq e^{-c\theta^2 \sigma_p^2}$  for  $|\theta| \leq \pi$  and a constant  $c > 0$  independent of  $p$ , and let  $\sigma_p = o(p)$ ,  $m_p \rightarrow \infty$ ,  $m_p \sigma_p^2 / p^2 \rightarrow t \in (0, \infty)$ . With  $\theta_t$  the unit-circle heat kernel with Fourier coefficients  $e^{-2\pi^2 t j^2}$  and  $q_{p,t}$  its discretization,  $q_{p,t}(x) \propto \theta_t(x/p)$ ,*

$$\|\mu_p^{*m_p} - q_{p,t}\|_{\text{TV}} \rightarrow 0, \quad \log p - H(\mu_p^{*m_p}) \rightarrow \int_0^1 \theta_t(u) \log \theta_t(u) du.$$

*Resolution.* The conjecture is false as stated: the characteristic-function bound does not exclude rare macroscopic jumps, and a refuting family, a lazy-walk core of  $\lfloor \sqrt{p} \rfloor$  steps carrying jumps of size  $\lfloor p/3 \rfloor$  with probability  $\lfloor \sqrt{p} \rfloor \lfloor p/3 \rfloor^{-2}$ , satisfies every hypothesis with  $c = 1/6$  while its wrap limit is the heat kernel convolved with a compound-Poisson jump law, the mode-three coefficient tending to  $e^{-6\pi^2 t}$  against the asserted  $e^{-18\pi^2 t}$ ; an independent transform computation here reproduces the jump prediction to six decimals. A Lindeberg condition at scale  $p$ , or a uniform  $o(p)$  support bound, restores the intended law and is the recorded repair.

**Conjecture 230** (Fixed- $m$  entropy-power law on prime cycles). *For every fixed integer  $m \geq 2$  and every mesoscopic sequence  $\mu_p$ ,*

$$\liminf_{p \rightarrow \infty} [H(\mu_p^{*m}) - H(\mu_p)] \geq \frac{1}{2} \log m,$$

*and asymptotic equality under the hypotheses of Conjecture 228 forces the same Gaussian limit.*

**Conjecture 231** (Uniform pre-wrap entropy monotonicity). *Let  $\mu_p$ , after an affine automorphism, have integer log-concave lifts with variances  $\sigma_p^2$  and  $H(\mu_p) \rightarrow \infty$ , and let  $M_p \rightarrow \infty$  satisfy  $\log M_p = o(H(\mu_p))$  and  $M_p \sigma_p^2 / p^2 \rightarrow 0$ . Then, uniformly for  $1 \leq m \leq M_p$ ,*

$$H(\mu_p^{*(m+1)}) - H(\mu_p^{*m}) \geq \frac{1}{2} \log \frac{m+1}{m} - o(1/m).$$

## 50 Information dimension of random-free graphons

**Conjecture 232** (Exact-dimensional entropy identity). *Let  $W$  be random-free with purified row space  $(\Omega_W, d_W, \rho)$ . If  $\rho$  is exact-dimensional of finite dimension  $s$  and satisfies a uniform local doubling bound, then*

$$\frac{H(G(n, W))}{n \log n} \longrightarrow s.$$

**Conjecture 233** ( $L^1$  multifractal average-dimension law). *For random-free  $W$ , define  $D_\varepsilon(x) = \log(1/\rho(B(x, \varepsilon))) / \log(1/\varepsilon)$  on the purified row space. If  $\rho$  satisfies a uniform local doubling bound and  $D_\varepsilon \rightarrow d(\cdot)$  in  $L^1(\rho)$ , then*

$$\frac{H(G(n, W))}{n \log n} \longrightarrow \int d(x) d\rho(x).$$

**Conjecture 234** (Second-order graph entropy constant). *If the purified row space of a random-free graphon is a compact  $s$ -dimensional  $C^2$  metric manifold, boundary allowed, and  $\rho$  has a positive  $C^1$  density in local charts, then*

$$\kappa(W) = \lim_{n \rightarrow \infty} \frac{H(G(n, W)) - sn \log n}{n}$$

*exists and is finite.*

**Conjecture 235** (Equipartition for random-free samples). *Under the hypotheses of Conjecture 232, let  $G_n \sim G(n, W)$  and  $I_n = -\log \Pr(G_n)$ . Then  $I_n / (n \log n) \rightarrow s$  in probability.*

**Conjecture 236** (Sparse-observation dimension law). *Reveal each potential edge of  $G(n, W)$  independently with probability  $q_n$ , condition on the mask, and let  $O_n$  be the revealed bits. Under the hypotheses of Conjecture 233, if  $nq_n \rightarrow \infty$ , then*

$$\frac{H(O_n \mid \text{mask})}{n \log(nq_n)} \longrightarrow \int d(x) d\rho(x).$$

## Part XIII

# Flux torsion, quantum merge monotonicity, and effective string geometry

*Added to the deposit: 15 August 2026.*

The nineteen conjectures below extend the string-geometric programmes of Part X in five directions. A flux programme (Conjectures 237 to 240) lifts the discrete saturation and threshold laws of Conjectures 177 and 178 from three-forms to every odd degree and opens the integral torsion of the complete flux: full degree-wise shape laws for the prime-torsion multiplicities. A mirror programme (Conjectures 241 to 245) runs the merge order of Part X through genuinely quantum invariants, the mirror map, the genus-zero Gopakumar–Vafa numbers [75, 76], and the conifold radius in the flat coordinate, asserting strict monotonicity at every level. A stability programme (Conjectures 246 to 248) counts Bridgeland walls [81, 79], sharpens the  $\Gamma$ -constant of the Brill–Noether construction of [80], and asserts a tame atlas for wall arrangements in the spirit of definable Hodge theory [82]. An effective-geometry programme (Conjectures 249 to 253) posits polynomial condition numbers, in an algebraic gauge of distance to the discriminant, for Ricci-flat metrics near degeneration [83, 84], balanced-metric and Hermitian–Yang–Mills convergence [85, 88, 86, 87], matter spectral gaps [89, 90], and certified heterotic Yukawa couplings [91, 92]. A degeneration programme (Conjectures 254 and 255) closes with special-Lagrangian realization at infinite distance [77, 78] and a Laplace-tower rigidity law [93, 94]. The proved layer is Proposition 37: the classical monotonicity of degree and discriminant scale under merges, and the exact Chern floors of the five threefold configurations. Since deposit it has grown three times, by Proposition 39, the shifted torsion duality, by Proposition 30, whose complete-flux normal form resolves Conjecture 240 true and proves the interval-support clause of Conjecture 239, and by Proposition 38, the subcritical direction of Conjecture 238. One identity is recorded as an uncounted remark rather than a conjecture: the insertion-power formula of Remark 1, whose source construction [73] computes an infinite family of higher operations for the torus that could not be compared in full through the present channel, with the attribution recorded where it occurs. Conjectures 237 and 238 are stated in a random model; Conjectures 249 to 255 assert laws, the price of aiming at effective theorems in geometric analysis; Conjecture 255 is the part’s closest to prior work, since the small-eigenvalue asymptotics of the induced Kähler degeneration metrics are now a theorem of [95] and only the Ricci-flat, integrally LMHS-indexed content is claimed; and Conjecture 246 is possibly accessible from the existing wall geometry of [81].

Throughout, the flux notation of Part X is retained:  $E_n(R) = \Lambda^* R^n$ , and for  $H \in \Lambda^r \mathbb{Z}^n$  of odd degree  $r$ ,  $d_H = H \wedge -$ , with  $b_k(H; R)$  the cohomology dimensions and  $g_{n,r}$  the degree-wise profile on the Zariski-open locus of maximal wedge ranks, so that  $g_{n,3} = g_n$ . For the complete

flux  $H_{\Delta}^{(n)}$ ,  $t_{p,k}(n)$  counts the  $p$ -primary cyclic summands of the integral cohomology in degree  $k$ , and  $T_p(n) = \sum_k t_{p,k}(n)$ . A dimension  $n$  is  $r$ -essential when  $g_{n,r}$  is not  $(1+t)$  times the profile one dimension lower, so that a form ignoring one coordinate cannot saturate. On the mirror side, configurations  $\mathbf{d}$  and merges are as in Part X;  $D(\mathbf{d}) = \prod_i d_i$ ,  $\Lambda(\mathbf{d}) = \prod_i d_i^{d_i}$ ; for the five one-parameter projective complete-intersection Calabi–Yau threefolds (5), (4, 2), (3, 3), (3, 2, 2), (2, 2, 2, 2), the mirror map  $q_{\mathbf{d}}(z) = z + O(z^2)$ , the genus-zero Gopakumar–Vafa invariants  $N_{\mathbf{d}}(\beta)$ , and the A-model Yukawa coupling  $K_{\mathbf{d}}(q) = D(\mathbf{d}) + \sum_{\beta \geq 1} N_{\mathbf{d}}(\beta) \beta^3 q^{\beta} / (1 - q^{\beta})$  are the classical mirror data [75], and  $q_c(\mathbf{d})$  is the smallest positive singularity of  $K_{\mathbf{d}}$ , the conifold point in the flat coordinate.

**Proposition 37** (Merge monotonicity and Chern floors). (i) Under every merge  $d_i, d_j \mapsto d_i + d_j - 1$ , the degree  $D$  strictly decreases and the scale  $\Lambda$  strictly increases; in particular the algebraic conifold point  $z_c = \Lambda^{-1}$  strictly decreases. (ii) For a complete-intersection Calabi–Yau threefold of multidegree  $\mathbf{d}$  in  $\mathbb{CP}^{k-1}$ ,  $k = \sum_i d_i$ , one has  $c_2(X) \cdot H = \frac{1}{2}(\sum_i d_i^2 - k)D$ , so the ratios  $\text{td}_2(X) \cdot H / H^3$  for the five threefold configurations (5), (4, 2), (3, 3), (3, 2, 2), (2, 2, 2, 2) are 5/6, 7/12, 1/2, 5/12, 1/3.

## 51 Odd flux and torsion shapes

**Conjecture 237** (Odd-degree discrete saturation). *Fix an odd  $r \geq 3$  and a finite symmetric  $V \subset \mathbb{Z} \setminus \{0\}$  with  $\gcd(V) = 1$ , and let every coefficient of  $H_n \in \Lambda^r \mathbb{Z}^n$  be independent and uniform on  $V$ . Then  $\Pr[(b_k(H_n; \mathbb{Q}))_k = g_{n,r}] = 1 - O(e^{-cn})$  for some  $c = c(r, V) > 0$ .*

**Conjecture 238** (Odd coverage-scale threshold). *Fix odd  $r \geq 3$ , include each coordinate  $r$ -subset independently with probability  $p_n$ , and give included monomials independent coefficients from a fixed finite symmetric nonzero set. Along  $r$ -essential dimensions,  $p_n n^{r-1} / \log n \rightarrow \infty$  implies saturation probability tending to one.*

**Proposition 38** (Odd coverage threshold, zero direction). *Fix an odd  $r \geq 3$  and an  $r$ -essential dimension  $n$ . If  $p_n n^{r-1} / \log n \rightarrow 0$  then the saturation probability of Conjecture 238 tends to zero.*

**Proposition 39** (Shifted torsion duality). *For every prime  $p$  and all  $n, k$ ,  $t_{p,k}(n) = t_{p,n+3-k}(n)$ .*

**Conjecture 239** (Torsion shape law for the complete flux). *For every prime  $p$  and every  $n$ , every interior positive triple of  $k \mapsto t_{p,k}(n)$  is strictly log-concave.*

**Conjecture 240** (Post-onset torsion persistence, resolved true). *For every prime  $p$ ,  $T_p(n+1) > T_p(n)$  for every  $n \geq 4p-1$ .*

*Resolution.* The conjecture is true for every prime, not only conditionally at  $p = 2$ : Proposition 30 gives  $T_p(2g+2) = 2T_p(2g+1)$  from the null vector in even dimension and  $T_{g+1} - 2T_g = \sum_{r \geq 1} \binom{g}{r-1} 2^{g+1-r} A_r > 0$  from Pascal’s rule in odd dimension, both strict after onset. An independent recomputation confirms strict growth for every prime  $p \leq 11$  through dimension 150.

**Remark 1** (Insertion powers of the complete flux). The integral twisting theory of [73] attaches to a three-form its insertion powers, odd-degree forms driving an extended differential in the manner of Kraines’ higher Massey powers [74]. Exact enumeration of the signed regular families



on  $2m + 1$  labels gives, for  $m \leq 4$ , the identity that the  $m$ -th insertion power of the complete three-form is  $(-1)^{m-1}(2m + 1)^{m-1}$  times the complete  $(2m + 1)$ -form, a Cayley-like coefficient collapsing a sum of 76,545 terms at  $m = 4$ . Since [73] computes an infinite family of higher operations for the torus that could not be compared in full through the present channel, the identity is recorded here as a verified calibration observation with its attribution unresolved, and is not counted among the conjectures.

## 52 Quantum monotonicity along the merge order

**Conjecture 241** (Normalized mirror-map contraction). *For complete-intersection Calabi–Yau  $n$ -folds with  $n \geq 3$ , write  $\hat{q}_{\mathbf{d}}(x) = \Lambda(\mathbf{d}) q_{\mathbf{d}}(x/\Lambda(\mathbf{d})) = x + \sum_{m \geq 2} c_m(\mathbf{d})x^m$ . Under every merge,  $c_m(\mathbf{d}') \leq c_m(\mathbf{d})$  for every  $m \geq 2$ , with strict inequality for every  $m \geq 3$ .*

**Conjecture 242** (Gopakumar–Vafa amplification). *For the five one-parameter threefold configurations and every merge edge,  $A_\beta = N_{\mathbf{d}'}(\beta)/N_{\mathbf{d}}(\beta)$  is strictly increasing in  $\beta \geq 1$ .*

**Conjecture 243** (Monotone conifold approach). *For each of the five configurations,  $R_\beta = N(\beta + 1)/N(\beta)$  is strictly increasing, and  $R_\beta \rightarrow 1/q_c$ ; equivalently the invariants are strictly log-convex from degree one with ratio limit the conifold growth constant.*

**Conjecture 244** (Merge monotonicity of the conifold constant). *For every merge edge among the five configurations,  $q_c(\mathbf{d}') < q_c(\mathbf{d})$ .*

**Conjecture 245** (Conifold-normalized Yukawa monotonicity). *For every merge edge among the five configurations and every  $0 < x < 1$ ,  $K_{\mathbf{d}'}(xq_c(\mathbf{d}'))/D(\mathbf{d}') > K_{\mathbf{d}}(xq_c(\mathbf{d}))/D(\mathbf{d})$ .*

## 53 Stability walls and sharp constants

**Conjecture 246** (Quadratic wall complexity). *Let  $X$  be a Picard-rank-one Calabi–Yau threefold with a constructed component  $\text{Stab}^\dagger(X)$  satisfying the support property,  $K$  a compact set in a fundamental domain for the  $\widetilde{\text{GL}}_2^+(\mathbb{R})$  action, and  $\|\cdot\|$  a Euclidean norm on the numerical charge lattice. Then there is  $C_K$  with  $N_K(v) \leq C_K(1 + \|v\|)^2$  for every primitive charge  $v$ , where  $N_K(v)$  counts numerical walls for  $v$  meeting  $K$ .*

**Conjecture 247** (Sharp  $\Gamma$  for Brill–Noether stability). *For a Picard-rank-one complete-intersection Calabi–Yau threefold covered by the Brill–Noether construction of [80], with  $\varepsilon_*$  the supremum of  $\varepsilon$  for which their condition  $\text{BG3}(\varepsilon)$  holds, the infimum of  $\gamma$  for which  $\Gamma = \gamma H^2 - \text{td}_2(X)$  satisfies the  $\Gamma$ -twisted Bayer–Macrì–Toda ( $\Gamma$ -BMT) stability condition is exactly  $\max\{4/(H^3\varepsilon_*), \text{td}_2(X) \cdot H/H^3\}$ .*

**Conjecture 248** (Tame wall atlas). *For  $X$  as in Conjecture 246 and compact  $K$ , fix a finite effective  $\mathbb{R}_{\text{an}, \text{exp}}$  presentation of the central charge and the numerical wall equations. Let formula format record length, quantifier depth, polynomial degree and logarithmic coefficient height in that presentation. There is  $C = C(X, K)$  such that the union  $W_Q$  of numerical walls for primitive charges with  $\|v\| \leq Q$  has a defining formula of format at most  $Q^C$ , and  $K \setminus W_Q$  has a compatible definable stratification into  $O(Q^C)$  strata and chambers.*

## 54 Effective geometry of Calabi–Yau families

**Gauge convention.** Two positive gauges  $\delta$  and  $\delta'$  on the smooth locus of a fixed compact semialgebraic chart are *power-equivalent* if, after truncation to  $(0, 1]$ , there are positive constants  $c, C, a, b$  with  $c\delta^a \leq \delta' \leq C\delta^b$ . Every polynomial estimate below is understood up to this equivalence. The gauge  $\delta_{\text{alg}}$  is represented by a truncated norm of generators of the reduced discriminant ideal. For a family of bundle pairs,  $\delta_{\text{pair}}$  is represented by the minimum of  $\delta_{\text{alg}}$  and a fixed positive stability margin that vanishes exactly on the chosen chamber boundary.

**Conjecture 249** (Polynomial conditioning of Ricci-flat metrics). *For a projective polarised Calabi–Yau family over a compact semialgebraic chart, with  $\delta_{\text{alg}}$  the truncated norm of generators of the reduced discriminant ideal, the volume-normalised Ricci-flat metrics satisfy, for every derivative order  $\ell$ , bounds  $\|\nabla^\ell \text{Rm}\|_\infty \leq C_\ell \delta_{\text{alg}}^{-A_\ell}$ , with diameter bounded by  $C\delta_{\text{alg}}^{-A}$  and, in the noncollapsing case, injectivity radius bounded below by  $C^{-1}\delta_{\text{alg}}^A$ , the same polynomial form holding for the regularity scale in collapsing strata away from the canonical collapsing set.*

**Conjecture 250** (Discriminant-effective balanced convergence). *In the setting of Conjecture 249, the level- $k$  balanced metrics satisfy  $\|g_{s,k} - g_s\|_{C^\ell(g_s)} \leq C_\ell \delta_{\text{alg}}(s)^{-A_\ell} k^{-1}$  whenever  $k \geq C_\ell \delta_{\text{alg}}(s)^{-B_\ell}$ , with the corresponding  $k^{-N-1}$  bounds after subtracting  $N$  Bergman expansion terms.*

**Conjecture 251** (Discriminant-effective Hermitian–Yang–Mills convergence). *For an algebraic family of polarised Calabi–Yau threefolds with slope-stable bundles on a constant-stability chamber, with  $\delta_{\text{pair}}$  as in the gauge convention, the level- $k$  balanced-bundle approximations converge to the Hermitian–Yang–Mills connection in Coulomb gauge with  $\|A_{s,k} - A_s\|_{C^\ell} \leq C_\ell \delta_{\text{pair}}^{-A_\ell} k^{-1}$  whenever  $k \geq C_\ell \delta_{\text{pair}}^{-B_\ell}$ .*

**Conjecture 252** (Polynomial matter spectral gap). *In the setting of Conjecture 251, for a representation and form degree with locally constant bundle cohomology, the smallest positive eigenvalue of the bundle-valued Dolbeault Laplacian satisfies  $\lambda_+(s) \geq C^{-1} \delta_{\text{pair}}(s)^A$ , simultaneously for any fixed finite list of representations and degrees.*

**Conjecture 253** (Certified heterotic Yukawa convergence). *Under Conjectures 249 to 252, the level- $k$  physical heterotic Yukawa couplings satisfy  $|Y_{abc}^{(k)}/Y_{abc} - 1| \leq C \delta_{\text{pair}}^{-A} \delta_Y^{-A} k^{-1}$  for  $k \geq C \delta_{\text{pair}}^{-B} \delta_Y^{-B}$ , where  $\delta_Y$  is the power-equivalence class of a truncated algebraic gauge for the zero locus of the corresponding holomorphic coupling. An absolute error bound of the same form holds with no  $\delta_Y$  factor.*

## 55 Degenerations and towers

**Conjecture 254** (Saturation-index special-Lagrangian realization). *For a semistable one-parameter Calabi–Yau threefold degeneration of type II, III, or IV with unipotent monodromy, let  $L_{\text{orb}}$  be the nilpotent-orbit lattice of asymptotically light D3 charges of common limiting phase,  $L_{\text{sat}}$  its rational saturation in integral homology, and  $I_\tau$  the exponent of  $L_{\text{sat}}/L_{\text{orb}}$ . Every primitive charge generating an extremal ray of the saturated light cone with the required positive limiting phase has a multiple  $m\gamma$ , with  $m$  dividing  $I_\tau$ , admitting a calibrated special-Lagrangian representative for all sufficiently deep parameters.*

**Conjecture 255** (LMHS-controlled Laplace towers). *For a semistable one-parameter polarised Calabi–Yau degeneration at infinite Weil–Petersson distance, fix a normalisation of  $d_{\text{WP}}$  and volume-normalise the Ricci-flat metrics. The vanishing scalar Laplace spectrum admits a finite partition into tower types  $a$  with rational exponents  $\alpha_a$  such that: (i) the list of exponents depends only on the isomorphism class of the integral limiting mixed Hodge structure and the combinatorial dual-intersection and essential-skeleton data. (ii) for each tower there is a positive normalizer  $N_a(t) = e^{2\alpha_a d_{\text{WP}}(t)} \ell_a(t)$ , with  $\ell_a$  slowly varying in  $e^{d_{\text{WP}}(t)}$ , such that for every fixed  $j$  the quantity  $N_a(t) \lambda_{a,j}(t)$  converges to the  $j$ -th eigenvalue of a canonical weighted Laplace operator on the collapsed base geometry. (iii) with eigenvalues labelled in increasing order at each parameter, every fixed-rank eigenvalue branch tending to zero belongs eventually to one of these towers. The limiting operator may depend on the canonical metric data of the collapsed base and is not asserted to be determined by the limiting mixed Hodge structure alone.*

## Part XIV

# Convexity, defect, and dispersion laws for projective invariants

*Added to the deposit: 19 August 2026.*

The fifteen conjectures below take three of the oldest invariants of projective algebraic geometry, the middle cohomology of a complete intersection, the graded Betti table of an embedded surface, and the Hodge decomposition of a hypersurface, and conjecture the discrete shape of each as its defining degrees move. They form three programmes. A convexity programme (Conjectures 256 to 261) studies the degree-normalized primitive middle Betti number of smooth complete intersections on the full multidegree lattice, asserting a coherent discrete-convexity package: coordinate monotonicity and convexity, supermodular absolute interaction with log-submodular relative interaction, Schur-convexity at fixed degree sum, and a strict decrease under every codimension-raising split that preserves the adjunction coefficient. A syzygy programme (Conjectures 262 to 266) studies the two nontrivial rows of the Betti tables of  $\mathbf{P}^1 \times \mathbf{P}^1$  under Segre–Veronese embeddings, replacing global shape claims by a controlled-defect law: the nonlinear row is strictly log-concave, while the linear row can fail log-concavity at most once, only past its mode, only where the nonlinear row is already present, and only within one step of that row’s mode. A dispersion programme (Conjectures 267 to 270) studies the normalized primitive Hodge profiles of smooth hypersurfaces as the degree grows, asserting that the known drift toward the Eulerian profile is monotone in four canonical stochastic orders: likelihood-ratio flattening, central-window escape, convex-order spread, and total-variation distance. The proved layer is five propositions: Proposition 40 proves all six convexity laws for curves from the exact formula  $B = \sum_i d_i - c - 2 + 2/D$ , Proposition 42 pins the two syzygy rows to the Hilbert series by an exact antidiagonal identity, Proposition 43 proves all four dispersion laws for surfaces from an explicit rational function of the degree, Proposition 41 proves the coordinate monotonicity and convexity laws of Conjectures 256 and 257 for hypersurfaces in every dimension from a closed form, and Proposition 44 derives Conjectures 268 and 269 from the likelihood-ratio flattening of Conjecture 267; in addition, for odd  $n$  the Calabi–Yau slice of Conjecture 261 is already a corollary of the merge monotonicity of Proposition 17. Two negative controls are retained in

place: the unnormalized primitive Betti number fails Schur-convexity and has no split direction already for curves and surfaces, and coordinatewise monotone convergence to the Eulerian limit fails at dimension twelve, so the normalization of the convexity programme and the total-variation formulation of the dispersion programme are both forced. Conjectures 262 to 266 rest on the thirteen complete published tables, the entire deposited corpus, and are resting on a small evidence base; Conjecture 262 sits closest to the published unimodality conjecture of [127] and is the part's closest to prior work; Conjectures 267 to 270 assert laws at every degree and dimension, and the Eulerian limit itself is exact in the abelian setting of [125] and asymptotic in the toric setting of [124], so only the monotone interpolation is claimed as new.

For a smooth complex complete intersection  $X^n(\mathbf{d}) \subset \mathbf{P}^{n+c}$  of dimension  $n \geq 1$ , codimension  $c \geq 1$ , and multidegree  $\mathbf{d} = (d_1, \dots, d_c)$  with every  $d_i \geq 2$ , write  $D(\mathbf{d}) = \prod_i d_i$ ; the Lefschetz theorem confines the new topology to the middle degree, and the primitive middle Betti number is  $b_n^{\text{prim}} = (-1)^n(\chi(X) - (n+1))$  with  $\chi$  evaluated exactly by the Chern-class formula  $\chi = D[H^n](1+H)^{n+c+1}/\prod_i(1+d_iH)$  [120, 121]. The invariant of the convexity programme is the *degree-normalized primitive complexity*  $B_{n,c}(\mathbf{d}) = b_n^{\text{prim}}/D(\mathbf{d})$ , and  $e_i$  denotes a coordinate step in the multidegree lattice. For the Segre–Veronese embedding of  $\mathbf{P}^1 \times \mathbf{P}^1$  by  $\mathcal{O}(a, b)$  with  $a, b \geq 2$ , in characteristic zero, the graded Betti numbers  $\beta_{p,p+q}(a, b)$  of the homogeneous coordinate ring vanish outside  $q = 0, 1, 2$ ; write  $\Delta_p^{(q)} = \beta_{p,p+q}^2 - \beta_{p-1,p-1+q}\beta_{p+1,p+1+q}$  for the log-concavity determinant at an interior index of the  $q$ th row, call  $p$  a *defect* of the row when  $\Delta_p^{(q)} < 0$ , and let  $m_q$  be the least index at which the row attains its maximum. For a smooth hypersurface  $X_d^n \subset \mathbf{P}^{n+1}$  with  $n \geq 2$ ,  $d \geq 3$ , the primitive Hodge numbers are  $h_p(n, d) = [t^{(p+1)d-(n+2)}](1+t+\dots+t^{d-2})^{n+2}$  by Griffiths' formula [123]; write  $\pi_{n,d}$  for the normalized profile  $h_p/\sum_j h_j$  on  $\{0, \dots, n\}$ , symmetric about  $n/2$ , and  $\varepsilon_n(p) = A(n+1, p)/(n+1)!$  for the Eulerian profile, with  $A$  the Eulerian numbers. A profile  $P$  precedes  $Q$  in *convex order* when  $\mathbf{E}\varphi(P) \leq \mathbf{E}\varphi(Q)$  for every convex  $\varphi$ , and  $\|P - Q\|_{\text{TV}} = \frac{1}{2} \sum_p |P(p) - Q(p)|$ .

**Proposition 40** (Curve calibration for the convexity programme). For  $n = 1$  one has  $B_{1,c}(\mathbf{d}) = \sum_i d_i - c - 2 + 2/D$ , and all six laws of Conjectures 256 to 261 hold strictly for curves.

**Proposition 41** (Hypersurface calibration for the convexity programme). For a smooth hypersurface ( $c = 1$ ) of degree  $d \geq 2$  in  $\mathbf{P}^{n+1}$ , write  $x = d - 1$ ; then  $B_{n,1}(d) = f_n(x)$  with  $f_n(x) = (x^{n+2} + (-1)^n x)/(x+1)^2$ , and  $f_n$  is strictly increasing and strictly convex on  $x \geq 1$  for every  $n \geq 1$ . In particular Conjectures 256 and 257 hold at codimension one in every dimension.

**Proposition 42** (Antidiagonal calibration for the syzygy programme). Let  $N+1 = (a+1)(b+1)$  and  $c_j = [t^j](1-t)^{N+1} \sum_{k \geq 0} (ka+1)(kb+1)t^k$ . Then  $\beta_{j-1,j}(a, b) - \beta_{j-2,j}(a, b) = (-1)^{j-1} c_j$  for every  $j \geq 2$ , the resolution has length  $N - 2$ , and  $\beta_{N-2,N} = (a-1)(b-1)$ .

**Proposition 43** (Surface calibration for the dispersion programme). For  $n = 2$  set  $x(d) = \pi_{2,d}(0) = \frac{(d-2)(d-3)}{6(d^2-3d+3)}$ . Then  $x(d)$  is strictly increasing in  $d \geq 3$  with  $x(d) < \frac{1}{6}$  and  $x(d) \rightarrow \frac{1}{6}$ , and consequently all four laws of Conjectures 267 to 270 hold for surfaces, strictly for  $d \geq 4$ .

**Proposition 44** (Dispersion consequences of likelihood-ratio flattening). Conjecture 267 implies Conjectures 268 and 269.

## 56 Normalized primitive cohomology of complete intersections

**Conjecture 256** (Coordinate monotonicity of normalized complexity). For every  $n \geq 1$ ,  $c \geq 1$ , every  $\mathbf{d} \in \mathbf{Z}_{\geq 2}^c$ , and every coordinate  $i$ ,  $B_{n,c}(\mathbf{d} + e_i) > B_{n,c}(\mathbf{d})$ .

**Conjecture 257** (Strict discrete convexity in each degree). *Under the same hypotheses,  $B_{n,c}(\mathbf{d} + 2e_i) - 2B_{n,c}(\mathbf{d} + e_i) + B_{n,c}(\mathbf{d}) > 0$ .*

**Conjecture 258** (Additive supermodularity across degrees). *For  $i \neq j$ ,  $B(\mathbf{d} + e_i + e_j) - B(\mathbf{d} + e_i) - B(\mathbf{d} + e_j) + B(\mathbf{d}) > 0$ .*

**Conjecture 259** (Strict log-submodularity). *Whenever the four values are positive,  $B(\mathbf{d} + e_i + e_j) B(\mathbf{d}) < B(\mathbf{d} + e_i) B(\mathbf{d} + e_j)$ .*

**Conjecture 260** (Schur-convexity at fixed degree sum). *If  $\mathbf{d}$  majorizes  $\mathbf{d}'$  in  $\mathbf{Z}_{\geq 2}^c$  with equal coordinate sums, then  $B_{n,c}(\mathbf{d}) \geq B_{n,c}(\mathbf{d}')$ , strictly unless the vectors agree up to permutation.*

**Conjecture 261** (Adjunction-preserving splits decrease complexity). *For  $d_i = d \geq 3$  and  $a, b \geq 2$  with  $a + b = d + 1$ , replacing the degree  $d$  by the pair  $(a, b)$  gives  $B_{n,c+1}(\dots, a, b, \dots) < B_{n,c}(\dots, d, \dots)$ .*

## 57 Controlled defects in Segre–Veronese syzygies

**Conjecture 262** (Strict log-concavity of the nonlinear row). *For every  $a, b \geq 2$  and every interior index  $p$  of the positive support of the  $q = 2$  row,  $\Delta_p^{(2)} > 0$ .*

**Conjecture 263** (The linear row is log-concave through its mode). *For every  $a, b \geq 2$  and every interior index  $p \leq m_1$  of the  $q = 1$  row,  $\Delta_p^{(1)} > 0$ .*

**Conjecture 264** (At most one defect in the linear row). *For every  $a, b \geq 2$ , the set  $\{p : \Delta_p^{(1)} < 0\}$  has at most one element.*

**Conjecture 265** (Defects occur only inside the nonlinear overlap). *For every  $a, b \geq 2$ , if  $\Delta_p^{(1)} < 0$  then  $\beta_{p,p+2}(a, b) > 0$ .*

**Conjecture 266** (Defect-mode locking). *For every  $a, b \geq 2$ , if the linear row has a defect at  $p_*$  then  $p_* \in \{m_2, m_2 + 1\}$ .*

## 58 Monotone dispersion of hypersurface Hodge profiles

**Conjecture 267** (Likelihood-ratio flattening in the degree). *For every  $n \geq 2$ ,  $d \geq 3$ , and  $0 \leq p < q \leq \lfloor n/2 \rfloor$ ,  $h_p(n, d+1) h_q(n, d) \geq h_p(n, d) h_q(n, d+1)$ .*

**Conjecture 268** (Central-window escape). *For every  $n \geq 2$ ,  $d \geq 3$ , and  $1 \leq k \leq \lfloor n/2 \rfloor$ ,  $\sum_{p=k}^{n-k} \pi_{n,d+1}(p) \leq \sum_{p=k}^{n-k} \pi_{n,d}(p)$ .*

**Conjecture 269** (Convex-order dispersion). *For every  $n \geq 2$  and  $d \geq 3$ , the profile  $\pi_{n,d}$  precedes  $\pi_{n,d+1}$  in convex order.*

**Conjecture 270** (Monotone total-variation convergence to the Eulerian profile). *For every  $n \geq 2$  and  $d \geq 3$ ,  $\|\pi_{n,d+1} - \varepsilon_n\|_{\text{TV}} \leq \|\pi_{n,d} - \varepsilon_n\|_{\text{TV}}$ , with equality only at  $(n, d) = (3, 3)$ .*

**Significance.** The normalized profiles drift toward the Eulerian distribution, exactly the law of descents of a random permutation of  $n + 1$  letters; the conjecture upgrades that drift to monotone convergence in total variation from the first degree onward, a finite-degree error principle that one verified threshold would convert into bounds at every larger degree. Convergence alone forces no monotonicity, and the coordinatewise strengthening is false, a retained control: at  $n = 12$  the  $p = 5$  mass crosses its Eulerian limit between degrees six and seven and overshoots, so the cancellation across positions asserted here is the genuine content. The Eulerian limit is exact for hypersurfaces in abelian varieties [125], asymptotic in the toric setting [124], and feeds effective arithmetic applications [126]; only the monotone interpolation is claimed. A first decisive theorem would be a one-crossing law between  $\pi_{n,d}$  and  $\varepsilon_n$ . In tests on the corpus described in the stress-test section, the distance never increased across 1,682 grid and 2,730 extension transitions, the sole equality being the cubic–quartic threefold coincidence. A counterexample would be a degree step moving the profile away from the Eulerian law.

## Part XV

# Stochastic order, curvature, and dependence laws for self-avoiding walks

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The fifteen conjectures below concern the uniform measure on  $n$ -step self-avoiding walks on the hypercubic lattice, the classical model whose most basic monotonicity questions, including whether the mean-square displacement grows with the length, remain open [135, 133]. They form four programmes. A curvature programme (Conjectures 271 to 273) asserts exact all-length curvature laws for three count sequences: the square-lattice counts  $c_n$  are strictly log-concave in steps of two with curvature relaxing monotonically to zero, sharpening the ratio analysis of Hammersley’s programme carried out by Guttmann and Jensen [131]; every radial-moment sum  $M_q(n)$  obeys the same parity log-concavity at every real order  $q \geq 0$ ; and the bridge counts  $b_n$  obey the opposite law, strict log-convexity from  $n = 5$  onward, so that walk ratios would descend to  $\mu^2$  while bridge ratios ascend to  $\mu$ . A transport programme (Conjectures 274 to 278) asserts stochastic-order growth in the length: likelihood-ratio growth of the bridge height and of the coordinate span, coordinatewise monotone couplings of the absolute endpoint vector and of the span vector under  $n \rightarrow n + 2$ , and stochastic growth of the contact count; the endpoint coupling strictly implies the open mean-square-displacement monotonicity along each parity class. A dependence programme (Conjectures 279 to 282) assigns signs to four joint laws at every fixed length: the two spans are negatively quadrant dependent, radius and terminal atmosphere positively, radius and contacts negatively, and contacts and atmosphere negatively, an extension–compaction dependence structure for a measure with no known FKG-type property. A phase-and-dimension programme (Conjectures 283 to 285) states a partition-curvature trichotomy across the collapse transition whose three constants are the established predictions of [136, 137, 138], and two strict cross-dimensional laws: the probability that a nonbacktracking walk is self-avoiding rises with the dimension at every fixed length, and every positive radial moment falls. The proved layer is three calibrations: Proposition 45 proves the extreme-order regime of both moment laws from a two-walk extremal count and a spanning-axes inequality  $c_n(d + 1) > (1 + 1/d) c_n(d)$ ;



Proposition 46 computes the top four diagonals of the bridge-height array in closed form and proves the top corner of the likelihood-ratio law at every length; and Proposition 47 computes the survival probabilities  $q_{d,n}$  for  $n \leq 7$  as explicit rational functions of  $d$  and proves Conjecture 284 there in every dimension at once; and Proposition 48 proves Conjecture 285 at length two for every dimension and every positive moment order. Four negative controls are retained in place: the absolute endpoint coordinates are not negatively quadrant dependent already at  $n = 3$ , so the span law of Conjecture 279 is not a symmetry artifact; the full radial distributions cross between adjacent dimensions, so the moment formulation of Conjecture 285 is forced; bridge log-convexity fails exactly at  $n = 2$  and  $n = 4$ , so the threshold of Conjecture 273 is sharp; and the finite-size roots of the curvature diagnostic of Conjecture 283 oscillate below  $n \approx 12$ , so that statement is asymptotic only. Conjecture 271 is the part's closest to prior work: its statistic and its asymptotics are explicitly Guttman–Jensen's [131], and only the exact strict inequalities and the second-level monotonicity are proposed as new; Conjecture 283 is a consolidation, since each of its three regimes is separately the standard prediction and the new content is the uniform sequence-level formulation.

An  $n$ -step self-avoiding walk on  $\mathbf{Z}^d$  is a nearest-neighbour path  $\omega = (\omega_0, \dots, \omega_n)$ ,  $\omega_0 = 0$ , with distinct vertices;  $c_n(d)$  is their number,  $c_n = c_n(2)$ , and  $U_n$  is the uniform measure on the square-lattice walks, whose connective constant is  $\mu = \lim c_n^{1/n} \approx 2.63815853$ . Write  $\omega_n = (X_n, Y_n)$ ,  $R_n^2 = |\omega_n|^2$ ,  $M_q(n) = \sum_{\omega} |\omega_n|^q$ , and let  $S_x, S_y$  be the coordinate spans  $\max_i x_i - \min_i x_i$  and its  $y$ -analogue. The terminal atmosphere  $A_n \in \{0, 1, 2, 3\}$  is the number of unvisited neighbours of  $\omega_n$ ; since every  $(n+1)$ -step walk extends a unique  $n$ -step walk,  $\sum_{\omega} A_n(\omega) = c_{n+1}$ , so  $\mathbf{E}[A_n] = c_{n+1}/c_n$  exactly, the identity underlying atmospheric sampling methods [140, 139]. The contact count  $C_n$  is the number of nonbonded nearest-neighbour pairs of visited vertices, with partition function  $Z_n(u) = \sum_{\omega} u^{C_n(\omega)}$  for the interacting walk [141, 142]. A bridge is a walk with  $x_0 = 0$ ,  $x_i \geq 1$  for  $i \geq 1$ , and  $x_i \leq x_n$  for all  $i$ ;  $b_n$  counts them,  $B(n, h)$  counts those of height  $x_n = h$ , and  $b_1, \dots, b_{10} = 1, 3, 7, 17, 41, 101, 251, 631, 1591, 4029$ , with  $b_m b_n \leq b_{m+n}$  and  $b_n^{1/n} \rightarrow \mu$  by Hammersley–Welsh unfolding [134]. A uniformly random  $n$ -step nonbacktracking walk on  $\mathbf{Z}^d$  is self-avoiding with probability  $q_{d,n} = c_n(d)/(2d(2d-1)^{n-1})$  [143]. For laws  $P, Q$  on  $\mathbf{Z}$  or  $\mathbf{Z}^2$ ,  $P \leq_{\text{st}} Q$  means  $Q$  dominates  $P$  on upper sets, equivalently a coupling exists with the  $Q$ -variable coordinatewise at least the  $P$ -variable;  $\leq_{\text{lr}}$  is likelihood-ratio order, the total positivity of the two-row array; and  $(X, Y)$  is positively (negatively) quadrant dependent when  $\mathbf{P}(X \geq a, Y \geq b)$  is at least (at most)  $\mathbf{P}(X \geq a)\mathbf{P}(Y \geq b)$  for all thresholds. All finite verifications below are exact: integer enumeration, integer cross-products, and integer maximum flows for the coupling laws, with the full joint laws enumerated through  $n = 21$  ( $c_{21} = 2,408,806,028$  walks at the top length), bridges through  $n = 23$ , dimensions through  $d = 10$ , and published counts through  $n = 79$  [132].

**Proposition 45** (Extreme-order calibration for the moment laws). (i) For all  $n \geq 2$  and  $d \geq 1$ ,  $c_n(d+1) > (1 + 1/d)c_n(d)$ . (ii) For every  $n \geq 3$  there is  $q_0(n)$  such that  $M_q(n)^2 > M_q(n-2)M_q(n+2)$  for all real  $q \geq q_0(n)$ . (iii) For every  $n \geq 2$  and  $d \geq 2$  there is  $q_1(n, d)$  such that  $\mathbf{E}_{d+1}[R_n^q] < \mathbf{E}_d[R_n^q]$  for all real  $q \geq q_1(n, d)$ .

**Proposition 46** (Top-corner calibration for the bridge programme). For all admissible  $n$ ,  $B(n, n) = 1$ ,  $B(n, n-1) = 2(n-1)$ ,  $B(n, n-2) = 2(n-2)^2$ , and  $B(n, n-3) = \frac{2}{3}(n-3)(2n^2 - 12n + 19)$ ; consequently the top two adjacent cross-products of Conjecture 274 hold at every length:

$$\begin{aligned} B(n, n-1)B(n+1, n) - B(n, n)B(n+1, n-1) &= 2(n-1)(n+1), \\ B(n, n-2)B(n+1, n-1) - B(n, n-1)B(n+1, n-2) &= \frac{4}{3}(n-2)(n-1)(n^2 - n - 3), \end{aligned}$$

both positive for  $n \geq 2$  and  $n \geq 3$  respectively.

**Proposition 47** (Survival calibration in every dimension). For all  $d \geq 2$ ,

$$1 - q_{d,4} = \frac{2(d-1)}{(2d-1)^3}, \quad 1 - q_{d,5} = \frac{2(d-1)(4d-3)}{(2d-1)^4}, \quad 1 - q_{d,6} = \frac{8(d-1)(3d^2-2d-2)}{(2d-1)^5},$$

$$1 - q_{d,7} = \frac{2(d-1)(32d^3-28d^2-46d+41)}{(2d-1)^6},$$

and each  $q_{d,n}$ ,  $4 \leq n \leq 7$ , is strictly increasing in  $d$  on all of  $d \geq 2$ ; Conjecture 284 therefore holds for  $n \leq 7$  in every dimension.

**Proposition 48** (Radial moments at length two). For  $n = 2$ , every  $d \geq 2$ , and every real  $q > 0$ ,  $\mathbf{E}_{U_2^{(d)}}[|\omega_2|^q] = 2^{q/2} + (2^q - 2^{q/2})/(2d-1)$ , strictly decreasing in  $d$ ; in particular Conjecture 285 holds at  $n = 2$ .

## 59 Curvature laws for counts, radial moments, and bridges

**Conjecture 271** (Parity log-concavity and curvature relaxation of walk counts). For  $\rho_n = c_{n+2}c_{n-2}/c_n^2$  and every  $n \geq 2$ :  $0 < \rho_n < 1$  and  $\rho_n < \rho_{n+2} < 1$ .

**Conjecture 272** (Radial-moment parity log-concavity at every order). For every real  $q \geq 0$  and every  $n \geq 3$ ,  $M_q(n)^2 > M_q(n-2)M_q(n+2)$ .

**Conjecture 273** (Eventual strict log-convexity of bridge counts). For every  $n \geq 5$ ,  $b_n^2 < b_{n-1}b_{n+1}$ .

## 60 Monotone transport in the length

**Conjecture 274** (Bridge height rises in likelihood-ratio order). For every  $n \geq 1$  the bridge height satisfies  $H_n \leq_{\text{lr}} H_{n+1}$ : for all  $h_1 < h_2$ ,  $B(n, h_2)B(n+1, h_1) \leq B(n, h_1)B(n+1, h_2)$ .

**Conjecture 275** (Monotone coupling of the absolute endpoint). For every  $n \geq 0$  there is a coupling of  $U_n$  and  $U_{n+2}$  under which  $(|X_n|, |Y_n|) \leq (|X_{n+2}|, |Y_{n+2}|)$  coordinatewise almost surely.

**Conjecture 276** (Monotone coupling of the span vector). For every  $n \geq 0$  there is a coupling under which  $(S_x, S_y)$  at length  $n$  is coordinatewise dominated by  $(S_x, S_y)$  at length  $n+2$  almost surely.

**Conjecture 277** (Span likelihood-ratio expansion). For every  $n \geq 0$ ,  $S_{x,n} \leq_{\text{lr}} S_{x,n+2}$ .

**Conjecture 278** (Stochastic growth of the contact count). For every  $n \geq 2$ ,  $C_n \leq_{\text{st}} C_{n+1}$ .

## 61 Dependence laws at fixed length

**Conjecture 279** (Negative quadrant dependence of the two spans). For every  $n \geq 1$ ,  $(S_x, S_y)$  is negatively quadrant dependent under  $U_n$ :  $\mathbf{P}(S_x \geq a, S_y \geq b) \leq \mathbf{P}(S_x \geq a)\mathbf{P}(S_y \geq b)$  for all  $a, b$ .

**Conjecture 280** (Positive quadrant dependence of radius and atmosphere). For every  $n \geq 1$ ,  $(R_n^2, A_n)$  is positively quadrant dependent under  $U_n$ .

**Conjecture 281** (Negative quadrant dependence of radius and contacts). *For every  $n \geq 1$ ,  $(R_n^2, C_n)$  is negatively quadrant dependent under  $U_n$ .*

**Conjecture 282** (Negative quadrant dependence of contacts and atmosphere). *For every  $n \geq 1$ ,  $(C_n, A_n)$  is negatively quadrant dependent under  $U_n$ .*

## 62 Phase and dimension laws

**Conjecture 283** (The collapse trichotomy of partition curvature). *Let  $K_n(u) = \log Z_{n+2}(u) - 2 \log Z_n(u) + \log Z_{n-2}(u)$  and let  $u_\theta$  be the square-lattice collapse point,  $u_\theta \approx 1.9474$  [142]. Along either parity class:  $n^2 K_n(u) \rightarrow -11/8$  for  $0 < u < u_\theta$ ;  $n^2 K_n(u_\theta) \rightarrow -4/7$ ; and  $n^{3/2} K_n(u) \rightarrow \sigma(u)$  with  $\sigma(u) > 0$  for  $u > u_\theta$ .*

**Conjecture 284** (Self-avoidance survival rises with dimension). *For every  $n \geq 4$  and  $d \geq 2$ ,  $q_{d+1,n} > q_{d,n}$ .*

**Conjecture 285** (All radial moments fall with dimension). *For every  $n \geq 2$ , every real  $q > 0$ , and every  $d \geq 2$ ,  $\mathbf{E}_{U_n^{(d+1)}}[|\omega_n|^q] < \mathbf{E}_{U_n^{(d)}}[|\omega_n|^q]$ .*

## Part XVI

# Budgeted irreversibility and the spectral geometry of relaxation

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The nine conjectures below concern a single organizing question of non-equilibrium dynamics: when a relaxation process is allowed only a finite budget of irreversibility, how is that budget spectrally allocated? They form four programmes. A current-fluctuation law (Conjecture 286) asserts that a reversible shortcut added to a driven ring at fixed activity strictly amplifies the standardized skewness of the winding current, against the mixing intuition, sharpening the cumulant inequalities of [151, 152] into a monotonicity law. A frontier programme (Conjectures 287 to 291) studies the linear diffusion  $dX_t = -(S + A)X_t dt + \sqrt{2S} dW_t$  with fixed Gaussian steady state, whose entropy-production rate is the exact quadratic  $\Sigma(A) = \text{tr}(A^\top S^{-1}A)$  [145], and conjectures the complete architecture of the budgeted-gap frontier: nested mode recruitment ordered by the reversible spectrum, piecewise-convex arcs with downward marginal-gain resets, exact two-mode support at small budget, a rank-two extremizer for the oscillation frequency, and a pairwise formula for the cheapest exceptional point, the thermodynamic analogue of the nearest-defective-matrix problem [150]; the unlimited endpoint is the theorem of Lelièvre, Nier, and Pavliotis [144], and the spectral-perturbation bound of [146] and the dissipation-coherence decomposition of [147] bound the same objects, so Conjecture 290 is closely related to the latter work. A coherent-acceleration programme (Conjectures 292 and 293) transfers the frontier architecture to primitive Lindblad dynamics with detailed balance, where a stationary Hamiltonian is a skew perturbation in the Gelfand–Naimark–Segal (GNS) geometry [154] and coherent gap enhancement is the theorem of Li and Lu [153]: sequential operator-mode recruitment under the commutator-accessibility constraint, and saturation of the classical envelope exactly on

$\mathfrak{su}(2)$ -closed mode systems. A preparation-geometry programme (Conjecture 294) turns from rates to eigenvector geometry in the anomalous-relaxation setting of the Markovian Mpemba effect [155, 156]: the measure of preparations that robustly cancel the slow mode should collapse in inverse proportion to the biorthogonal condition number as a Liouvillian exceptional point is approached, quantifying the antagonism between exceptional-point acceleration [157, 158] and robustness [159]. The proved layer is three calibrations: Proposition 49 solves the two-mode frontier in closed form, identifies the second-slowest mode as the unique best infinitesimal partner of the slowest, and gives the exact pair costs used throughout; Proposition 50 proves that the classical frontier is an upper envelope for every coherently driven detailed-balance Lindbladian and that the unital qubit attains it exactly; and Proposition 51 proves the basin-collapse law exactly in the qubit model, with the sphere prefactor equal to one. Four negative findings are retained as controls, and they shape the part: the Markov analogues of the exceptional-point frontier laws were tested at optimizer level and discarded, since on a three-state ring the budgeted-gap frontier reaches its supremum only on the unidirectional boundary at infinite entropy production, on rings of four and five states the slow pair's real part continues to improve past the exceptional point rather than pinning, on the four-state instance the slow sector turns complex strictly before the frontier's plateau, and the fixed-trace pinning that motivated those claims is an artifact of three-state spectra; gap optimizers generically sit at defective matrices [149] and at exceptional points of Liouvillians [148], so only the budgeted, structure-revealing refinements below are claimed as new; and a proposed resonance criterion for entangling coherent acceleration was tested in exact second-order arithmetic and discarded, since in every nondegenerate product instance the best local Hamiltonian already attains the full optimum, the commutator structure forcing every entangling transition out of a simple product slow mode to carry a strictly larger spectral denominator than its local counterpart, while the claimed advantage appears only through slow-mode degeneracies outside the criterion's own hypotheses; and a proposed capacity principle equating the number of slow modes simultaneously cancellable by unitary preparation with the maximal rank of the tangent overlap map was refuted on a generic qutrit instance, where the slow coordinate's range over the fixed-spectrum orbit omits zero entirely, the spectral offset between the stationary state and the orbit dominating the orbit's fluctuation, so feasibility of Mpemba preparation on an orbit is an affine range condition invisible to the tangent geometry, constructive steering protocols [160] notwithstanding, and Conjecture 294 carries that condition as an explicit hypothesis. Conjectures 287, 288, 292, and 293 assert laws for every dimension.

For a continuous-time generator  $Q$  on a finite connected graph with stationary law  $\pi$ , the activity is  $\kappa = \sum_i \pi_i \sum_{j \neq i} q_{ij}$  and the entropy-production rate is  $\sigma = \frac{1}{2} \sum_{i \neq j} (\pi_i q_{ij} - \pi_j q_{ji}) \log(\pi_i q_{ij} / \pi_j q_{ji})$ . On the homogeneous  $N$ -ring with affinity  $F$ , rates  $e^{\pm F/(2N)}$ , the winding current  $J_t$  has long-time cumulant rates  $c_m$  obtained from the tilted generator, and  $S_3 = |c_3|/c_2^{3/2}$  is its standardized skewness. For the linear diffusion above,  $S = S^\top > 0$  is diagonal with simple spectrum  $s_1 < \dots < s_n$  after an orthogonal change of basis,  $A^\top = -A$ , the stationary covariance is the identity,  $r(A) = \min_j \operatorname{Re} \lambda_j(S + A)$ ,  $\Omega(A) = \max_j |\operatorname{Im} \lambda_j(S + A)|$ , and the frontier is  $r_*(\Sigma) = \sup\{r(A) : \Sigma(A) \leq \Sigma\}$  with  $\Sigma(A) = \sum_{i < j} a_{ij}^2(1/s_i + 1/s_j)$ ; since the real parts sum to  $\operatorname{tr} S$ , always  $r(A) \leq \bar{s} = \operatorname{tr} S/n$ , and  $\bar{s}$  is attained at finite budget [144]. A mode  $i$  is *active* for  $A$  when the  $i$ th row of  $A$  is nonzero. For a primitive Lindblad dissipator  $\mathcal{D}$  in detailed balance with respect to  $\rho_*$ , the operator  $K = -\mathcal{D}$  is positive self-adjoint in the GNS inner product with eigenmodes  $KF_j = s_j F_j$  on the traceless sector, a stationary Hamiltonian ( $[H, \rho_*] = 0$ ) acts as the skew-adjoint  $\mathcal{A}_H = -i[H, \cdot]$  with matrix elements  $a_{jk}(H)$  and coherent budget  $\Sigma_H = \sum_{j < k} |a_{jk}(H)|^2(1/s_j + 1/s_k)$ , and  $r_{\text{OU}}$  denotes the frontier of the diagonal

spectrum  $\{s_j\}$ . For a density matrix  $\rho_0$  with simple spectrum,  $\mathcal{O}_{\rho_0}$  is its unitary orbit, the left Liouvillian eigenmodes  $\ell_1, \ell_2, \dots$  are ordered from slow to fast and split into real functionals,  $T_\rho : K \mapsto (\langle \ell_\alpha, i[K, \rho] \rangle)_\alpha$  is the tangent overlap map with real rank  $\text{rank } T_\rho$ , and for a family  $\mathcal{L}(h)$  with simple slow eigenvalue the biorthogonal condition number is  $\kappa_{\text{eig}}(h) = \|\ell_1(h)\| \|r_1(h)\|$  under  $\langle \ell_1, r_1 \rangle = 1$ ; the  $\varepsilon$ -robust basin on an orbit  $\mathcal{O}$  is  $\mathcal{B}_\varepsilon(h) = \{\rho \in \mathcal{O} : |\langle \ell_1(h), \rho - \rho_* \rangle| \leq \varepsilon\}$  with the invariant orbit measure  $\mu_{\mathcal{O}}$ .

**Proposition 49** (Pair calibration for the frontier programme). (i) A perturbation supported on the pair  $(i, j)$  leaves  $\text{span}\{e_i, e_j\}$  invariant with eigenvalues  $\frac{s_i + s_j}{2} \pm \sqrt{(\frac{s_j - s_i}{2})^2 - a_{ij}^2}$ ; the block reaches its exceptional point at cost  $\Sigma_{ij}^{\text{EP}} = \frac{(s_j - s_i)^2}{4}(1/s_i + 1/s_j)$  and generates frequency  $\Omega_{ij}(\Sigma) = (\Sigma/(1/s_i + 1/s_j) - (\frac{s_j - s_i}{2})_+^{1/2})$  at cost  $\Sigma$ . (ii) The small-budget rate gain of coupling the slowest mode to mode  $j$  is  $\gamma_{1j} = s_1 s_j / ((s_j - s_1)(s_j + s_1))$  per unit entropy production, strictly decreasing in  $s_j$ ; the second-slowest mode is the unique best infinitesimal partner. (iii) For  $n = 2$  the frontier is

$$r_*(\Sigma) = \frac{s_1 + s_2}{2} - \sqrt{\left(\frac{s_2 - s_1}{2}\right)^2 - \frac{\Sigma}{1/s_1 + 1/s_2}} \quad (\Sigma \leq \Sigma_{12}^{\text{EP}}), \quad r_*(\Sigma) = \bar{s} \quad (\Sigma \geq \Sigma_{12}^{\text{EP}}),$$

continuous, strictly increasing and strictly convex on the arc, with marginal gain diverging at the threshold and resetting to zero beyond it; all laws of Conjectures 287 to 291 hold for  $n = 2$ .

**Proposition 50** (Envelope calibration for the coherent programme). (i) For every primitive detailed-balance dissipator with traceless-sector spectrum  $\{s_j\}$  and every stationary Hamiltonian  $H$ ,

$$\text{gap}(-\mathcal{D} - \mathcal{A}_H) \leq r_{\text{OU}}(\Sigma_H).$$

(ii) For the unital qubit dissipator with Bloch decay rates  $(s_1, s_2, s_3)$  and  $H = \frac{1}{2} h \cdot \sigma$ , the traceless Liouvillian spectrum equals  $-\text{spec}(S + A(h))$  with  $A(h)$  the cross-product matrix of  $h$ , every  $3 \times 3$  skew matrix is realized, and the envelope is attained exactly.

**Proposition 51** (Basin calibration for the preparation programme). Let  $\mu$  be the uniform measure on the unit sphere of Bloch vectors and let the slow left functional of the family  $B(h) = S + A(h)$ ,  $S = \text{diag}(s_1, s_2, s_3)$ ,  $A(h)$  the rotation block  $a_{12} = -h$ , be normalized by  $\langle \ell_1, r_1 \rangle = 1$  with  $\|r_1\| = 1$ , so  $\kappa_{\text{eig}}(h) = \|\ell_1(h)\|$ . Then for every  $0 < \varepsilon \leq \kappa_{\text{eig}}(h)$  and every  $h$  below the exceptional point  $h_c = (s_2 - s_1)/2$ ,

$$\mu\{x : |\langle \ell_1(h), x \rangle| \leq \varepsilon\} = \frac{\varepsilon}{\kappa_{\text{eig}}(h)},$$

exactly, and  $\kappa_{\text{eig}}(h) = \sqrt{h_c/2}(h_c - h)^{-1/2}(1 + o(1))$  as  $h \uparrow h_c$ , so the basin measure collapses at the exact rate  $\varepsilon \sqrt{2(h_c - h)/h_c}$ .

## 63 A current-fluctuation law for equilibrium shortcuts

**Conjecture 286** (Reversible shortcuts amplify winding skewness). *For every even  $N \geq 4$  and affinity  $F \neq 0$ , add to the homogeneous driven  $N$ -ring a reversible antipodal chord of strength  $g$  carrying no winding increment, rescaling all rates to keep the activity fixed. Then the standardized skewness  $S_3(g)$  of the long-time winding current is strictly increasing in  $g \geq 0$ .*

## 64 The finite-budget frontier of linear diffusions

**Conjecture 287** (Sequential spectral recruitment). *For every diagonal  $S$  with simple spectrum there are thresholds  $0 = \Sigma_0 < \Sigma_1 \leq \Sigma_2 \leq \dots$  and a choice of maximizer  $A(\Sigma)$  of the frontier at every budget whose active-mode set is  $\{1, \dots, m\}$  with  $m$  nondecreasing in  $\Sigma$ : no mode is recruited before every slower mode.*

**Conjecture 288** (Convex arcs with marginal-gain resets). *For simple  $S$  the frontier  $r_*(\Sigma)$  is continuous and piecewise real-analytic; on each maximal interval with constant active set it is convex; the one-sided marginal gain jumps only downward at arc boundaries; and  $r_*$  reaches the ceiling  $\bar{s}$  at a finite budget and is constant beyond.*

**Conjecture 289** (Finite two-mode radius). *For simple  $S$  there exists  $\Sigma_1 > 0$  such that for every  $0 < \Sigma < \Sigma_1$ , every global maximizer of the frontier, up to sign and rotations inside exact degeneracies, is supported on the two slowest modes;  $\Sigma_1$  is the first budget at which the two-mode branch collides or loses optimality to a branch containing mode three.*

**Conjecture 290** (Rank-two oscillation extremizer). *For every diagonal  $S > 0$  and budget  $\Sigma$ ,*

$$\sup_{\Sigma(A) \leq \Sigma} \Omega(A) = \max_{i < j} \Omega_{ij}(\Sigma),$$

*with  $\Omega_{ij}$  the pair formula of Proposition 49(i): no coupling of three or more modes generates a larger spectral frequency than the best single rotation block.*

**Conjecture 291** (Cheapest exceptional point is a pair). *For simple  $S$ , the infimum of  $\Sigma(A)$  over antisymmetric  $A$  making  $S + A$  defective equals  $\min_{i < j} \Sigma_{ij}^{\text{EP}}$ , attained by the single block of Proposition 49(i) on a minimizing pair.*

## 65 Coherent acceleration of dissipative quantum dynamics

**Conjecture 292** (Recruitment under commutator accessibility). *For a primitive detailed-balance dissipator with simple traceless spectrum, every small-budget maximizer of  $\text{gap}(-\mathcal{D} - \mathcal{A}_H)$  over stationary Hamiltonians of coherent budget  $\Sigma_H$  concentrates, as  $\Sigma_H \rightarrow 0$ , on the accessible direction maximizing the restricted quotient  $\sum_{j>1} |a_{1j}(H)|^2 / (s_j - s_1)$  over  $\Sigma_H$ ; and there is a maximizing selection whose active dissipative modes are nested and nondecreasing in the budget, with changes only at spectral collisions or accessibility branch crossings.*

**Conjecture 293** (Envelope saturation is Lie-algebraic). *For a primitive detailed-balance Lindbladian, the envelope of Proposition 50 is attained on a nontrivial budget interval if and only if the dissipative eigenmodes active in the classical optimizer close under commutation into a direct sum of three-dimensional  $\mathfrak{su}(2)$ -type modules together with spectators, so that the unrestricted skew optimizer is realized by a Hamiltonian commutator.*

## 66 The geometry of anomalous relaxation

**Conjecture 294** (Exceptional points collapse robust Mpemba basins). *Let  $\mathcal{L}(h)$  approach a second-order exceptional point  $h_c$  in its slow sector from the diagonalizable side, normalize the slow eigenvectors by  $\|r_1\| = 1$  and  $\langle \ell_1, r_1 \rangle = 1$  so that  $\kappa_{\text{eig}}(h) = \|\ell_1(h)\|$  and the basin*



$\mathcal{B}_\varepsilon(h) = \{\rho \in \mathcal{O} : |\langle \ell_1(h), \rho - \rho_* \rangle| \leq \varepsilon\}$  is fixed in scale, and suppose the slow functional vanishes transversally on the fixed smooth orbit  $\mathcal{O}$ , uniformly near the exceptional point: the tangential gradient of the unit-normalized slow functional is bounded above and below by positive constants along its zero set on  $\mathcal{O}$ , and that zero set is nonempty with hypersurface measure bounded below, uniformly for  $h$  near  $h_c$ . Then there are constants  $0 < c < C$  with

$$c \frac{\varepsilon}{\kappa_{\text{eig}}(h)} \leq \mu_{\mathcal{O}}(\mathcal{B}_\varepsilon(h)) \leq C \frac{\varepsilon}{\kappa_{\text{eig}}(h)}$$

for all sufficiently small  $\varepsilon$  and all  $h$  near  $h_c$ : the measure of preparations that robustly cancel the slow mode collapses in exact inverse proportion to the biorthogonal condition number.

## Part XVII

# Inverse problems and design laws for multivariate extremes

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The thirteen conjectures below concern a single organizing question of extreme-value theory: how much of the dependence structure of rare events can be recovered from partial, thinned, or redesigned observation, and at what cost in stability. They form five programmes. A geometric programme (Conjectures 295 to 298) studies the extremal coefficient  $\theta_m(\Gamma)$  of the Hüsler–Reiss family [161, 162] as a functional on variogram matrices: an isoperimetric recruitment staircase at fixed total variogram energy with a proved symmetry-breaking boundary, the regular simplex maximal for three variables and at weak dependence but strictly beaten below it by configurations carrying fewer active components, a pairwise closure law at weak dependence sharpening the consistency inequalities of Schlather and Tawn [163] into an asymptotic equivalence, a scale-elasticity bound of one half paired with entrywise monotonicity whose conjunction yields a dimension-free sparsification law for resistance variograms, strengthening the variogram-level guarantees of [164] to the extremal coefficient itself, and a positive-curvature law at complete dependence whose convex-body shadow is the Wills functional [165]. A cluster-tomography programme (Conjectures 299 and 300) turns the missing-data results for subsampled sequences [169, 170, 171] into an inverse theory: the Bernoulli-thinning response curve should identify the entire limiting cluster-size law through its generating function, and the two-parameter systematic-plus-Bernoulli response should identify the cluster support exactly up to the equivalence generated by its modular occupancy data. A topological programme (Conjectures 301 and 302) makes the topology of extreme directions an inferential target: a bottleneck-consistency law for persistent homology of intermediate-order angular samples at the rate  $b(U(n/k)) + (\log k/k)^{1/s}$ , composing the diagram-estimation rates of [175] with a finite-threshold angular bias, and a leakage trichotomy locating exactly when hidden regular variation [176, 177] is topologically visible after support filtering. A tail-tomography programme (Conjectures 303 and 304) quantifies the inverse problem of angular reconstruction from one-sided projection tails: a severe stability barrier, no Hölder inverse on measures and only a logarithmic modulus on Hölder classes, the analytic-kernel phenomenon of moment problems [180] instantiated for the fractional transforms whose injectivity dichotomy is recorded in Proposition 55 following [178, 179], and a parameter-count identifiability threshold for  $s$ -atomic angular measures at exactly  $m = sd - 1$  generic fractional projections. A design

programme (Conjectures 305 to 307) redesigns block maxima themselves: Poissonized block sizes should remove the universal  $V^2/(2n)$  finite-block correction and the corresponding  $n^{-1}$  estimator bias documented in the block-maxima second-order theory [181, 182, 183], factorial-cumulant matching should extend the removal to a full ladder of orders, and exact second-order cancellation in a Hall family [184] should create a cusp at which threshold adaptation, whose logarithmic penalty away from the cusp is the theorem of Carpentier and Kim [185], cannot be uniformly oracle-optimal, with critical window  $|c| \asymp n^{-\beta/(\alpha+4\beta)}$ . The proved layer is four calibrations. Proposition 52 gives the exact Gaussian-orthant formula for  $\theta_m$  used throughout, proves the elasticity bound in the pairwise case, and proves that every collinear configuration has vanishing second curvature, the easy direction of Conjecture 298. Proposition 53 proves the cluster-level thinning algebra: the Bernoulli response formula, the systematic random-phase formula, the identifiability of the occupancy multisets from the joint response, the recovery of the entire difference multiset from occupancy data by Möbius inversion, and the closure of occupancy equivalence under dilation. Proposition 55 proves the integer-fractional injectivity dichotomy for the projection transforms, with an explicit two-atom versus three-atom pair invisible at  $\alpha = 2$  and separated at  $\alpha = 3/2$ . Proposition 56 proves the exact Poissonization identity for Pareto seeds and constructs, for every order  $r$ , finitely supported non-Poisson integer block-size laws whose factorial cumulants match the Poisson law through order  $r$ , with the exact probability-generating-function identity showing that the finite-block error then falls to order  $n^{-r}$ . Five negative findings are retained as controls, and they shape the part. The deposited isoperimetry claim, that the regular simplex maximizes  $\theta_m$  at fixed energy for every  $m$ , is refuted for  $m \geq 4$ : the width identity of Proposition 52(iv) makes configurations with fewer active components strictly better away from weak dependence, with exact orthant-probability instances at  $m = 4$ , scale 0.12, and  $m = 5$ , scale 0.35, where the four-active configuration beats the simplex by  $3.5 \times 10^{-3}$  and is found by unrestricted optimization, so Conjecture 295 keeps the simplex law at three variables and states the recruitment staircase above. A proposed modular reconstruction claim, that the occupancy multisets over all moduli determine a finite cluster support up to translation and reflection, is refuted: the six-mark pair  $\{0, 1, 4, 10, 12, 17\}$  and  $\{0, 1, 8, 11, 13, 17\}$  of Bloom [172], the classical counterexample to Piccard’s theorem on difference multisets [173, 174], shares every occupancy multiset, by finite verification below the diameter and trivially above it, so occupancy tomography can never distinguish this pair; the repaired Conjecture 300 asks instead for determination within size five and for the two-member bound on equivalence classes, over the exhaustive verification that all ten occupancy collisions among the 6437 canonical sets of size at most six through diameter eighteen and the 3312 of size at most seven through diameter fourteen are homometric pairs, while the pair  $\{0, 1, 3, 8, 10, 14\}$  versus  $\{0, 1, 7, 9, 11, 14\}$  shows the occupancy data strictly refines the difference multiset. The subsampling formulas themselves, the Bernoulli and systematic extremal-index responses at the cluster level, are known in substance for subsampled stationary sequences [169, 170, 171] and are therefore recorded as calibration, not conjecture; what is claimed as new in Conjectures 299 and 300 is the uniform process-level transport and the inverse-theory content. A multiplicity observed at the identifiability threshold of Conjecture 304 on one random instance was exposed as an artifact of a near-coincident atom pair, ill-conditioning inflating rounding-level copies of the single true solution, and the atom-separation hypothesis of the conjecture records that boundary. A second multiplicity is genuine and refutes global uniqueness at the counting boundary itself: at three atoms in three dimensions, two of six well-separated random instances admit a second exact interior mixture at  $m = sd - 1$ , verified to forty-digit residuals, and both become unique

one measurement higher, so the global clause of Conjecture 304 begins at  $m = sd$  and the counting-boundary refutation is recorded in place. Conjectures 295, 296, 297, 301, 302, and 305 to 307 assert laws for every dimension or every sample size; Conjecture 297 is closely related to the sparsification guarantees of [164], and Conjecture 307 carries one against the adaptation lower bounds of [185]. Since deposit, Conjecture 298 has been resolved true by Proposition 54.

A variogram matrix  $\Gamma$  on  $m$  sites is symmetric with zero diagonal and conditionally negative definite, so that  $\Gamma_{ij} = \|x_i - x_j\|^2$  for centered points  $x_1, \dots, x_m$  in some  $\mathbb{R}^k$ ; the Hüsler–Reiss max-stable vector with variogram  $\Gamma$  [161] has extremal coefficient  $\theta_m(\Gamma) \in [1, m]$ , the exponent function at  $\mathbf{1}$ , with spectral representation  $\theta_m(\Gamma) = \mathbb{E} \max_i \exp(\langle Z, x_i \rangle - \|x_i\|^2/2)$  for a standard Gaussian  $Z$ , pairwise coefficients  $\theta_{ij} = 2\Phi(\sqrt{\Gamma_{ij}}/2)$ , energy  $\mathcal{S}(\Gamma) = \sum_{i < j} \Gamma_{ij}$ , and scale elasticity  $\eta(t) = d \log \theta_m(t\Gamma) / d \log t$ . For a stationary regularly varying sequence whose exceedances converge to a compound-Poisson cluster process [166] with cluster size  $N$ ,  $\mathbb{E}N < \infty$ , the extremal index is  $\theta = 1/\mathbb{E}N$  [167, 168], and when the limiting cluster has a finite support  $S \subset \mathbb{Z}$ , defined up to translation, the occupancy counts are  $n_{m,r}(S) = |\{s \in S : s \equiv r \pmod{m}\}|$ , the occupancy data of  $S$  is the family of unordered multisets  $\mathcal{N}_m(S) = \{n_{m,0}(S), \dots, n_{m,m-1}(S)\}$  over all  $m \geq 2$ , two sets are occupancy equivalent when their occupancy data agree, and homometric when their difference multisets agree. A  $d$ -variate regularly varying vector has angular measure  $H$  on the simplex  $\Delta_{d-1}$ ; the fractional projection transform is  $(T_\alpha H)(a) = \int (a \cdot w)^\alpha H(dw)$  for directions  $a \in (0, \infty)^d$ ; and for block maxima,  $F$  lies in a max-domain with exponent function  $V$  and second-order rate  $A(n)$ , a block of size  $n$  has exceedance probability  $q_n(x) = \bar{F}(a_n x + b_n)$ , and a randomized block size  $N_n$  has factorial cumulants  $\kappa_r^{(F)}$  defined by  $\log \mathbb{E}(1+u)^{N_n} = \sum_{r \geq 1} \kappa_r^{(F)} u^r / r!$ , so that  $\log \mathbb{E}(1-q)^{N_n} = \sum_{r \geq 1} \kappa_r^{(F)} (-q)^r / r!$  exactly.

**Proposition 52** (Tilting calibration for the dependence programme). (i) For every admissible  $\Gamma$  realized by centered points  $x_1, \dots, x_m$ ,

$$\theta_m(\Gamma) = \sum_{i=1}^m \Phi_{m-1} \left( \left( \frac{\Gamma_{ij}}{2} \right)_{j \neq i}; \left( \frac{\Gamma_{ij} + \Gamma_{ik} - \Gamma_{jk}}{2} \right)_{j,k \neq i} \right),$$

where  $\Phi_{m-1}(\cdot | C)$  is the centered Gaussian orthant probability with covariance  $C$ . (ii) For  $m = 2$  the elasticity satisfies  $0 \leq \eta(t) \leq \sup_{x>0} x\varphi(x)/(2\Phi(x)) = 0.1472\dots < 1/2$  for all  $t > 0$ . (iii) If the points  $x_i$  are collinear, then with  $M = \max_i \langle Z, x_i \rangle$  and  $I$  the almost-surely unique argmax,  $\mathbb{E}[M^2 - \|x_I\|^2] = 0$ : every collinear configuration has vanishing second curvature. (iv) For  $3 \leq k \leq m$  let the  $k$ -active configuration place  $k$  points at the vertices of a centered regular simplex and  $m - k$  points at the origin, with energy  $S$ . Its Gaussian width is

$$W_k(m, S) = \mathbb{E} \max_i \langle Z, x_i \rangle = \sqrt{\frac{S}{m}} \frac{e_k}{\sqrt{k-1}},$$

where  $e_k$  is the expected maximum of  $k$  independent standard Gaussians; and  $e_3/\sqrt{2} = 0.59841\dots$  exceeds  $e_4/\sqrt{3} = 0.59431\dots$  and  $e_5/2 = 0.58148\dots$ , so for  $m \in \{4, 5\}$  the triangle configuration  $k = 3$  has strictly larger width than the full simplex  $k = m$ , and hence strictly larger  $\theta_m$  at fixed energy for all sufficiently small scale.

## 67 The geometry of Hüsler–Reiss dependence

**Conjecture 295** (Isoperimetry and symmetry breaking at fixed energy). (a) For  $m = 3$  and every  $S > 0$ , among all admissible variograms with energy  $\mathcal{S}(\Gamma) = S$ , the equidistant variogram

$\Gamma^{\text{eq}}$  with all off-diagonal entries  $S/\binom{m}{2}$ , realized by a regular triangle, uniquely maximizes  $\theta_3(\Gamma)$ , up to relabeling. (b) For every  $m \geq 4$  and every  $S$ , the maximum of  $\theta_m$  at energy  $S$  is attained by a  $k$ -active configuration of Proposition 52(iv),  $k$  components at the vertices of a centered regular simplex and  $m - k$  coincident at the centroid, for some  $3 \leq k \leq m$ , uniquely up to relabeling; the maximizing  $k$  is nondecreasing in  $S$ , equals 3 for all sufficiently small  $S$ , equals  $m$  for all sufficiently large  $S$ , and passes through every intermediate value, so the fixed-energy optimizer recruits one component at a time through a staircase of symmetry-breaking transitions.

**Conjecture 296** (Pairwise closure at weak dependence). *Fix an admissible  $\Gamma$  with strictly positive off-diagonal entries such that no three-or-higher-way Gaussian large-deviation saddle shares the exact rate of a contributing pairwise saddle. Then, with  $\delta_{ij}(t) = 2 - \theta_{ij}(t\Gamma) = 2[1 - \Phi(\sqrt{t\Gamma_{ij}}/2)]$ ,*

$$m - \theta_m(t\Gamma) \sim \sum_{i < j} \delta_{ij}(t), \quad t \rightarrow \infty.$$

**Conjecture 297** (Scale elasticity and monotone dependence). *For every admissible nonzero  $\Gamma$ : (a)  $0 \leq \eta(t) \leq \frac{1}{2}$  for all  $t > 0$ ; (b) if  $\Gamma'$  is admissible and  $\Gamma_{ij} \leq \Gamma'_{ij}$  entrywise, then  $\theta_m(\Gamma) \leq \theta_m(\Gamma')$ . Consequently, if  $L, \tilde{L}$  are connected graph Laplacians with  $(1 - \varepsilon)L \preceq \tilde{L} \preceq (1 + \varepsilon)L$  on  $\mathbf{1}^\perp$  and  $\Gamma(L)$  is the resistance variogram  $\Gamma_{ij}(L) = (e_i - e_j)^\top L^\dagger (e_i - e_j)$ , then*

$$(1 + \varepsilon)^{-1/2} \leq \frac{\theta_m(\Gamma(\tilde{L}))}{\theta_m(\Gamma(L))} \leq (1 - \varepsilon)^{-1/2}.$$

**Conjecture 298** (Positive curvature at complete dependence, resolved true). *For centered points  $x_1, \dots, x_m$  realizing  $\Gamma$ , with  $M = \max_i \langle Z, x_i \rangle$  and  $I$  the argmax, as  $t \downarrow 0$ ,*

$$\theta_m(t\Gamma) = 1 + \sqrt{t} \mathbb{E} M + t q(\Gamma) + o(t), \quad q(\Gamma) = \frac{1}{2} \mathbb{E}[M^2 - \|x_I\|^2],$$

*and  $q(\Gamma) \geq 0$  always, with  $q(\Gamma) = 0$  if and only if, after deleting non-exposed points, the convex hull of the  $x_i$  is a point or a segment.*

*Resolution.* True, by Proposition 54 below: the expansion holds together with the non-negativity of  $q(\Gamma)$  and the equality classification, the positivity following from a Gaussian integration-by-parts identity.

## 68 Cluster inverse problems under thinning

**Proposition 53** (Thinning calibration for the cluster programme). Let the limiting cluster have size  $N$  with generating function  $G_N$  and, where stated, finite support  $S \subset \mathbb{Z}$ . At the level of the limiting compound-Poisson cluster process: (i) independent retention with probability  $p$  gives the response  $\theta(p) = \theta(1 - G_N(1 - p))/p$ ; (ii) retention of one residue class modulo  $m$  with uniform random phase gives  $\theta_m^{\text{sys}} = \theta \mathbb{E} R_m(S)$ , where  $R_m(S)$  is the number of occupied residues; (iii) the joint response  $Q_{m,S}(p) = \frac{1}{m} \sum_{r=0}^{m-1} [1 - (1 - p)^{n_{m,r}(S)}]$ , known for  $p$  in any infinite subset of  $(0, 1)$ , determines the occupancy multiset  $\mathcal{N}_m(S)$ ; (iv) the occupancy data  $\{\mathcal{N}_m(S) : m \geq 2\}$  determines the difference multiset of  $S$ ; (v) occupancy equivalence is closed under dilation:  $\mathcal{N}_m(tS) = \mathcal{N}_{m/\gcd(t,m)}(S)$  for every  $t \geq 1$ , so  $S \sim S'$  implies  $tS \sim tS'$ .

**Proposition 54** (Resolution of the complete-dependence curvature law). Conjecture 298 holds in full: the displayed expansion is valid,  $q(\Gamma) \geq 0$  for every admissible  $\Gamma$ , and  $q(\Gamma) = 0$  if and only if the convex hull of the exposed points is a point or a segment.

*Proof.* Write  $\varepsilon = \sqrt{t}$ ,  $G_i = \langle Z, x_i \rangle$ ,  $M = \max_i G_i$  with almost-surely unique  $\operatorname{argmax} I$  after deleting non-exposed and duplicate points, and  $V_\varepsilon = \max_i (\varepsilon G_i - \frac{\varepsilon^2}{2} \|x_i\|^2)$ , so  $\theta_m(t\Gamma) = \mathbb{E} e^{V_\varepsilon}$ . From  $|V_\varepsilon - \varepsilon M| \leq \frac{\varepsilon^2}{2} \max_i \|x_i\|^2$  and uniform integrability of  $e^{\varepsilon M}$  for small  $\varepsilon$ , dominated expansion gives  $\theta_m(t\Gamma) = 1 + \varepsilon \mathbb{E} M + \frac{\varepsilon^2}{2} \mathbb{E}[M^2 - \|x_I\|^2] + o(\varepsilon^2)$ , the asserted expansion with  $q(\Gamma) = \frac{1}{2} \mathbb{E}[M^2 - \|x_I\|^2]$ . Now  $M = h_K(Z)$  is the support function of  $K = \operatorname{conv}\{x_i\}$ ; the  $x_i$  are centered, so  $0 = \frac{1}{m} \sum_i x_i$  lies in the relative interior of  $K$  and  $M \geq 0$ , while  $M$  is convex and 1-homogeneous with  $\nabla M = x_I$  almost everywhere. Gaussian integration by parts and Euler's identity  $\langle z, \nabla M \rangle = M$  give

$$\mathbb{E}[M^2] = \mathbb{E}\langle Z, M \nabla M \rangle = \mathbb{E}\|\nabla M\|^2 + \mathbb{E}[M \Delta M],$$

where  $\Delta M \geq 0$  is the curvature measure of the convex function  $M$ ; since  $M \geq 0$  the last term is nonnegative and  $\|\nabla M\|^2 = \|x_I\|^2$  almost everywhere, so  $q(\Gamma) \geq 0$ . Equality forces  $M = 0$  on  $\operatorname{supp} \Delta M$ ; if  $\dim K \geq 2$  then 0 lies in the relative interior of  $K$ , so  $M > 0$  at every nonzero point of the affine span of  $K$ , on which the nonzero curvature measure  $\Delta M$  is supported, and the term is strictly positive. Hence  $q(\Gamma) = 0$  iff  $\dim K \leq 1$ , and for a point or a segment the curvature measure sits where  $M = 0$ , giving  $q = 0$ .  $\square$

**Conjecture 299** (Bernoulli-thinning spectroscopy of clusters). *Let  $(X_t)$  be stationary and regularly varying, with exceedances converging to a compound-Poisson cluster process with cluster size  $N$ ,  $\mathbb{E} N < \infty$ , and extremal index  $\theta = 1/\mathbb{E} N$ , under mixing and anti-clustering conditions. Retain each observation independently with probability  $p$  and compress the time axis. Then the retained sequence satisfies the same hypotheses with extremal index  $\theta(p) = \theta(1 - G_N(1 - p))/p$ , uniformly for  $p$  in compact subsets of  $(0, 1]$ ; consequently the response curve  $p \mapsto \theta(p)$  identifies the entire limiting cluster-size law.*

**Conjecture 300** (Occupancy tomography of cluster supports). *(a) Every finite  $S \subset \mathbb{Z}$  with  $|S| \leq 5$  is determined by its occupancy data up to translation and reflection. (b) For every finite size, each occupancy-equivalence class of finite integer sets contains at most two translation-reflection classes.*

## 69 The topology of extreme directions

**Conjecture 301** (Consistency of persistent angular homology). *Let  $X$  be  $d$ -variate regularly varying whose angular law  $H$  is supported on a compact  $C^2$   $s$ -dimensional manifold  $M$  of positive reach in the open simplex, with density bounded above and below. Let  $A_{n,k}$  be the angular components of the  $k = k_n$  largest radii,  $k \rightarrow \infty$ ,  $k/n \rightarrow 0$ , and suppose the conditional angular law beyond radius  $t$  couples to  $H$  with Hausdorff bias at most  $b(t) \rightarrow 0$ . Then for every homological degree  $q$ ,*

$$d_B(\operatorname{Dgm}_q(\operatorname{VR}(A_{n,k})), \operatorname{Dgm}_q(\operatorname{VR}(M))) = O_p\left(b(U(n/k)) + \left(\frac{\log k}{k}\right)^{1/s}\right),$$

with  $U$  the radial quantile function and  $d_B$  the bottleneck distance.

**Conjecture 302** (A leakage trichotomy for hidden extremes). *Let  $X = R_0 W_0 + R_1 W_1$  with independent components:  $R_0$  regularly varying with index  $\alpha_0$ ,  $R_1$  with index  $\alpha_1 > \alpha_0$ , and angular laws  $W_0 \sim H_0$ ,  $W_1 \sim H_1$  supported on compact sets  $M_0$ ,  $M_1$  separated in the simplex,  $H_1$  as in Conjecture 301. Remove the angular neighborhood  $N_{\delta_n}(M_0)$ , rerank the surviving radii, and let  $L_n$  be the ratio of the primary to the hidden mechanism's contribution to the surviving*



exceedance intensity at the hidden threshold. Let  $\Lambda_n$  be the expected number of surviving leaked primary exceedances in the reranked top- $k_n$  sample, and suppose the leaked support  $M_{\text{leak}}$  is persistence-separated from  $M_1$ ,  $d_B(\text{Dgm}_q(M_1 \cup M_{\text{leak}}), \text{Dgm}_q(M_1)) \geq c_0 > 0$ . Then: if  $\Lambda_n \rightarrow 0$ , the persistence diagrams of the surviving angular sample recover those of  $M_1$  at the rate of Conjecture 301; if  $\Lambda_n \rightarrow \infty$  they do not; and for  $\Lambda_n \rightarrow \lambda \in (0, \infty)$  the surviving leaked primaries are asymptotically Poisson- $\lambda$  and the probability of recovery within any fixed bottleneck tolerance below the hidden feature scale converges to a limit  $p(\lambda) \in (0, 1)$ , so feature detection switches on the leaked-count scale  $\Lambda_n \asymp 1$ , that is  $L_n \asymp 1/k_n$ .

## 70 Tail tomography and its stability

**Proposition 55** (Moment calibration for the tomography programme). (i) If  $\alpha > 0$  is not an integer, then  $T_\alpha H$  restricted to any open subset of  $(0, \infty)^d$  determines the finite measure  $H$  on  $\Delta_{d-1}$  uniquely. (ii) If  $\alpha = q \in \mathbb{N}_0$ , then  $T_q H$  depends only on the moments of  $H$  of degree at most  $q$ , and injectivity fails; explicitly, for  $d = 2$ ,  $q = 2$ , the measures

$$H_1 = \frac{1}{2}\delta_{0.2} + \frac{1}{2}\delta_{0.8}, \quad H_2 = 0.28125\delta_{0.1} + 0.4375\delta_{0.5} + 0.28125\delta_{0.9}$$

satisfy  $T_2 H_1 = T_2 H_2$  identically while  $T_{3/2} H_1 \neq T_{3/2} H_2$ .

**Conjecture 303** (The stability barrier of fractional tail tomography). Fix noninteger  $\alpha > 0$ ,  $0 < \eta < 1$ , and the direction box  $A_\eta = \{a : \eta \leq a_i \leq \eta^{-1}\}$ . (a) There exist no  $C, \beta > 0$  with  $W_1(H_1, H_2) \leq C \|T_\alpha H_1 - T_\alpha H_2\|_{L^\infty(A_\eta)}^\beta$  for all probability measures  $H_1, H_2$  on  $\Delta_{d-1}$ . (b) On the class of angular densities lying in a Hölder ball  $\{f : \|f\|_{C^s} \leq L, f \geq c > 0\}$ , the inverse admits a logarithmic modulus of continuity:  $W_1(H_1, H_2) \lesssim (\log(1/\varepsilon))^{-\kappa_s}$  with  $\varepsilon$  the transform error, for some  $\kappa_s > 0$  depending on  $s$ , and no polynomial modulus  $\varepsilon^\beta$  is valid on the class.

**Conjecture 304** (A parameter-count threshold for sparse tail tomography). Let  $H = \sum_{r=1}^s \pi_r \delta_{w_r}$  with distinct atoms in the open simplex  $\Delta_{d-1}^\circ$ , weights  $\pi_r > 0$  summing to one, and let  $p = sd - 1$  be the parameter count. Fix noninteger  $\alpha > 0$  and observe  $y_j = (T_\alpha H)(a_j)$  at generic positive directions  $a_1, \dots, a_m$ . Then, outside a proper real-analytic exceptional set of atom configurations and designs, and quantitatively away from atom collisions and the simplex boundary: for  $m < p$  identifiability fails on a positive-dimensional fiber; for  $m = p$  the Jacobian has full rank and the fiber is finite, so the mixture is locally rigid but need not be unique; and for every  $m \geq sd$  the measurement vector determines  $H$  globally up to permutation of atoms, with local inverse modulus controlled by atom separation and boundary distance.

## 71 Randomized block designs

**Proposition 56** (Design calibration for the block programme). (i) If  $\bar{F}(x) = x^{-\alpha}$  and  $N_n \sim \text{Poisson}(n)$ , then  $\Pr(M_{N_n}/n^{1/\alpha} \leq x) = e^{-x^{-\alpha}}$  exactly for every  $n$ . (ii) For any block-size law,  $\log \mathbb{E}(1 - q)^{N_n} = \sum_{r \geq 1} \kappa_r^{(F)} (-q)^r / r!$ , so matching factorial cumulants of the Poisson law through order  $r$  cancels the sampling corrections through order  $q^r$ . (iii) For every  $r \geq 2$  there is a law  $\xi$  on a finite subset of  $\{0, 1, 2, \dots\}$ , not Poisson, whose cumulants match those of Poisson(1) through order  $r$ ; the convolution  $N_n = \xi_1 + \dots + \xi_n$  then has  $\mathbb{E} N_n = n$ ,  $\text{Var } N_n = n$ ,  $\kappa_j^{(F)}(N_n) = 0$  for  $2 \leq j \leq r$ ,  $\kappa_{r+1}^{(F)}(N_n) = O(n)$ , and for the Pareto seed

$$\sup_x |\Pr(M_{N_n}/n^{1/\alpha} \leq x) - e^{-x^{-\alpha}}| = O(n^{-r})$$



uniformly on compact  $x$ -sets. For  $r = 3$  the law on  $\{0, 1, 3\}$  with weights  $(\frac{1}{3}, \frac{1}{2}, \frac{1}{6})$  and for  $r = 4$  the law on  $\{0, 1, 2, 4\}$  with weights  $(\frac{3}{8}, \frac{1}{3}, \frac{1}{4}, \frac{1}{24})$  are such laws.

**Conjecture 305** (Poissonized blocks remove the universal bias). *Let  $F$  lie in a max-domain of attraction with a uniform second-order expansion  $n\bar{F}(a_nx + b_n) = V(x) + A(n)B(x) + o(A(n)) + o(n^{-1})$  on compact interior sets. Then Poissonized block maxima satisfy  $\log \Pr(M_{N_n} \leq a_nx + b_n) = -V(x) - A(n)B(x) + o(A(n)) + o(n^{-1})$ , the universal  $V(x)^2/(2n)$  correction of fixed blocks being absent; and for generalized-extreme-value likelihood estimation from many independent blocks, Poissonization leaves the first-order asymptotic variance unchanged while removing the universal  $n^{-1}$  bias contribution, so that when  $A(n) = o(n^{-1})$  the estimator bias improves from order  $n^{-1}$  to the intrinsic second order.*

**Conjecture 306** (The factorial-cumulant design ladder). *Fix  $r \geq 2$  and let  $N_n$  be an  $r$ -th ladder law as in Proposition 56(iii). Then in every max-domain with the second-order expansion of Conjecture 305, block maxima with exposure  $N_n$  satisfy the expansion with sampling-mechanism corrections of order  $O(n^{-r})$  in place of  $V^2/(2n)$ ; and generalized-extreme-value likelihood estimation from many independent randomized blocks retains the first-order limit law and variance of deterministic blocks while its sampling-mechanism bias falls to  $O(n^{-r}) + o(A(n))$ .*

**Conjecture 307** (An adaptation cusp at second-order cancellation). *In a smooth Hall family  $\bar{F}_\lambda(x) = x^{-\alpha}[1 + c(\lambda)x^{-\beta} + d(\lambda)x^{-2\beta} + o(x^{-2\beta})]$  with  $c(\lambda_0) = 0$ ,  $c'(\lambda_0) \neq 0$ ,  $d(\lambda_0) \neq 0$ , the oracle threshold for Hill-type estimation moves from  $k \asymp n^{2\beta/(\alpha+2\beta)}$  generically to  $k \asymp n^{4\beta/(\alpha+4\beta)}$  at cancellation, the two regimes meeting at the critical window  $|c(\lambda)| \asymp n^{-\beta/(\alpha+4\beta)}$ ; and no data-driven threshold selector is uniformly oracle-rate optimal, with the same constants, over both  $|c(\lambda)| \leq Cn^{-\beta/(\alpha+4\beta)}$  and  $|c(\lambda)| \gg n^{-\beta/(\alpha+4\beta)}$ : an adaptation penalty, at least logarithmic or a nontrivial constant, is unavoidable in the cusp.*

## Part XVIII

# Extremal, randomized, and inverse laws for persistent topology

*Added to the deposit: 20 August 2026.*

The twelve conjectures below concern a single organizing question of topological data analysis: which parts of a persistence computation are laws rather than artifacts, sharp constants instead of worst-case bounds, invertible summaries instead of lossy signatures, and calibrated randomness instead of parameter sensitivity. They form five programmes. An extremal-and-sampling programme (Conjectures 308 to 310) concerns degree-one Vietoris–Rips persistence: a circle isoperimetry asserting that among  $n$  points on a Euclidean circle the regular polygon uniquely maximizes the multiplicative persistence ratio, the chordal, every- $n$ , ratio-level refinement of the geodesic persistence-set extremality of Gómez and Mémoli [188]; a saturation law asserting that the classical factor two of Vietoris–Rips stability [189] is asymptotically attained by greedy landmarking of smooth convex curves; and a two-sided Johnson–Lindenstrauss threshold [190, 191] for the simultaneous preservation of the loop ratios of orthogonal hexagonal channels under Gaussian projection, whose upper half is known in substance [192, 193] and whose lower half would be the first matching bound for a topological statistic. A statistical programme (Conjectures 311

and 312) makes the geometry of Wasserstein means of diagrams [194, 195] quantitative: an optional feature of prevalence  $q$  appears in the mean at exactly  $q$  times its conditional persistence up to exponentially small matching leakage, sharpening the shrinkage phenomenon visible in the probabilistic Fréchet means of [196] and the grouping calculus of [197], and empirical means use the intended latent grouping up to  $CNm e^{-c\gamma^2/\sigma^2}$ . A Mapper programme (Conjectures 313 and 314) builds on an exact phase calculus proved here: randomizing the phase of a uniform one-dimensional cover [198] makes the survival probability of an isolated Reeb cycle the universal piecewise-linear function  $F_\alpha(\ell/h)$  with an explicit kernel, the phase-averaged Betti curve of a chain of cycles is exactly additive with the spans recoverable from its kinks, and the dithered midpoint estimator of a branch critical value is exactly unbiased with the quantization variance  $s^2/12$  of [199]; the conjectures transfer this calculus to general tame Reeb graphs with a stable inversion, and to discretized and sampled Mapper with mesh-order corrections. A tomography programme (Conjecture 315) concerns rectangle-decomposable two-parameter modules [200], for which the fibered barcode over all lines is a complete invariant [201]: a proposition proved here sharpens completeness to a single generic direction with the line parameterization retained, the deposited three-direction claim being recorded as calibration, and the conjecture asserts the quantitative sequel, that three fixed generic directions give a locally bi-Lipschitz invariant with polynomially controlled constant and that the direction profile of the matching distance is never needle-like, so uniformly random slopes see near-maximal discrepancy; it is closely related to the local metric equivalence of Vipond [202], proved with a universal constant over all lines, and the Lipschitz foliation bounds of [203] behind deterministic approximation schemes [204, 205]. A finite-summary programme (Conjectures 316 to 319) asks how small a complete or faithful summary can be: a restricted-isometry law for random Fourier sketches [206] of the persistence-weighted Gaussian kernel [207], bi-Lipschitz to the Wasserstein metric on separated bounded-cardinality classes, which the known non-embeddability theorems [208, 209] do not forbid because they require collisions or unbounded cardinality, with restricted-class embeddings known in spirit [210]; a Whitney-count law [211] asserting that  $4m + 1$  generic evaluations of the Gaussian persistence surface, whose continuum injectivity is known [212] while grid images are not injective [213], form a complete invariant of  $m$ -point diagrams; a moment law asserting that the first  $4m$  moments of the aggregate landscape [214] determine a generic  $m$ -bar diagram, a compressed instance of the finite algebraic coordinate systems of [215, 216]; and an average-case kinetic law asserting that random affine motion of  $n$  planar points produces only  $O(n^2 \log n)$  expected changes of the degree-one persistence pairing, far below the quartic worst-case count that vineyard algorithms [217, 218, 219, 220] are built to survive. The proved layer is three calibrations. Proposition 57 proves the exact survival law with kernel  $F_\alpha(x) = \min(1, \max(0, (x - 1)/(1 - \alpha)))$ , the exact additivity and kink inversion of the phase-averaged Betti curve for chains of isolated cycles, and the exact dither law for branch critical values. Proposition 58 proves that one generic direction of the parameterized fiber field determines a generic rectangle-decomposable module, by tracking the interval pairs of the fiber barcodes through the parallel family and reading each rectangle off the envelope of its track. Proposition 59 proves the exact closed form of the aggregate-landscape moments and the one-bar case of the moment law, a single bar being determined by its first two moments. Three negative findings are retained as controls, and they shape the part. The unrestricted planar version of the extremality claim is refuted: a six-point perturbation search found planar configurations whose multiplicative loop persistence exceeds the regular hexagon's by the factor 1.052, exact to the arithmetic of the boundary-matrix reduction, so the circle hypothesis of Conjecture 308 is essential and is recorded with the refuting

instance; this failure of unrestricted extremality appears to be undocumented. The deposited three-direction tomography claim is true but not sharp, one direction sufficing by Proposition 58, and is recorded as calibration. A deposited majority-threshold claim for barycenter features, that an optional feature appears in the mean exactly when its prevalence exceeds one half, is refuted by the flat grouping algebra itself, the mean feature shrinking continuously with prevalence instead, and Conjecture 311 carries the corrected law. The deposited linear stable-inversion clause of Conjecture 313 is refuted at coalescing spans: the two-cycle span multisets  $\{\ell, \ell\}$  and  $\{\ell - \delta, \ell + \delta\}$  have bottleneck distance  $\delta$ , yet on the affine middle branch of the survival kernel the symmetric perturbation cancels, so the phase-averaged Betti curves differ only in  $O(\delta)$ -wide neighborhoods of the moving kinks and  $\|S_G - S_{G'}\|_{L^2(dh/h)} = \Theta(\delta^{3/2})$ , whence no bound  $\delta \leq C\eta$  can hold; the repaired clause requires the span values to be pairwise separated, near-collisions admitting only a Hölder inverse of exponent  $2/3$ , the constant  $\eta/\delta^{3/2} \rightarrow 6.51$  at  $\alpha = 0.3$ ,  $\ell = 0.5$  found by exact integration. Conjectures 309 to 312, 313, 314, and 316 to 319 assert asymptotic or every-instance laws; Conjecture 308 is closely related to [188], and Conjecture 315 is closely related to [202] and [203].

For a finite metric space  $X$ , the Vietoris–Rips complex at scale  $r$  has as simplices the subsets of diameter at most  $r$ ; homology is over the two-element field, computed here by boundary-matrix reduction of the full two-skeleton, which suffices in degree one. For a degree-one bar  $(b, d)$  with  $b > 0$  the multiplicative persistence is  $d/b$ , and  $\Pi_1^\times(X)$  is its maximum over bars, set to one if none exists;  $d_B$  and  $W_2$  are the bottleneck and two-Wasserstein distances with diagonal matching. For a regular  $n$ -gon inscribed in the unit circle the dominant degree-one bar has  $b_n = 2 \sin(\pi/n)$  and  $d_n = 2 \sin(\lceil n/3 \rceil \pi/n)$  [187, 186], so  $\Pi_1^\times = \sin(\lceil n/3 \rceil \pi/n) / \sin(\pi/n)$ , reproduced exactly by the present implementation for  $n \leq 12$ . A uniform one-dimensional Mapper cover has interval length  $h$ , overlap fraction  $\alpha \in (0, 1)$ , step  $s = h(1 - \alpha)$ , intervals  $[ks - U, ks - U + h]$ , and uniformly random phase  $U$  on  $[0, s)$ ; the Mapper graph is the simple nerve of the connected components of the interval preimages, and an isolated cycle of a tame Reeb graph is a pair of arcs joining two branch points whose filter values differ by the span  $\ell$  (its degree-one extended-persistence bar length), with all other critical values farther than the cover scale. A rectangle-decomposable module is a finite direct sum of interval modules supported on axis-parallel rectangles; for a positive-slope direction  $v$ , the fiber field assigns to each line parallel to  $v$ , indexed by its transverse intercept  $\tau$ , the multiset of parameter intervals cut by the rectangles, and endpoints are generic when the finitely many coincidence conditions among side coordinates and crossing values fail. The Gaussian persistence surface of a diagram  $D$  is  $\rho_D(z) = \sum_{p \in D} \text{pers}(p) \exp(-\|z - p\|^2 / 2\sigma^2)$ ; the aggregate landscape is  $A_D(t) = \sum_i \max\{0, \min(t - b_i, d_i - t)\}$  with moments  $M_r(D) = \int t^r A_D(t) dt$ ; and for points moving affinely in the plane, a vineyard event is a time at which the multiset of (birth edge, death triangle) pairs of the degree-one reduction changes.

**Proposition 57** (Phase calculus for one-dimensional Mapper). Fix  $\alpha \in (0, 1)$  and the uniform random phase. (i) For an isolated cycle of span  $\ell$ ,

$$\Pr(\beta_1(M_{h,U}) = 1) = F_\alpha(\ell/h), \quad F_\alpha(x) = \min\left(1, \max\left(0, \frac{x-1}{1-\alpha}\right)\right),$$

and  $\beta_1 = 0$  otherwise: the survival probability is a universal piecewise-linear function of  $\ell/h$ , zero for  $h \geq \ell$  and one for  $h \leq \ell/(2 - \alpha)$ . (ii) For a series chain of isolated cycles with spans  $\ell_1, \dots, \ell_r$ , the phase-averaged Betti curve satisfies  $S(h) = \mathbb{E}_U \beta_1(M_{h,U}) = \sum_j F_\alpha(\ell_j/h)$  exactly for every  $h > 0$ , and the multiset  $\{\ell_j\}$  is recovered from  $S$ : its kinks in  $h$  occur exactly at the points  $\ell_j$  and  $\ell_j/(2 - \alpha)$ , and descending induction on the largest kink identifies every span with multiplicity.

(iii) For an isolated branch critical value  $c$ , the midpoint estimator  $\hat{c}(U) = L(U) - s/2$ , with  $L(U)$  the lower endpoint of the first interval whose preimage has the post-branch component count, satisfies  $\hat{c} - c \sim \text{Unif}(0, s] - s/2$ , so  $\mathbb{E}\hat{c} = c$  exactly and  $\text{Var } \hat{c} = s^2/12$ .

**Proposition 58** (One direction determines a rectangle module). Let  $M = \bigoplus_{i=1}^m \mathbf{k}_{R_i}$  be rectangle-decomposable with generic endpoints, and fix any direction  $v$  of positive slope. Then the fiber field of  $M$  along  $v$ , with the line parameterization retained, determines the multiset  $\{R_1, \dots, R_m\}$ .

**Proposition 59** (Moment calibration for aggregate landscapes). Write bar  $i$  as center  $c_i = (b_i + d_i)/2$  and half-width  $a_i = (d_i - b_i)/2$ . (i) For every  $r \geq 0$ ,

$$M_r(D) = \sum_{i=1}^m \sum_{\substack{0 \leq j \leq r \\ j \text{ even}}} \binom{r}{j} c_i^{r-j} \frac{2a_i^{j+2}}{(j+1)(j+2)}.$$

(ii) For  $m = 1$ , the first two moments determine the bar:  $M_0 = a^2$  and  $M_1 = c a^2$ , so  $a = \sqrt{M_0}$  and  $c = M_1/M_0$ ; in particular the first  $4m$  moments of Conjecture 318 suffice at  $m = 1$  with two to spare. (iii) The second distributional derivative  $A_D'' = \sum_i (\delta_{b_i} - 2\delta_{c_i} + \delta_{d_i})$  is a signed measure with at most  $3m$  atoms whose moments obey  $\mu_0 = \mu_1 = 0$  and  $\mu_k = k(k-1)M_{k-2}$  for  $k \geq 2$ ; hence for a diagram with  $3m$  distinct nodes the first  $6m - 2$  aggregate moments  $M_0, \dots, M_{6m-3}$  determine the diagram, by Prony reconstruction of the nodes from  $\mu_2, \dots, \mu_{6m-1}$  and the pairing  $b_i + d_i = 2c_i$  of endpoints to centers by the weights  $+1$  and  $-2$ .

## 72 Extremal and sampling laws for Vietoris–Rips persistence

**Conjecture 308** (Circle isoperimetry for multiplicative persistence). *For every  $n \geq 4$  and every set  $X$  of  $n$  distinct points on the Euclidean unit circle,*

$$\Pi_1^\times(X) \leq \frac{\sin(\lceil n/3 \rceil \pi/n)}{\sin(\pi/n)},$$

*with equality exactly for the regular  $n$ -gon, up to isometries of the circle.*

**Conjecture 309** (Saturation of the stability factor two). *Let  $\Gamma \subset \mathbb{R}^2$  be a  $C^3$  strictly convex closed curve, let  $L_m$  be the first  $m$  points of a farthest-point sampling of  $\Gamma$  from a generic seed, and let  $\rho_m = d_H(L_m, \Gamma)$ . Then*

$$\lim_{m \rightarrow \infty} \frac{d_B(\text{Dgm}_1(L_m), \text{Dgm}_1(\Gamma))}{\rho_m} = 2.$$

**Conjecture 310** (A two-sided projection threshold for loop ratios). *For  $q \geq 2$  let  $H_1, \dots, H_q$  be regular hexagons in mutually orthogonal coordinate two-planes of  $\mathbb{R}^{2q}$ , let  $A$  be an  $m \times 2q$  Gaussian matrix with independent  $N(0, 1/m)$  entries, and for fixed small  $\varepsilon > 0$  let  $\mathcal{E}_{q,m,\varepsilon}$  be the event that every channel ratio  $\Pi_1^\times(AH_j)/\Pi_1^\times(H_j)$  lies within  $\varepsilon$  of one. Then there are constants  $0 < c < C$  with*

$$m \leq c\varepsilon^{-2} \log q \implies \Pr(\mathcal{E}_{q,m,\varepsilon}) \rightarrow 0, \quad m \geq C\varepsilon^{-2} \log q \implies \Pr(\mathcal{E}_{q,m,\varepsilon}) \rightarrow 1,$$

*as  $q \rightarrow \infty$ .*

## 73 The statistical geometry of diagram means

**Conjecture 311** (Prevalence shrinkage with exponential leakage). *Let random diagrams consist of a background diagram plus, with probability  $q$ , one signal point, the signal cluster separated from the background and the diagonal by a matching margin  $\gamma$ , with conditional perturbations centered subgaussian of scale  $\sigma \ll \gamma$ . Then the population  $W_2$  Fréchet mean has a unique signal point whose diagonal coordinate satisfies  $\bar{U} = \mathbb{E}[U \mid \text{present}] + O(e^{-c\gamma^2/\sigma^2})$  and whose distance to the diagonal satisfies*

$$\bar{V} = q \mathbb{E}[V \mid \text{present}] + O(e^{-c\gamma^2/\sigma^2}),$$

*uniformly over bounded separated model classes.*

**Conjecture 312** (Exponential stability of latent groupings). *Let  $D_1, \dots, D_N$  be independent diagrams from a latent model with  $m$  feature clusters, matching margin  $\gamma$  between clusters and to the diagonal, subgaussian scale  $\sigma$ , and prevalences bounded away from zero. Then*

$$\Pr\{\text{some optimal mean grouping differs from the latent grouping}\} \leq CNm e^{-c\gamma^2/\sigma^2},$$

*and on the complementary event the empirical  $W_2$  Fréchet mean is unique, its coordinates obey the flat projection formulas, and featurewise central-limit behavior holds at rate  $N^{-1/2}$ .*

## 74 A phase calculus for Mapper

**Conjecture 313** (Stable survival tomography of Reeb graphs). *Let  $(G, f)$  be a tame Reeb graph whose  $r$  degree-one extended-persistence bars have lengths (spans)  $\ell_1, \dots, \ell_r$ , each below the separation ( $\ell_j < \gamma$ ), and pairwise critical-value separation at least  $\gamma > 0$ , and let  $S_G(h) = \mathbb{E}_U \beta_1(M_{h,U})$  be the phase-averaged Betti curve at overlap  $\alpha$ . Then (a)  $S_G(h) = \sum_j F_\alpha(\ell_j/h)$  for every  $h < \gamma$ , with the kernel of Proposition 57; and (b) the inversion is stable away from the collision stratum: if two tame Reeb graphs with  $\gamma$ -separated features and distinct span values pairwise separated by at least  $\beta > 0$  have curves with  $\|S_G - S_{G'}\|_{L^2((0,\gamma), dh/h)} \leq \eta$ , then the equal-cardinality span multisets are within bottleneck distance  $C(\gamma, \alpha, r, \beta) \eta$  of each other; the span separation is essential, coalescing spans admitting only a Hölder inverse of exponent  $2/3$ .*

**Conjecture 314** (The dither law survives discretization). *For a Reeb graph discretized at filter mesh  $\eta = o(h)$ , or sampled densely enough that its Mapper is that of the underlying graph, the midpoint estimator of an isolated branch critical value satisfies*

$$\mathbb{E}\hat{c} - c = O(\eta), \quad \text{Var}(\hat{c}) = \frac{s^2}{12} + O(s\eta + \eta^2),$$

*uniformly over admissible discretizations, with constants depending only on the local branch geometry.*

## 75 Rectangle tomography

**Conjecture 315** (Three-direction stability and random-slope observability). *Fix a box, a maximum count  $m$ , and  $\delta > 0$ , and consider rectangle-decomposable modules whose corner coordinates, side lengths, and pairwise separations are at least  $\delta$ , and whose corners stay at distance at least  $\delta$  from the coincidence loci of  $v_1, v_2, v_3$ , the configurations at which two rectangles*

cut a common fiber-interval endpoint along some  $v_j$ . (a) For generic fixed positive-slope directions  $v_1, v_2, v_3$ , the invariant  $d_3(M, N) = \max_j \sup_{L \parallel v_j} d_B(\text{Dgm}(M_L), \text{Dgm}(N_L))$  is locally bi-Lipschitz equivalent to the minimal corner displacement  $d_{\text{rect}}$ , with constants polynomial in  $m$  and  $1/\delta$ . (b) The weighted direction profile  $g(\theta)$  of the matching distance between two such modules is Lipschitz relative to its own maximum on compact slope ranges,  $\text{Lip } g \leq C \|g\|_\infty$  with  $C$  polynomial in  $m$ , the box size,  $1/\delta$ , and the slope margin, so its near-maximal set has measure bounded below and  $K$  uniformly random slopes recover a  $(1 - \varepsilon)$ -approximation of the matching distance with failure probability exponentially small in  $K$ .

## 76 Finite summaries and dynamic complexity

**Conjecture 316** (A restricted isometry for persistence sketches). *On the class of diagrams with at most  $m$  points, persistence and pairwise separation at least  $\delta$ , in a box of diameter  $R$ , let  $\Phi_M$  be the random Fourier sketch of the persistence-weighted Gaussian embedding at bandwidth comparable to  $\delta^{-1}$  with  $M$  frequencies. Then there are class constants  $0 < c_0 < C_0$ , polynomial in  $m, R, 1/\delta$ , such that with probability  $1 - \eta$ ,*

$$c_0(1 - \varepsilon) W_2(D, E) \leq \|\Phi_M(D) - \Phi_M(E)\|_2 \leq C_0(1 + \varepsilon) W_2(D, E)$$

for all  $D, E$  in the class, provided  $M \geq C\varepsilon^{-2}m \log(mR/(\delta\eta))$ .

**Conjecture 317** (Whitney-count evaluations of persistence surfaces). *Fix  $\sigma > 0$  and the class of diagrams with exactly  $m$  points, pairwise  $\delta$ -separated and  $\delta$ -far from the diagonal, in a compact box. Choose  $q \geq 4m + 1$  evaluation points independently from any distribution with positive density on a neighborhood of the box. Then, almost surely in the evaluations, the map  $D \mapsto (\rho_D(z_1), \dots, \rho_D(z_q))$  is injective on the class, and locally bi-Lipschitz off the discriminant.*

**Conjecture 318** (Aggregate-landscape moment determinacy). *For a Zariski-open, full-measure set of  $m$ -bar diagrams, the first  $4m$  moments  $M_0, \dots, M_{4m-1}$  of the aggregate landscape determine the diagram uniquely up to bar permutation.*

**Conjecture 319** (Near-quadratic vineyard complexity in the mean). *Let  $n$  planar points move affinely,  $x_i(t) = a_i + tv_i$  on  $[0, 1]$ , with  $a_i, v_i$  independent standard Gaussians, and let  $N_1(n)$  be the number of times the degree-one Vietoris–Rips persistence pairing changes. Then  $\mathbb{E}N_1(n) = O(n^2 \log n)$ , and  $N_1(n) = O(n^2 \log^2 n)$  with probability  $1 - n^{-c}$ .*

## Part XIX

# Temporal irreversibility and spectral selection in nonequilibrium dynamics

*Added to the deposit: 22 August 2026.*

The fifteen conjectures below study how much of the thermodynamic arrow of time survives when a stationary nonequilibrium process is observed only through two time points. For an irreducible continuous-time Markov chain the complete trajectory over an interval of length  $t$  carries a path-space entropy production  $\sigma t$ ; reducing the trajectory to its endpoints keeps only



the *lag irreversibility*  $L(t)$ , the Kullback–Leibler divergence between the stationary two-time law and its time reverse, and data processing gives  $0 \leq L(t) \leq \sigma t$ . The organizing object is the *temporal sampling efficiency*  $\eta(t) = L(t)/(\sigma t) \in [0, 1]$ , the fraction of dissipation still visible at lag  $t$ . The conjectures form three programmes. A temporal-and-spectral programme (Conjectures 320 to 326) asserts monotone loss of efficiency, spectral bounds on the temporal support and the integrated budget of the arrow of time, an irreversibility-specific relaxation exponent that doubles the slowest dissipative mode, its generic coincidence with the ordinary spectral gap together with an exceptional invisible-slow-mode phenomenon, and the discrete mode selection performed by deterministic coarse-graining. A unicyclic-and-winding programme (Conjectures 327 to 331) sharpens these on a single driven cycle, where endpoint sampling loses exactly the integer winding number, giving a strict efficiency law, an amplitude cap, a weak-affinity wrapped-Brownian scaling limit, a ballistic winding-aliasing horizon, and a single Péclet crossover unifying the two. A Gaussian programme (Conjectures 332 to 334) transfers the peak-horizon, integrated-budget, and mode-doubling laws to multivariate Ornstein–Uhlenbeck processes, where every quantity is an explicit function of the drift and diffusion matrices. The proved layer is Proposition 60: the endpoint data-processing bounds, the short-time expansion  $L(t) = \sigma t + O(t^2)$  giving  $\eta(t) \rightarrow 1$ , and the exact Gaussian lag-KL formula. Conjectures 323, 325, 326, and 334 assert long-time exponential laws; Conjectures 329 to 331 are scaling-limit statements; Conjecture 324 is a genericity statement carrying its own exceptional set. The central observable is the lag irreversibility of Merino Negrete, Maldonado, and Salgado-García [221], and the closest spectral precedent is the finite-time cross-correlation asymmetry of Van Vu, Vo, and Saito [223]; the normalization by path entropy production, the temporal horizons and integrated budgets, the doubled irreversibility exponent and its coarse-grained selection, and the winding scaling laws are what is proposed as new. Since deposit, Conjectures 323, 325, and 334, the discrete and Gaussian irreversibility-doubling and coarse-grained mode-selection laws, have been resolved true by Propositions 61 and 68 from the quadratic equivalence of lag KL with the antisymmetric endpoint flux, and Conjecture 329, the wrapped-Brownian limit, by Proposition 65. Since deposit, Conjectures 320 and 327, the efficiency-monotonicity and unicyclic strict-monotonicity laws, have been resolved false by Proposition 63: a homogeneous ten-state ring has non-monotone efficiency, reviving by more than two orders of magnitude across an interior lag interval through winding aliasing. Since deposit, Conjecture 326, the hidden-mode exponential-separation law, has been resolved false by Proposition 62: a product of two driven three-cycles has an oscillatory dominant mode with exact zeros, so the coarse-to-full ratio returns to one at infinitely many times and the  $\liminf$  separation exponent is zero. Since deposit, Conjecture 331, the universal diffusive-to-ballistic Péclet crossover, has been resolved false as stated by Proposition 66: the ballistic first-crossing coefficient depends on the edge affinity, so no single Péclet function recovers Conjecture 330 for every fixed affinity, though a one-variable crossover survives in the weak-edge-bias regime. Since deposit, Conjecture 328, the sharp unicyclic amplitude cap  $\sup_t \gamma L(t)/\sigma \leq \frac{1}{2}$ , has been resolved false by Proposition 64: an explicit near-decomposable seven-state ring reaches  $0.643 > \frac{1}{2}$ , so the constant  $\frac{1}{2}$  is not universal. Since deposit, Conjectures 332 and 333, the Gaussian peak-horizon and spectral-area bounds, have likewise been resolved false by Proposition 67: a five-dimensional Jordan-block Ornstein–Uhlenbeck drift with  $g_A = 1$  has its lag-KL peak at  $t_* \approx 1.069$  and normalized area  $\approx 1.266$ , so the spectral abscissa alone bounds neither quantity for a strongly nonnormal drift.

Let  $X_t$  be an irreducible continuous-time Markov chain on  $\{1, \dots, n\}$  with row generator  $Q = (q_{ij})$ , stationary law  $\pi > 0$ , and bidirected support ( $q_{ij} > 0 \Leftrightarrow q_{ji} > 0$ ), so the steady-state

entropy production is finite. Write  $P_t = e^{tQ}$ , the stationary endpoint law  $F_t(i, j) = \pi_i P_t(i, j)$ , its time reverse  $F_t^\top$ , the lag irreversibility

$$L(t) = D_{\text{KL}}(F_t \| F_t^\top) = \sum_{i,j} F_t(i, j) \log \frac{F_t(i, j)}{F_t(j, i)},$$

and the entropy-production rate  $\sigma = \sum_{i \neq j} \pi_i q_{ij} \log(\pi_i q_{ij} / \pi_j q_{ji})$ , so that  $\eta(t) = L(t) / (\sigma t)$ . Let  $q_{ij}^* = \pi_j q_{ji} / \pi_i$  be the reversed generator,  $\bar{Q} = (Q + Q^*) / 2$  the additive reversibilization,  $\gamma = \lambda_2(-\bar{Q}) > 0$  its reversible  $L^2(\pi)$  spectral gap, and  $g = \min\{-\Re \lambda : \lambda \in \text{spec}(Q) \setminus \{0\}\}$  the ordinary spectral-abscissa gap. Write  $A_t = F_t - F_t^\top = \Pi e^{tQ} - e^{tQ^\top} \Pi$  with  $\Pi = \text{diag}(\pi)$ , and let  $\alpha_{\text{irr}}$  be the smallest  $-\Re \lambda > 0$  over generator eigenmodes whose coefficient in  $A_t$  is nonzero (on generalized eigenspaces for Jordan blocks, the polynomial factors not affecting the rate). For a deterministic map  $f : \{1, \dots, n\} \rightarrow \{1, \dots, m\}$  let  $F_t^f(a, b) = \sum_{f(i)=a, f(j)=b} F_t(i, j)$ ,  $L_f(t) = D_{\text{KL}}(F_t^f \| (F_t^f)^\top)$ , and  $\alpha_{\text{irr}}(f)$  the smallest such rate surviving the block summation. For the homogeneous  $N$ -cycle with clockwise/counterclockwise rates  $p_N = e^{A/(2N)}$ ,  $q_N = e^{-A/(2N)}$  and time  $t_N = N^2 \tau / (p_N + q_N)$ , the wrapped Gaussian on  $x \in [0, 1)$  is  $w_{A, \tau}(x) = \sum_{k \in \mathbb{Z}} (2\pi\tau)^{-1/2} \exp[-(x + k - A\tau/2)^2 / (2\tau)]$ , with reverse  $w_{-A, \tau}$ ; at fixed edge affinity  $a \neq 0$  write  $p = e^{a/2}$ ,  $q = e^{-a/2}$ ,  $v = p - q$ , the Péclet number  $\text{Pe} = N(p - q) / (p + q)$ , and  $t_c(N, \cdot) = \inf\{t : \eta_N(t) \leq c\}$ . For the Ornstein–Uhlenbeck process  $dX_t = AX_t dt + \sqrt{2D} dW_t$  with  $A$  Hurwitz and  $D \succ 0$ , the stationary covariance solves  $AC + CA^\top + 2D = 0$ , the two-time law is Gaussian with cross-covariance  $Ce^{A^\top t}$ ,  $L_{\text{OU}}(t)$  is the KL divergence to its block-swapped reverse,  $\sigma_{\text{OU}} = \text{tr}[(A + DC^{-1})^\top D^{-1}(A + DC^{-1})C]$ ,  $g_A = \min_\lambda(-\Re \lambda(A))$ ,  $H(t) = e^{At}C - Ce^{A^\top t}$ , and  $\alpha_H$  the slowest exponential rate in  $H(t)$ .

**Proposition 60** (Endpoint bounds, short-time efficiency, and the Gaussian lag KL). For every chain in the stated class,  $0 \leq L(t) \leq \sigma t$ , and as  $t \downarrow 0$  single-jump paths dominate, giving  $L(t) = \sigma t + O(t^2)$  and  $\eta(t) \rightarrow 1$ ; as  $t \rightarrow \infty$ ,  $F_t \rightarrow \pi \otimes \pi$  and  $L(t) \rightarrow 0$ . For the Ornstein–Uhlenbeck process the two-time covariance and its block swap have equal determinants, so

$$L_{\text{OU}}(t) = \frac{1}{2} [\text{tr}(\Sigma_t^{R-1} \Sigma_t) - 2d].$$

## 77 Temporal irreversibility and spectral selection

**Conjecture 320** (Temporal sampling-efficiency monotonicity, resolved false). *For every finite irreducible bidirected chain with  $\sigma > 0$ , the efficiency  $\eta(t) = L(t) / (\sigma t)$  is non-increasing on  $(0, \infty)$ , and strictly decreasing for a generic nonequilibrium chain.*

*Resolution.* The conjecture is false: on the homogeneous 10-cycle with edge affinity 1 the efficiency  $\eta$  falls to  $9.3 \times 10^{-6}$  near half a winding and then rises above  $1.2 \times 10^{-3}$ , verified at eighty digits (Proposition 63), so it is not monotone.

**Conjecture 321** (Two-gap-time location bound). *If  $t_*$  is any global maximizer of  $L(t)$ , then  $\gamma t_* \leq 2$ .*

**Conjecture 322** (Spectral integrated-budget bound). *For every chain in the class,  $\int_0^\infty L(t) dt \leq \frac{2\sigma}{\gamma^2}$ .*

**Conjecture 323** (Irreversibility-gap doubling law, resolved true). *The long-time envelope of lag  $KL$  decays at twice the slowest irreversible rate:*

$$\liminf_{t \rightarrow \infty} -\frac{1}{2t} \log L(t) = \alpha_{\text{irr}},$$

and generically  $L(t) = t^m e^{-2\alpha_{\text{irr}} t} [c(t) + o(1)]$  with  $m \geq 0$  a Jordan multiplicity and  $c$  bounded, nonnegative, and not identically zero.

**Conjecture 324** (Generic equality of spectral and irreversibility gaps). *Fix a strongly connected bidirected graph and vary its positive rates. Outside a measure-zero analytic exceptional set, and excluding detailed balance,  $\alpha_{\text{irr}} = g$ ; strict inequality  $\alpha_{\text{irr}} > g$  occurs exactly when every mode at real-part gap  $g$  contributes zero to  $A_t$ .*

**Conjecture 325** (Coarse-grained irreversibility mode-selection law, resolved true). *For every deterministic partition with  $L_f \not\equiv 0$ ,*

$$\liminf_{t \rightarrow \infty} -\frac{1}{2t} \log L_f(t) = \alpha_{\text{irr}}(f).$$

*Resolution.* Now proved, Proposition 61: the exponent identity holds unconditionally, and the refined law holds with power  $m = 2r$  (twice the Jordan-polynomial degree) without genericity.

**Proposition 61** (Irreversibility-gap doubling and coarse-grained mode selection). Let  $Q$  be an irreducible bidirected generator with stationary  $\Pi$ , endpoint law  $F_t = \Pi e^{tQ}$ , and antisymmetric part  $A_t = F_t - F_t^\top \not\equiv 0$ . For all large  $t$  there are  $0 < c < C$  with  $c\|A_t\|_F^2 \leq L(t) \leq C\|A_t\|_F^2$ ;  $A_t$  is a finite exponential polynomial; and with  $\alpha_{\text{irr}} = \min\{-\Re \lambda : \text{surviving in } A_t\}$  and  $r$  the largest polynomial degree among surviving terms of real part  $-\alpha_{\text{irr}}$ ,  $\liminf_{t \rightarrow \infty} -\frac{1}{2t} \log L(t) = \alpha_{\text{irr}}$  and  $L(t) = t^{2r} e^{-2\alpha_{\text{irr}} t} [c(t) + o(1)]$  with  $c$  a nonnegative bounded trigonometric polynomial, not identically zero. The same holds with  $L_f$ ,  $\alpha_{\text{irr}}(f)$  for every deterministic partition with  $L_f \not\equiv 0$ . This proves Conjectures 323 and 325.

*Proof.* Grouping the two orderings,  $L(t) = \sum_{i < j} (F_{ij} - F_{ji}) \log(F_{ij}/F_{ji}) = \sum_{i < j} (F_{ij} - F_{ji})^2 / \xi_{ij}$  with  $\xi_{ij}$  between  $F_{ij}$  and  $F_{ji}$ ; since  $F_{ij}, F_{ji} \rightarrow \pi_i \pi_j > 0$  the bound  $c\|A_t\|_F^2 \leq L \leq C\|A_t\|_F^2$  follows. The Jordan expansion makes  $A_t$  an exponential polynomial; dividing by  $t^r e^{-\alpha_{\text{irr}} t}$  leaves a nonzero matrix trigonometric polynomial plus  $o(1)$ , giving  $\liminf(-t^{-1} \log \|A_t\|_F) = \alpha_{\text{irr}}$  and the squared leading term. Block summation is a linear map with limits  $\pi_A \pi_B > 0$ , so the same argument gives  $\alpha_{\text{irr}}(f)$ .  $\square$

**Conjecture 326** (Hidden-mode exponential separation, resolved false). *If  $\alpha_{\text{irr}}(f) > \alpha_{\text{irr}}$ , then  $\liminf_{t \rightarrow \infty} -\frac{1}{2t} \log[L_f(t)/L(t)] = \alpha_{\text{irr}}(f) - \alpha_{\text{irr}}$ ; if the two exponents coincide with nonzero leading coefficients,  $L_f(t)/L(t)$  approaches a nonzero bounded envelope.*

*Resolution.* The conjecture is false, by Proposition 62: a product of two driven three-cycles has coarse-to-full ratio returning to one at infinitely many times, so the liminf separation exponent is zero.

**Proposition 62** (Failure of hidden-mode exponential separation). There is a product of two independent homogeneous driven three-cycles, a slow factor with  $(p_s, q_s) = (1, 0.01)$  and a fast factor with  $p_f + q_f = \frac{16}{15}$  and  $p_f - q_f = 0.99\sqrt{11/10}$ , for which the coarse (fast-factor) exponent  $\alpha_{\text{irr}}(f) = 1.6$  exceeds the full exponent  $\alpha_{\text{irr}} = 1.515$ , yet  $\liminf_{t \rightarrow \infty} -\frac{1}{2t} \log[L_f(t)/L(t)] = 0$ . Hence Conjecture 326 is false.

*Proof.* For a homogeneous driven three-cycle the nonzero generator eigenvalues are  $-\alpha \pm i\omega$  with  $\alpha = \frac{3}{2}(p+q)$  and  $\omega = \frac{\sqrt{3}}{2}(p-q)$ , so the antisymmetric endpoint flux is proportional to  $e^{-\alpha t} \sin(\omega t)$  and  $L(t) \asymp e^{-2\alpha t} \sin^2(\omega t)$  vanishes exactly at  $t = n\pi/\omega$ . For the product chain independence gives  $L = L_s + L_f$ , and observing only the fast factor gives coarse irreversibility  $L_f$ . At  $t_n = n\pi/\omega_s$  the slow flux vanishes,  $L_s(t_n) = 0$ , while  $L_f(t_n) > 0$  because  $\omega_f/\omega_s = \sqrt{11/10}$  is irrational, so  $L_f(t_n)/L(t_n) = 1$  and  $-\frac{1}{2t_n} \log[L_f/L] = 0$ . Since  $t_n \rightarrow \infty$  the liminf is zero, not the gap 0.085; the values are reproduced in the verification script.  $\square$

## 78 Unicyclic networks and winding aliasing

**Conjecture 327** (Unicyclic strict monotonicity, resolved false). *For every finite bidirected unicyclic chain with nonzero total cycle affinity,  $\eta'(t) < 0$  for all  $t > 0$ .*

*Resolution.* The conjecture is false on the homogeneous ring itself, by Proposition 63.

**Proposition 63** (A homogeneous-ring counterexample to efficiency monotonicity). On the homogeneous continuous-time  $N$ -cycle with clockwise rate  $p = e^{a/2}$ , counterclockwise rate  $q = e^{-a/2}$ , uniform stationary law, and  $\sigma = (p-q)a$ , the efficiency  $\eta(t) = L(t)/(\sigma t)$  is not monotone. For  $N = 10$  and  $a = 1$ , writing  $u = (p-q)t/N$ , an eighty-digit Fourier diagonalization of the circulant generator gives  $\eta = 9.31 \times 10^{-6}$  at  $u = 0.54$  and  $1.22 \times 10^{-3}$  at  $u = 0.68$ , an increase by a factor above 130. Hence Conjectures 320 and 327 both fail.

*Proof.* The endpoint law is circulant,  $P_t(k) = \frac{1}{N} \sum_m e^{\lambda_m t} \omega^{-mk}$  with  $\omega = e^{2\pi i/N}$  and  $\lambda_m = p(\omega^m - 1) + q(\omega^{-m} - 1)$ , and  $L(t) = \sum_k P_t(k) \log(P_t(k)/P_t(-k))$ ; the values follow at eighty digits and are reproduced in the verification script. Near half a winding the forward and reverse wrapped endpoint means nearly coincide, collapsing  $\eta$ , then separate before diffusion mixes the cycle, so  $\eta$  revives; the first-crossing horizons of Conjectures 330 and 331 are unaffected.  $\square$

**Conjecture 328** (Sharp unicyclic spectral-amplitude cap, resolved false). *For every nonequilibrium unicyclic chain,  $\sup_{t>0} \frac{\gamma L(t)}{\sigma} \leq \frac{1}{2}$ .*

*Resolution.* The conjecture is false: Proposition 64 exhibits an explicit near-decomposable seven-state ring with  $\gamma L/\sigma \geq 0.643 > \frac{1}{2}$ , verified to fifty digits, so no universal cap at  $\frac{1}{2}$  holds; the correct sharp constant, if finite, remains open, and a near-decomposable regime is where it is approached.

**Proposition 64** (A near-decomposable amplitude-cap counterexample). There is a bidirected seven-state ring, with one weak edge three orders of magnitude below the others, whose reversible gap  $\gamma$ , entropy-production rate  $\sigma$ , and maximal lag irreversibility satisfy  $\gamma \sup_t L(t)/\sigma = 0.643$  to three places. Hence the sharp cap  $\frac{1}{2}$  of Conjecture 328 fails.

*Proof.* The rates are recorded in the verification script; the reversible gap and entropy-production rate are computed from their definitions and the lag irreversibility from the exact endpoint law  $F_t = \Pi e^{tQ}$  on a fine grid, with an independent high-precision recomputation of the stationary law, the spectral gap, the entropy-production rate, and  $\sup_t L(t)$  agreeing with the double-precision value to fifty digits. The resulting ratio is  $0.643 > \frac{1}{2}$ . The gap  $\gamma \approx 0.89$  is of order one, so the excess is not a conditioning artifact of a vanishing denominator.  $\square$

**Conjecture 329** (Wrapped-Brownian irreversibility limit, resolved true). *For each fixed  $A \neq 0$  and  $\tau > 0$ ,*

$$\eta_N(t_N) \longrightarrow \mathcal{E}(A, \tau) := \frac{2}{A^2 \tau} \int_0^1 w_{A, \tau}(x) \log \frac{w_{A, \tau}(x)}{w_{-A, \tau}(x)} dx,$$

*locally uniformly on compacta of  $(A, \tau) \in (\mathbb{R} \setminus \{0\}) \times (0, \infty)$ .*

*Resolution.* Now proved, Proposition 65.

**Proposition 65** (Wrapped-Brownian irreversibility limit). *On the homogeneous  $N$ -cycle with  $p_N = e^{A/(2N)}$ ,  $q_N = e^{-A/(2N)}$ ,  $t_N = N^2 \tau / (p_N + q_N)$ ,  $\eta_N(t_N) \rightarrow \mathcal{E}(A, \tau)$  locally uniformly on compacta of  $(A, \tau) \in (\mathbb{R} \setminus \{0\}) \times (0, \infty)$ . This proves Conjecture 329.*

*Proof.* The mode- $m$  Fourier coefficient  $\exp(t_N[p_N(\omega^m - 1) + q_N(\omega^{-m} - 1)]) \rightarrow e^{-2\pi^2 m^2 \tau + i\pi m A \tau}$ , so the wrapped lattice density converges uniformly to the wrapped Gaussian  $w_{A, \tau}$ , which is bounded below; continuity of the KL integrand and  $\sigma_N t_N \rightarrow A^2 \tau / 2$  give the limit, uniformly on compacta.  $\square$

**Conjecture 330** (Ballistic winding-aliasing horizon). *For fixed edge affinity  $a \neq 0$  and level  $c \in (0, 1)$  there is a finite positive  $u_c(a)$  with  $|v| t_c(N, a) / N \rightarrow u_c(a)$  as  $N \rightarrow \infty$ , so fixed-bias endpoint irreversibility is lost on the ballistic winding timescale  $N/|v|$ , not the diffusive mixing timescale.*

**Conjecture 331** (Diffusive-to-ballistic irreversibility crossover, resolved false). *For every fixed  $c \in (0, 1)$  there is a universal function  $\Phi_c$  with  $t_c = \frac{N^2}{p+q} \Phi_c(\text{Pe})[1+o(1)]$  in the joint large- $N$  scaling, where  $\Phi_c(z) \rightarrow \Phi_c(0) \in (0, \infty)$  as  $z \rightarrow 0$  and  $\Phi_c(z) \sim u_c/|z|$  as  $|z| \rightarrow \infty$ , so Conjectures 329 and 330 are its weak- and strong-drift limits.*

*Resolution.* The conjecture is false as stated, by Proposition 66: the ballistic first-crossing coefficient  $u_c(a)$  of Conjecture 330 depends on the edge affinity, so no single universal  $\Phi_c(\text{Pe})$  has Conjecture 330 as its strong-drift limit for every fixed affinity; a one-variable Péclet crossover survives in the weak-edge-bias regime  $a_N \rightarrow 0$ .

**Proposition 66** (Affinity-dependence of the ballistic crossover coefficient). *The ballistic first-crossing coefficient  $u_c(a)$  of Conjecture 330 depends nontrivially on the edge affinity  $a$ , so no single universal  $\Phi_c(\text{Pe})$  with  $\Phi_c(z) \sim u_c/|z|$  recovers Conjecture 330 for every fixed affinity. With the Skellam rate function  $I_{u,a}(d) = \frac{u}{v} J_{p,q}(dv/u)$ ,  $p = e^{a/2}$ ,  $q = e^{-a/2}$ ,  $v = p - q$ , the level- $\frac{1}{2}$  coefficient solving  $\min\{1, I_{u,a}(1-u)/(au)\} = \frac{1}{2}$  equals 0.2923, 0.2894, 0.2806, 0.2576, 0.2354 at  $a = 0.2, 0.5, 1, 2, 3$ , with  $u_{1/2}(a) \rightarrow \frac{1}{2(1+\sqrt{1/2})} = 0.2929$  as  $a \downarrow 0$  and  $u_{1/2}(a) \rightarrow 0$  as  $a \rightarrow \infty$ . Hence Conjecture 331 is false as a universal crossover.*

*Proof.* Since  $\text{Pe} \asymp |v|N/(p+q)$ , fixing  $a$  and letting  $N \rightarrow \infty$  drives  $\text{Pe} \rightarrow \infty$ , and matching  $t_c = \frac{N^2}{p+q} \Phi_c(\text{Pe})$  to the ballistic law  $t_c \sim u_c(a)N/|v|$  forces the tail constant  $u_c$  to equal  $u_c(a)$  for every  $a$ . The fluctuation relation gives  $I_{u,a}(-u) = au$ ; as  $a \downarrow 0$  the Skellam rate has the Gaussian form  $I_{u,a}(1-u) = a(1-2u)^2/(4u) + o(a)$ , giving  $u_c(a) \rightarrow 1/(2(1+\sqrt{c}))$ , while as  $a \rightarrow \infty$  backward jumps vanish and  $I_{u,a}(1-u) \rightarrow (1-u) \log \frac{1-u}{u} - 1 + 2u$  against the linear denominator  $au$ , forcing  $u_c(a) \rightarrow 0$ . The values are reproduced in the verification script.  $\square$

## 79 Nonequilibrium Ornstein–Uhlenbeck dynamics

**Conjecture 332** (OU one-relaxation-time peak horizon, resolved false). *If  $\sigma_{\text{OU}} > 0$  and  $t_*$  is a global maximizer of  $L_{\text{OU}}(t)$ , then  $g_A t_* \leq 1$ .*

*Resolution.* The bound is false: Proposition 67 exhibits a five-dimensional nonnormal drift, a single Jordan block with every eigenvalue  $-1$  (so  $g_A = 1$ ) and isotropic diffusion, whose lag-KL peak lies at  $t_* = 1.069\dots$ , giving  $g_A t_* = 1.069 > 1$ ; the earlier stress test sampled only drifts with negative-definite symmetric part and so missed the strong nonnormal transient amplification that pushes the horizon past one drift-relaxation time, and a repaired bound must control nonnormality (a numerical-abscissa or hypocoercive constant), not the spectral abscissa alone.

**Conjecture 333** (OU spectral-area bound, resolved false). *For every stable nonequilibrium OU process,  $\int_0^\infty L_{\text{OU}}(t) dt \leq \frac{\sigma_{\text{OU}}}{g_A^2}$ .*

*Resolution.* The bound is false: for the same five-dimensional Jordan-block drift of Proposition 67,  $g_A = 1$  and  $\sigma_{\text{OU}} = 32$  while  $\int_0^\infty L_{\text{OU}} dt = 40.50\dots$ , so the normalized area is  $1.266 > 1$ ; as with Conjecture 332, only a nonnormality-corrected budget can hold.

**Proposition 67** (A nonnormal Ornstein–Uhlenbeck counterexample to the peak-horizon and spectral-area bounds). Let  $A = -I_5 + 2N$  with  $N$  the  $5 \times 5$  nilpotent first-superdiagonal shift and  $D = I_5$ , so  $g_A = 1$ . The stationary covariance solving  $AC + CA^\top + 2D = 0$  has integer entries,  $\sigma_{\text{OU}} = 32$ , and the Gaussian lag KL  $L_{\text{OU}}$  attains its maximum at  $t_* = 1.0693\dots$  with  $g_A t_* > 1$ , and  $\int_0^\infty L_{\text{OU}} dt = 40.497\dots$ , so  $g_A^2 \int L_{\text{OU}} dt / \sigma_{\text{OU}} = 1.2655\dots > 1$ . Hence Conjectures 332 and 333 both fail.

*Proof.* The covariance  $C$  is obtained by solving the Lyapunov equation; substitution confirms  $AC + CA^\top + 2I_5 = 0$  with zero residual, and  $\sigma_{\text{OU}} = \text{tr}[(A + DC^{-1})^\top D^{-1}(A + DC^{-1})C] = 32$ . Using the Jordan-polynomial matrix exponential  $e^{At}$  and the trace formula of Proposition 60,  $L_{\text{OU}}$  rises from  $L_{\text{OU}}(t) = \sigma_{\text{OU}}t + O(t^2)$ , is still increasing at  $t = 1$  ( $L'_{\text{OU}}(1) \approx 1.60 > 0$ ), peaks at  $t_* = 1.0693\dots$ , and has integral  $40.497\dots$ ; all quantities are reproduced in the verification script, and the margins (6.9% and 26.5%) far exceed any conditioning artifact.  $\square$

**Conjecture 334** (OU antisymmetric-covariance doubling law, resolved true). *For every stable OU process with  $\sigma_{\text{OU}} > 0$ ,*

$$\liminf_{t \rightarrow \infty} -\frac{1}{2t} \log L_{\text{OU}}(t) = \alpha_H,$$

*so the long-time Gaussian arrow of time decays at twice the slowest time-antisymmetric covariance mode.*

*Resolution.* Now proved, Proposition 68.

**Proposition 68** (Gaussian antisymmetric-covariance doubling). For a stable Ornstein–Uhlenbeck process with stationary covariance  $C \succ 0$  and  $\sigma_{\text{OU}} > 0$ , the two-time covariance tends to  $C \oplus C \succ 0$  with time-antisymmetric part  $H(t) = e^{At}C - Ce^{A^\top t} \not\equiv 0$ , and  $c\|H(t)\|_F^2 \leq L_{\text{OU}}(t) \leq C'\|H(t)\|_F^2$  for large  $t$ ; since  $H$  is an exponential polynomial with slowest rate  $\alpha_H$ ,  $\liminf_{t \rightarrow \infty} -\frac{1}{2t} \log L_{\text{OU}}(t) = \alpha_H$ . This proves Conjecture 334.

*Proof.* Gaussian KL between the two-time law and its reversal vanishes to second order at the positive-definite limit  $C \oplus C$ , so it is bounded above and below by  $\|H(t)\|_F^2$ ;  $H$  is an exponential polynomial and the  $\liminf$  argument of Proposition 61 applies.  $\square$



## Part XX

# Log-concavity laws for exactly solved random conformal geometry

*Added to the deposit: 22 August 2026.*

The two conjectures below concern the shape of two exactly solved objects in random conformal geometry: the DOZZ three-point structure constant of Liouville conformal field theory, and the modulus law of the Brownian annulus. Each asserts strict log-concavity of an exact special-function density along its natural spectral or scale coordinate. Both objects are given by closed forms—the Barnes-function representation of the DOZZ constant [229] and the exact modulus transform of the Brownian annulus [230]—so both conjectures are exactly and finitely checkable on any grid of parameters. Since deposit, Conjecture 336, the annulus modular log-concavity, has been resolved false by Proposition 70: at large boundary-length ratio the logarithmic-modulus curvature turns positive, so log-concavity fails in the tail, while the DOZZ law of Conjecture 335 stands.

For Liouville conformal field theory fix  $b > 0$ , write  $Q = b + b^{-1}$ ,  $\gamma(y) = \Gamma(y)/\Gamma(1 - y)$ , and let  $\mu > 0$  be the cosmological constant. The Barnes function  $\Upsilon_b$  is defined for  $0 < \Re x < Q$  by

$$\log \Upsilon_b(x) = \int_0^\infty \frac{dt}{t} \left[ \left( \frac{Q}{2} - x \right)^2 e^{-t} - \frac{\sinh^2((\frac{Q}{2} - x)t/2)}{\sinh(bt/2) \sinh(t/2b)} \right],$$

and extends to an entire function. The DOZZ structure constant for real charges  $\alpha_1, \alpha_2, \alpha_3$  with  $\bar{\alpha} = \alpha_1 + \alpha_2 + \alpha_3$  is

$$C_b(\alpha_1, \alpha_2, \alpha_3) = [\pi \mu \gamma(b^2) b^{2-2b^2}]^{(Q-\bar{\alpha})/b} \frac{\Upsilon'_b(0) \prod_{i=1}^3 \Upsilon_b(2\alpha_i)}{\Upsilon_b(\bar{\alpha} - Q) \prod_{i=1}^3 \Upsilon_b(\bar{\alpha} - 2\alpha_i)},$$

and the principal series is parameterized by internal momentum  $P \geq 0$  through  $\alpha = \frac{Q}{2} + iP$ . For an equal external charge  $a$  write  $M_a(P) = |C_b(a, a, \frac{Q}{2} + iP)|$ . For the Brownian annulus, following Ang, Remy, and Sun [230], let  $\eta$  be the Dedekind eta function, and conditional on the boundary-length ratio  $x > 0$  let  $X_\tau$  be the positive random variable with Mellin transform

$$\mathbb{E}[X_\tau^{it}] = \frac{2\pi t}{3 \sinh(2\pi t/3)} e^{-2\pi \tau t^2/3}, \quad t \in \mathbb{R},$$

and density  $\rho_\tau$ ; the modulus  $\tau > 0$  then has density  $p_x(\tau) \propto \eta(2i\tau) \rho_\tau(x)$ , the constant of proportionality depending only on  $x$ .

**Proposition 69** (Curvature reductions and boundary symmetry).  $\Upsilon_b$  is entire with simple zeros exactly at  $-b\mathbb{N}_0 - b^{-1}\mathbb{N}_0$  and at  $Q + b\mathbb{N}_0 + b^{-1}\mathbb{N}_0$ ; in particular  $\Upsilon_b(Q) = 0$ , so  $M_a(0) = 0$  and the log-concavity of  $M_a$  is a statement on  $P > 0$ . The prefactor modulus and the momentum-independent factor  $|\Upsilon_b(2a)|$  are constant in  $P$ , so

$$\frac{d^2}{dP^2} \log M_a(P) = \frac{d^2}{dP^2} \left[ \log |\Upsilon_b(Q + 2iP)| - 2 \log |\Upsilon_b(\frac{Q}{2} + iP)| - 2 \log |\Upsilon_b(2a - \frac{Q}{2} + iP)| \right].$$

For the Brownian annulus,  $\log p_x(e^s) = \log \eta(2ie^s) + \log \rho_{e^s}(x) + \kappa(x)$  with  $\kappa$  independent of  $s$ , so  $d^2 \log p_x(e^s)/ds^2$  is independent of the normalization, and  $p_{1/x}(\tau) = x^2 p_x(\tau)$ , so that curvature is invariant under  $x \mapsto 1/x$ .

## 80 Spectral and modular log-concavity

**Conjecture 335** (DOZZ spectral log-concavity). *For every  $b > 0$  and every real external charge  $a$  with  $\frac{Q}{3} \leq a < \frac{Q}{2}$ , the symmetric DOZZ spectral modulus is strictly log-concave in the internal momentum:*

$$\frac{d^2}{dP^2} \log |C_b(a, a, \frac{Q}{2} + iP)| < 0 \quad \text{for all } P > 0.$$

**Conjecture 336** (Brownian-annulus modular log-concavity, resolved false). *For the Brownian annulus and every boundary-length ratio  $x > 0$ , the modulus density is log-concave in the logarithmic modulus:*

$$s \mapsto \log p_x(e^s) \quad \text{is strictly concave on } \mathbb{R}.$$

*Resolution.* The conjecture is false at large boundary ratio, by Proposition 70.

**Proposition 70** (Failure of annulus log-concavity at large boundary ratio). For the Brownian-annulus modulus density  $p_x(\tau) \propto \eta(2i\tau)\rho_\tau(x)$ , the logarithmic-modulus curvature  $d^2 \log p_x(e^s)/ds^2$  is positive for sufficiently large boundary ratio  $x$ . At  $x = e^5$ ,  $\tau = 0.30$  an independent Fourier inversion of the exact modulus transform, multiplied by the exact eta factor, gives curvature  $+0.3798$ ; the value grows to  $+0.90$  at  $x = e^7$  and to the positive limit  $\mathcal{C}_\infty(0.30) = 0.9746$  as  $x \rightarrow \infty$ . Hence Conjecture 336 is false.

*Proof.* From the Mellin transform  $\mathbb{E}[X_\tau^{it}] = \frac{2\pi t}{3 \sinh(2\pi t/3)} e^{-2\pi t^2/3}$ ,  $\log X_\tau = Z + G_\tau$  with  $Z$  centred logistic of scale  $\frac{2}{3}$  and  $G_\tau \sim N(0, \frac{4\pi\tau}{3})$  independent. Since  $f_Z(z) = \frac{3}{8} \operatorname{sech}^2(\frac{3z}{4}) \sim \frac{3}{2} e^{-3z/2}$  as  $z \rightarrow \infty$ , Gaussian averaging gives  $g_\tau(y) \sim \frac{3}{2} e^{-3y/2} \mathbb{E}[e^{3G_\tau/2}] = \frac{3}{2} e^{-3y/2} e^{3\pi\tau/2}$  locally uniformly in  $\tau$ , and likewise after two  $\tau$ -derivatives. With  $\eta(2i\tau)$  real and  $\log \eta(2i\tau) = -\pi\tau/6 + \sum_{n \geq 1} \log(1 - e^{-4\pi\tau n})$ , and writing  $x_n = 4\pi n\tau$ ,  $q_n = e^{-x_n}$ , the curvature of  $\log p_x(e^s)$  in  $s = \log \tau$  tends as  $x \rightarrow \infty$  to  $\mathcal{C}_\infty(\tau) = \frac{4\pi}{3}\tau + \sum_{n \geq 1} (\frac{x_n q_n}{1 - q_n} - \frac{x_n^2 q_n}{(1 - q_n)^2}) > 0$ , in which the  $\frac{4\pi}{3}\tau$  term is the logistic-Gaussian tail contribution  $(-\frac{\pi}{6} + \frac{3\pi}{2})\tau$  and the sum is the eta  $q$ -series, which does not vanish at finite  $\tau$  and cannot be dropped. At  $\tau = 0.30$  this gives  $\mathcal{C}_\infty = 0.9746$ . The finite value at  $x = e^5$ ,  $\tau = 0.30$  is obtained by direct high-precision Fourier inversion of the exact transform, stable under refinement of the step, and is reproduced in the verification script.  $\square$

## Part XXI

# Bold conjectures in theoretical fluid dynamics

*Added to the deposit: 22 August 2026.*

The fifteen conjectures below are sharp, falsifiable statements about the small-scale and singular behaviour of the incompressible Euler, Navier–Stokes, Boussinesq, generalized surface quasi-geostrophic, and point-vortex systems. They fall in six groups: two-dimensional small-scale creation (Conjectures 337 to 339), enhanced dissipation and hypoelliptic mixing (Conjectures 340 and 341), the geometry and regularity of three-dimensional Euler and Navier–Stokes singularities (Conjectures 342 to 344), band-limited kinematics and rapid rotation (Conjectures 345 and 346), stratified and active-scalar singularities (Conjectures 347 to 349), and point-vortex collapse

and non-Beltramization (Conjectures 350 and 351). Each is a determinate law with a stated mechanism, a nearest-literature boundary, a first decisive theorem, and a route to disproof, and each is deliberately bold: a proof or a counterexample would reorganize how one reads the corresponding small-scale or singular problem. The proved layer is Proposition 71: the exact three-vortex self-similar collapse anchor, the degenerate one-drift enhanced-dissipation exponent, the homogeneity-reduced generalized-SQG neck reduction, and the exact Beltrami Lamb-vector identity with Frenet dilation invariance. These are statements about continuum solutions—blow-up signatures, vanishing-viscosity and large-bandwidth limits, and long-time or asymptotic laws—so they lie and every one; the accompanying diagnostics test reduced models, exact anchors, or prerequisite scalings, never the full law. Several carry attributions where a sharp exponent or anchor is due to earlier work: the degenerate-shear exponent of Albritton, Beekie, and Novack [238] (Conjecture 341), the Lorentz-endpoint boundary of Phuc [252] (Conjecture 344), and the self-similar three-vortex collapse of Aref [250] (Conjecture 350); Conjecture 347 is where the part's novelty is most in question, since at equal viscosity and diffusivity its critical-layer anomaly may reduce to known inviscid repeated-root asymptotics times the Kelvin diffusion factor, and only its sharp generic coefficient is claimed as new.

Throughout,  $u$  is a divergence-free velocity field on the flat torus or on  $\mathbb{R}^d$ ,  $\omega = \nabla \times u$  (the scalar curl in two dimensions) its vorticity,  $S = \frac{1}{2}(\nabla u + \nabla u^\top)$  the strain,  $\xi = \omega/|\omega|$  the vorticity direction, and  $\hat{f}(k)$  the Fourier coefficients. Objects specific to a single conjecture are defined in its statement.

**Proposition 71** (Exact anchors and reductions). (i) For three planar point vortices with circulations  $\Gamma = (\Gamma_1, \Gamma_2, \Gamma_3)$  satisfying  $\Gamma_1\Gamma_2 + \Gamma_2\Gamma_3 + \Gamma_3\Gamma_1 = 0$  and vanishing angular impulse, the motion is exactly self-similar with a finite-time collapse or expansion; the symmetric choice  $\Gamma = (1, 1, -\frac{1}{2})$  satisfies the collapse condition. (ii) For the degenerate one-drift shear generator  $X_0 = y^m \partial_x$ ,  $X_1 = \partial_y$  with dissipation  $-\nu X_1^* X_1$ , the enhanced-dissipation gap obeys  $\text{gap}(L_\nu) \asymp \nu^{m/(m+2)}$  as  $\nu \downarrow 0$ . (iii) The homogeneity-reduced neck equation  $\dot{d} = -d^{1-\beta}$ ,  $0 < \beta \leq 1$ , has  $d(t) = [\beta(T-t)]^{1/\beta}$ , so  $d(t) \sim a_\beta(T-t)^{1/\beta}$  and, under the reciprocal-curvature ansatz  $\kappa_{\max} \sim b_\beta(T-t)^{-1/\beta}$ , the product  $d\kappa_{\max}$  tends to a finite nonzero constant. (iv) A Beltrami field ( $u \times \omega \equiv 0$ ), in particular the ABC field, has identically vanishing Lamb vector; and the dimensionless pair (total curvature, scale-invariant torsion variation) of a  $C^4$  space curve is invariant under spatial dilation.

## 81 Small-scale creation in two-dimensional Euler and Navier–Stokes

**Conjecture 337** (Bulk-enchrophy spectral front). *Let  $\omega$  be a smooth mean-zero solution of the 2D Euler equation on  $\mathbb{T}^2$  and, for  $q \in (0, 1)$ , let  $K_q(t) = \inf\{K \geq 0 : \sum_{|k| \leq K} |\hat{\omega}(k, t)|^2 \geq q \|\omega(t)\|_2^2\}$  be the radius carrying a fraction  $q$  of the enstrophy. For every  $0 < q < q' < 1$  there are constants  $C, c$  depending only on  $q, q'$  such that  $K_q(t) \leq CK_{q'}(0) \exp(c\|\omega_0\|_\infty t)$  for all  $t \geq 0$ .*

**Conjecture 338** (Super-exponential gradient sparsity). *For  $\eta \in (0, 1)$  let  $G_\eta(t) = \{x : |\nabla \omega(x, t)| \geq \eta \|\nabla \omega(t)\|_\infty\}$ . If a smooth 2D Euler solution has times  $t_n \rightarrow \infty$  with  $\log \|\nabla \omega(t_n)\|_\infty / t_n \rightarrow \infty$ , then  $|G_\eta(t_n)| \rightarrow 0$  for every fixed  $\eta \in (0, 1)$ .*

**Conjecture 339** (Vanishing-viscosity global branch selection). *On  $\mathbb{T}^2$  fix mean-zero vorticity and initial palinstrophy  $P(\omega_0) = P_0 > 0$ . For viscosity  $\nu > 0$  and horizon  $T$  let  $D_{\nu, T} =$*

$\nu \int_0^T \|\nabla \omega(t)\|_2^2 dt$  and  $M_{\nu,T}$  its supremum under the palinstrophy constraint. Fix  $\tau \in (0, \infty)$ , let  $\nu_j \rightarrow 0$  and  $T_j \sqrt{P_0} \rightarrow \tau$ , and take relative near-maximizers  $D_{\nu_j, T_j}(\omega_{0,j}) \geq (1 - \varepsilon_j) M_{\nu_j, T_j}$  with  $\varepsilon_j \rightarrow 0$ . With  $\ell_j = \|\omega_{0,j}\|_2 / \|\nabla \omega_{0,j}\|_2$  and  $W_j(y) = P_0^{-1/2} \omega_{0,j}(x_j + \ell_j y)$ , every concentration limit is, after torus isometries, reflection, and sign, the same nontrivial profile  $W_\tau$ ; and, when the normalized objectives are equicoercive, they  $\Gamma$ -converge at fixed  $\tau$  to a variational problem whose maximizer is unique modulo these symmetries.

## 82 Enhanced dissipation and hypoelliptic mixing

**Conjecture 340** (Reeb-graph spectral desert). *Let  $H$  be a Morse Hamiltonian on a compact surface whose regular level sets form finitely many cells with quadratic saddles, and let  $L_\nu = -\nabla^\perp H \cdot \nabla + \nu \Delta$  on mean-zero scalars. In every fixed bounded window of imaginary frequency, as  $\nu \rightarrow 0$  the spectrum splits into slow eigenvalues  $\lambda_j(\nu) = -\nu \mu_j + o(\nu)$ , where  $\mu_j$  are the eigenvalues of the effective weighted diffusion on the Reeb graph of  $H$ , and remaining eigenvalues with  $\Re \lambda \leq -c_H \nu^{1/2}$ ; in particular no eigenvalue branch decays at a rate  $\nu^\alpha$  with  $\frac{1}{2} < \alpha < 1$ .*

**Conjecture 341** (Hörmander-step enhanced-dissipation universality). *Consider  $\partial_t f + X_0 f = -\nu \sum_{j=1}^r X_j^* X_j f$  on a compact manifold with smooth invariant volume,  $X_0$  volume-preserving, in a one-drift equiregular Hörmander class where the missing drift directions are generated by Lie words containing exactly one  $X_0$  and repeated commutators with the diffusive fields. Let  $m$  be the maximal number of diffusive commutators needed to span the tangent space. If there is no nonconstant common first integral, then  $\text{gap}(L_\nu) \asymp \nu^{m/(m+2)}$  as  $\nu \rightarrow 0$ .*

## 83 Geometry and regularity of three-dimensional singularities

**Conjecture 342** (Frenet-frame completion barrier). *Suppose a Type-I Euler singularity at  $T$  is carried by a connected high-vorticity tube  $R_t$  containing a maximum-vorticity point, with radius comparable to the length  $L(t)$  of a  $C^4$  central vortex-line segment  $\gamma_t$  of uniform chord-arc constant and nonvanishing curvature, so that Frenet torsion  $\tau$  and  $\partial_s \tau$  are defined. With  $K(t) = \int \kappa ds$  and  $\text{Tor}(t) = \int L(t) |\partial_s \tau| / (1 + L(t) |\tau|) ds$ , one has  $\limsup_{t \rightarrow T} [K(t) + \text{Tor}(t)] = \infty$ .*

**Conjecture 343** (Type-I stretching coherence). *Let  $\Omega(t) = \|\omega(t)\|_\infty$ ,  $A_\lambda(t) = \{x : |\omega(x, t)| \geq \lambda \Omega(t)\}$ ,  $\alpha = \xi \cdot S\xi$ , and let  $\bar{\alpha}_\lambda, V_\lambda$  be the  $|\omega|^2$ -weighted mean and standard deviation of  $\alpha$  over  $A_\lambda$ . If a Type-I singularity satisfies  $0 < c \leq (T - t)\Omega(t) \leq C$ , then for every  $\lambda$  sufficiently close to 1 there is a sequence  $t_n \rightarrow T$  with  $(T - t_n)\bar{\alpha}_\lambda(t_n) \rightarrow 1$  and  $(T - t_n)V_\lambda(t_n) \rightarrow 0$ .*

**Conjecture 344** (Weak-Lorentz projected-Lamb regularity criterion). *Let  $u$  be a Leray–Hopf solution of the 3D Navier–Stokes equations on  $\mathbb{R}^3$ , or on  $\mathbb{T}^3$  in the Galilean frame with mean velocity zero, let  $\omega = \nabla \times u$ ,  $P$  the Leray projector, and  $\Lambda = P(u \times \omega)$  the projected Lamb vector. If  $\Lambda \in L^2((0, T); L_x^{3/2, \infty})$ , then  $u$  is smooth on  $(0, T]$ .*

## 84 Band-limited kinematics and rapid rotation

**Conjecture 345** (Helicity-defect law for stretching extremals). *Let  $V_N$  be divergence-free mean-zero fields on  $\mathbb{T}^3$  with Fourier support  $|k| \leq N$ . Fix admissible kinetic energy  $K$  and enstrophy  $E$ , put relative helicity  $\rho = H/(2\sqrt{KE})$ , and let  $M_N(\rho; K, E) = \sup |\int \omega \cdot S\omega|$  over  $V_N$  at these*

constraints and helicity  $\rho$ . Whenever  $\rho$  and 0 are both uniformly feasible as  $N \rightarrow \infty$  and the enstrophy-to-energy ratio  $E/K$  is bounded away from its first-shell minimum, so that zero-sum stretching triads persist and  $M_N(0; K, E) > 0$  uniformly,  $M_N(\rho; K, E)/M_N(0; K, E) \rightarrow (1 - \rho^2)^{3/2}$  locally uniformly for  $\rho$  in compact subsets of  $(-1, 1)$ .

**Conjecture 346** (Diophantine small-divisor rotation law). *For  $K \geq 1$  let  $\delta(K)$  be the minimum nonzero inertial-wave triad detuning among triads with maximal wavenumber at most  $K$  on an anisotropic torus, and assume the sharp arithmetic law  $\delta(K) = K^{-\tau+o(1)}$ . For fixed data of finite Sobolev regularity  $H^s$  whose Fourier coefficients obey a polynomial lower envelope  $|\hat{u}(k)| \gtrsim |k|^{-s'}$  along the wavenumbers of the triads realizing  $\delta(K_*)$  (a frequency-rich nondegeneracy, which also excludes a single inertial-wave mode, for which  $(u \cdot \nabla)u \equiv 0$ ; for  $C^\infty$  data the super-polynomial Fourier decay empties this condition, so the lower envelope forces finite regularity and only the upper bound  $E(\Omega, \nu) \lesssim \epsilon_*^{1+o(1)}$  survives without it) and  $O(1)$  times let  $E(\Omega, \nu)$  be the error between rotating 3D Navier–Stokes at rotation  $\Omega$  and its exact-resonant normal form in a fixed lower Sobolev norm, with a strictly positive data constant. In the joint regime  $K_* = (\Omega/\nu)^{1/(\tau+2)} \rightarrow \infty$  and  $\epsilon_* = (\Omega^2 \nu^\tau)^{-1/(\tau+2)} \rightarrow 0$ , one has  $E(\Omega, \nu) = \epsilon_*^{1+o(1)}$  up to Sobolev-loss, subpower, and data constants, and the  $\Omega$  power cannot be uniformly improved over tori realizing the same sharp  $\tau$ .*

## 85 Stratified and active-scalar singularities

**Conjecture 347** (Critical-Richardson viscous log anomaly). *For a fixed Kelvin ray  $(k, \eta)$ ,  $k \neq 0$ , in the linearized equally viscous and diffusive 2D Boussinesq system about Couette flow at Richardson number  $\text{Ri} = \frac{1}{4}$ , let  $\hat{\omega}_{k,\eta}(t)$  be the vorticity amplitude along the sheared frequency  $(k, \eta - kt)$ . For generic data with nonzero projection onto the repeated critical indicial mode,  $\sup_{1 \leq t \leq c(\nu k^2)^{-1/3}} |\hat{\omega}_{k,\eta}(t)|/(1+t)^{1/2} \sim C_{k,\eta} \log(1/\nu)$  as  $\nu \rightarrow 0$ , with  $C_{k,\eta} > 0$ ; for every fixed  $\text{Ri} > \frac{1}{4}$  the analogous normalized supremum is  $O(1)$ .*

**Conjecture 348** (Generalized-SQG neck-curvature similarity law). *Use  $u = \nabla^\perp \Lambda^{-2+\beta} \theta$  with  $0 < \beta \leq 1$ . Suppose a smooth generalized-SQG patch boundary develops a single symmetric self-similar neck singularity at  $T$ , with minimum opposing-boundary distance  $d(t)$  and maximum curvature  $\kappa_{\max}(t)$  in the same similarity core. Then  $d(t) \sim a_\beta (T-t)^{1/\beta}$ ,  $\kappa_{\max}(t) \sim b_\beta (T-t)^{-1/\beta}$ , and  $d(t)\kappa_{\max}(t) \rightarrow c_\beta \in (0, \infty)$ .*

**Conjecture 349** (Boussinesq source-core co-location and alignment). *On  $\mathbb{T}^2$  let  $\theta^\circ = \theta - \langle \theta \rangle$ ,  $\ell_\theta = \|\theta^\circ\|_\infty / \|\nabla \theta\|_\infty$ ,  $A_\lambda^\omega = \{|\omega| \geq \lambda \|\omega\|_\infty\}$ , and  $A_\lambda^\theta = \{|\partial_x \theta| \geq \lambda \|\partial_x \theta\|_\infty\}$ . If a smooth inviscid Boussinesq solution develops an isolated finite-time singularity at  $x_*$ , then, intersecting both intense sets with a fixed ball isolating the core, for every  $\lambda$  close to 1 the distance between the localized sets divided by  $\ell_\theta$  tends to zero; moreover, along a sequence in the core with distinct strain eigenvalues,  $\nabla \theta$  aligns with the most compressive strain eigenvector.*

## 86 Point-vortex collapse and non-Beltramization

**Conjecture 350** (Minimal-admissible point-vortex collapse core). *Fix nonzero circulations for which at least one collapse-admissible subset exists—a subset being admissible if its isolated subsystem has a nontrivial self-similar finite-time collapse—and let  $m_*$  be the least admissible cardinality. Let  $\mathcal{C}$  be the set of initial configurations undergoing a finite-time collision, with the resonance loci where two distinct minimal cores collapse simultaneously removed. Then the union*

of collision strata whose fastest-collapsing core contains exactly  $m_*$  vortices is relatively open and dense in  $\mathcal{C}$ , while strata with a strictly larger fastest core lie in positive-codimension substrata; and along every collision in this relatively generic set, similarity rescaling of the fastest core converges to an isolated self-similar  $m_*$ -vortex collapse with every spectator asymptotically farther than the core diameter.

**Conjecture 351** (High-vorticity non-Beltramization barrier). *On  $\mathbb{R}^3$  with finite energy, or on  $\mathbb{T}^3$  in the zero-mean Galilean frame, let  $C_\lambda(t)$  be the connected component of  $\{|\omega| \geq \lambda\Omega(t)\}$  containing a maximum-vorticity point, and  $B_\lambda(t) = \int_{C_\lambda} |u \times \omega|^2 / \int_{C_\lambda} |u|^2 |\omega|^2$  when the denominator is nonzero. Any Type-I Euler singularity with  $0 < c \leq (T-t)\Omega(t) \leq C$  satisfies  $\limsup_{t \rightarrow T} B_\lambda(t) > 0$  for every  $\lambda$  sufficiently close to 1.*

## Part XXII

# Metric geometry and spectral convexity in random conformal geometry

*Added to the deposit: 22 August 2026.*

The thirteen conjectures below complete the random-conformal-geometry bundle whose two exactly solved shape members—strict log-concavity of the DOZZ structure constant and of the Brownian-annulus modulus—are Conjectures 335 and 336 of Part XX. They fall in three groups: the metric geometry of Liouville quantum gravity (LQG), covering geodesic stars, the radial metric residual, metric-ball volume and boundary, space-filling SLE, and distance profiles (Conjectures 352 to 359); the spectral convexity of Liouville conformal field theory, covering monotone likelihood ratio and log-concavity of sphere and torus blocks and DOZZ-weighted spectral densities (Conjectures 360 to 363); and quantum-area nesting in conformal loop ensembles (Conjecture 364). Each is a determinate, sharp, falsifiable law with a stated mechanism, a nearest-literature boundary, and a first decisive theorem. The metric-geometry statements are continuum-limit laws (Hausdorff dimensions, scaling exponents, mixing and functional central limit theorems); the CFT statements are exact special-function shape properties whose descendant contributions are here tested only in the global ( $SL(2)$  or global-torus) block limit, a reduced-model check and a verification target. The proved layer is Proposition 72: the exact scaling triangle relating the boundary-length, roughness, and branching exponents, and the anchor reductions at the Brownian point  $\gamma = \sqrt{8/3}$ . Several statements are sharp interpolations of recognized open problems and attribute to the works whose anchors they extend.

Fix  $\gamma \in (0, 2)$ , write  $Q = \frac{2}{\gamma} + \frac{\gamma}{2}$ , let  $h$  be a Gaussian free field variant,  $\mu_h$  the  $\gamma$ -LQG area measure, and  $D_h$  the  $\gamma$ -LQG metric, with Hausdorff dimension  $d_\gamma = \dim_{\mathbb{H}}(\mathbb{C}, D_h)$  and  $\xi = \gamma/d_\gamma$ ; the metric obeys Weyl scaling  $D_{h+c} = e^{\xi c} D_h$  and the area obeys  $\mu_{h+c} = e^{\gamma c} \mu_h$ , and  $d_{\sqrt{8/3}} = 4$ . For Liouville CFT write  $b > 0$ ,  $Q_b = b + b^{-1}$ , central charge  $c = 1 + 6Q_b^2$ , principal-series weights  $\Delta(P) = \frac{Q_b^2}{4} + P^2$ , sphere OPE blocks  $\mathcal{F}_P(z)$  and torus blocks  $\mathcal{F}_{a,P}^{\text{tor}}(q)$  in the standard normalizations, and  $C_b$  for the DOZZ structure constant of Part XX. Objects specific to a single conjecture are defined in its statement.

**Proposition 72** (Scaling triangle and Brownian anchors). Set  $p_\gamma = d_\gamma/2$ ,  $H_\gamma = 2/d_\gamma$ , and  $\alpha_\gamma = 1 + 2/d_\gamma$ . Then  $p_\gamma = 1/H_\gamma = 1/(\alpha_\gamma - 1)$ , so a single scalar controls the metric boundary-



length dimension, the distance-profile roughness exponent, and the perimeter branching index. At the Brownian point  $d_\gamma = 4$  these are  $2, \frac{1}{2}, \frac{3}{2}$ , the known values. The affine star spectrum  $s_k(\gamma) = 2 + \frac{3-k}{2}(d_\gamma - 2)$  satisfies  $s_1 = d_\gamma$ ,  $s_3 = 2$  for every  $d_\gamma$ , and  $s_k = 5 - k$  at  $d_\gamma = 4$ ; and  $s_k \rightarrow 2$  as  $d_\gamma \downarrow 2$ .

## 87 Metric geometry of Liouville quantum gravity

**Conjecture 352** (Affine LQG geodesic-star spectrum). *For a point  $z$  call  $z$  a  $k$ -star point if there are  $r > 0$  and unit-speed  $D_h$ -geodesics  $\eta_1, \dots, \eta_k$  from  $z$  with disjoint interiors on  $(0, r]$ ; let  $S_k$  be the set of such points. With  $s_k(\gamma) = 2 + \frac{3-k}{2}(d_\gamma - 2)$ : if  $s_k(\gamma) > 0$  then almost surely  $\dim_{D_h}^{D_h} S_k = s_k(\gamma)$ ; if  $s_k(\gamma) < 0$  then  $S_k = \emptyset$  almost surely; at the threshold  $s_k = 0$  the set has  $D_h$ -dimension at most zero.*

**Conjecture 353** (Exponential scale-mixing of the radial metric residual). *Let  $h$  be a whole-plane GFF with  $h_1(0) = 0$ ,  $h_r(0)$  its circle average, and  $X_t = \log[e^{\xi Q t - \xi h_{e^{-t}}(0)} D_h(0, \partial B(0, e^{-t}))]$ , a process stationary in  $t$  by scale covariance. Its strong-mixing coefficient satisfies  $\alpha_X(s) \leq C e^{-cs}$ , and  $\sup_t \Pr[\sup_{0 \leq u \leq 1} |X_{t+u}| > x] \leq C e^{-cx}$ ; consequently  $\sup_{0 \leq u \leq T} |X_u| = O(\log T)$  almost surely and the radial log-distance process has the Brownian functional scaling limit and law of the iterated logarithm.*

**Conjecture 354** (Quantum-cone metric-ball shell functional CLT). *For a rooted LQG surface let  $V(r) = \mu_h(B_{D_h}(z, r))$  and  $Y_j^{(r)} = -\log[V(re^{-(j+1)})/V(re^{-j})]$ . There is a stationary inward-scale version  $(Y_j^{\text{cone}})$  for the rooted  $\gamma$ -quantum-cone tangent law that is exponentially  $\alpha$ -mixing with  $\mathbb{E}Y_0^{\text{cone}} = d_\gamma$  and finite positive asymptotic variance  $\sigma_\gamma^2$ , with  $\frac{1}{\sqrt{n}} \sum_{j < nt} (Y_j^{\text{cone}} - d_\gamma) \Rightarrow \sigma_\gamma B_t$  in  $D([0, 1])$ ; and for any finite-volume surface rooted at a quantum-typical point,  $(Y_j^{(r)})_{j \in J} \Rightarrow (Y_j^{\text{cone}})_{j \in J}$  as  $r \downarrow 0$ .*

**Conjecture 355** (Pointwise sfSLE metric Hölder exponent). *Let  $\eta$  be a whole-plane space-filling SLE $_{16/\gamma^2}$  independent of  $h$ , parameterized by LQG area,  $\mu_h(\eta([s, t])) = t - s$ . For Lebesgue-almost every  $t$ , almost surely  $\lim_{\varepsilon \downarrow 0} \log \text{diam}_{D_h} \eta([t - \varepsilon, t + \varepsilon]) / \log \varepsilon = 1/d_\gamma$ .*

**Conjecture 356** (Stable-CSBP perimeter universality). *For every  $\gamma \in (0, 2)$ , filled metric balls in a whole-plane or quantum-cone  $\gamma$ -LQG surface admit a canonical scalar outer-boundary length  $L_h^\gamma(r)$ , measurable from the metric-measure surface, with  $L_{h+c}^\gamma(e^{\xi c} r) = e^{\gamma c/2} L_h^\gamma(r)$ . Under reverse metric-radius exploration, after the natural normalization and excursion/Doob conditioning, the perimeter process is the conditioned/time-reversed stable continuous-state branching process of index  $\alpha_\gamma = 1 + 2/d_\gamma$ .*

**Conjecture 357** (Linear confluence response). *On a unit-area  $\gamma$ -LQG sphere sample  $z, x, y$  independently from  $\mu_h$ . On  $\{D_h(x, y) \leq \varepsilon\}$  let  $b$  be the terminal point of the common initial segment of the geodesics from  $z$  to  $x, y$ ,  $\Delta = \max\{D_h(b, x), D_h(b, y)\}$ , and  $C_\varepsilon = \Delta/D_h(x, y)$ . Conditioned on  $D_h(x, y) \leq \varepsilon$ ,  $C_\varepsilon$  converges in law as  $\varepsilon \downarrow 0$  to a nondegenerate  $C_\gamma \in (0, \infty)$  with finite moments in a neighbourhood of zero.*

**Conjecture 358** (Filled metric-ball outer boundary dimension). *Let  $h$  be a whole-plane GFF and  $B_{D_h}^{\infty, \bullet}(0, r)$  the filled metric ball targeted at infinity. For every deterministic  $r > 0$ , almost surely  $\dim_{D_h}^{D_h} \partial B_{D_h}^{\infty, \bullet}(0, r) = d_\gamma/2$ .*

**Conjecture 359** (Fractional-Brownian tangent of distance profiles). *Let  $\mathcal{D}$  be a finite-boundary-length  $\gamma$ -LQG disk,  $z$  sampled from  $\mu_h$ , and  $X(u) = D_h(z, \partial(u))$  the distance to the boundary point at length coordinate  $u$ . At boundary-length-almost every  $u_0$  there is  $\sigma_\gamma > 0$  with  $\varepsilon^{-H_\gamma}[X(u_0 + \varepsilon t) - X(u_0)] \Rightarrow \sigma_\gamma B_{H_\gamma}(t)$ ,  $H_\gamma = 2/d_\gamma$ , in local uniform topology, where  $B_H$  is two-sided fractional Brownian motion.*

## 88 Spectral convexity in Liouville conformal field theory

**Conjecture 360** (Sphere spectral monotone likelihood ratio). *Fix  $b > 0$ , four equal real external charges  $Q_b/3 \leq a < Q_b/2$ , and  $0 < z_1 < z_2 \leq \frac{1}{2}$ . The OPE-normalized sphere block  $\mathcal{F}_P(z)$  is nonzero on this domain and  $E = P^2 \mapsto |\mathcal{F}_{\sqrt{E}}(z_2)|^2 / |\mathcal{F}_{\sqrt{E}}(z_1)|^2$  is strictly increasing; equivalently  $\partial_E \partial_z \log |\mathcal{F}_{\sqrt{E}}(z)|^2 > 0$ .*

**Conjecture 361** (Four-point spectral log-concavity). *For  $b > 0$ ,  $Q_b/3 \leq a < Q_b/2$ ,  $0 < z \leq \frac{1}{2}$ , and  $P > 0$ , the principal-series magnitude  $I_z(P) = |C_b(a, a, \frac{Q_b}{2} + iP)|^2 |\mathcal{F}_P(z)|^2$  is positive and  $\frac{d^2}{dP^2} \log I_z(P) < 0$ .*

**Conjecture 362** (Torus thermal spectral monotone likelihood ratio). *Fix  $b > 0$  and a real external charge  $Q_b/3 \leq a < Q_b/2$ , and use the full torus block  $\mathcal{F}_{a,P}^{\text{tor}}(q) = q^{\Delta(P) - c/24} \mathcal{H}_{a,P}^{\text{tor}}(q)$  with  $q = e^{-2\pi\beta}$ . For  $\beta_2 > \beta_1$  the blocks are nonzero for  $P > 0$  and  $E = P^2 \mapsto |\mathcal{F}_{a,\sqrt{E}}^{\text{tor}}(q_2)|^2 / |\mathcal{F}_{a,\sqrt{E}}^{\text{tor}}(q_1)|^2$  is strictly decreasing.*

**Conjecture 363** (Multipoint joint spectral log-concavity of the modulus). *For  $n \geq 5$ , real moduli  $\mathbf{z}$  in a positive OPE cell, equal physical external charges  $Q_b/3 \leq a < Q_b/2$ , and internal momenta  $\mathbf{P} = (P_1, \dots, P_{n-3})$  on the principal series, let  $K_{\mathbf{z}}(\mathbf{P})$  be the standard-normalized comb sewing kernel (product of DOZZ vertices and the multipoint block) and  $\Phi_{\mathbf{z}}(\mathbf{P}) = \log |K_{\mathbf{z}}(\mathbf{P})|$ . Then  $K_{\mathbf{z}} \neq 0$  and  $\nabla_{\mathbf{P}}^2 \Phi_{\mathbf{z}} < 0$  for every  $P_j > 0$ .*

## 89 Nesting in conformal loop ensembles

**Conjecture 364** (Quantum-cone nesting area-ratio functional CLT). *Let  $\kappa \in (8/3, 4) \cup (4, 8)$  with  $\gamma = \sqrt{\kappa}$  for  $\kappa < 4$  and  $\gamma = 4/\sqrt{\kappa}$  for  $\kappa > 4$ . For nested  $\text{CLE}_\kappa$  around the marked point of the matched  $\gamma$ -quantum-cone tangent law, with  $U_n$  the successive marked components,  $A_n = \mu_h(U_n)$ , and  $Y_n = -\log(A_{n+1}/A_n)$ : the sequence  $(Y_n)$  admits a stationary version that is exponentially  $\alpha$ -mixing with analytic scaled cumulant generating function near zero and positive asymptotic variance, so  $\frac{1}{\sqrt{n}} \sum_{j < nt} (Y_j - v_{\gamma,\kappa}) \Rightarrow \sigma_{\gamma,\kappa} B_t$  in  $D([0, 1])$ ; and finite-volume deep-nesting increments converge to this stationary law.*

## Part XXIII

# Hydrodynamic limits of interacting particle systems

*Added to the deposit: 22 August 2026.*

The fifteen conjectures below propose hydrodynamic scaling limits for interacting particle systems whose microscopic rules carry a new structural ingredient: an endogenous local jump index, refreshed or fatigued environments, non-Markovian renewal clocks, singular reaction surfaces, finite crossing budgets, shared common noise, odd multi-species transport, and moving or storing membranes. They fall in five groups (Conjectures 365 to 367, 368 to 370, 371 to 373, 374 to 376, and 377 to 379). Each states a microscopic model, the conjectured limiting equation, its conserved quantities, the critical scaling that makes the new mechanism survive, and a route to disproof. These are scaling-limit statements about continuum PDE or SPDE, and every one; the diagnostics test event-count scalings, conservation, reduced linearizations, or exact ODE reductions, never the limit theorem. The proved layer is Proposition 73: exact mass conservation of the limiting equations, the fatigue mobility-budget identity, the perturbative self-arrest residual, and the critical event-count exponents. Four statements have explicit high overlaps with prior work, since a closely related continuum object or one-particle input appeared in 2025–2026 work: the Markov-switching Robin equation of Colantoni [270] (Conjecture 374), the scalar vorticity/odd-diffusion exclusion of De Carlo, Gabrielli, and Gonçalves [271] (Conjecture 375), the critical-percolation random-walk scaling limit of Dankovic, Markering, Miller, and Yuan [273] (Conjecture 377), and the finite-reservoir Wentzell theorem of Franco, Franco, and Gonçalves [275] (Conjecture 379); for these the novel content is the exact microscopic derivation and the specific interior/two-sided/multi-species structure, not the limiting object itself.

Throughout,  $\eta$  is an exclusion configuration on the discrete torus  $\mathbb{T}_N$  or on  $\mathbb{R}$ ,  $\rho$  the limiting density, and time is diffusively scaled unless stated otherwise. Objects specific to a single conjecture are defined in its statement.

**Proposition 73** (Conservation, mobility budget, self-arrest residual, and critical counts). (i) Each limiting equation displayed below conserves the relevant total mass (for multi-species models, each color mass): integrating the gain–loss or divergence form and interchanging dummy variables cancels the flux. (ii) For the fatigue ODE  $\dot{a} = c(1 - a)^p$  with  $c = 2\rho(1 - \rho) > 0$ , the conductance reaches zero in finite time iff  $p < 1$ , with  $t_{\text{hit}} = 1/(c(1 - p))$ , while the integrated mobility  $\int_0^\infty (1 - a)^p dt = 1/c$  is finite for every  $p$ . (iii) For the occupation-deepened trap system linearized about  $\rho = \bar{\rho}$  with  $b = 0$ , the first-order perturbation solves  $\partial_t v = e^{-a\bar{\rho}t} \Delta v$ , so  $v(t) = e^{\tau(t)\Delta} v_0$  with  $\tau(t) = (1 - e^{-a\bar{\rho}t})/(a\bar{\rho})$  and  $v_\infty = e^{\Delta/(a\bar{\rho})} v_0$ , generally nonzero. (iv) A codimension-one reaction surface at generator rate  $N^\theta$  contributes at order  $N^{\theta-1}$ , critical at  $\theta = 1$ ; and a membrane of step  $N^a$ , move rate  $N^{-b}$ , and slow conductance  $\alpha/N$  has order-one Brownian motion, healing length, and slow activity per dwell simultaneously iff  $b = 2a$  and  $a = \frac{1}{2}$ .

## 90 Density-adaptive and refreshed exclusion hydrodynamics

**Conjecture 365** (Density-adaptive Lévy exclusion). On  $\mathbb{T}_N$  let  $r_x^N(\eta)$  be the mean occupation in a block of radius  $\ell_N$  with  $\ell_N \rightarrow \infty$ ,  $\ell_N/N \rightarrow 0$ , and  $\log N/\sqrt{\ell_N} \rightarrow 0$ , and let  $\alpha : [0, 1] \rightarrow [\alpha_-, \alpha_+] \subset (0, 2)$  be  $C^1$ . A particle at  $x$  jumps to a vacant  $y$  at rate  $N^{\alpha(r_x^N)} c(\alpha(r_x^N)) |y - x|_N^{-1-\alpha(r_x^N)}$ . For smooth profiles bounded away from 0, 1 the empirical density converges to the conservative variable-order gain–loss equation

$$\partial_t \rho(u) = \text{PV} \int_{\mathbb{T}} [K_{\alpha(\rho(v))}(u - v) \rho(v) (1 - \rho(u)) - K_{\alpha(\rho(u))}(v - u) \rho(u) (1 - \rho(v))] dv.$$

**Conjecture 366** (Refreshing heavy traps: crossover trichotomy). Start from a reversible heavy-trap exclusion model with static anomalous clock  $\theta_N$  and index  $\beta \in (0, 1)$ , refresh each trap at

rate  $q_N$  independently of the particles, and set  $\lambda_N = q_N \theta_N$ . If  $\lambda_N \rightarrow 0$  the static fractional-kinetics limit persists; if  $\lambda_N \rightarrow \lambda \in (0, \infty)$  the hydrodynamics is governed by a model-dependent renewal/Bernstein symbol  $\psi_\lambda(s)$  with  $\psi_0(s) = cs^\beta$  and  $\psi_\lambda(s) \sim m_\lambda s$  as  $s \downarrow 0$ ; if  $\lambda_N \rightarrow \infty$  the limit is ordinary diffusion after effective-diffusivity renormalization.

**Conjecture 367** (Critical sparse shortcuts add a teleportation operator). *Let nearest-neighbor SSEP run on  $\mathbb{T}_N$  at rate  $N^2$ , and add  $M_N$  macroscopic shortcut edges ringing at rate  $\kappa$ , with  $M_N/N \rightarrow \lambda/2$  and endpoint-pair measure converging to uniform. The empirical density converges to  $\partial_t \rho = D \partial_{uu} \rho + \kappa \lambda (\bar{\rho} - \rho)$ , with  $\bar{\rho} = \int_{\mathbb{T}} \rho$ ; for a nonuniform endpoint kernel the nonlocal term becomes the corresponding conservative jump operator.*

## 91 Renewal exclusion, fatigue, and boundary memory

**Conjecture 368** (Crowding-renormalized successful-reset renewal exclusion). *Let a particle of age  $a$  attempt jumps with hazard  $\beta/(a + a_0)$ ,  $0 < \beta < 1$ , resetting its age only after a successful jump. In dimension  $d \geq 2$ , linearizing about a homogeneous equilibrium  $\rho_*$ , on the clock  $t_{N,*} = N^{2/\gamma(\rho_*)}$  with  $\gamma(\rho_*) = \beta(1 - \rho_*)$  the density perturbation converges to the fractional diffusion  $\partial_t^{\gamma(\rho_*)} u = D(\rho_*) \Delta u$ ; a genuinely nonconstant profile admits no single global polynomial clock, requiring a variable-order-in-time renewal kernel.*

**Conjecture 369** (Activity-fatigued conductances). *Attach damage  $a_b(t) \in [0, 1]$  to each bond, exchanging symmetrically at rate  $N^2(1 - a_b)^p$ , with  $a_b$  increased by  $N^{-2}$  at each successful crossing (a crossing budget of order  $N^2$ ). Writing  $g(a) = (1 - a)^p$ , the damage field then evolves macroscopically as  $\partial_t a = 2g(a)\rho(1 - \rho)$ , this law derived from the increment rule rather than imposed. The coupled hydrodynamic limit is  $\partial_t \rho = \partial_u(g(a)\partial_u \rho)$ ,  $\partial_t a = 2g(a)\rho(1 - \rho)$ , with finite-time loss of ellipticity at  $p < 1$  but finite integrated mobility for every  $p$ , so some smooth nonconstant data retain nonconstant long-time limits.*

**Conjecture 370** (Common heavy-tailed boundary clock: random Stieltjes boundary law). *Let a single heavy-tailed renewal process control all left-boundary updates, scaled so that  $N^{-1}R_N(N^2t) \Rightarrow \kappa_L E_t^L$  with  $E^L$  an inverse- $\gamma$ -stable clock, and analogously on the right. The bulk solves  $\partial_t \rho = \partial_{uu} \rho$ , but the boundary fluxes are singular random Stieltjes measures: in weak form  $d\langle \rho_t, G \rangle = \langle \rho_t, G'' \rangle dt + \kappa_L G(0)[\alpha - \rho_t(0)]dE_t^L + \kappa_R G(1)[\beta_R - \rho_t(1)]dE_t^R$ , remaining random after averaging over the clocks.*

## 92 Catalysts, mixed clocks, and crossing budgets

**Conjecture 371** (Codimension-one nonlinear catalyst transmission law). *Let  $\Sigma_N$  approximate a smooth hypersurface  $\Sigma \subset \mathbb{T}^d$  with bulk SSEP at rate  $N^2$  and catalyst birth/death flips at generator rate  $N^\theta$  with reaction  $R(r) = b(r)(1 - r) - d(r)r$ . The critical exponent is  $\theta = 1$ , at which  $\partial_t \rho = \Delta \rho$  off  $\Sigma$  with continuous trace and derivative jump  $[\partial_n \rho]_\Sigma = -R(\rho|_\Sigma)$ ; for  $\theta < 1$  the catalyst disappears and for  $\theta > 1$  a stable zero of  $R$  pins the trace.*

**Conjecture 372** (Mixed-clock two-species Maxwell–Stefan hydrodynamics). *Let species  $A$  use exponential attempt clocks and species  $B$  use i.i.d. heavy-tailed attempt clocks of index  $\gamma \in (0, 1)$ , reset after every attempt. Observed on time  $N^{2/\gamma}$  with  $A$ 's microscopic rate  $N^{2-2/\gamma}$ , both species stay hydrodynamically visible, and around a homogeneous composition the linearized law is the*

*mixed-order Maxwell–Stefan system  $s\widehat{\delta a} - \delta a_0 = -k^2(D_*\widehat{\delta})_A$ ,  $s^\gamma\widehat{\delta b} - s^{\gamma-1}\delta b_0 = -k^2(D_*\widehat{\delta})_B$  with non-diagonal exclusion cross-diffusion  $D_*$ .*

**Conjecture 373** (Finite crossing-budget membrane: endogenous stopping-time transition). *Give a critical slow bond (microscopic rate  $\alpha/N$ ) a budget  $B_N = \lfloor \kappa N \rfloor$  of successful exchanges, after which its rate is zero forever. With unsigned local activity  $A(r_-, r_+) = r_-(1 - r_+) + r_+(1 - r_-)$ , the normalized success count satisfies  $S_N(t)/N \rightarrow \alpha \int_0^t A(\rho(s, 0^-), \rho(s, 0^+)) ds$ , and the density obeys the critical Robin transmission law before the exhaustion time  $\tau = \inf\{t : \alpha \int_0^t A \geq \kappa\}$  and reflecting conditions after, so the PDE changes type at an endogenous stopping time.*

## 93 Common noise, odd transport, and self-generated arrest

**Conjecture 374** (Telegraph-modulated critical membrane). *Let a single two-state Markov chain  $Z_t \in \{0, 1\}$ , converging on diffusive time to a finite-rate telegraph process, multiply a critical slow bond's rate,  $\alpha Z_t/N$ . Conditional on  $Z$ , the density converges to the heat equation with time-dependent Robin coefficient  $\alpha Z_t$ , so the empirical density converges jointly with  $Z$  to a random hydrodynamic limit; the fast-switch regime converges to the averaged Robin law.*

**Conjecture 375** (Multi-species chiral-vorticity odd cross-diffusion). *Use at least two conserved colors plus vacancies, with symmetric color-vacancy exchanges and plaquette rotations at rates  $c_\pm(\xi) = N^2 c_0[1 \pm \varepsilon \Phi(\mathcal{N}(\xi))]$  depending only on the rotation-invariant plaquette composition  $\mathcal{N}(\xi)$ , so grand-canonical product color measures are invariant. The diffusive hydrodynamics is  $\partial_t r_i = \nabla \cdot \sum_j [D_{ij}^S(r) \nabla r_j + D_{ij}^O(r) R \nabla r_j]$  with  $R$  rotation by  $\pi/2$  and a composition-dependent odd cross matrix  $D^O$  that changes sign with  $\varepsilon$  and affects individual density equations.*

**Conjecture 376** (Perturbative self-generated arrest). *Attach trap depth  $E_x(t) \geq 0$  with symmetric bond rate  $N^2 \exp[-(E_x + E_{x+1})/2]$  and  $dE_x/dt = a\eta_t(x) - bE_x$ . The finite-time limit is  $\partial_t \rho = \partial_u(e^{-\varepsilon} \partial_u \rho)$ ,  $\partial_t e = a\rho - be$ ; with  $b = 0$ , linearizing about  $\rho = \bar{\rho}$  gives residual  $v_\infty = e^{\Delta/(a\bar{\rho})} v_0$ , so for small smooth mean-zero data the full nonlinear system converges to the nonconstant limit  $\rho_\infty = \bar{\rho} + \varepsilon e^{\Delta/(a\bar{\rho})} v_0 + O(\varepsilon^2)$ .*

## 94 Critical geometry and moving membranes

**Conjecture 377** (Critical-percolation exclusion transfer). *Let  $(V_N, c_N, \mu_N)$  be the rescaled critical triangular-percolation graph sequence for which simple random walk converges quenched to the semigroup  $P_t^\omega$  of  $\text{CLE}_6$  Brownian motion on the limiting resistance space, with time normalization  $a_N$ . Run SSEP on  $V_N$  with the same conductances and time change. If the initial color field converges to  $\rho_0 \mu^\omega$ , then quenched  $\pi_t^N \Rightarrow \rho_t \mu^\omega$  with  $\rho_t = P_t^\omega \rho_0$ , so exclusion produces no additional transport exponent beyond the one-particle normalization.*

**Conjecture 378** (Critically mobile slow membrane). *Let a distinguished slow bond of rate  $\alpha/N$  move by microscopic distance  $N^a$  at rate  $N^{-b}$ . At the uniquely matched scaling  $a = \frac{1}{2}$ ,  $b = 1$  (step  $\sqrt{N}$ , move rate  $N^{-1}$ , conductance  $\alpha/N$ ), the membrane path converges to Brownian motion and the density converges to a nontrivial random singular-interface heat flow conditional on that Brownian path; the membrane is not annealed away, and the exact moving-Robin generator is a sub-conjecture.*

**Conjecture 379** (Interior two-sided capacitor membrane). *Split a one-dimensional system into left and right SSEP bulks and an interior storage node of capacity  $\lfloor \kappa N \rfloor$ , conserving total mass, with side-to-storage rates of order  $N$ . The limit is  $\mu_t = \rho_- du + \rho_+ du + m(t)\delta_0$  with heat equations in the bulks, nonlinear vertex fluxes  $F_\pm(r, m) = a_\pm r(1 - m/\kappa) - b_\pm(m/\kappa)(1 - r)$ , boundary conditions  $-\partial_{n_\pm} \rho_\pm = F_\pm$ , and  $\dot{m} = F_- + F_+$ , conserving  $\int \rho_- + \int \rho_+ + m$ .*

## Part XXIV

# Fixed-traffic circulation laws for nonequilibrium Markov chains

*Added to the deposit: 22 August 2026.*

The four conjectures below concern a controlled family of finite-state continuous-time Markov chains—fixed-traffic circulation rays—in which the stationary law, every undirected stationary edge traffic, and every escape rate are held fixed while an irreversible stationary circulation of strength  $\alpha$  is increased. The ray therefore varies time-reversal asymmetry without touching the equilibrium landscape or the local activity. Two conjectures concern visited-set statistics (Conjectures 380, 382, 383) and one the spectral geometry of the generator (Conjecture 381). Each is a determinate law with a stated mechanism, a nearest-literature boundary, a first decisive theorem, and a route to disproof. The proved layer is Proposition 74: the endpoint spectral bounds, the exact even-analytic dependence of visited-set laws on  $\alpha$ , and an explicit high-precision counterexample showing that the natural stronger companion—whole-ray monotonicity of the fast real spectral edge—is *false*, so only the endpoint bound survives there. Conjecture 381 is the strongest of the four; its own stronger global-unimodality form is also false and is recorded as a control. The visited-set laws are exact finite-state statements, tested here by exact augmented-chain propagation rather than trajectory sampling, but with the alternating-derivative hierarchy of Conjecture 380 checked only to finite order.

Let the state space be  $\{1, \dots, n\}$  with stationary law  $\pi > 0$ . For each undirected edge fix a symmetric traffic  $T_{ij} = T_{ji} > 0$  and an antisymmetric divergence-free current  $J_{ij} = -J_{ji}$ ,  $\sum_j J_{ij} = 0$ , with  $|J_{ij}| < T_{ij}$ . The circulation ray has off-diagonal rates  $L_{\alpha,ij} = (T_{ij} + \alpha J_{ij})/(2\pi_i)$ ,  $0 \leq \alpha \leq 1$ , and zero row sums; then  $\pi$  is stationary, the undirected traffic  $q_{ij}(\alpha) + q_{ji}(\alpha) = T_{ij}$  and the escape rate  $r_i = \frac{1}{2\pi_i} \sum_j T_{ij}$  are  $\alpha$ -independent. Writing  $H_\alpha = -D^{1/2} L_\alpha D^{-1/2}$  with  $D = \text{diag}(\pi)$  and restricting off the stationary mode gives  $G_\alpha = K + \alpha C$  with  $K = K^* \succ 0$  and  $C^* = -C$ . Let  $h(\alpha) = \max\{\Re \lambda : \lambda \in \sigma(-L_\alpha) \setminus \{0\}\}$  and  $g(\alpha) = \min\{\Re \lambda : \lambda \in \sigma(-L_\alpha) \setminus \{0\}\}$  be the fast and slow real spectral edges. Let  $V_t = \{X_s : 0 \leq s \leq t\}$  be the visited set,  $w_i^{(\gamma)} = \pi_i^\gamma / \sum_j \pi_j^\gamma$  escort weights, and  $U_{\gamma,t} = \sum_i w_i^{(\gamma)} \mathbf{1}_{\{\tau_i > t\}}$  the unexplored escort mass, with  $x = \alpha^2$ .

**Proposition 74** (Endpoint bounds, even dependence, and the failure of fast-edge ray monotonicity). For every ray,  $\Re \lambda(G_\alpha) \in [\lambda_{\min}(K), \lambda_{\max}(K)]$ , so  $h(\alpha) \leq h(0) = \lambda_{\max}(K)$  and  $g(\alpha) \geq g(0) = \lambda_{\min}(K)$ . Under a stationary start, path reversal maps the law of  $L_\alpha$  to that of  $L_{-\alpha}$  and fixes every visited-set statistic, so  $V_t^{(\alpha)} \stackrel{d}{=} V_t^{(-\alpha)}$  and all finite-state visited-set probabilities are even analytic functions of  $\alpha$ , whence  $x = \alpha^2$  is the natural variable. The endpoint bound does *not* extend to whole-ray monotonicity of the fast edge: there is a bidirected



six-state ray on which  $h(\alpha)$  strictly increases over an interior interval by more than 0.6, verified to 40 digits, so the companion statement  $h(\beta) \leq h(\alpha)$  for  $\alpha \leq \beta$  is false.

## 95 Visited-set and spectral circulation laws

**Conjecture 380** (Escort unexplored-mass alternating-derivative hierarchy). *For every integer  $k \geq 1$ , integer  $r \geq 0$ , time  $t \geq 0$ , and  $0 \leq \gamma \leq 1$ , with  $x = \alpha^2$ ,*

$$(-1)^r \frac{d^r}{dx^r} \mathbb{E}_{\sqrt{x}}[U_{\gamma,t}^k] \geq 0$$

*throughout the admissible physical interval.*

**Conjecture 381** (One-maximum principle for the slow spectral edge). *The slow real spectral edge  $g(\alpha)$  has at most one strict interior local maximum on  $(0, 1)$ .*

**Conjecture 382** (Unicyclic range stochastic acceleration). *For every bidirected chain whose undirected support is connected of cycle rank one, with  $X_0 \sim \pi$  and  $R_t = |V_t|$ : for every  $m$ ,  $t \geq 0$ , and  $0 \leq \alpha \leq \beta \leq 1$ ,*

$$\Pr_{\beta}(R_t \geq m) \geq \Pr_{\alpha}(R_t \geq m).$$

**Conjecture 383** (Pure-cycle stationary-mass coverage acceleration). *For a simple-cycle support with  $X_0 \sim \pi$  and  $M_t = \pi(V_t)$ : for every  $z \in [0, 1]$ ,  $t \geq 0$ , and  $0 \leq \alpha \leq \beta \leq 1$ ,*

$$\Pr_{\beta}(M_t \geq z) \geq \Pr_{\alpha}(M_t \geq z).$$

## Part XXV

# Cross-shadow universality

*Added to the deposit: 24 August 2026.*

The five conjectures below share one principle: two limiting observables read off the same discrete randomness decouple because they probe distinct resolution, frequency, or representation-theoretic sectors. The proved layer is Proposition 75.

**Proposition 75** (Cross-shadow anchor identities). (i) For an integral matrix  $A_n$  with  $\det A_n \neq 0$ ,  $\mathbb{Z}^n/A_n\mathbb{Z}^n$  is finite of order  $|\det A_n|$ . (ii) For a conjugation-invariant word measure  $\mu_{w,N}$  on  $S_N$ ,  $\chi^2(\mu_{w,N}||U_N) = \sum_{\rho \neq \mathbf{1}} |\mathbb{E}_w \chi_{\rho}|^2$ . (iii) Every  $\mathbb{Q}_p$  root of a Littlewood polynomial is a  $p$ -adic unit. (iv) The  $\mathbb{F}_p$  singularity probability of a balanced random matrix is  $c_p = 1 - \prod_{j \geq 1} (1 - p^{-j}) = 1/p + O(p^{-2})$ , so  $c_p - 1/p$  is summable. (v)  $t^{-1/2}|\text{core}_e(\lambda_t)|$  and  $(4/\pi)\Gamma_e^{\text{Ros}}$  differ only by the factor  $\pi/4$ .

*Proof.* (i) Smith normal form  $A_n = U \text{diag}(d_i)V$ ,  $U, V \in \text{GL}_n(\mathbb{Z})$ , gives  $\mathbb{Z}^n/A_n\mathbb{Z}^n \cong \bigoplus \mathbb{Z}/d_i$  of order  $\prod |d_i| = |\det A_n|$ . (ii) With  $f = N!\mu_{w,N}$ , expanding  $f - 1$  in the orthonormal character basis gives coefficient  $\mathbb{E}_w \chi_{\rho}$  at  $\rho \neq \mathbf{1}$ , and Parseval gives the identity. (iii) If  $|x|_p < 1$  the unit constant term dominates and if  $|x|_p > 1$  the leading term dominates, so any root has  $|x|_p = 1$ . (iv) Expand the product. (v) Rostam's definition.  $\square$

**Conjecture 384** (Local spectral–Smith factorization). *For a symmetric integral Wigner matrix  $A_n$  (centered entries, edge-universality moments, odd-prime anti-concentration) with rescaled real edge process  $\mathcal{E}_n$  and  $p$ -primary paired cokernel  $C_{n,p}$ , and every fixed finite set  $P$  of odd primes,  $(\mathcal{E}_n, (C_{n,p})_{p \in P}) \Rightarrow \mathcal{E}_{\text{GOE}} \otimes \bigotimes_{p \in P} \mu_p^{\text{sym}}$ , and likewise with any fixed finite collection of microscopic bulk and edge windows; the  $p = 2$  analogue is a secondary statement.*

**Conjecture 385** (Random-integer multiplicative universality of determinants). *For  $B_n$  i.i.d. Bernoulli- $\frac{1}{2}$  in  $\{0, 1\}$  and  $D_n = |\det B_n| \geq 3$ ,  $(\omega(D_n) - \log \log D_n) / \sqrt{\log \log D_n} \Rightarrow N(0, 1)$  and  $(\log Q_{i,n} / \log D_n)_i \Rightarrow PD(1)$  in the ranked simplex.*

**Conjecture 386** (Plancherel modular-charge/Airy factorization). *For poissonized Plancherel  $\lambda_t$ , fixed  $e \geq 2$ , Rostam-normalized charge  $Y_t^{(e)}$ , and upper-edge process  $\mathcal{A}_t$ ,  $(Y_t^{(e)}, \mathcal{A}_t) \Rightarrow G_e \otimes \mathcal{A}_{\text{GUE}}$ ; consequently  $(\frac{\pi}{4\sqrt{t}} |\text{core}_e(\lambda_t)|, t^{-1/6}(\lambda_{t,1} - 2\sqrt{t})) \Rightarrow \Gamma_e^{\text{Ros}} \otimes F_2$ .*

**Conjecture 387** (Stable-spectrum saturation of word measures). *For  $w \in F_r$  with primitive abelianization but non-primitive in  $F_r$ , with  $\beta_{\text{st}}(w) = \inf_{\lambda \neq \emptyset} \liminf_N \frac{-\log |\mathbb{E}_w \chi_{\lambda[N]}|}{\log N}$ ,  $\chi^2(\mu_{w,N} \| U_N) = N^{-2\beta_{\text{st}}(w) + o(1)}$  and  $D_{\text{KL}}(\mu_{w,N} \| U_N) = N^{-2\beta_{\text{st}}(w) + o(1)}$ .*

**Conjecture 388** (Adelic root-process factorization). *For a random Littlewood polynomial  $F_n$  with real near-unit root process  $\mathcal{R}_n$  and  $p$ -adic root process  $\mathcal{Z}_{p,n}$  on  $\mathbb{Z}_p^\times$ , and every fixed finite set  $P$  of odd primes,  $(\mathcal{R}_n, (\mathcal{Z}_{p,n})_{p \in P}) \Rightarrow \nu_\infty \otimes \bigotimes_{p \in P} \nu_p$ ; the prime 2 is excluded.*

## Part XXVI

# Entropy and rare-event laws for random algebraic and topological structures

*Added to the deposit: 29 August 2026.*

The five conjectures below place deep algebraic and topological invariants on canonical random ensembles and assert entropy, threshold, and rare-event laws. The proved layer is Proposition 76.

**Proposition 76** (Anchor identities and reductions). (i) At a two-face tropical breakpoint of  $P_{\beta,N}(z) = \sum_k e^{\beta a_k} z^k G_{k,N}$ , after rescaling, every inactive coefficient carries a factor  $e^{-\beta \Delta_k} \leq e^{-\beta \delta}$ , and the zeros of the dominant pencil  $G_{r,N} + \zeta^m G_{s,N}$  are the  $m$ -th-root lift of the complex spherical ensemble. (ii) Matroid Kazhdan–Lusztig polynomials are multiplicative under direct sums, forests give  $P \equiv 1$ , and the cycle  $C_k$  contributes  $U_{k-1,k}$ . (iii) For a diagram with  $c$  crossings on  $m$  strands,  $\log Z(u, v) = \log \sum_{i,j} \dim_{\mathbb{F}_2} K h^{i,j} e^{ui+vj}$  is convex and bounded by  $C(c+m)(1+|u|+|v|)$ . (iv) The Weil–Petersson short-geodesic Poisson mean is  $\Lambda(\varepsilon) = \varepsilon^2/4 + O(\varepsilon^4)$ ; for each fixed  $k$ ,  $\varepsilon^{-2k} \Pr(\ell_k^{\text{hom}} \leq \varepsilon) \rightarrow 1/(4^k k!)$  in the large-genus then shrinking-window limit; and  $\pi \text{cap}(\ell)/\ell \rightarrow 1$ . (v) For a rational homology sphere,  $\text{rank}_{\mathbb{F}_2} \widehat{HF}(Y) \geq |H_1(Y; \mathbb{Z})|$ .

*Proof.* (i) Divide by  $e^{\beta \Phi(x^*)}$ ;  $\det(G_r + \zeta^m G_s) = \det(G_s) \prod_i (\zeta^m - \lambda_i)$  with  $\lambda_i \in \text{spec}(-G_s^{-1} G_r)$ , the (rotation-invariant) spherical ensemble. (ii) Multiplicativity and  $P_{B_n} \equiv 1$  are Elias–Proudfoot–Wakefield; a bridge is a coloop and the block decomposition is classical. (iii) Log-sum-exp convexity; the resolution cube has  $2^c$  vertices with at most  $c+m$  circles each, and gradings are  $O(c+m)$ . (iv)  $\lambda(\ell) = \sum_{r \geq 1} \ell^{2r-1}/(2r)!$  integrates to  $\varepsilon^2/4 + \varepsilon^4/96 + \dots$ ; Mirzakhani–Petri

give the Poisson law, and each fixed short multicurve has connected complement asymptotically almost surely, so its classes are homologically independent and the  $k$ th tail is asymptotic to  $\Lambda(\varepsilon)^k/k!$ . (v)  $\widehat{HF}$  splits over  $|H_1|$   $\text{Spin}^c$  structures, each summand of Euler characteristic  $\pm 1$ , hence nonzero over  $\mathbb{F}_2$ .  $\square$

**Conjecture 389** (Double-scaling tropical shadowing). *In the setting of Proposition 76(i), if  $\beta_N \delta / \log N \rightarrow \infty$ , then on every compact annular sector within one  $m$ -th-root branch, after removing a boundary layer of vanishing limiting intensity, the zeros of the full matrix polynomial match the zeros of  $G_{r,N} + \zeta^m G_{s,N}$  with maximal displacement  $o_{\mathbb{P}}(N^{-1/2})$ ; hence the unfolded selected zero process has the spherical-ensemble branch-lift local limit.*

**Conjecture 390** (Kazhdan–Lusztig entropy percolation). *For  $G_n \sim G(n, c/n)$  and  $H_n(c) = n^{-1} \log P_{M(G_n)}(1)$ , there is a deterministic  $H_{\text{KL}}: (0, \infty) \rightarrow [0, \infty)$  with  $H_n(c) \rightarrow H_{\text{KL}}(c)$  in probability for every fixed  $c \neq 1$ ,  $H_{\text{KL}}(c) = 0$  for  $0 < c < 1$ ,  $H_{\text{KL}}(c) > 0$  for  $c > 1$ , and  $H_n(1) \rightarrow 0$  in probability.*

**Conjecture 391** (Khovanov free energy of random braid closures). *For  $m \geq 3$  and a finitely supported  $\mu$  on  $B_m$ , aperiodic on the symmetric-group quotient and with nonelementary image in  $B_m/Z(B_m)$  acting on the  $m$ -punctured disc, let  $L_n$  be the closure of  $\beta_n = X_1 \cdots X_n$  and  $Z_n(u, v) = \sum_{i,j} \dim_{\mathbb{F}_2} Kh^{i,j}(L_n; \mathbb{F}_2) e^{ui+vj}$  in normalized oriented bigradings. For every fixed  $(u, v)$  there is a deterministic finite  $F_\mu(u, v)$  with  $n^{-1} \log Z_n(u, v) \rightarrow F_\mu(u, v)$  almost surely and in  $L^1$ , and  $F_\mu$  is convex.*

**Conjecture 392** (Two-sided thin-handle Hodge successive-minima equivalence). *For Weil–Petersson random  $X_g$ , let  $m_j(J(X_g))$  be the Hodge successive minima and define*

$$\ell_j^{\text{hom}}(X_g) = \inf\{L : \text{rank}_{\mathbb{Z}} \langle [\gamma] : \gamma \text{ primitive nonseparating simple closed geodesic, } \ell(\gamma) \leq L \rangle \geq j\}.$$

*For every fixed  $k \geq 1$  and every  $\eta > 0$ ,*

$$\lim_{\varepsilon \downarrow 0} \lim_{g \rightarrow \infty} \mathbb{P} \left( \max_{1 \leq j \leq k} \left| \frac{\pi m_j(J(X_g))^2}{\ell_j^{\text{hom}}(X_g)} - 1 \right| > \eta \mid \min\{\pi m_k(J(X_g))^2, \ell_k^{\text{hom}}(X_g)\} \leq \varepsilon \right) = 0,$$

*and consequently  $\varepsilon^{-2k} \mathbb{P}(\pi m_k(J(X_g))^2 \leq \varepsilon) \rightarrow 1/(4^k k!)$  in the same sequential limit.*

**Conjecture 393** (Floer cancellation saturation for random Heegaard splittings). *For  $g \geq 2$  and finitely supported  $\mu$  generating the genus- $g$  mapping class group as a semigroup, let  $Y_n = H_g \cup_{\phi_n} H_g$  with  $\phi_n = X_1 \cdots X_n$  and  $Q_n = \{b_1(Y_n) = 0\}$ . Conditional on  $Q_n$ , the excess Floer entropy  $E_n = \log \text{rank}_{\mathbb{F}_2} \widehat{HF}(Y_n) - \log |H_1(Y_n; \mathbb{Z})| \geq 0$  satisfies  $E_n/n \rightarrow 0$  in probability; equivalently, whenever the homological torsion exponent  $\alpha_\mu$  exists in probability on  $Q_n$ ,  $n^{-1} \log \text{rank}_{\mathbb{F}_2} \widehat{HF}(Y_n) \rightarrow \alpha_\mu$ .*

## Part XXVII

# Random Hamiltonian cocycle entropy

*Added to the deposit: 30 August 2026.*

**Proposition 77** (Deterministic and lazy-walk calibrations). (i) For a point-mass driving law  $\delta_\phi$  on the Hamiltonian diffeomorphisms of a closed symplectic surface, all-class finite-bar barcode entropy equals  $h_{\text{top}}(\phi)$ . (ii) For  $\mu = (1-p)\delta_{\text{id}} + p\delta_\phi$ , both quenched random topological entropy and all-class finite-bar entropy equal  $p h_{\text{top}}(\phi)$  almost surely.

**Conjecture 394** (Quenched random Floer barcode entropy). *Let  $(\Sigma, \omega)$  be a closed symplectic surface, let  $\mu$  be finitely supported on  $\text{Ham}(\Sigma, \omega)$ , and put  $\Phi_n = X_n \circ \cdots \circ X_1$  for i.i.d.  $X_i \sim \mu$ . If  $b_\varepsilon^{\text{fin}}(\psi)$  counts bars of finite length greater than  $\varepsilon$  in the unpinned absolute fixed-point Floer barcode over  $\mathbb{F}_2$ , summed over every free homotopy class and extended to degenerate maps by the lower-semicontinuous  $C^\infty$  perturbation  $\liminf$ , and  $h_{\text{top}}^q(\mu)$  is the quenched random topological entropy defined by the sample-path Bowen metric, then almost surely*

$$\lim_{\varepsilon \downarrow 0} \limsup_{n \rightarrow \infty} \frac{1}{n} \log \max\{1, b_\varepsilon^{\text{fin}}(\Phi_n)\} = h_{\text{top}}^q(\mu).$$

## Part XXVIII

# Cut-rank laws for graph states, forests, and random width

*Added to the deposit: 30 August 2026.*

For  $G = (V, E)$  and  $S \subseteq V$ , put  $\rho_G(S) = \text{rank}_{\mathbb{F}_2} A_G[S, V \setminus S]$  and  $Z_G(x, y) = \sum_{S \subseteq V} x^{|S|} y^{\rho_G(S)}$ .

**Conjecture 395** (Quenched balanced-cut nullity universality). *For the loopless complete graph on  $2m + s$  vertices whose unordered pairs receive independent uniform labels in  $\mathbb{F}_q$  (zero meaning no edge),  $s \in \{0, 1\}$ , the empirical nullity law over all  $m$ -vertex cuts converges in probability in total variation to  $\pi_{q,s}(d) = q^{-d(d+s)} P_{q,\infty} / (P_{q,d} P_{q,d+s})$ .*

**Conjecture 396** (Dependence-sensitive extreme balanced-cut defect). *With*

$$D_{\max}(G) = \max_{|S|=m} (m - \rho_G(S)),$$

*one has*

$$\frac{D_{\max}(G)}{\sqrt{(2m+s) \log_q 2}} \longrightarrow 1$$

*in probability.*

**Conjecture 397** (Poisson law for random graph-state distance). *For  $G \sim G(n, 1/2)$ , the count of nonempty  $A$  with  $|A| + |\text{Odd}_G(A) \setminus A| \leq \ell$  converges in total variation to  $\text{Poisson}(\Lambda_n(\ell))$  whenever its exact mean  $\Lambda_n(\ell)$  stays bounded away from zero and infinity.*

**Conjecture 398** (Typical rigidity of the labelled cut-rank function). *With high probability, every graph on the same labels with the same function  $S \mapsto \rho_G(S)$  is locally equivalent to  $G$ , without relabelling.*

**Conjecture 399** (Positive-rank-tail log-concavity). *For connected  $G$ , if  $c_r = \#\{S : \rho_G(S) = r\}$ , then  $c_r^2 \geq c_{r-1} c_{r+1}$  for  $2 \leq r < \max_S \rho_G(S)$ .*

**Conjecture 400** (Full log-concavity for percolated forests). *For every forest  $F$ , the sequence  $a_j = \#\{B \subseteq E(F) : \nu((V(F), B)) = j\}$  is log-concave.*

**Conjecture 401** (Hurwitz stability of the percolated-matching polynomial). *Every zero of  $\sum_{B \subseteq E(F)} z^{\nu((V(F), B))}$  has negative real part for every forest  $F$ .*

**Conjecture 402** (Path–star stochastic extremality under bond noise). *For every  $n$ -vertex tree  $T$ ,  $p \in (0, 1)$ , and  $k$ ,  $\Pr(\nu((S_n)_p) \geq k) \leq \Pr(\nu(T_p) \geq k) \leq \Pr(\nu((P_n)_p) \geq k)$ , with the stated extremizers unique.*

**Conjecture 403** (Fixed-leaf broom minimum). *For every fixed  $p \in (0, 1)$ , among  $n$ -vertex trees with  $L$  leaves, the subdivided star broom  $B_{n,L}$  stochastically minimizes  $\nu(T_p)$ , and equality in distribution holds only for the broom.*

**Conjecture 404** (Typical tree reconstruction from the entanglement enumerator). *A uniform labelled tree is, with probability tending to one, determined up to isomorphism by  $Z_T(x, y)$ .*

**Conjecture 405** (Rank-width density of random regular graph states). *For fixed  $d \geq 3$ ,  $\text{rw}(G_{n,d})/n \rightarrow \kappa_d \in (0, 1/3]$  in probability along  $dn$  even.*

**Conjecture 406** (Cubic branch-width density and kernel transfer). *As even  $N \rightarrow \infty$ , uniform simple cubic graphs satisfy  $\text{bw}(H_N)/N \rightarrow \beta_3 > 0$ . Conditional on a cubic configuration multigraph, let  $H_N^*$  be any simple graph obtained by  $o(N)$  edge insertions and deletions. Subdivide all but  $o(N)$  edges at least once, allow further subdivisions, and take arbitrary one-vertex sums with trees; uniformly over these choices, the resulting  $S_N$  satisfies  $\text{rw}(S_N) = \text{bw}(H_N^*) + o(N)$ . Hence in the weakly supercritical window  $\text{rw}(G(n, (1 + \varepsilon)/n))/(\varepsilon^3 n) \rightarrow 4\beta_3/3$ .*

**Conjecture 407** (Rank-width surface tension for off-critical percolated grids). *For fixed  $p \neq 1/2$ ,  $\text{rw}(Q_L(p))/L \rightarrow \tau_{\text{rw}}(p)$  in probability, with phase values 0 below  $1/2$ , positive above  $1/2$ , and  $\tau_{\text{rw}}(1) = 1$ .*

**Conjecture 408** (Saturation of local-complementation orbit entropy). *For  $G \sim G(n, 1/2)$ ,  $n^{-1} \log_2 |\mathcal{O}_{\text{LC}}(G)| \rightarrow \log_2 3$  in probability.*

**Conjecture 409** (Rank-width density curve for sparse Erdős–Rényi graph states). *For fixed  $c > 0$ ,  $\text{rw}(G(n, c/n))/n \rightarrow \kappa_{\text{ER}}(c)$  in probability, with  $\kappa_{\text{ER}}(c) = 0$  for  $c \leq 1$  and positive for  $c > 1$ .*

## Part XXIX

# Arithmetic games, prime continuants and Hessian shadows

*Added to the deposit: 2 September 2026.*

This part states fifteen conjectures on prime continuants of typical real numbers, Hausdorff dimensions of continued-fraction sets, Legendre-symbol and random Toeplitz and Hankel games, and plane quartics with their Hessians and duals over finite fields. The proved layer is Proposition 78. Throughout,  $x \in (0, 1)$  is irrational with continued fraction  $x = [0; a_1, a_2, \dots]$ ,

convergents  $p_n/q_n$ , and  $\mu$  is Lebesgue measure; for an odd prime  $p$ ,  $\chi_p$  is the Legendre symbol and we write  $m = (p-1)/2$ ,

$$H_p = (\chi_p(i+j))_{1 \leq i,j \leq m}, \quad T_p = (\chi_p(i-j))_{1 \leq i,j \leq m},$$

and  $v(A)$  for the value of the zero-sum game with payoff matrix  $A$  to the row player, both players using mixed strategies. For  $p \equiv 1 \pmod{4}$ ,  $\varepsilon_p > 1$  denotes the fundamental unit of  $\mathbb{Q}(\sqrt{p})$ ; it has norm  $-1$ . The Dedekind function is  $\Psi(k) = k \prod_{\ell|k} (1 + 1/\ell)$  and the Lévy constant is  $L = \pi^2/(12 \log 2)$ , so that  $\log q_n(x) \sim nL$  for  $\mu$ -almost every  $x$ .

**Proposition 78** (Local laws, symmetric games, exact Legendre values and quartic identities). (i) For every integer  $k \geq 2$  and  $\mu$ -almost every  $x$ , the set  $\{n : k \mid q_n(x)\}$  has density  $1/\Psi(k)$ . For every prime  $p$  and almost every  $x$ , the set  $\{n : p \nmid q_n(x)q_{n+1}(x)\}$  has density  $(p-1)/(p+1)$ , and the set  $\{n : p \mid q_n(x), p \mid q_{n+2}(x)\}$  has density  $g_p/(p+1)$ , where  $g_p = \sum_{j \geq 1} \log_2(1 + 1/(jp(jp+2)))$  is the Gauss measure of  $\{x : p \mid a_1(x)\}$ . (ii) If  $A$  is a circulant matrix, then  $v(A)$  equals the mean of the entries of  $A$ ; if  $A$  is antisymmetric, then  $v(A) = 0$ ; and for any sequence  $(\xi_k)$  the Toeplitz matrix  $(\xi_{i-j})_{1 \leq i,j \leq m}$  and the Hankel matrix  $(\xi_{i+j-m-1})_{1 \leq i,j \leq m}$  have the same value. (iii) For  $p \in \{17, 41, 73, 89, 97, 113, 137, 193, 233, 241, 281, 313, 337, 409, 433, 449, 457, 521, 569, 601, 617\}$  one has  $v(H_p) = -y/x$ , where  $\varepsilon_p = x + y\sqrt{p}$ ; for  $p \in \{5, 13\}$  one has  $v(H_p) = -y/x$  with  $\varepsilon_p = (x + y\sqrt{p})/2$ , namely  $v(H_5) = -1$  and  $v(H_{13}) = -1/3$ ; for  $p = 401$  the unit is  $\varepsilon_p^5$  and for  $p = 577$  it is  $\varepsilon_p^7$ . For  $p \in \{5, 13, 29, 37, 61, 101, 109, 157, 181, 277, 349, 373, 421, 541, 613\}$  one has  $v(T_p) = +y/x$ , where  $\varepsilon_p^3 = x + y\sqrt{p}$ ; for  $p = 229$  the unit is  $\varepsilon_p^9$ . In every listed case  $x + y\sqrt{p}$  has norm  $-1$ , and an optimal strategy common to both players has full support  $\{1, \dots, m\}$  for  $T_p$  and support  $\{1, \dots, m-1\}$  for  $H_p$ . (iv) Let  $q$  be an odd prime power and  $F \in \mathbb{F}_q[X, Y, Z]$  a quartic form defining a smooth curve. Then

$$\sum_{P \in \mathbb{P}^2(\mathbb{F}_q)} \chi_q(F(P)) = qt(F),$$

where  $\chi_q$  is the quadratic character of  $\mathbb{F}_q$  and  $t(F)$  is the trace of the Frobenius on the  $E_7$  root lattice of the degree-two del Pezzo surface  $w^2 = F$ ; in particular  $t(F) \in \mathbb{Z}$  and  $|t(F)| \leq 7$ . (v) Let  $C$  be a smooth plane quartic over  $\mathbb{F}_q$ ,  $q$  a power of a prime larger than three so that the Gauss map is separable and  $C$  is reflexive, all of whose flexes are ordinary and whose 28 bitangents are ordinary (no hyperflex). Write  $b_s$  for the number of  $\mathbb{F}_q$ -rational bitangents whose two points of tangency are  $\mathbb{F}_q$ -rational and  $b_c$  for the number whose two points of tangency are swapped by the Frobenius. Then the dual curve satisfies  $\#C^\vee(\mathbb{F}_q) = \#C(\mathbb{F}_q) - b_s + b_c$ .

*Proof.* (i) Write  $r_n = q_n/q_{n-1} \in \mathbb{P}^1(\mathbb{F}_p)$  for the projective class of the column  $(q_{n-1}, q_n)$  modulo  $p$ ; since consecutive continuants are coprime,  $r_n$  is well defined, and the recursion  $q_{n+1} = a_{n+1}q_n + q_{n-1}$  gives  $r_{n+1} = a_{n+1} + 1/r_n$ . Each map  $r \mapsto a + 1/r$  is a bijection of  $\mathbb{P}^1(\mathbb{F}_p)$ , so the product of the Gauss measure with the uniform measure on  $\mathbb{P}^1(\mathbb{F}_p)$  is invariant under the skew product  $(x, r) \mapsto (Gx, a_1(x) + 1/r)$ ,  $G$  the Gauss map; this skew product is ergodic, which is the content of the equidistribution theorems of Moeckel and Jager and Liardet for  $(q_{n-1}, q_n)$  modulo  $p$ . The event  $p \mid q_n$  is  $r_n = 0$ , of stationary probability  $1/(p+1)$ , and for a general modulus  $k$  the same argument on  $\mathbb{P}^1(\mathbb{Z}/k)$ , which has  $\Psi(k)$  points, gives  $1/\Psi(k)$ ; the ergodicity of the skew product for composite  $k$  is part of the cited theorems. Since  $r_n = 0$  forces  $r_{n+1} = \infty$ , the events  $p \mid q_n$  and  $p \mid q_{n+1}$  are disjoint, whence the density  $1 - 2/(p+1)$ . Finally  $r_n = 0$  forces  $r_{n+2} = a_{n+2}$ , so  $p \mid q_n$  and  $p \mid q_{n+2}$  together mean  $r_n = 0$  and  $p \mid a_1(G^{n+1}x)$ ; the



integrand  $\mathbf{1}_{r=0} \mathbf{1}_{p|a_2(x)}$  has product-measure integral  $g_p/(p+1)$ , and the ergodic theorem for the skew product gives the density. (ii) Every row and every column of a circulant matrix is a permutation of the first, so the uniform strategy of either player makes every pure reply of the other player yield the mean entry; both guarantees coincide. Since  $v(-A^\top) = -v(A)$  for every  $A$ , if  $A^\top = -A$  then  $v(A) = v(-A^\top) = -v(A)$ , so  $v(A) = 0$ . Reversing the order of the columns turns  $(\xi_{i-j})$  into  $(\xi_{i+j-m-1})$  and does not change the value. (iii) Each matrix is symmetric. For the listed  $p$ , the linear system  $\sum_i x_i A_{ij} = v$  for  $j$  in the stated support,  $\sum_i x_i = 1$ ,  $x_i = 0$  off the support, has a unique rational solution, which is nonnegative and satisfies  $\sum_i x_i A_{ij} = v$  for every column  $j$ , including the column  $m$  omitted from the support of  $H_p$ . Thus  $x$  guarantees  $v$  to the row player, and by symmetry of  $A$  the same vector guarantees  $v$  to the column player, so  $v(A) = v$ . The rational numbers  $v$  so obtained are the stated unit coordinate ratios; the exact rational arithmetic is reproduced in the stress-test section. (iv) Let  $X$  be the surface  $w^2 = F(X, Y, Z)$  in the weighted projective space  $\mathbb{P}(1, 1, 1, 2)$ . Since  $F$  is smooth,  $X$  is a smooth del Pezzo surface of degree two, the double cover of the plane branched along  $C = \{F = 0\}$ , and  $\#X(\mathbb{F}_q) = \sum_P (1 + \chi_q(F(P)))$  because  $F(\lambda P) = \lambda^4 F(P)$ . For a smooth rational surface Weil's formula gives  $\#X(\mathbb{F}_q) = q^2 + q \operatorname{tr}(\operatorname{Frob} | \operatorname{Pic} \bar{X}) + 1$ . The Picard lattice has rank eight and splits orthogonally as  $\mathbb{Z}K_X \oplus K_X^\perp$  with  $K_X^\perp \cong E_7(-1)$ , the Frobenius fixing  $K_X$  and acting on  $K_X^\perp$  through the Weyl group  $W(E_7)$ . Hence  $\operatorname{tr} = 1 + t(F)$  and the character sum equals  $\#X(\mathbb{F}_q) - (q^2 + q + 1) = qt(F)$ ; an orthogonal transformation of a rank-seven lattice has trace at most seven in absolute value. (v) The Gauss map  $C \rightarrow C^\vee$ ,  $P \mapsto T_P C$ , is the normalisation of  $C^\vee$  and is defined over  $\mathbb{F}_q$ . Under the ordinarieness hypotheses the singular points of  $C^\vee$  are the 24 cusps, images of the flexes, each with one branch, and the 28 nodes, images of the bitangents, each with two branches through the two points of tangency. Away from the nodes the Gauss map is a bijection between  $\mathbb{F}_q$ -points, and a rational point of  $C$  maps to a rational point of  $C^\vee$ . A rational node whose branches are rational receives two rational points of  $C$ , a rational node with conjugate branches receives none, and a non-rational node receives no rational point since its branches are not rational. Counting gives the identity.  $\square$

**Conjecture 410** (Prime continuant law). (i) For  $\mu$ -almost every  $x$ ,

$$\#\{n \leq N : q_n(x) \text{ prime}\} = (2 \log 2 + o(1)) \log N,$$

and the same holds for the numerators  $p_n(x)$ . (ii)  $\mu\{x : q_N(x) \text{ prime}\} = (2 \log 2 + o(1))/N$  as  $N \rightarrow \infty$ . (iii) For  $x$  uniformly distributed, the point process  $\sum_{n: q_n(x) \text{ prime}} \delta_{\log n - T}$  converges in law, as  $T \rightarrow \infty$ , to the Poisson process on  $\mathbb{R}$  of intensity  $2 \log 2$ . (iv) The count in (i), for  $x$  uniformly distributed, has Gaussian fluctuations of variance  $(2 \log 2 + o(1)) \log N$ .

**Conjecture 411** (Prime-pair correlations along the convergents). (i)  $\mu\{x : q_n(x), q_{n+1}(x) \text{ both prime}\} = (24 \log^2 2 / \pi^2 + o(1)) / n^2$ , that is, primality at consecutive indices is negatively correlated by the exact factor  $6/\pi^2$  relative to the product of the two marginal probabilities of Conjecture 410. (ii) The same asymptotic holds for  $\mu\{x : p_n(x), q_n(x) \text{ both prime}\}$ . (iii) At lag two the correlation factor is

$$C_2 = \prod_p \left(1 - \frac{1}{p^2} + \frac{g_p(p+1)}{p^2}\right) = 0.91957 \dots,$$

with  $g_p$  as in Proposition 78(i), so that  $\mu\{x : q_n, q_{n+2} \text{ both prime}\} \sim C_2 (2 \log 2)^2 / n^2$ . (iv) For almost every  $x$  and every fixed lag  $h \geq 1$ , only finitely many  $n$  have  $q_n(x)$  and  $q_{n+h}(x)$  both prime, and only finitely many  $n$  have  $p_n(x)$  and  $q_n(x)$  both prime.

**Conjecture 412** (Squarefree and totient densities of continuants). *For  $\mu$ -almost every  $x$ ,*

$$\lim_{N \rightarrow \infty} \frac{1}{N} \# \{n \leq N : q_n(x) \text{ squarefree}\} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n \leq N} \frac{\varphi(q_n(x))}{q_n(x)} = \prod_p \left(1 - \frac{1}{p(p+1)}\right) = 0.70445\dots,$$

*and more generally, for every multiplicative  $f$  with  $f(p^k) = 1 + O(p^{-k})$ ,  $N^{-1} \sum_{n \leq N} f(q_n(x))$  converges to  $\prod_p (1 + \sum_{k \geq 1} (f(p^k) - f(p^{k-1}))/\Psi(p^k))$ .*

**Conjecture 413** (Erdős–Kac law along the convergents). *For  $\mu$ -almost every  $x$  and every real  $t$ ,*

$$\lim_{N \rightarrow \infty} \frac{1}{N} \# \left\{ n \leq N : \frac{\omega(q_n(x)) - \log n}{\sqrt{\log n}} \leq t \right\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-u^2/2} du,$$

*where  $\omega$  counts distinct prime divisors; the centring  $\log n$  may be replaced by  $\sum_{p \leq q_n} 1/(p+1)$ , which differs from  $\log \log q_n$  by a bounded amount.*

**Conjecture 414** (Dimension of the prime-continuant sets). (i) *The set  $\{x \in (0, 1) : q_n(x) \text{ is prime for every } n \geq 1\}$  has Hausdorff dimension exactly  $1/2$ .* (ii) *The set  $\{x \in (0, 1) : q_n(x) \text{ is composite for every } n \geq 1\}$  has Hausdorff dimension 1.*

**Conjecture 415** (Digit sets in arithmetic progressions). *For integers  $1 \leq a \leq m$  let  $E_{a,m}$  be the set of  $x \in (0, 1)$  all of whose partial quotients lie in  $a + m\mathbb{N}_0 = \{a, a+m, a+2m, \dots\}$ . Then, uniformly for  $a/m$  in any compact subinterval of  $(0, 1]$ ,*

$$\dim_{\text{H}} E_{a,m} = \frac{1}{2} + \frac{1}{2m} - \frac{\log m}{2m^2} - \frac{\psi_{\Gamma}(a/m)}{2m^2} + O\left(\frac{\log^2 m}{m^3}\right),$$

*where  $\psi_{\Gamma}$  is the digamma function; in particular  $\dim_{\text{H}} E_{m,m} = \frac{1}{2} + \frac{1}{2m} - \frac{\log m}{2m^2} + \frac{\gamma}{2m^2} + O(m^{-3} \log^2 m)$  with Euler's constant  $\gamma$ .*

**Conjecture 416** (Continuant avoidance dimension). *For a prime  $p$  let  $A_p = \{x \in (0, 1) : p \nmid q_n(x) \text{ for every } n \geq 1\}$ . Then*

$$\dim_{\text{H}} A_p = 1 - \frac{6 \log 2}{\pi^2} \cdot \frac{1}{p} + O\left(\frac{1}{p^2}\right),$$

*and  $\mu\{x : p \nmid q_n(x) \text{ for all } n \leq N\} = \exp(-(\rho_p + o(1))N)$  with  $\rho_p = (p+1)^{-1}(1 + O(1/p))$ .*

**Conjecture 417** (Unit law for the Legendre half-box games). *Let  $p \equiv 1 \pmod{4}$  be prime. (i) The linear system  $\sum_{i=1}^{m-1} x_i \chi_p(i+j) = v$  ( $1 \leq j \leq m-1$ ),  $\sum_{i < m} x_i = 1$ , is nonsingular and its value is  $v = -y/x$  for a unit  $x + y\sqrt{p}$  of norm  $-1$  in  $\mathbb{Z}[\sqrt{p}]$  (a solution of  $x^2 - py^2 = -1$ ) or, when  $p \equiv 5 \pmod{8}$ , possibly a unit  $(x + y\sqrt{p})/2$  of norm  $-1$  in the maximal order; the unit is an odd power  $\varepsilon_p^e$  of the fundamental unit, and  $e \neq 1$  only if  $p$  is of the form  $k^2 + 1$  or  $k^2 + 4$ . (ii) The full linear system  $\sum_{i=1}^m x_i \chi_p(i-j) = v$  ( $1 \leq j \leq m$ ),  $\sum_i x_i = 1$ , has value  $v = +y/x$  with  $x + y\sqrt{p} = \varepsilon_p^{3e}$  when  $p \equiv 5 \pmod{8}$ , and  $v = -y/x$  with the same unit as in (i) when  $p \equiv 1 \pmod{8}$ , in which case the solution always has a negative coordinate. (iii) Whenever the solution in (i) (respectively (ii)) is nonnegative it is the unique optimal strategy of both players and  $v(H_p)$  (respectively  $v(T_p)$ ) equals the unit ratio exactly; this happens for a set of primes  $p \equiv 1 \pmod{8}$  of positive relative density in case (i) and for a set of primes  $p \equiv 5 \pmod{8}$  of positive relative density in case (ii). (iv)  $\sqrt{p}v(H_p) \rightarrow -1$  as  $p \rightarrow \infty$  through primes  $p \equiv 1 \pmod{8}$ , and  $\sqrt{p}v(T_p) \rightarrow +1$  as  $p \rightarrow \infty$  through primes  $p \equiv 5 \pmod{8}$ ; in the unit regime of (iii) this is automatic, since  $py^2/x^2 = 1 + 1/x^2$  or  $1 + 4/x^2$ .*

**Conjecture 418** (Least odd residue law in the complementary classes). *Let  $\ell(p)$  denote the least odd prime that is a quadratic residue modulo  $p$ . (i) As  $p \rightarrow \infty$  through primes  $p \equiv 5 \pmod{8}$  with  $\ell(p) \in \{3, 5\}$ ,  $\sqrt{p}v(H_p) \rightarrow -1$ ; as  $p \rightarrow \infty$  through primes  $p \equiv 1 \pmod{8}$  with  $\ell(p) \in \{3, 5\}$ ,  $\sqrt{p}v(T_p) \rightarrow +1$ . (ii) For every prime  $\ell \geq 7$ , along primes  $p \equiv 5 \pmod{8}$  with  $\ell(p) = \ell$  one has  $\limsup \sqrt{p}v(H_p) < 0$  and  $\liminf \sqrt{p}v(H_p) > -1$ , and along primes  $p \equiv 1 \pmod{8}$  with  $\ell(p) = \ell$  one has  $\liminf \sqrt{p}v(T_p) > 0$  and  $\limsup \sqrt{p}v(T_p) < 1$ ; consequently neither  $\sqrt{p}v(H_p)$  over  $p \equiv 5 \pmod{8}$  nor  $\sqrt{p}v(T_p)$  over  $p \equiv 1 \pmod{8}$  converges.*

**Conjecture 419** (Brownian shift games). *Let  $(\xi_k)_{k \in \mathbb{Z}}$  be independent random variables with mean zero, variance one and finite fourth moment, and let  $K^{(m)} = (\xi_{i+j})_{1 \leq i, j \leq m}$ ; by Proposition 78(ii) the Toeplitz game  $(\xi_{i-j})_{1 \leq i, j \leq m}$  has the same value as the Hankel game of the shifted sequence, so it suffices to treat  $K^{(m)}$ . Then  $\sqrt{m}v(K^{(m)})$  converges in law to a nondegenerate symmetric limit  $V$  that does not depend on the law of  $\xi$ , and  $V$  is the value of the continuum game*

$$V = \lim_{R \rightarrow \infty} \sup_f \inf_{y \in [0, 1]} \int_0^1 f(x) dW(x + y),$$

where  $W$  is a standard Brownian motion on  $[0, 2]$ , the supremum runs over probability densities  $f \in L^2[0, 1]$  with  $\|f\|_2 \leq R$ , and the assertion includes that each capped supremum of infima is almost surely finite and that the caps increase to an almost surely finite nondegenerate limit. In contrast, for independent  $\xi$  arranged in a circulant matrix  $\sqrt{m}v$  converges to a standard Gaussian, and for a full matrix of independent entries  $m v$  is tight.

**Conjecture 420** (Hessian big monodromy over finite fields). *For a smooth plane quartic  $C$  over  $\mathbb{F}_q$  let  $H_C$  be its Hessian sextic and write  $a_C = (q + 1 - \#C(\mathbb{F}_q))/\sqrt{q}$ ,  $a_H = (q + 1 - \#H_C(\mathbb{F}_q))/\sqrt{q}$ . As  $q \rightarrow \infty$  through prime powers with characteristic at least five, for  $C$  uniformly random among smooth quartics with smooth Hessian, the pair  $(a_C, a_H)$  converges in law to  $(\text{tr } U_6, \text{tr } U_{20})$  with  $U_6 \in \text{USp}(6)$  and  $U_{20} \in \text{USp}(20)$  independent and Haar distributed; the number of  $\mathbb{F}_q$ -rational flexes converges to the fixed-point count of a uniform element of the symmetric group  $S_{24}$ , independently of the pair. The same holds for smooth curves of every degree  $d \geq 4$  in characteristic larger than  $d$ , with the Hessian of degree  $3(d - 2)$ , genus  $(3d - 7)(3d - 8)/2$ , and the flexes numbering  $3d(d - 2)$ .*

**Conjecture 421** (Sato–Tate independence of a curve and its Hessian). *Let  $C$  be a smooth plane quartic over  $\mathbb{Q}$  whose Hessian  $H_C$  is smooth and such that the Mumford–Tate groups of the Jacobians of  $C$  and of  $H_C$  are  $\text{GSp}(6)$  and  $\text{GSp}(20)$ , with no nonzero homomorphism between the two Jacobians over  $\mathbb{Q}$ . Write  $a_p(C) = p + 1 - \#C(\mathbb{F}_p)$ . Then, as  $p \rightarrow \infty$  through primes of good reduction, the pairs  $(a_p(C)/\sqrt{p}, a_p(H_C)/\sqrt{p})$  are equidistributed with respect to the product of the trace laws of  $\text{USp}(6)$  and  $\text{USp}(20)$ .*

**Conjecture 422** (Tangency law and dual-curve point counts). *As  $q \rightarrow \infty$  with characteristic at least five, for a random smooth plane quartic over  $\mathbb{F}_q$  the triple  $(r, b_s, b_c)$ , where  $r$  is the number of  $\mathbb{F}_q$ -rational bitangents and  $b_s, b_c$  are as in Proposition 78(v), converges in law to  $(F, B, F - B)$ , where  $F$  is the number of fixed points of a Haar-random element of  $\text{Sp}(6, \mathbb{F}_2)$  on the 28 odd theta characteristics and, conditionally on  $F$ ,  $B$  is binomial with parameters  $F$  and  $\frac{1}{2}$ . Consequently  $\#C^\vee(\mathbb{F}_q) - \#C(\mathbb{F}_q)$  converges in law to  $F - 2B$ , a symmetric integer law with variance one. Jointly, the number of rational bitangents whose two lines on the del Pezzo double cover are rational converges to the count of fixed weights of a Haar-random element of  $W(E_7)$  in its minuscule 56-point action, with  $t(F)$  of Proposition 78(iv) converging to the reflection-representation trace of the same element.*

**Conjecture 423** (Positivity of the quarter-box Legendre games). *For a prime  $p \equiv 3 \pmod{4}$  let  $m = \lfloor p/4 \rfloor$  and  $Q_p = (\chi_p(i+j))_{1 \leq i,j \leq m}$ . Then  $v(Q_p) > 0$  for every prime  $p \equiv 3 \pmod{4}$  with  $p \geq 7$ , and there are constants  $0 < c_1 < c_2 < \infty$  with  $c_1 \leq \sqrt{m} v(Q_p) \leq c_2$  for all such  $p$ , while  $\sqrt{m} v(Q_p)$  does not converge. For  $p \equiv 1 \pmod{4}$  the sign of  $v(Q_p)$  is positive and negative for infinitely many  $p$  each.*

**Conjecture 424** (Vanishing law for the half box in the class 3 modulo 4). *For primes  $p \equiv 3 \pmod{4}$  one has  $\sqrt{p} v(H_p) \rightarrow 0$ ; more precisely  $v(H_p) = O(p^{-3/2})$  along  $p \equiv 7 \pmod{8}$ , while  $p^{3/2} v(H_p)$  is not bounded along  $p \equiv 3 \pmod{8}$ . Moreover  $v(H_p) \leq 0$  for every  $p \equiv 7 \pmod{8}$ , while along  $p \equiv 3 \pmod{8}$  the value is positive for infinitely many  $p$  and negative for infinitely many  $p$ .*

## Part XXX

# Forking without stable witnesses: quantum modules over graphs

*Added to the deposit: 3 September 2026.*

This part states fifteen conjectures on stable forking after its refutation, on modules over the quantum algebras of graphs, and on the scalar division rings of those modules. The proved layer is Proposition 79. Throughout,  $k$  is a countable field of characteristic different from two,  $q = -1$ , and  $\Gamma$  is a countable graph with edge relation  $E$ . The *quantum algebra* of  $\Gamma$  is the  $k$ -algebra

$$A_q(\Gamma) = k\langle x_v : v \in \Gamma \rangle / (x_u x_v + x_v x_u \ (uEv), \ x_u x_v - x_v x_u \ (u \neq v, \neg uEv)),$$

so that generators at adjacent vertices anticommute and all other pairs commute; the index  $q$  records that the same constructions make sense for any root of unity or any  $q \in k^\times$  once the edges are oriented.  $D_q(\Gamma)$  is its division ring of fractions (Proposition 79(i)). For  $1 \leq n \leq \omega$ ,  $M_q^{(n)}(\Gamma)$  is the two-sorted structure with a vertex sort carrying  $E$ , a vector sort  $V$  carrying the abelian group structure of a left  $D_q(\Gamma)$ -vector space of dimension  $n$  together with a unary function  $w \mapsto \lambda w$  for each  $\lambda \in k$ , and the function  $(v, w) \mapsto x_v w$  from the vertex sort times  $V$  to  $V$ ;  $M_q(\Gamma)$  is  $M_q^{(\omega)}(\Gamma)$  and  $T_q(\Gamma)$  is its complete theory. The scalars of  $D_q(\Gamma)$  are not a sort; the scalars named by the language are the  $k$ -linear combinations of products of generators at named vertices. We write  $\Gamma_{\text{bip}}$  for the random bipartite graph with its bipartition named and  $\Gamma_{\text{rand}}$  for the random graph. A theory has *stable forking* if it satisfies the conclusion of the stable forking conjecture for all sets  $A \subseteq B$ , and *stable forking over models* if it satisfies it whenever  $A$  is a model. For  $k \geq 1$ , a formula  $\varphi(x; y_1, \dots, y_k)$  has the  *$k$ -independence property* if there are tuples  $a_i^j$  ( $i < \omega$ ,  $1 \leq j \leq k$ ) and  $b_S$  ( $S \subseteq \omega^k$ ) with  $\varphi(b_S; a_{i_1}^1, \dots, a_{i_k}^k)$  if and only if  $(i_1, \dots, i_k) \in S$ ; a formula  $\varphi(x; y)$  is  *$k$ -dependent* if no partition of  $y$  into  $k$  blocks gives it the  $k$ -independence property. In a simple theory a formula is stable if and only if it is 1-dependent. When a formula  $\varphi(\bar{x}, \bar{b})$  with parameters  $\bar{b}$  is called stable, all of its parameters, vertices included, are displayed as variables and  $\varphi(\bar{x}; \bar{y})$  is required to have no order property; this is the notion of the stable forking conjecture. Where vertex parameters are instead treated as named constants, as in Proposition 79(iv) and Conjecture 436, this is said explicitly. The counterexample of Freitag and Mutchnik lives on a module over  $D_q(\Gamma_{\text{bip}})$ ; the conjectures below concern the structures  $M_q(\Gamma)$

in the language just fixed, which is expected to be interdefinable with theirs but is defined here independently, and the statements about the counterexample itself are stated for  $T_q(\Gamma_{\text{bip}})$ .

**Proposition 79** (Quantum algebras of graphs and their modules). (i) Fix a linear order on  $\Gamma$ . The ordered monomials  $x_{v_1}^{a_1} \cdots x_{v_n}^{a_n}$  with  $v_1 < \cdots < v_n$  form a  $k$ -basis of  $A_q(\Gamma)$ ;  $A_q(\Gamma)$  is a domain graded by the exponent vector in  $\mathbb{N}^{(\Gamma)}$ , an iterated Ore extension of  $k$  when  $\Gamma$  is finite, and a directed union of such algebras in general; it satisfies the left and right Ore conditions, so  $D_q(\Gamma)$  exists. An induced-subgraph embedding  $\Gamma' \rightarrow \Gamma$  induces embeddings  $A_q(\Gamma') \rightarrow A_q(\Gamma)$  and  $D_q(\Gamma') \rightarrow D_q(\Gamma)$ , and every automorphism of  $\Gamma$  extends to an automorphism of  $D_q(\Gamma)$ ; in particular  $D_q(\Gamma')$  embeds in  $D_q(\Gamma_{\text{rand}})$  for every countable  $\Gamma'$ . (ii) An element of  $A_q(\Gamma)$  is central if and only if each of its monomials is central, and the monomial  $x^a = \prod_v x_v^{a_v}$  is central if and only if  $\sum_{u: uEv} a_u$  is even for every vertex  $v$ , that is, the exponent vector reduced modulo two lies in the kernel of the adjacency matrix over  $\mathbb{F}_2$ . Hence for finite  $\Gamma$  of odd order the centre contains a monomial with some odd exponent, while for  $\Gamma_{\text{rand}}$  and  $\Gamma_{\text{bip}}$  the centre of  $A_q(\Gamma)$  is the polynomial ring  $k[x_v^2 : v \in \Gamma]$ . (iii) If the adjacency matrix of  $\Gamma$  has infinite rank over  $\mathbb{F}_2$ , in particular for  $\Gamma_{\text{rand}}$  and  $\Gamma_{\text{bip}}$ , then  $D_q(\Gamma)$  satisfies no polynomial identity and is infinite-dimensional over its centre; if moreover  $k$  has positive characteristic, then the theory of the division ring  $D_q(\Gamma)$  in the ring language is not simple. (iv) The reduct of any model of  $T_q(\Gamma)$  to the vector sort with each  $x_v$  ( $v \in \Gamma$ ) as a unary function symbol, without the vertex sort, is a module over  $A_q(\Gamma)$ ; its theory is stable and every formula in it is equivalent to a Boolean combination of positive primitive formulas. Consequently every formula of  $T_q(\Gamma)$  in which the vertex variables are replaced by named constants and no vertex quantifier occurs is stable. Moreover the module reducts of  $M_q^{(n)}(\Gamma)$  for  $1 \leq n \leq \omega$  are all elementarily equivalent, so infinite dimensionality over  $D_q(\Gamma)$  is not expressible in the module reduct. (v) Let  $\psi(w; u)$  be the formula  $\exists v (w = x_v u)$  and  $\theta(w; u)$  the formula  $\exists v \exists v' (w = x_v x_{v'} u)$ , both with vector variables  $w, u$  and no parameters. Then  $\psi$  has no order pattern of length three in any model of  $T_q(\Gamma)$ , hence is stable, for every  $\Gamma$ . For distinct vertices  $a, c$  and any nonzero vector  $b$ ,  $\theta(x_a b, -x_c^{-1} b)$  holds if and only if  $aEc$ . Consequently, if  $\Gamma$  contains for every  $n$  distinct vertices  $a_1, \dots, a_n, c_1, \dots, c_n$  with  $a_i E c_j$  if and only if  $i \leq j$ , as  $\Gamma_{\text{bip}}$  and  $\Gamma_{\text{rand}}$  do, then  $\theta$  has the order property in  $T_q(\Gamma)$ ; and if  $E$  has the independence property on distinct vertices of  $\Gamma$ , then so does  $\theta$  in  $T_q(\Gamma)$ . (vi) A supersimple theory without stable forking is not one-based. In particular the theory of the counterexample of Freitag and Mutchnik is not one-based.

*Proof.* (i) For finite  $\Gamma = \{v_1 < \cdots < v_n\}$ ,  $A_q(\Gamma)$  is the iterated Ore extension  $k[x_{v_1}][x_{v_2}; \sigma_2] \cdots [x_{v_n}; \sigma_n]$  where  $\sigma_j$  multiplies  $x_{v_i}$  by  $q$  or 1 according to adjacency; the Poincaré–Birkhoff–Witt basis, the domain property and the Noetherian property follow, and a Noetherian domain is Ore; the defining relations are symmetric in each pair, so no orientation of the edges is needed, and they are homogeneous, so the algebra is graded by the exponent vector. For countable  $\Gamma$  the algebra is the directed union of the algebras of its finite induced subgraphs, each embedded in the next by the basis description, and a directed union of Ore domains is an Ore domain because the Ore condition concerns two elements at a time. The functoriality and the automorphism extension follow from the universal property of Ore localisation, and the last claim from the universality of the random graph. (ii) Multiplying a monomial  $x^a$  by  $x_v$  on the left and on the right gives  $x_v x^a = (-1)^{\sum_{uEv} a_u} x^a x_v$ , so  $x_v$  commutes with a linear combination of distinct monomials if and only if it commutes with each, and  $x^a$  is central if and only if every such exponent sum is even. An alternating form on an odd-dimensional space over  $\mathbb{F}_2$  is singular, which gives the odd-order clause. For the random graph, given  $a$  with some odd entry  $a_u$ , let  $S$  be the set of vertices with

odd exponent; the extension property gives a vertex  $v$  adjacent to  $u$  and to no other element of  $S$ , so  $\sum_{u'Ev} a_{u'}$  is odd; for the random bipartite graph take  $v$  on the side opposite to  $u$ , adjacent to  $u$  and to no other vertex of  $S$  on the side of  $u$ , and note that  $v$  is adjacent to no vertex on its own side. Thus the central monomials are exactly those with all exponents even. (iii) If the adjacency form has rank  $2r$  on a finite induced subgraph  $H$ , a symplectic basis of the form on  $\mathbb{F}_2^H$  gives  $2r$  monomials  $y_1, z_1, \dots, y_r, z_r$  in the generators of  $H$ , with  $y_i z_i = -z_i y_i$  and all other pairs commuting, whose squares are central; the subalgebra they generate is a quantum affine space with these relations, a domain whose division ring of fractions has centre  $k(y_i^2, z_i^2 : i \leq r)$  and, by the basis of monomials with exponents 0 or 1, dimension  $4^r$  over it, the centre of the division ring of fractions being the field of fractions of the centre by Posner's theorem, so it is a tensor product of  $r$  quaternion algebras over its centre, of polynomial identity degree  $2^r$ . A ring satisfying a polynomial identity of degree  $d$  passes it to every subring, and a division ring finite-dimensional over its centre satisfies one, so unbounded  $r$  excludes both. In positive characteristic, a division ring whose theory is NTP<sub>2</sub>, in particular one whose theory is simple, is finite-dimensional over its centre by a theorem of Milliet, which gives the last claim. (iv) The reduct is a module over the ring  $A_q(\Gamma)$  in the usual language of modules; the Baur–Monk theorem gives elimination of quantifiers down to positive primitive formulas and stability. A formula of  $T_q(\Gamma)$  with vertex parameters and without vertex quantifiers is a formula of this reduct. For the elementary equivalence, two modules are elementarily equivalent when the indices  $|\varphi(M)/\psi(M)|$  agree for all pairs of positive primitive formulas  $\psi \rightarrow \varphi$  in one free variable, each index counted as 1, a finite number or  $\infty$ . A positive primitive subgroup is invariant under every endomorphism of the module, and for  $M = D_q(\Gamma)^{(n)}$  the right multiplications by column-finite matrices over  $D_q(\Gamma)$  are endomorphisms which carry any nonzero vector to any vector; hence the positive primitive subgroups of  $M$  in one variable are 0 and  $M$ . Positive primitive formulas commute with direct sums, so  $\varphi(D_q(\Gamma)^{(n)}) = 0$  if and only if  $\varphi(D_q(\Gamma)) = 0$ , and all indices agree across  $n$ , each being 1 or  $\infty$ . (v) We use that scalars act faithfully on nonzero vectors:  $db = d'b$  with  $b \neq 0$  forces  $d = d'$ , and  $x_v$  is invertible in  $D_q(\Gamma)$ . First a lemma: if  $x_s x_t^{-1} = x_{s'} x_{t'}^{-1}$  in  $D_q(\Gamma)$  for vertices  $s, t, s', t'$ , then  $(s, t) = (s', t')$  or  $s = t$  and  $s' = t'$ . Indeed  $x_s x_t^{-1} x_{t'} = x_{s'}$ , and  $x_t^{-1} x_{t'} = \varepsilon x_{t'} x_t^{-1}$  with  $\varepsilon = \pm 1$ , so  $\varepsilon x_s x_{t'} = x_{s'} x_t$  in  $A_q(\Gamma)$ ; both sides are homogeneous of degree one in each of their generators, so  $\{s, t'\} = \{s', t\}$  as multisets, which is the claim. Now suppose  $w_1, w_2, w_3, u_1, u_2, u_3$  satisfy  $\psi(w_i; u_j)$  exactly when  $i \leq j$ . From  $\psi(w_1; u_j)$  there are vertices  $v_j$  with  $w_1 = x_{v_j} u_j$ . If  $w_1 = 0$  then every  $u_j$  is 0 and  $\psi(w_3; u_2)$  and  $\psi(w_3; u_3)$  have the same truth value, a contradiction; so  $u_j = x_{v_j}^{-1} w_1$  with  $w_1 \neq 0$ . From  $\psi(w_2; u_2)$  and  $\psi(w_2; u_3)$  there are  $v', v''$  with  $x_{v'} x_{v_2}^{-1} = x_{v''} x_{v_3}^{-1}$ . By the lemma either  $v_2 = v_3$ , so  $u_2 = u_3$  and again  $\psi(w_3; u_2), \psi(w_3; u_3)$  agree, or  $v' = v_2$ , so  $w_2 = w_1$  and  $\psi(w_2; u_1)$  holds, a contradiction in both cases. A formula with no order pattern of length three is stable. The facts used, namely that each  $x_v$  acts bijectively on  $V$ , that scalars act faithfully on nonzero vectors, the lemma and the sign identity below, are expressed by sentences with vertex quantifiers that hold in  $M_q(\Gamma)$ , so they hold in every model. For  $\theta$ :  $\theta(x_a b, -x_c^{-1} b)$  says that  $x_a = -x_v x_{v'} x_c^{-1}$  for some  $v, v'$ , that is,  $x_a x_c = -x_v x_{v'}$  in  $A_q(\Gamma)$ ; by the grading  $\{v, v'\} = \{a, c\}$ , and  $(v, v') = (a, c)$  would give  $2x_a x_c = 0$ , impossible in a domain of characteristic different from two, so  $(v, v') = (c, a)$  and the condition is  $x_a x_c = -x_c x_a$ , which holds exactly when  $aEc$ . Given the vertices  $a_i, c_j$ , the vectors  $w_i = x_{a_i} b$  and  $u_j = -x_{c_j}^{-1} b$  satisfy  $\theta(w_i; u_j)$  exactly when  $i \leq j$ ; the independence property transfers in the same way. In  $\Gamma_{\text{bip}}$  and  $\Gamma_{\text{rand}}$  the required configurations exist by the extension axioms, with the  $a_i$  and the  $c_j$  on opposite sides in the bipartite case. (vi) Supersimple theories eliminate hyperimaginaries, and a one-based simple theory that eliminates hyperimaginaries has stable forking. Freitag and



Mutchnik prove that their theory is supersimple and lacks stable forking.  $\square$

**Conjecture 425** (Finite-rank stable forking). *Every supersimple theory of finite SU-rank has stable forking.*

**Conjecture 426** (Rank-one stable forking). *Let  $T$  be simple,  $A \subseteq B$ , and  $p \in S(B)$  a type of SU-rank one that forks over  $A$ . Then some stable formula in  $p$  forks over  $A$ . Granted Conjecture 433(i), the rank bound is sharp: the types there have SU-rank two and fork over their base without any stable forking formula.*

**Conjecture 427** (Stable forking for countably categorical simple theories). *Every  $\aleph_0$ -categorical simple theory has stable forking.*

**Conjecture 428** (The counterexample is two-dependent). (i) *For every countable graph  $\Gamma$  and every  $k \geq 1$ ,  $T_q(\Gamma)$  is  $k$ -dependent if and only if  $\text{Th}(\Gamma)$  is  $k$ -dependent.* (ii) *In particular  $T_q(\Gamma_{\text{bip}})$  and  $T_q(\Gamma_{\text{rand}})$  are 2-dependent and not NIP, so the stable forking conjecture fails in a 2-dependent supersimple theory.*

**Conjecture 429** (Transfer of dividing lines from graphs to quantum modules). *For every countable graph  $\Gamma$  the following hold.* (i)  *$T_q(\Gamma)$  is simple if and only if  $\text{Th}(\Gamma)$  is simple, and supersimple if and only if  $\text{Th}(\Gamma)$  is supersimple.* (ii)  *$T_q(\Gamma)$  is NSOP<sub>1</sub> if and only if  $\text{Th}(\Gamma)$  is NSOP<sub>1</sub>.* (iii)  *$T_q(\Gamma)$  is stable if and only if  $\text{Th}(\Gamma)$  is stable.* (iv) *If  $\Gamma$  is infinite and  $\text{Th}(\Gamma)$  is supersimple of SU-rank one, then  $T_q(\Gamma)$  is supersimple of SU-rank  $\omega$ , the type of a vector outside the span of the parameters having rank exactly  $\omega$ .*

**Conjecture 430** (Stable forking dividing line for quantum modules). *Let  $\Gamma$  be a countable graph with simple theory. Then  $T_q(\Gamma)$  has stable forking if and only if  $\text{Th}(\Gamma)$  is stable. Moreover, when  $\text{Th}(\Gamma)$  is unstable the failure is witnessed by the type of a single vector over a set consisting of finitely many vectors and finitely many vertices.*

**Conjecture 431** (Global linear disjointness is forking in every quantum module). *Let  $\Gamma$  be a countable graph with simple theory, and work in a monster model of  $T_q(\Gamma)$  with vertex sort  $\Gamma^*$ . For a set  $A$  write  $A_\Gamma$  for its vertex part,  $A_V$  for its vector part,  $\langle A \rangle$  for the  $D_q(\Gamma^*)$ -span of  $A_V$ , and  $\text{cl}(A)$  for the pair consisting of the vertex part of  $\text{acl}(A)$  and the  $D_q(\text{acl}(A)_\Gamma)$ -span of  $A_V$ . Then for all  $A, B, C$ ,*

$$A \vdash_C B \iff \langle AC \rangle \cap \langle BC \rangle = \langle C \rangle \text{ and } \text{cl}(AC)_\Gamma \vdash_{\text{cl}(C)_\Gamma} \text{cl}(BC)_\Gamma \text{ in } \text{Th}(\Gamma),$$

*where the last independence is forking independence in  $\text{Th}(\Gamma)$ ; and  $T_q(\Gamma)$  has quantifier elimination up to closure, in the sense that for tuples  $\bar{a}, \bar{a}'$  and a set  $C = \text{cl}(C)$ ,  $\text{tp}(\bar{a}/C) = \text{tp}(\bar{a}'/C)$  if and only if there is an isomorphism  $\text{cl}(\bar{a}C) \rightarrow \text{cl}(\bar{a}'C)$  of two-sorted structures over  $C$  carrying  $\bar{a}$  to  $\bar{a}'$  whose vertex part is an elementary map of  $\Gamma^*$ .*

**Conjecture 432** (Where one-basedness fails in the counterexample). *In  $T_q(\Gamma_{\text{bip}})$ : (i) for every tuple of vectors  $\bar{w}$  and every boundedly closed set  $C$  there is a finite tuple  $\bar{y}$  of vertices of  $\text{acl}(\bar{w}C)$  such that the canonical base of  $\text{tp}(\bar{w}/C)$  is contained in  $\text{bdd}(\bar{w}\bar{y})$ ; (ii) the theory is CM-trivial, so that it is 1-ample but not 2-ample.*

**Conjecture 433** (Failure of stable forking over models). (i) *In  $T_q(\Gamma_{\text{bip}})$  let  $M$  be a model,  $b$  a vector outside  $\langle M \rangle$ ,  $m \in M$  a vector, and  $y, y'$  distinct vertices not in  $\text{acl}(Mb)$ . Then the type of  $w = x_y x_{y'} b + m$  over  $Mb$  forks over  $M$  and contains no stable formula that forks over  $M$ ; in*

particular  $T_q(\Gamma_{\text{bip}})$  does not have stable forking over models. (ii) For every countable graph  $\Gamma$  whose theory is  $\text{NSOP}_1$  and unstable,  $T_q(\Gamma)$  fails the weak stable Kim-forking property: some type over a set  $B$  containing a model  $M$  Kim-forks over  $M$  while no stable formula over  $B$  in the type Kim-forks over  $M$ . In particular the proposal that every  $\text{NSOP}_1$  theory has the weak stable Kim-forking property fails already for simple theories.

**Conjecture 434** (Decidability and effective forking). *For  $k = \mathbb{Q}$ , the theories  $T_q(\Gamma_{\text{bip}})$  and  $T_q(\Gamma_{\text{rand}})$  are decidable; given a formula  $\varphi(x, \bar{y}, \bar{z})$  and tuples  $\bar{a}, \bar{b}$  from the computable presentation of the prime model, it is decidable whether  $\varphi(x, \bar{a}, \bar{b})$  forks over  $\bar{a}$ ; and both theories admit a computable normal form for types over finite sets, namely the isomorphism type of the closure of the tuple over the closure of the parameters in the sense of Conjecture 431.*

**Conjecture 435** (The theories of the scalar division rings are not simple). *For every field  $k$  of characteristic different from two and every countable graph  $\Gamma$  whose adjacency form has infinite rank over  $\mathbb{F}_2$ , the theory of the division ring  $D_q(\Gamma)$  in the ring language is not simple. In particular the theory of the scalars of the counterexample is not simple although the theory of the module is.*

**Conjecture 436** (Stable formulas and hidden vertices). *Let  $\Gamma$  be a countable graph with simple theory. Vertex parameters are treated throughout this statement as named constants, formulas are in vector variables, and stability is asserted for every partition of the vector variables into an object block and a parameter block. (i) Every formula of  $T_q(\Gamma)$  with at most one vertex quantifier is stable. (ii) For  $n \geq 2$  and integers  $a_1, \dots, a_n \geq 1$  of which at least two are odd, the formula  $\exists v_1 \dots \exists v_n (v_i \neq v_j \ (i < j) \wedge w = x_{v_1}^{a_1} \dots x_{v_n}^{a_n} u)$  has the order property in  $T_q(\Gamma)$  whenever  $\Gamma$  contains the half-graph configurations of Proposition 79(v). (iii) If  $\text{Th}(\Gamma)$  is unstable, then a formula is stable if and only if it is equivalent to a Boolean combination of formulas of the form  $\exists v (\chi(v, \bar{c}) \wedge \rho(\bar{w}, \bar{u}, x_v, \bar{x}_{\bar{c}}))$ , where  $\chi$  is a formula of the graph with vertex parameters  $\bar{c}$  and  $\rho$  is a positive primitive formula of the module reduct in which  $x_v$  and the generators at  $\bar{c}$  are the only scalars.*

**Conjecture 437** (Nonforking extensions do not restore stable witnesses). (i) *A simple theory  $T$  has stable forking if and only if for every  $A \subseteq B$  and every  $p \in S(B)$  forking over  $A$  there are  $C \supseteq B$  and a nonforking extension  $r \in S(C)$  of  $p$  containing a formula  $\varphi(x, c)$  with  $\varphi(x; y)$  stable that forks over  $A$ . (ii) In  $T_q(\Gamma_{\text{bip}})$ , with  $M, b, m, y, y'$  and  $w = x_y x_{y'} b + m$  as in Conjecture 433(i), no nonforking extension of the type of  $w$  over  $Mb$  contains a stable formula that forks over  $M$ .*

**Conjecture 438** (Definable groups in quantum modules). *Every group interpretable in  $T_q(\Gamma_{\text{bip}})$  or in  $T_q(\Gamma_{\text{rand}})$  is abelian-by-finite, and every infinite such group has a definable subgroup of finite index that is definably isomorphic to a quotient of a definable subgroup of  $V^n$  by a definable subgroup, for some  $n$ .*

**Conjecture 439** (Countable models of the counterexample). *For  $k = \mathbb{Q}$ , the structures  $M_q^{(n)}(\Gamma_{\text{bip}})$  for  $1 \leq n \leq \omega$  are pairwise elementarily equivalent and pairwise non-isomorphic, and every countable model of  $T_q(\Gamma_{\text{bip}})$  is isomorphic to one of them. In particular  $T_q(\Gamma_{\text{bip}})$  has exactly  $\aleph_0$  countable models, the prime model is  $M_q^{(1)}(\Gamma_{\text{bip}})$ , and the theory is not  $\aleph_0$ -categorical although its vertex sort is.*

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